SHIP-TO-SHORE MOVEMENT NTTP 3-02.1M/MCWP 3-31.5

EDITION MAY 2007

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JAMES F. AMOS
Lieutenant General, U.S. Marine Corps
Deputy Commandant for Combat Development

T. L. DAVISON
Acting
Navy Warfare Development Command

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PREFACE

NTTP 3-02.1M/MCWP 3-31.5, Ship-to-Shore Movement, discusses the doctrine, command relationships, tactics, techniques, and procedures (TTP) for planning and executing ship-to-shore movement during amphibious operations. It revises and updates the legacy publication (NWP 3-02.1/FMFM 1-8) and incorporates and expands on joint doctrine contained in JP 3-02, Joint Doctrine for Amphibious Operations, and other relevant publications and documents. It provides the detail required by amphibious task force (ATF) and landing force (LF) commanders and staffs to plan and conduct amphibious operations, specifically the ship-to-shore movement of any size LF.

Throughout this publication, references to other publications imply the effective edition. Unless otherwise stated, masculine nouns and pronouns do not refer exclusively to men.

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proposed change and its urgency. Information addressees should comment as appropriate. See accompanying sample for urgent change recommendation format.

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COMMANDER
SURFACE WARFARE DEVELOPMENT GROUP
2200 AMPHIBIOUS DRIVE
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COMMANDER
NAVY WARFARE DEVELOPMENT COMMAND
ATTN N5
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The following definitions apply to warnings, cautions, and notes found throughout this publication:



An operating procedure, practice, or condition that may result in injury or death if not carefully observed or followed.



An operating procedure, practice, or condition that may result in damage to equipment if not carefully observed or followed.



An operating procedure, practice, or condition that is essential to emphasize.

WORDING

The concept of word usage and intended meaning adhered to in this publication is:

1. "Shall" is used only when application of a procedure is mandatory.

2. "Should" is used only when application of a procedure is recommended.

3. "May" and "need not" are used only when application of a procedure is optional.

4. "Will" is used only to indicate futurity, never to indicate any degree of requirement for application of a procedure.

FM ORIGINATOR
TO *(Primary Review Authority)*
INFO COMNAVWARDEVCOM NEWPORT RI//
COMLANTFLT NORFOLK VA//
COMPACFLT PEARL HARBOR HI//
(Additional Commands as Appropriate)//
BT
CLASSIFICATION//N03510//
MSGID/GENADMIN/(*Organization ID*)//
SUBJ/URGENT CHANGE RECOMMENDATION FOR NTTP 3-02.1M/MCWP 3-31.5//
REF/A/DOC/NTTP 1-01//
POC/*(Command Representative)*//
RMKS/1. IAW REF A URGENT CHANGE IS RECOMMENDED FOR NTTP 3-02.1M/MCWP 3-31.5
2. PAGE _____ ART/PARA NO _____ LINE NO _____ FIG NO _____
3. PROPOSED NEW TEXT (*Include classification*)

4. JUSTIFICATION//
BT

Message provided for subject matter; ensure that actual message conforms to MTF requirements

Urgent Change Recommendation Message Format

DEPARTMENT OF THE NAVY
NAME OF ACTIVITY
STREET ADDRESS
CITY, STATE XXXX-XXXX

5219
Code/Serial No.
Date

From: *(Name, Grade or Title, Activity, Location)*
To: *(Primary Review Authority)*

Subject: ROUTINE CHANGE RECOMMENDATION TO NTTP 3-02.1M/MCWP 3-31.5 *(Edition, Change Number), Ship-To-Shore Movement*

Encl: *(List Attached Tables, Figures, etc.)*

1. The following changes are recommended for NTTP 3-02.1M/MCWP 3-31.5, *(Edition, Change Number)*:

a. CHANGE: (Page 1-1, Paragraph 1.1.1, Line 1)
Replace "...the ~~National Command Authority~~ President and Secretary of Defense establishes procedures for the..."
REASON: SECNAVINST ####, dated ####, instructing the term "National Command Authority" be replaced with "President and Secretary of Defense."

b. ADD: (Page 2-1, Paragraph 2.2, Line 4)
Add sentence at end of paragraph "See Figure 2-1."
REASON: Sentence will refer to enclosed illustration.
Add Figure 2-1 (see enclosure) where appropriate.
REASON: Enclosed figure helps clarify text in Paragraph 2.2.

c. DELETE: (Page 4-2, Paragraph 4.2.2, Line 3)
Remove "Navy Tactical Support Activity."
"...the Naval War College, ~~Navy Tactical Support Activity,~~ and the Navy Warfare Development Command are responsible for..."
REASON: Activity has been deactivated.

2. Point of contact for this action is *(Name, Grade or Title, Telephone, E-mail Address)*.

(SIGNATURE)
NAME

Copy to:
COMUSFLTFORCOM
COMPACFLT
COMNAVWARDEVCOM

Routine Change Recommendation Letter Format

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Executive Summary

EX.1 INTRODUCTION

This dual-service NTTP 3-02.1M and MCWP 3-31.5 is a revision and update of the legacy publication, NWP 3-02.1/FMFM 1-8. It provides the detail required by ATF and LF commanders and their staffs to plan and conduct amphibious operations. This publication describes:

1. The organization of the ATF and LF units to conduct the ship-to-shore movement of any type or size LF
2. The command relationships within and between these forces
3. The planning process to develop the landing plan
4. The execution of surfaceborne and airborne ship-to-shore movement.

NTTP 3-02.1M/MCWP 3-31.5 serves as a doctrinal guide for operational staffs, unit commanders, and school commands.

This NTTP/MCWP expands on current joint doctrine and governs the manner in which the Navy and Marine Corps execute movement ashore now and will execute such operations in the near future. It is in line with, and supplements, current joint amphibious doctrine contained in JP 3-02, Joint Doctrine for Amphibious Operations, and other relevant publications and documents.

EX.2 PUBLICATION ORGANIZATION

This NTTP/MCWP is organized along the same lines as the legacy NWP/FMFM, but has eliminated or updated outdated TTP. While this publication provides guidance for Navy and Marine Corps planners in the movement of personnel, equipment, and supplies from ships to the shore in support of the LF CONOPS, it also provides its users with the current joint and service references that address various aspects of ship-to-shore movement in greater detail and specificity.

Chapter 1 provides an overview of amphibious operations and the ship-to-shore movement concept. Chapter 2 discusses ATF and LF organization and command relationships. Chapter 3 discusses planning and operational organization to execute the plan, and Chapters 4 and 5 cover surfaceborne and airborne ship-to-shore movement, respectively.

The NTTP/MCWP also contains several appendixes. Specific chapter and appendix overviews are provided in Figure EX-1.

Chapter/ Appendix	Chapter/ Appendix Title	Brief Description of Chapter/Appendix Contents
1	Overview	Provides tactical planners with a ready reference and overview of characteristics and types of amphibious operations, the ship-to-shore movement concept, and highlights the ships, craft, and organizations participating in ship-to-shore movement operations. Terminology used in this chapter and throughout the publication is that found in JP 3-02, not the terminology that will eventually be used in ESG/ESF operations.
2	Organization and Command	Provides the Navy and Marine Corps organizational and command relationships for planning and executing ship-to-shore movement.
3	Planning	Covers the planning process for ship-to-shore movement and the manner in which the AO and landing area are organized to facilitate amphibious operations. Discusses the planning sequence for ATF and LF commanders to land troops, equipment, and supplies at prescribed times and places, and the organization to support the LF CONOPS ashore.
4	Conducting Surfaceborne Ship-to-Shore Movement	Discusses the doctrine, methodology, and C2 for LF debarkation and surfaceborne ship-to-shore movement. Also covers planning considerations that should be taken into account when preparing to conduct surfaceborne ship-to-shore movement operations.
5	Conducting Airborne Ship-to-Shore Movement	Discusses the doctrine, command relationships, delegation of authority, and the C2 organization for conducting LF debarkation and airborne ship-to-shore movement. Stresses importance of coordination with surfaceborne assault or action forces, explains the manner in which helicopters and rotary-wing aircraft may be employed, and the organizations that control vertical assault operations during ship-to-shore movement. This chapter contains a considerable amount of new terminology and procedures.
A	Landing Craft and Amphibious Vehicle Formations and Control Signals	Depicts various formations for landing craft and amphibious vehicles. Depicts hand and arm signals for controlling displacement landing craft and amphibious vehicles.
B	Identification Flags, Insignia, Markers, Lights, and Signals	Depicts and discusses flags, insignia, markers, and lights used in surfaceborne ship-to-shore movement to identify, control ships, landing craft, and waves assigned to a landing beach.
C	Landing Craft and Amphibious Vehicle Control Procedures	Describes the grid reference system of landing craft and wave control. Highlights difficulties inherent and inherent inaccuracies of the system, and introduces some alternative craft control procedures.
D	Salvage Operations	Discusses the assets and procedures to salvage landing craft during ship-to-shore operations.
E	Responsibilities for Loading, Stowing, and Offloading Landing Force Equipment	Covers, in general terms, Navy and LF responsibilities for loading, stowing, and offloading LF equipment.
F	The Tactical-Logistical Group	Covers the organization and functions of the TACLOG group and its relationship with the Navy control organization. Describes the manner in which the TACLOG group assists Navy control officers aboard ship in the ship-to-shore movement of troops, equipment, and supplies.
G	Landing Force Support Party	Describes the organization, functions, operational and command relationships, and the planning involved in establishing the LFSP as it executes initial CS and CSS for the LF during ship-to-shore movement.
H	The Helicopter Support Team	Provides indepth coverage of HST fundamentals, organization, and responsibilities, as well as the operational aspects of providing CSS to the helicopterborne unit.
I	Transformational Concepts, Equipment, and Capabilities	Provides a brief description and overview of some of the transformational concepts, equipment, and capabilities that will impact on future amphibious operations and movement ashore.
J	Planning Diagrams and Tables	Depicts various diagrams and tables originally contained in Chapter 3. Generally applicable to MEB- /MEF-level operations, but moved to preserve them for possible future use.

Figure EX-1. Publication Organization

EX.3 TRAINING AID

This section provides a training aid in the form of a PowerPoint presentation. The PowerPoint presentation provides an overview of this publication, including:

1. Summarization of the publication's organization by chapter and appendix

2. Methodology used in developing the publication.

A copy of the presentation is provided in the remainder of this section.

To view the PowerPoint presentation, click here.

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NTTP 3-02.1M/MCWP 3-31.5

Ship-to-Shore Movement



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Introduction

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- **Purpose:**
  To revise and update existing amphibious doctrine, to expand on current joint guidelines, providing Navy and Marine Corps commanders and their staffs with current details needed to plan and execute the movement ashore of personnel, equipment, and supplies in any form for an amphibious operation now or in the near term.
- **Publication Layout:**
  - **5 Chapters (including Overview, Organization and Command, Planning, Conducting Surfaceborne Ship-to-Shore Movement, and Conducting Airborne Ship-to-Shore Movement).**
  - **10 Appendixes (including Landing Craft and Amphibious Vehicle Formations and Control Signals; Identification Flags, Insignia, Markers, Lights, and Signals; Landing Craft and Amphibious Vehicle Control Procedures; Salvage Operations; Responsibilities for Loading, Stowing, and Offloading Landing Force Equipment; the Tactical-Logistical Group; Landing Force Support Party; the Helicopter Support Team; Transformational Concepts, Equipment, and Capabilities; and Planning Diagrams and Tables).**

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Chapter 1 Overview

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Ship-to-Shore Movement can be conducted via surfaceborne, airborne, or a combination of both. It may be initiated from over the horizon (OTH), near shore, or a combination of both.

Amphibious Task Force (ATF) - a Navy task organization formed to conduct amphibious operations.

Landing Force (LF) - a Marine Corps or Army task organization formed to conduct amphibious operations.

Amphibious Force (AF) - an ATF and an LF together with other forces that are trained, organized, and equipped for amphibious operations.

Amphibious Operation - a military operation launched from the sea by an AF, embarked in ships or craft with the primary purpose of introducing the LF ashore to accomplish the assigned mission.

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Phases of Amphibious Operations (PERMA)

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Phase	Description
Planning	A continuous process from receipt of the order initiating the amphibious operation until the termination of the operation. Results in: 1. Landing force (LF) concept of operations ashore 2. Landing plan 3. Loading plan 4. Amphibious force (AF) organization.
Embarkation	Period during which the LF is embarked in shipping. For MEB/MEF size operations, the LF may be organized into assault echelons (AEs) and assault follow-on echelons (AFOEs).
Rehearsal	Normally conducted during the movement phase to test feasibility and adequacy of the landing plan, timing of detailed operations, combat readiness of participating forces, and test communications.
Movement	Period when the AF departs embarkation/rehearsal area and proceeds to the amphibious objective area (AOA). May be organized into movement groups. Operations occurring during movement are: 1. Supporting operations 2. Advance force operations.
Assault	Begins when enough of the AF is in position in the landing area to initiate the ship-to-shore movement and terminates with mission accomplishment. Includes: 1. Supporting arms 2. Ship-to-shore movement 3. Logistics/combat service support (CSS) 4. Patient movement.

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Types of Amphibious Operations

- Amphibious Assault
- Withdrawals
- Demonstration
- Raids
- Military Operations Other than War (MOOTW) – Noncombatant Evacuation Operation (NEO), Humanitarian Assistance (HA), etc.



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Amphibious Ships

Ship Class	Troops	Aircraft	LCAC	LCU	LCM 8 (Not usually deployed)	AAV
LHA	1,903	CH-46, CH-53, AH-1, UH-1, AV-8B, RH-53, SH-60, MV-22	1	4	NA	12 (Not usually deployed on LHA)
LHD	2,102	CH-46, CH-53, AH-1, UH-1, AV-8B, RH-53, SH-60, MV-22	3	2	NA	12 (Not usually deployed on LHD)
LPD 4	9154/841 Nonflag/flag	CH-46, CH-53, AH-1, UH-1, AV-8B, SH-60	1 (Slave haven only)	1	4	26
LPD 17	699	CH-46, CH-53, AH-1, UH-1, SH-60, MV-22	2	1	4	14
LSD 41	504	CH-46, CH-53, AH-1, UH-1, SH-60	4 5 (Admin load)	3	10	15 (Admin load)
LSD 49	504	CH-46, CH-53, AH-1, UH-1, SH-60	2	1	4	15 (Admin load)

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Other Amphibious Transport Ships

Military Sealift Command (MSC)
The sea transportation component for the U.S. Transportation Command (USTRANSCOM). The MSC force consists of four programs: afloat prepositioning force (APF), the naval fleet auxiliary force (NFAF), special mission ships, and sealift ships. MSC can also activate and employ another group of ships, the ready reserve force (RRF).



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Vertical Assault Aircraft

- Vertical assault aircraft employed in ship-to-shore movement are organic assets to the LF and the supported commander primarily determines their employment.
- Types of vertical assault aircraft:

CH-53E



AH-1W



CH-46E



UH-1N



MV-22



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Landing Craft/Assault Vehicles

- Displacement and nondisplacement landing craft are used to land troops, equipment, and supplies.
- Types of craft:

LCU



LCAC



EFV



AAV



(AAVP - Personnel, C - Communications, R - Recovery)

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Special-Purpose Craft

- Special-purpose craft are employed by LF reconnaissance teams, sea-air-land teams (SEALs), and troop units to conduct advance force operations, clandestine assault support, and raids.
- Types of special-purpose craft:

LCM-8



CRRC



INLS



RIB



Causeway Ferry (CSP & CNSP)



Mk V (SOC)



Warping Tug (SLWT)



LARC



ABLTS



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Chapter 2
Organization and Command

Ship-to-Shore Movement Control:

- ATF commander responsible for control of ship-to-shore movement.
- Decentralized execution of the landing plan by subordinate commanders.
- Each subordinate ATF and LF commander maintains centralized control through their respective control groups.
- Central Control Officer (CCO) and Tactical Air Officer (TAO) plan and conduct the surfaceborne and airborne operations.
- The LF commander establishes two organizations: the Tactical-Logistical (TACLOG) group and Landing Force Support Party (LFSP) to facilitate and influence the ship-to-shore movement and keep advised progress of the assault.

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Task Organization - Determined by Mission

Small-Scale Amphibious Operations - a Marine Expeditionary Unit (MEU)-size LF deployed with an Expeditionary Strike Group (ESG) consisting of amphibious ships (LHA/D, LPD, and LSD), surface combatants, and submarines





Medium- to Large-Scale Amphibious Operations - a Marine Expeditionary Brigade (MEB) or Marine Expeditionary Force (MEF)-size LF deployed as part of an ATF/Expeditionary Strike Force (ESF)

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Amphibious Task Forces

ATF Commander and Staff

Responsible for LF debarkation until termination of the amphibious operation. In addition to amphibious and surface combatant ships, Navy amphibious forces are comprised of the following elements:

- **Naval Beach Group (NBG) Detachments:**
  - ***Beachmaster Unit (BMU)*** – facilitates landing and movement of troops, equipment, and supplies across the beach and evacuation of casualties and enemy prisoners of war (EPWs)
  - ***Amphibious Construction Battalion (PHIBCB)*** – provides causeways, barges, salvage, and camp support for MPF operations
  - ***Assault Craft Units (ACUs)*** – provide nondisplacement (landing craft, air cushion (LCAC)) and displacement (LCU) landing craft and support personnel.
- **Tactical Air Control Squadron (TACRON):**

  A component of the tactical air control group (TACGRU) that provides aircraft control and warning facilities afloat. TACRON operates the Navy tactical air control center (TACC), or tactical air direction center (TADC), to control all amphibious aircraft in the AO.

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Marine Air-Ground Task Force

Marine Air-Ground Task Force (MAGTF) is comprised of:



- **Command Element (CE)**
  - The core element and headquarters of the MAGTF
- **Ground Combat Element (GCE)**
  - Infantry organization that can vary in size from one or more Marine Divisions to a Regimental Landing Team (RLT), which can be further broken down into a Battalion Landing Team (BLT)
- **Aviation Combat Element (ACE)**
  - Marine Corps aviation consisting of helicopters and fixed-wing aircraft
- **Logistics Combat Element (LCE)**
  - Provides supply, maintenance, transportation, health, and other support services

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Size of MAGTF

- **Marine Expeditionary Unit – Special Operations Capable (MEU (SOC))**
  - Organized, trained, and equipped to provide the regional combatant commander or joint force commander (JFC) with an expeditionary force that is balanced, sustainable, flexible, responsive, expandable, and credible.
  - 2,200 Marines and Sailors.
  - 15-day sustainability.



- **MEB**
  - Tasked organized to a specific situation.
  - 8,000 to 18,000 Marines and Sailors.
  - 30-day sustainability.
- **MEF**
  - A MEF is the largest MAGTF and the Marine Corps principal warfighting organization, particularly for larger crises or contingencies.
  - 60-day sustainability.
- **Special-Purpose MAGTF (SPMAGTF)**
  - The SPMAGTF is a MAGTF organized, trained, and equipped with narrowly focused capabilities. Missions are limited in scope and duration.

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Command Relationships

- **Planning Relationships**
  - When the order initiating planning for the amphibious operation is received, commanders designated in the order are coequal in planning matters and decisions.
- **Execution of Ship-to-Shore Movement**
  - A supported/supporting command relationship is established between the ATF and LF commanders to ensure unity of effort in achieving operational objectives which may change dependant upon the phase of the amphibious operation.
  - Delegating operational control (OPCON) or tactical control (TACON) of AFs to the functional/service components that best supports and accomplishes the overall mission is another relationship that may be used during execution.

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Chapter 3
Planning

Amphibious Planning references: JP 3-02/MCWP 5-1

Marine Corps Planning Process Tenets:

- **Top-Down Planning –** commanders participate and drive the planning process, developing a two-way communication flow between staff and subordinates.
- **Single-Battle Concept** – allows the commander to focus all efforts of the AF elements on mission accomplishment wherever in the AO that may be.
- **Integrated Planning –** planners use integrated planning to consider all relevant factors, reduced omissions, and share information across all functions.

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Planning Methods

Planning Methods:

- **Deliberate Planning** – performed well in advance of expected military operations execution.
- **Rapid Response Planning Process (R2P2) or Crisis Action Planning (CAP)** – time-sensitive planning that occurs in response to a situation that may result in military operations.

Key Factors to Successful R2P2 or CAP:

1. **Planning Cells** (Crisis Action Team (CAT), Battle staff, Mission planning) – experienced planners already in place to efficiently respond to an alert order.
2. **Standard Operating Procedures (SOPs)** (Operations and Planning) – enable commands to accomplish familiar tasks rapidly without seeking higher-level guidance.
3. **Intelligence (INTEL)** – planners must stay abreast of latest intelligence.
4. **Information Management** – reduces volume of information to manageable pieces.

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Planning Process

- **Six-Step Process** (outlined in JP 3-02, MCWP 5-1)
  Used in deliberate and CAP for any size operation. The process follows a logical progression from receipt of order through development of operation plan (OPLAN), operation order (OPORD), or operation task (OPTASK):
  1. Mission analysis
  2. Course of action (COA) development
  3. COA war game
  4. COA comparison and decision
  5. Orders development
  6. Transition.
- **Naval Planning –** focuses on ship unloading and landing control.
- **LF Planning –** coordinated surface/air plan for organization and sequence of landing troops and equipment.

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Landing Serials

- **Serials** - an element or a group of elements (people, vehicles, equipment, supplies) within a series assigned a numerical or alphabetical designation to facilitate planning, scheduling, and control.
- **Serial Numbers** - a reference number assigned to each group of persons, equipment, vehicles, or supplies to be landed prior to general unloading.

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Troop and Equipment Movement Categories

1. **Scheduled Waves –** initial movement ashore via vertical lift, landing, or assault craft.
2. **On-call Waves –** waves standing by for immediate movement ashore.
3. **Nonscheduled Units –** remainder of serialized units.
4. **Prepositioned Emergency Supplies –** preloaded onto landing craft, amphibious assault vehicles (AAVs), or prestaged on air-capable ships.
5. **Remaining LF Supplies –** replenishment supplies and equipment.
6. **General Unloading –** nontime-critical items moved ashore as quickly as space/time permits after all serialized items are ashore.

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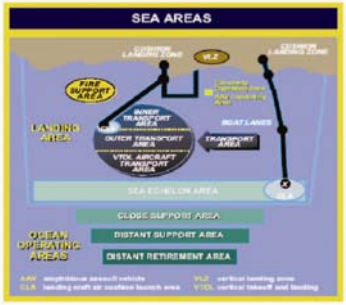
Planning Documents

ATF COMMANDER'S RESPONSIBILITY	LF COMMANDER'S RESPONSIBILITY
* Naval Landing Plan * Landing Craft Availability Plan * Landing Craft Employment Plan * Debarkation Schedule * Ship's Diagram * Pontoon Causeway Plan * Unloading Plan * Approach Schedule * Assault Wave Diagram * Landing Area Diagram * Transport Area Diagram * Beach Approach Diagram * Sea Echelon Area * Landing Control Plan * Amphibious Assault Bulk Fuel System and Offshore Petroleum Discharge System Plan	* LF Landing Plan * Amphibious Vehicle Availability Table * Landing Craft and Amphibious Vehicle Assignment Table * Landing Diagram * LF Serial Assignment Table * Landing Priority Table * LF Sequence Table * Assault Schedule * Amphibious Vehicle Employment Plan * Helicopter Availability Table * Helitearn Wave and Serial Assignment Table * Helicopter Enplaning Schedule * Helicopter Landing Diagram * HEALT * GCE Landing Plan * Consolidated Landing and Approach Plan * ACE and LF Aviation Landing Plan

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Sea Operating Areas



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Chapter 4
Surfaceborne Ship-to-Shore Movement

- **Discusses:**
  - **Doctrine**
  - **Methodology**
  - **Command and Control (C2) for LF debarkation and Surface Ship-to-Shore Movement**
- **Broken into three sections:**
  - **Planning**
  - **Execution**
  - **Communications**



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Planning Considerations for Surface Ship-to-Shore Movement

- Environmental conditions:
  - Weather
  - Sea state
  - Surf index
  - Beach hydrography
- Threat environment and intelligence surveillance and reconnaissance (ISR):
  - Identify possible or desired:
    - Hostile force locations
    - Mine locations
    - Obstruction locations
    - Reconnaissance force size and location
    - INTEL requirements
    - Target destruction requirements
    - Electronic warfare (EW) procedures

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Planning Considerations for Surface Ship-to-Shore Movement

- Mine Countermeasures (MCM) Coordination:
  - Available MCM assets.
  - Lane clearance.
  - Beach and surf zone clearance.
- ATF Unit Readiness:
  - Surface craft availability.
- Plans Versus Opposing Force Readiness/Effectiveness:
  - Use surface-only assault vice combined.
  - Use emission control (EMCON).

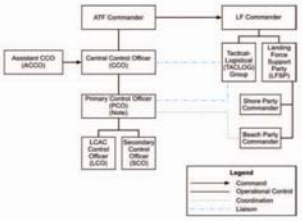
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Execution of Surface Ship-to-Shore Movement

Control Organization



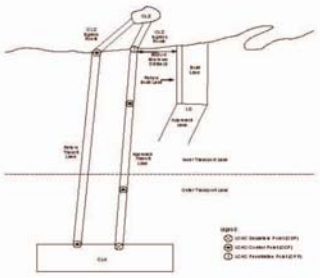


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Execution of Surface Ship-to-Shore Movement

Control Areas



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Controlling Authorities During Execution of Ship-to-Shore Movement

- **CCO**
  - Plans and supervises the ship-to-shore movement usually embarked on central control ship
- **Primary Control Officer (PCO)**
  - Embarked on a primary control ship (PCS) for each colored beach to control ship-to-shore movement of landing craft and AAVs
- **LCAC Control Officer (LCO)**
  - Embarked on a landing craft, air cushion control ship (LCS) for each colored beach to assist the PCO and control ship-to-shore movement of LCAC

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Communications for Surface Ship-to-Shore Movement



- Control Ship Coordination Net – used by CCO
- Primary Control Net – used by PCS, secondary control ship (SCS)
- Displacement Craft/AAV and LCU Control
  - Alfa Net – used by PCS for launch/recovery from ship
  - Bravo Net – used by "(Color) Beat," Beach Party Team (BPT), for beaching/unloading at beach
- Nondisplacement Craft/LCAC Control
  - LCAC Ops Net – used by LCS
  - CLZ Net – used by "(Color) Beat," BPT, for beaching/unloading at the beach

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Chapter 5
Airborne Ship-to-Shore Movement

Focuses and is broken down into the following major sections:

- Planning/Doctrine
- Command Relationships
- Delegation Authority
- C2 Relationships associated with Airborne Ship-to-Shore Movement



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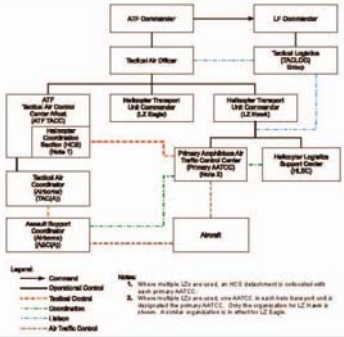
Planning Considerations for Airborne Ship-to-Shore Movement

- Planning Considerations:
  - Available airborne assets, capabilities/limitations
  - Available air-capable ships
  - Requirement for supporting arms
  - Weather, sea state
- Helicopter Support Requirements:
  - Fuel and rearm platforms
  - Established forward arming and refueling points (FARPs)

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Amphibious Air Control



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Amphibious Air Control

TACC:

- Controls all aircraft in designated amphibious airspace, including functioning as the Air Resource Element Coordinator (AREC) to properly allocate air resources
- Coordinates with Air Defense Commander (ADC) and other warfare commanders to identify and deconflict airspace
- Made up of 5 sections:
  - Air Traffic Control Section (ATCS)
  - Helicopter Coordination Section (HCS)
  - Air Defense Coordination Section (ADCS)
  - Air Support Control Section (ASCS)
  - Plans and Supports Section

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Amphibious Air Control

TADC:

- Subordinate to the TACC
- Used during multiple LHA/LHDs as part of an amphibious operation
- May also be used to control Close Air Support (CAS) missions for TACC during high tempo operations

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Amphibious Air Control

•HCS:

- Coordinates/monitors all vertical lift operations in amphibious airspace.
- Typical communications include:
  - Helicopter Request (HR) Nets
  - Helicopter Admin (HA) Nets
  - Helicopter Command (HC) Nets.



• Amphibious Air Traffic Control Center (AATCC):

- It is part of the HCS in TACC.
- Each LHD/LHA has an AATCC that controls launch/recovery operations within a radius of their ship prior to handover to the TACC.

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Transitioning Air Control Ashore During Ship-to-Shore Movement

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- During large-scale amphibious operations, control of aircraft is moved ashore from the Navy TACC to the Marine Corps TACC incrementally as more Marine Corps assets are moved ashore.
- The following positions/control organizations are established ashore in stages:
  - Forward Air Controller (FAC)
  - Tactical Air Control Party (TACP)
  - Direct Air Support Center (DASC)
  - Supporting Arms Liaison Team (SALT)
  - Firepower Control Team (FCT)
  - Air Support Liaison Team (ASLT)
  - Assault Support Coordinator (Airborne) (ASC(A))
  - Tactical Air Coordinator (Airborne) (TAC(A))
  - Forward Air Controller (Airborne) (FAC(A))
  - Tactical Air Operations Center (TAOC).

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Appendixes

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Ten Appendixes:

- Appendix A - Landing Craft and Amphibious Vehicle Formations and Control Signals
- Appendix B - Identification Flags, Insignia, Markers, Lights, and Signals
- Appendix C - Landing Craft and Amphibious Vehicle Control Procedures
- Appendix D - Salvage Operations
- Appendix E - Responsibilities for Loading, Stowing, and Offloading Landing Force Equipment
- Appendix F - The Tactical-Logistical Group
- Appendix G - Landing Force Support Party
- Appendix H - The Helicopter Support Team
- Appendix I - Transformational Concepts, Equipment, and Capabilities
- Appendix J - Planning Diagrams and Tables

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Appendix I Transformational Technologies

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- Sea Basing – a fundamental concept to Sea Power 21, which supports the Marine Corps Ship-to-Objective Maneuver (STOM) concept.
- The Sea Basing conceptualizes the ability of expeditionary forces to conduct amphibious operations from the security of a seaward area and provide support from the sea base instead of establishing logistics ashore.
- Maritime Prepositioning Force (Future) (MPF(F))
  - Sea-based operations that increase the security and mobility of operating forces by providing supplemental support, equipment, and troops from a secure seaward area.
  - Also allows for a sea-based staging area for at-sea arrival, assembly of units, and selective offloading of equipment.

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Appendix I Transformational Technologies (cont'd)

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Key Marine Corps Concepts: EMW, OMFTS, and STOM



EMW – Expeditionary Maneuver Warfare
OMFTS – Operational Maneuver From the Sea

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Appendix I Transformational Technologies (cont'd)

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High-Speed Vessel (HSV)



- Delivered in 2003
- Currently in use by the Army and Navy on an experimental basis
- Aluminum-hulled catamaran that has a maximum speed of 45+ knots and sustained operating speeds of 30+ knots
- Helicopter landing area, enhanced C2 suite, roll-on/roll-off (RO/RO) cargo and vehicle stowage areas
- Versatility – able to rapidly reconfigure the ship to meet various missions by installing modular suites that are tailored to meet specific mission areas

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Appendix I Transformational Technologies (cont'd)

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Expeditionary Fighting Vehicle (EFV)



- Formerly referred to as the AAAV, a key element to Marine Corps STOM concept
- Like the AAV, several variants will exist (personnel and communications)
- Increased speed; command, control, communications, computers, and intelligence (C4I); and weapons capabilities over AAV
- 25+ nm launch range

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CHAPTER 1

Overview

1.1 PURPOSE

This NTTP serves as a guide for Navy and Marine Corps commanders and staffs, unit commanders, and school commands. It discusses the doctrine, command relationships, and TTP in planning and executing ship-to-shore movement during amphibious operations. While MPF operations and MSC ships are often integral to amphibious operations and ship-to-shore movement, MPF and MSC operations are not discussed in detail in this publication. Detailed information on such operations is found in JP 3-02; NWP 3-02.21, MSC Support of Amphibious Operations; NTTP 3-02.3/MCWP 3-32, Maritime Prepositioning Force Operations; and NTRP 3-02.1.2, Naval Beach Group Support Element Operations.

This publication describes:

1. The organization of ATF and the LF units to conduct the ship-to shore movement of any type or size LF
2. Command relationships within and between these forces
3. The planning process to develop the landing plan
4. The conduct of surfaceborne and airborne ship-to-shore movement.

Chapter 1 provides tactical planners with a ready reference and an overview of the characteristics and types of amphibious operations, and the ship-to-shore movement concept. It also highlights the ships, craft, and organizations participating in ship-to-shore movement operations.

However, because the command relationships, staff structure and composition, C2, and specific missions are still being developed, throughout this publication terminology specific to the ESF and ESG concepts is not used. Terminology and definitions associated with amphibious operations are found in JP 1-02, Department of Defense Dictionary of Military and Associated Terms; JP 3-02; and MCRP 5-12C, Marine Corps Supplement to the Department of Defense Dictionary of Military and Associated Terms. For purposes of this publication, the following definitions apply:

1. An ATF is a Navy task organization formed to conduct amphibious operations.
2. An LF is a Marine Corps or Army task organization formed to conduct amphibious operations.
3. An AF is an ATF and an LF together with other forces that are trained, organized, and equipped for amphibious operations.

1.2 AMPHIBIOUS OPERATIONS BACKGROUND AND OVERVIEW

An amphibious operation is a military operation launched from the sea by an AF embarked in ships or craft with the primary purpose of introducing an LF ashore to accomplish the assigned mission. Such operations include the following phases: planning, embarkation, rehearsal, movement, and assault. A brief discussion of the five phases is provided in sequence in this section and in Figure 1-1. However, depending on the nature of the operation, the location, readiness, and availability of naval and LF assets, and the amount of time available to organize, plan, and execute, the phases do not always occur in the aforementioned order. Generally, forward-deployed AFs use the sequence: embarkation, movement, planning, rehearsal, and assault. A more detailed description of each phase is contained in JP 3-02. For detailed information on the embarkation phase, see JP 3-02.2, Joint Doctrine for Amphibious Embarkation; MCRP 4-11.3G, Unit Embarkation Handbook; and MCRP 4-11C, Combat Cargo Operations Handbook.

Upon receipt of the order initiating the amphibious operation, naval and LF planning begins. This order may come in the form of a WARNORD, an alert order, a planning order, an OPORD, or from a combination of these orders. The order normally specifies the purpose of the mission, command relationships, and scope of the action to be taken. Additional information on the contents of the order initiating the amphibious operation and a detailed discussion of support relationships are contained in Chapter 2 of this publication and Chapter II of JP 3-02.

When an order is received directing a mission that requires ship-to-shore movement, planning begins with the LF CONOPS. This concept outlines the intentions of the LF commander regarding how operations ashore will be conducted. It includes the organization for the landing and the CONOPS ashore. The CONOPS ashore is the supported commander's tactical plan for the LF to accomplish the assigned mission. It determines which LF units are required at the various landing locations within the landing area. Further, the CONOPS ashore is used to guide the assignment of LF units to amphibious ships, landing craft, and aircraft for ship-to-shore movement. This concept is examined by appropriate commanders or their staff representatives for supportability and approved by the supporting commander prior to the commencement of detailed planning. Chapter 3 of this publication and Chapter IV of JP 3-02 contain detailed discussions of the amphibious planning process.

With an approved CONOPS ashore, LF and naval requirements for mission accomplishment are consolidated and compared with the AF means and assets available. If necessary, additional assets are requested from appropriate higher authority. Should additional means and assets not be available, the CONOPS is adjusted accordingly, or another is developed.

The landing plan is prepared after completing the final allocation of assets. This plan is composed of naval and LF documents which delineate the guidance and instructions to execute surfaceborne and airborne ship-to-shore movement. They are covered in detail in Chapter 3.

When the landing plan is completed, embarkation planning begins. The assignment of personnel, equipment, and supplies to amphibious shipping and their sequence for embarkation constitute the loading plan. The loading plan is derived from and totally supports the landing plan. JP 3-02.2 contains additional information on embarkation planning.

Phase	Description
Planning	A continuous process from receipt of the order initiating the amphibious operation to terminate the operation. Results in: 1. LF concept of operations ashore 2. Landing plan 3. Loading plan 4. AF organization.
Embarkation	Period during which the LF is embarked in shipping. For MEB- /MEF-sized operations, the LF may be organized into AEs and AFOEs.
Rehearsal	Normally conducted during the movement phase to test feasibility and adequacy of the landing plan, timing of detailed operations, combat readiness of participating forces, and test communications.
Movement	Period when the AF departs embarkation/rehearsal area and proceeds to the AOA. May be organized into movement groups. Operations occurring during movement are: 1. Supporting operations 2. Advance force operations.
Assault	Begins when enough of the AF is in position in the landing area to initiate the ship-to-shore movement and terminates with mission accomplishment. This phase includes: 1. Supporting arms 2. Ship-to-shore movement (Note 1) 3. Logistics/CSS (Note 1) 4. Patient movement. (Note 2)
Notes: 1. These areas are discussed in detail in subsequent chapters and appendixes. 2. This area is discussed in NWP 4-02 and in greater detail in NTTP 4-02.2.	

Figure 1-1. Planning, Embarkation, Rehearsal, Movement, and Action: Phases of an Amphibious Operation

The AF transits from embarkation points to the AOA after the LF embarks. A rehearsal to test the landing plan is conducted during the movement phase, if feasible and time permits. During movement in support of MEB- / MEF-sized operations, the ATF may be organized into groups consisting of amphibious shipping with escorts for protection and combat logistics forces for sustainability. In such larger-scale operations, the LF is organized into echelons: the AE and the AFOE. Transport and movement groups, and the AE and the AFOE are discussed in Chapter 2.

The AOA is a geographical area delineated in the order initiating the amphibious operation for C2. The objectives to be secured are contained within this order. The area must be of sufficient size to ensure accomplishment of the mission and must provide sufficient area for conducting necessary sea, air, and land operations.

Supporting, preassault, and advance force operations may be conducted while the AE is en route.

The action phase commences when sufficient AF assets are positioned in the landing area to permit the commencement of ship-to-shore movement. The landing area is the part of the objective area within which the LF comes ashore. It includes the beach, approaches to the beach, transport areas, FSAs, airspace occupied by close supporting aircraft, and the land included in the LF's advance inland to the initial objective(s).

1.2.1 Characteristics of Amphibious Operations

Amphibious operations encompass a wide variety of missions that support the commander's campaign or OPLAN. The essential characteristics inherent in amphibious operations are discussed in Paragraphs 1.2.1.1 through 1.2.1.8.

1.2.1.1 Integration Between Navy and Landing Forces

An amphibious operation requires close coordination among the naval forces, the LF, and other designated forces. Ordinarily joint or combined in nature, an amphibious operation is typified by integrating forces trained, organized, and equipped for disparate functions.

1.2.1.2 Rapid Buildup of Combat Power From the Sea to Shore

An essential requirement in an amphibious assault is the uninterrupted projection of combat power ashore to support the CONOPS. Naval forces should also provide continuous support for forces ashore.

1.2.1.3 Task-Organized Forces

AFs are task organized based on the mission, and are capable of carrying out multiple missions to support joint, allied, and coalition amphibious operations. The ATF's C2 capabilities and those of the embarked LF facilitate the accomplishment of multiple missions as well as joint and multinational force integration.

1.2.1.4 Unity of Effort and Operational Coherence

The complexity of amphibious operations and the vulnerability of forces so engaged require exceptional unity of effort and operational coherence. This includes the full integration of organic assets as well as those of other joint or multinational forces.

1.2.1.5 Readiness

AFs are immediately available to respond to smaller-level contingencies. By maintaining proficiency and overall warfighting readiness, AFs can provide a wide range of services that support peacetime and combat operations.

1.2.1.6 Flexibility

The flexibility inherent in AFs permits political leaders and commanders to shift focus, reconfigure, and realign forces to handle a variety of contingencies by providing a wide range of weapons systems, military options, and logistic or administrative capabilities. For MEB- /MEF-sized operations, AFs are viable in carrying out forcible entry and air interdiction operations. Smaller scale AFs are fully capable of conducting NEOs, disaster relief, shows of force, maritime interdiction, and HA. Additionally, they can exercise sea control and the C2 capabilities of these forces can provide an operating platform for a joint HQ element to support or complement diplomatic efforts.

1.2.1.7 Self-Sustainment

AFs are capable of operating in forward areas without significant land-based supply infrastructure. UNREP and on-station replacement or rotation of personnel and ships allow such operations to continue indefinitely. To further enhance the sustainability of amphibious operations, forward logistics sites may be established.

1.2.1.8 Mobility

AFs, through strategic and tactical mobility, have the ability to monitor a situation passively, remain on station for a sustained period, respond to a crisis rapidly, and deploy in combat. Their mobility enables these forces to respond from OTH. If diplomatic, political, or economic measures succeed, AFs can be quickly withdrawn without further action ashore. They can also respond to indications of impending crises by relocating rapidly, usually independent of fixed logistics. In combat, the ability to position these forces provides commanders with a tactical and operational advantage.

1.2.2 Types of Amphibious Operations

Amphibious operations encompass assaults, withdrawals, demonstrations, raids, and other operations in a permissive, uncertain, or hostile environment. These operations are covered in greater detail in Chapter I of JP 3-02.

1.2.2.1 Amphibious Assault

An amphibious assault involves establishing a force on a hostile or potentially hostile shore. Moreover, the organic capabilities of a MEB- or MEF-sized AF, including fire support, logistics, and mobility, can facilitate access to a crisis area by forcible entry. JP 3-02 and JP 3-18, Joint Doctrine for Forcible Entry Operations, provide additional large-scale amphibious assault information.

1.2.2.2 Amphibious Withdrawal

An amphibious withdrawal is the extraction of forces by sea in naval ships or craft from a hostile or potentially hostile shore.

1.2.2.3 Amphibious Demonstration

An amphibious demonstration is conducted for the purpose of deceiving the enemy by a show of force with the expectation of deluding the enemy into an unfavorable COA.

1.2.2.4 Amphibious Raid

An amphibious raid is a swift incursion into or temporary occupation of an objective followed by a planned withdrawal.

1.2.2.5 Other Amphibious Operations

In addition to projecting power, other operations conducted by AFs include NEOs, HA, or civil support operations.

1.3 SHIP-TO-SHORE MOVEMENT CONCEPT

Ship-to-shore movement is that portion of the assault phase of an amphibious operation that includes the deployment of the LF from amphibious shipping to designated landing areas. Its objective is to land troops, equipment, and supplies at prescribed times and places and in the organization for landing necessary to support the LF CONOPS ashore. Ship-to-shore movement is surfaceborne, airborne, or a combination of both. It may be initiated from OTH (beyond the opposing force's visual and ground-based radar range), nearshore, or a combination of both. It commences when the commander supporting LF operations orders execution by signaling, "Land the landing force!" This is usually the ATF commander. It concludes when all assault shipping is unloaded. Unloading operations take place in two periods. The initial period in combat or opposed conditions is generally tactical and provides for the rapid buildup of combat power ashore and quick response to LF tactical and logistical requirements. In unopposed or noncombat conditions, the initial unloading period is administrative. As the sea-basing concept matures, additional LF functions will remain afloat. However, the general unloading period remains primarily logistical and emphasizes the rapid unloading of appropriately identified personnel and materiel remaining in the amphibious shipping to support LF operations ashore.

1.4 LANDING MEANS

The ship-to-shore movement of troops, equipment, and supplies ultimately depends on the landing means available. These include amphibious ships, MSC ships, amphibious vehicles, landing craft, and vertical lift aircraft. The general characteristics of these are described in the following publications:

1. NWP 11-1.01, Characteristics and Capabilities of U.S. Navy Combatant Ships
2. NWP 11-2, Characteristics and Capabilities of U.S. Navy Auxiliaries and MSC Ships
3. NTRP 3-02.1.2
4. NWP 3-02.21
5. NTTP 3-02.6 series applicable CTMs
6. NTRP 3-02.6 series applicable CTPs.

Figure 1-2 provides a comparison of amphibious ship troop, vertical lift aircraft, landing craft, and assault craft capabilities.

Ship Class	Troops	Aircraft	LCAC	or LCU	or EFV	or AAV
LHA	1,903	CH-46E, CH-53D/E, AH-1, UH-1, AV-8B, RH-53, SH-60, MV-22 (Note 1)	1	4	TBD	12 (Note 2)
LHD	2,104	CH-46E, CH-53D/E, AH-1, UH-1, AV-8B, RH-53, SH-60, MV-22 (Note 1)	3	2	TBD	12 (Note 2)
LPD 4	855 (Note 3)	CH-46E, CH-53D/E, AH-1, UH-1, AV-8B, SH-60	1	1	TBD	26
LPD 17	799	CH-46E, CH-53D/E, AH-1, UH-1, SH-60, MV-22 (Note 1)	2	1	TBD	14
LSD 41	504	CH-46E, CH-53D/E, AH-1, UH-1, SH-60	4 (Note 4)	3	TBD	15 (Note 5)
LSD 49	508	CH-46E, CH-53D/E, AH-1, UH-1, SH-60	2	1	TBD	15 (Note 5)

Notes:
1. MV-22 is discussed in Section 1.4.
2. Tactical employment of AAVs is not normally conducted from an LHA or LHD.
3. Flag-configured LPD 4 carries approximately 60 fewer LF troops.
4. Five LCAC can be carried with vehicle ramp raised.
5. Administrative load.

General Note: Tradeoffs in landing craft space available, but not at one-for-one ratio.

Figure 1-2. Amphibious Ship Troop, Vertical Lift Aircraft, Landing Craft, and Assault Craft Capabilities

1.4.1 Amphibious Ships

Amphibious ships are specifically designed to embark, transport, land, and support the LF in assault and a variety of other amphibious operations. They are capable of being loaded and unloaded by naval personnel without external assistance.

The following paragraphs describe amphibious ship classes.

1.4.1.1 LHA

LHAs combine the operational capabilities of several types of amphibious ships in a single hull. This class has the primary mission of AMW. In addition to embarked commanders and their staffs, embarked units may include the following:

1. GCEs
2. ACEs
3. AMCM detachments
4. Fleet support units, SEALs
5. TACRONs/TACGRUs
6. BMUs
7. PHIBCB
8. ACU detachments
9. FSTs/medical augmentation program personnel.

Embarked units may also include LCEs, AAVs, vertical lift aircraft, LCUs and/or LCAC, and their assigned crews. The ship cannot embark all of these simultaneously; therefore, the number and type of units embarked generally depend on the nature of the mission and/or the composition of the AF.

The LHA has vertical lift aircraft and V/STOL aircraft operating facilities greater than the LHD, a well deck capacity greater than LPD and the LSD, and a substantial vehicle and cargo capacity. The LHA can support embarked commanders and their staffs in any size AMW operation. It is an excellent joint C2 platform with facilities for a JCC, JIC, ATF TACC, SACC, TACLOG group, AATCC, HLSC, and central control of surfaceborne and airborne ship-to-shore movement. When augmented with an FST, an LHA can be designated to serve as a primary CRTS. A doctor and a dentist are assigned, and facilities include 4 medical and 2 dental ORs, 69 dedicated beds, and a frozen blood capability.

The LHA has a secondary mission of power projection and a tertiary mission of sea control through the launch and recovery of the LF's ACE, and its aircraft to include the fixed-wing AV-8 Harrier V/STOL aircraft and a mix of the UH-1N, AH-1W, CH-46E, and the CH-53D/E rotary-wing aircraft.

Additional detailed information on LHA class capabilities is contained in NWP 1-11.01, Characteristics and Capabilities of U.S. Navy Combatant Ships (U), and NTRP 3-02.6.12, LHA 1 Class Tactical Publication.

1.4.1.2 LHD

The LHD's primary mission is AMW, and it is the Navy's largest amphibious ship. Similar to the LHA in design, mission, and capabilities, the LHD's significant improvements over the LHA include increased AV-8 support capabilities, a redesigned well deck that can accommodate three LCAC, expanded medical facilities, and upgraded C2 capabilities. However, the vehicle stowage capacity of the LHD is less than that of the LHA.

When augmented with an FST, an LHD can be designated to serve as a primary CRTS. A doctor and a dentist are assigned to the LHD, and facilities include 6 medical ORs, 4 dental ORs, 41 ward and 23 intensive care beds, and a frozen blood capability.

LHD class ships can support embarked commanders for any size strike force or AMW operations. The LHD can also support alternate force/sector commanders.

NWP 1-11.01 and NTTP 3-20.6.13, LHD 1 Class Tactical Publication, contain additional detailed information on LHD class capabilities.

1.4.1.3 LPD 4 Class

LPD 4 Class ships transport troops and equipment for amphibious operations and land them in the assault area by means of helicopters in vertical assault or via landing craft or AAVs carried in the ship's well deck. The LPD is a variation of the LSD concept with increased troop and vehicle capacity, extensive ammunition and cargo stowage capabilities, and a smaller well deck. The LPD 4 Class can launch and recover Navy and Marine Corps helicopters, including the UH-1N, AH-1W, CH-46E, CH-53E, and AV-8 fixed-wing aircraft. Aircraft refueling and rearming can be conducted on the flight deck. All ships of the class, except LPD 4, have an expandable hangar and can provide service and maintenance to embarked helicopters.

Limited interface testing has demonstrated that an unloaded deep skirt-configured LCAC can effectively enter and exit the LPD 4 Class well deck. However, increased craft height reduces the clearance between outboard cabin bulkheads and ship's catwalks, and between the bow thrusters and ship's overhead ducting that may result in craft or support ship damage. Until ongoing interface testing is completed, it is strongly recommended that LPD 4 Class ships not be used as operational platforms for deep skirt-configured LCAC. Should operational necessity require recovering deep skirt-configured LCAC, special attention should be given to the cautions and procedures contained in NAVSEA S9LCA-AA-SSM-010, Safe Engineering and Operations (SEAOPS) Manual for Landing Craft, Air Cushion (LCAC), Volume III, Appendix D.

Flag-configured LPDs (LPD 7 through 13) provide C2 facilities for Navy and Marine Corps commanders and their staffs for AMW operations. These ships of this class can support a JCC, JIC, ATF TACC, and a SACC. Flag and nonflag-configured LPDs are equipped with TACLOG communication spaces for coordinating troop movement and logistics. The LPD 4 Class can function as a PCS for surfaceborne ship-to-shore movement.

The LPD 4 Class has limited medical facilities, but can be designated as a secondary CRTS when augmented with medical and surgical personnel, and is equipped with the appropriate HSS capabilities. A doctor and a dentist are assigned, and facilities include 1 medical and dental OR, and a 6- or 12-bed ward.

NWP 1-11.01 and NTRP 3-20.6.14, LPD 4 Class Tactical Publication, contain additional information on the LPD 4 Class and its capabilities.

1.4.1.4 LPD 17 Class

The LPD 17 Class ship's primary mission is AMW, and it is the Navy's newest amphibious ship. The capabilities of the LPD 17 Class include a state-of-the art C2 suite, substantially increased vehicle lift capacity, large flight deck, and advanced ship survivability features that enhance its ability to operate in the littoral environment. The LPD 17 Class provides AFs with enhanced operational flexibility. It can operate as part of an AF organized to accomplish a broad range of military objectives; or as an element of a "split AF" in which the LPD 17 is detached and operates independently to support low-risk amphibious operations.

The well deck in the LPD 17 supports two LCAC or one LCU. Its aviation facilities are designed with two primary landing spots and a permanent hangar sized to provide organizational-level support for embarked aircraft. It can launch and recover Navy and Marine Corps aircraft, including the UH-1N, AH-1W, CH-46E, CH-53E, and the AV-8 fixed-wing aircraft. When the expanded spots configuration is certified, the flight deck will support simultaneous operation of four CH-46s; the flight deck will also support four MV-22 Osprey aircraft (two folded and two spread).

As a secondary mission, the LPD 17 is equipped to function as a secondary CRTS when designated and augmented with medical and surgical personnel. A doctor and a dentist are embarked with a 24-bed ward and 1 OR.

Additional information on LPD 17 Class capabilities is contained in NTRP 3-20.6.21, LPD 17 Class Tactical Publication.

1.4.1.5 LSD 41 Class

The mission of the LSD 41 Class is to transport and launch AAVs and landing craft with its crews and embarked personnel in an amphibious operation. This ship class can embark up to four LCAC or three LCUs in the well deck. With its portable ramp removed, the ship can carry five LCAC. It is the primary support and operating platform for LCAC and may be used as the PCS or LCAC control ship. LSD 41 Class ships can also provide limited docking and repair services as a boat haven for small ships and craft.

LSD 41 has two primary helicopter spots, and can handle Navy and Marine Corps helicopters currently in the inventory. The ship has no helicopter hangar. Aircraft fueling and rearming can be conducted on the flight deck.

LSD 41 is not designed to embark commanders and its staffs, but is equipped with TACLOG communication spaces for coordinating troop movement and logistics.

LSD 41 Class ships have a doctor and dentist assigned, but have limited medical facilities. These ships have two dental ORs and a medical OR. LSD 41 can be designated to serve as a secondary CRTS when augmented with medical and surgical personnel and equipped with the appropriate HSS capabilities.

Additional information on LSD 41 Class capabilities is contained in NWP 1-11.01 and NTRP 3-20.6.17, LSD 41 Class Tactical Publication.

1.4.1.6 LSD 49 Class

The LSD 49 Class is designed to transport and land troops and their equipment by means of landing craft, AAVs, and helicopters. It may be used as the PCS or LCAC control ship. It differs from the LSD 41 Class in that it has significantly expanded cargo and ammunition stowage facilities. Its reduced well deck capacity precludes carrying more than two LCAC, and this class has two helicopter spots.

Additional information on LSD 49 Class capabilities is contained in NWP 1-11.01 and NTRP 3-20.6.17.

1.4.2 Casualty Receiving and Treatment Ship

The ESG commander designates specific amphibious ships as primary CRTSs to provide Level II HSS to the LF during expeditionary operations. Large-deck amphibious ships (LHAs, LHDs) are normally assigned as primary

CRTSs. FSTs are operationally assigned on TAD orders to large-deck amphibious assault ships to provide Level II surgical support to support an ESG with embarked MAGTF. The ESG commander may designate other amphibious assault ships as secondary CRTSs. At a minimum, a secondary CRTS should have the capability to receive and treat casualties, provided appropriate medical materiel and personnel are available to provide resuscitative care. Ships normally designated as secondary CRTSs include LPD and LSD class ships.

1.4.3 Military Sealift Command

MSC is the sea transportation component for USTRANSCOM. MSC's mission is to provide ocean transportation of equipment, fuel, supplies, and ammunition to sustain U.S. forces worldwide during peacetime and in war for as long as operational requirements dictate. The command operates ships that provide combat logistics support to U.S. ships at sea; special mission support to U.S. government agencies; prepositioning of U.S. military supplies and equipment at sea; and ocean transportation of DOD cargo in peacetime and war.

1.4.4. Military Sealift Command Ships

MSC owns or charters a number of ships that can support and have a significant impact on large-scale (MEB-/MEF-sized) amphibious operations. Assigned by MSC, they may be MSC nucleus fleet ships, contract-operated MSC ships, MSC-controlled time or voyage chartered commercial ships, or MSC-controlled ships allocated by the MARAD to MSC. The MSC force consists of four programs: APF, the NFAF, special mission ships, and sealift ships. MSC can also activate and employ another group of ships: the RRF. These ships provide the following capabilities: hospital support, troop transport, tankers, heavy lift, cargo (breakbulk, container, and RO-RO), special purpose (T-AVB), and T-ACS. Once activated, OPCON of MSC ships resides with a numbered fleet commander or CJTF.

Additional information on MSC support for amphibious operations and ship-to-shore movement is contained in NWP 3-02.21.

1.4.4.1 Afloat Prepositioning Force

MSC's APF is an essential element in the power projection triad: sea shield, sea strike, and seabasing. As a key element of seabasing, prepositioning ships make it possible to deploy on short notice the vital equipment, fuel, and supplies to initially support U.S. military forces whenever needed. The program includes long-term chartered commercial vessels, activated RRF ships, and U.S. government-owned ships. All are crewed by CIVMARs provided by companies under contract by MSC. Prepositioning ships are loaded with combat equipment for the Army, Air Force, Marine Corps, and Navy, as well as fuel for the DLA. APF ships are positioned in strategic areas around the world and are divided into three separate elements: the combat prepositioning force, the MPF, and the logistics prepositioning force.

1.4.4.1.1 Combat Prepositioning Force

Combat prepositioning force ships provide quick-response delivery of Army equipment for ground forces from strategic locations. The majority of this force consists of LMSR vessels loaded with combat equipment. Other combat prepositioning force ships carry ammunition as well as sustaining and support cargo such as purification units, food, and initial CS equipment.

1.4.4.1.2 Maritime Prepositioning Force

MPS can offload instream with organic cargo-handling equipment and landing craft if port facilities are unavailable. In either situation, a permissive environment is required. The three MPSRONs, which may depart their normal AOs to support contingencies, are:

1. MPSRON ONE is responsible for TACON of MPS usually located in the Mediterranean and eastern Atlantic Ocean.

2. MPSRON TWO has TACON of MPS usually located in or near the island of Diego Garcia in the Indian Ocean.

3. MPSRON THREE has TACON of MPS usually located in the Western Pacific Ocean near the islands of Guam and Saipan.

Detailed information on MPF operations and Navy cargo handling is found in NTRP 3-02.1.2 and NTTP 3-02.3M/MCWP 3-32, Maritime Prepositioning Force Operations.

1.4.4.1.3 Logistics Prepositioning Force

The logistics prepositioning force consists of ships that carry Air Force ammunition. Other ships in this element include MCDS vessels that carry ordnance for the Navy and have the capability to operate as shuttle replenishment ships for Navy strike groups. Ships in this force also carry DLA petroleum products for contingency use and include OPDS tankers.

T-AVBs provide dedicated sealift for movement of an aviation IMA to support the rapid deployment of Marine fixed- and rotary-wing aircraft units. More specifically, in large-scale operations, the IMA supports the MAGTF's ACE, which includes a predesignated mix of aircraft. These maintenance facilities are packaged in mobile containers and include operational work centers and ready access supply stores. In situations where the IMA is moved ashore, the T-AVB can be used in a common-user sealift mode to provide a resupply capability in a conventional container or RO/RO configuration. The T-AVBs are assigned to MSC, but are maintained and operated under contract by MARAD in a ROS.

1.4.4.2 Naval Fleet Auxiliary Force

NFAF ships are MSC-administered ships crewed by CIVMARs. NFAF ships and their basic functions are:

1. T-ATFs conduct towing and salvage operations.

2. T-AOEs conduct rapid replenishment of oil, ammunition, and dry and refrigerated stores.

3. T-AOs provide UNREP of fuel to Navy ships at sea and jet fuel for aircraft.

4. T-AFSs provide UNREP of all types of supplies, ranging from repair parts to fresh food and clothing.

5. T-AEs conduct ammunition transfer operations through a combination of at sea, alongside line transfers, and vertical lifts.

6. T-AKEs are replacing the T-AO, T-AE, and T-AFS.

7. T-AH's primary mission is to provide rapid, and flexible and mobile acute medical care to Marine, Army, and Air Force units deployed ashore, and naval units afloat. The secondary mission is to provide disaster assistance or humanitarian relief operations. The second Geneva Convention contains specific provisions relating to the unique HSS mission of hospital ships under the laws of armed conflict. Because hospital ships may be employed in situations other than intensive combat, the type of practice within these specialties can change. Expeditionary HSS units, especially hospital ships and fleet hospitals/expeditionary medical facilities, may be increasingly employed in protracted operations that require extensive community hospital functions. The MTF CO/OIC should determine the extent of practice modification basing the decision on the facility's equipment, supplies, and personnel, and on the population receiving care. Details are provided in NTTP 4-02.6, Hospital Ships. There are 2 T-AHs, and each contains 12 ORs and a 1,000-bed hospital facility. Normally kept in ROS, when called into action, they can be ready to sail in 5 days with a crew of 70 CIVMARs and more than 1,200 military medical personnel.

1.4.4.3 Special Mission Ships

MSC's special mission ships provide operating platforms and services for unique U.S. government and federal government missions. These ships include oceanographic and hydrographic survey ships, a cable-laying ship, missile range implementation ships for missile flight data collection and tracking, ocean underwater surveillance ships, and some chartered vessels for unique U.S. government operations, including deep water SAR missions and Navy submarine test support escort.

Both CIVMARs and contractor-employed mariners operate special mission vessels. Embarked military personnel, civilian scientists, and related technicians conduct technical work, research, and communications.

Recent additions to MSC's special missions program are the command ships, USS MOUNT WHITNEY (LCC 20) and USS CORONADO (AGF 3). Navigation, deck, engineering, laundry, and galley services are provided by MSC CIVMARs, while the remainder of the crew is military. A Navy captain commands these ships.

1.4.4.4 Sealift Program

The mission of MSC's sealift program is to provide high-quality, efficient, and effective ocean transportation for the DOD and other U.S. government agencies. The program is divided into three project offices: tankers, dry cargo, and surge.

1.4.4.4.1 Tankers

MSC works closely with the DFSC to transport petroleum products to DOD stowage and distribution facilities around the world, as well as to deliver fuel to MSC oilers and other Navy fleet oilers at sea.

1.4.4.4.2 Dry Cargo

DOD's dry cargo is shipped via U.S.-flagged commercial ships. Cargo ships under charter to MSC carry approximately 20 percent of this cargo.

1.4.4.4.3 Surge

Surge sealift includes three key resources that are kept in ROS until needed for a crisis or contingency. Surge ships include FSSs, LMSRs, and RRF ships, which are described as follows:

1. FSS are government-owned USNS RO/RO ships assigned to MSC and operated under contract by merchant shipping companies. These are the fastest ships in the world (traveling at speeds of up to 30 knots). They are ideally suited to carry tanks, helicopters, and other military vehicles and supplies. The eight ships of this class (T-AKR) together can lift nearly the equivalent of a full Army mechanized division. These ships can be ready to sail in 96 hours.

2. MSC's LMSRs have 380,000 ft^2 of cargo space, can achieve speeds of up to 24 knots, and can be ready to sail in 96 hours. LMSRs are equipped with board ramps and cranes to assist in loading oversize cargo, including helicopters, M1A1 tanks, and armored personnel carriers.

3. The RRF is an element of the NDRF. This fleet of nearly 48 ships is maintained by MARAD in peacetime. When activated in wartime or in response to other contingencies such as humanitarian operations or military exercises, the RRF comes under the control of MSC. RRF ships are crewless, but are maintained in class with certificates as approved by the ABS and the USCG. They are normally kept in 4-, 5-, 10-, or 20-day ROS. RRF ships include RO/RO ships, heavy-lift ships, crane ships, breakbulk ships, tankers, and other ships not readily available in the U.S. commercial market.

1.4.5 Vertical Assault Aircraft

Vertical assault aircraft employed in ship-to-shore movement are organic to the LF, and their employment is primarily determined by the MAGTF commander. They are used for troop and equipment transport, escort, and C2 during vertical assault ship-to-shore movement. LF vertical assault aircraft may be transported in and operated from amphibious ships, as discussed in Sections 1.4.1 through 1.4.1.6. Flight deck certification status for each ship class specifying aircraft types, restrictions on day or night operations, and support facilities available is contained in NAVAIR 00-80T-106, LHA/LHD NATOPS Manual; or APP 2(F)/MPP 2(F), Helicopter Operations From Ships Other Than Aircraft Carriers (HOSTAC). The types of aircraft employed by a MAGTF when embarked as the LF are described in Paragraphs 1.4.5.1 through 1.4.5.5.

1.4.5.1 CH-53D Sea Stallion and CH-53E Super Sea Stallion

Heavy lift helicopters are designed to transport personnel, supplies, and equipment in support of amphibious and shore operations. Additional information on CH-53D/E operations is contained in MCWP 3-2, Aviation Operations.

1.4.5.2 CH-46E Sea Knight

The CH-46E is a medium lift assault helicopter primarily used to move troops, and medium cargo and equipment. Additional information on CH-46E operations is contained in MCWP 3-2.

1.4.5.3 MV-22 Osprey

The Osprey is a medium lift assault aircraft primarily used for the transport of troops, equipment, and supplies to and from amphibious ships and land bases. The Osprey is a multi-engine, dual-piloted, self-deployable, VTOL

tilt-rotor aircraft designed for assault support across the range of military operations. It is replacing the CH-46E and CH-53D medium lift helicopters.

1.4.5.4 AH-1W Super Cobra

The AH-1W provides escort, fire support, and fire support coordination to the LF during amphibious assaults and subsequent operations ashore. Additional information is contained in MCWP 3-2.

1.4.5.5 UH-1N Iroquois (Huey)

The UH-1N is a light lift C2 aircraft that can carry a maximum of seven troops. Additional information is contained in MCWP 3-2.

1.4.5.5.1 Tactical Advantages

Tactical advantages for the helicopter include:

1. Ability to ascend and descend vertically into small unprepared areas for loading and unloading troops, equipment, and supplies
2. Delivering fresh troops in organized units to the objective
3. Increasing the LF's flexibility and mobility
4. Increasing the depth of the battlefield
5. Providing speed significantly greater than that of ground transportation in moving around the battlefield
6. Ability to provide rapid MEDEVAC or CASEVAC with minimum additional shock or trauma to injured personnel.

1.4.5.5.2 Tactical Disadvantages

Tactical disadvantages in employing vertical lift aircraft in ship-to-shore movement include:

1. Ability of low visibility, high winds, icing, or other severe weather conditions to limit or eliminate aircraft operations
2. Intensive maintenance requirements
3. The requirement for precise C2 for deconfliction and coordination with other air operations and supporting arms.

1.4.6 Landing Craft

Displacement and nondisplacement landing craft are used to land troops, equipment, and supplies. The capabilities and operating criteria of these landing craft are discussed in detail in NTRP 3-02.1.2; NWP 3-02.12/MCRP 3-31.1A, Employment of Landing Craft Air Cushion (LCAC); MCRP 3-31B, Amphibious Ships

and Landing Craft Data Book; LCAC SEAOPS; COMNAVSURFLANTINST/COMNAVSURFPACINST 3840.1B, Joint Surf Manual; and COMNAVSURFLANTINST/COMNAVSURFPACINST 3340.3, Wet Well Operations Manual. Brief descriptions of each craft are included below.

1.4.6.1 LCU

The LCU is a highly versatile displacement craft designed to beach where hydrographic and weather conditions permit, unload/load, and retract while performing its mission to land heavy vehicles, equipment, personnel, and cargo in an amphibious operation. LCUs transport wheeled and tracked vehicles, general cargo, and personnel from ship to shore, shore to ship, shore to shore, and in resupply, backload or recovery operations. LCUs have also been adapted for other uses such as salvage operation and ferry boats for vehicles and passengers, including evacuees during NEOs. These craft are usually preloaded and lifted to the landing area in LHA, LHD, LSD or LPD well decks.

1.4.6.2 LCM 8

The LCM 8 is another displacement craft whose stated mission is to land personnel, supplies, and equipment on the beach in an amphibious assault or to operate in support of MPF operations. It is also used for lighter and utility work in harbors. However, because the craft is no longer routinely deployed in amphibious shipping, the LCM 8's primary mission is now supporting MPF operations, conducting ammunition transfers, and serving as a SAR platform.

1.4.6.3 Landing Craft Air Cushion

The LCAC is a high-speed nondisplacement landing craft that complements the Marine Corps' rotary-wing aircraft (and ultimately the MV-22 tilt-rotor aircraft) in the conduct of ship-to-shore movement from OTH. The LCAC transports equipment, personnel, cargo, and weapons systems through the SZ and across the beach to landing points beyond the HWM and inland in a variety of environmental conditions. The combined effects of seas, ambient temperature, and craft load are considered in LCAC mission planning. Weather and environmental conditions can affect LCAC operations, but are of lesser concern than for other ship-to-shore delivery options. At OTH ranges of 12 to 100 nm, load and SWH permitting, LCAC offer a method to attain tactical surprise. In addition to supporting ship-to-shore movement during HA, IO, and support of military deception operations. These craft are preloaded and lifted to the landing area in LHA, LHD, LPD, or LSD well decks.

1.4.7 Amphibious Assault Vehicles

AAVs, employed from well deck ships, are organic to the LF. They are described in Paragraphs 1.4.7.1 through 1.4.7.3.

1.4.7.1 Amphibious Assault Vehicle Personnel Carrier

The AAVP7A1 is an armored assault full-tracked landing vehicle. During ship-to-shore movement, this vehicle provides protected transport through rough water and the SZ to the beach for a crew of 3 and up to 21 combat-loaded Marines. It can then carry troops to inland objectives after coming ashore. The AAVP7A1 provides a forcible entry amphibious capability that is unique to the Marine Corps. It can be fitted with an enhanced appliqué armor kit or sandwich-plated steel armor with a layer of Kevlar underneath to protect troops from high-caliber weapons fire. Its own firepower consists of a M2 50-caliber machinegun and a Mk 19 grenade launcher. The

vehicle can move at speeds of up to 45 miles per hour on land and 8 knots at sea. The Marine Corps plans to replace the AAV with the EFV. The EFV is discussed in Appendix I.

1.4.7.2 Amphibious Assault Vehicle Command and Control

The AAVC7A1 has the same basic characteristics as the AAVP7A1, but is designed to provide a mobile regimental or battalion tactical command post platform. The system consists of five radio operator stations, three staff stations, and two master stations. The command communication system contains equipment to provide external secure radio transmission between each AAVC7A1 vehicle and other vehicles and radios. It is armed with a 7.62-mm machinegun and can carry a crew of 3 with a commander and his staff of up to 10 personnel.

1.4.7.3 Amphibious Assault Vehicle Recovery

The AAVR7A1 also has the same basic characteristics as the AAVP7A1. It is designed to recover similar or smaller sized vehicles, on land only, and carries basic maintenance equipment to provide field support maintenance to vehicles in the field. This equipment includes a generator, air compressor, welder, hydraulic crane, and a winch. It is armed with a M240G machinegun and can carry a crew of five.

1.4.8 Expeditionary Fighting Vehicle

The EFV is the Marine Corps' replacement for the AAV7A1 and the AAV7C1. It is considered the keystone to the STOM and EMW concepts and will be the primary means of tactical mobility for the Marine rifle squad during the conduct of amphibious operations, including ship-to-shore movement, and subsequent ground combat operations ashore.

The EFV is a self-deploying, high water speed, armored and fully tracked amphibious/infantry combat vehicle capable of transporting troops from OTH to inland objectives. While providing the speed and maneuvering capabilities to operate with the main battle tank (M1A1 Abrams) on land, the EFV can negotiate current obstacles to the LF (oceans, lakes, and rivers) as high speed avenues of approach and maneuver. It is operated by a crew of 3 and can carry up to 17 Marines with their combat equipment (assault load).

There are two versions of the EFV: the EFV (P) and the EFV (C). They are organized into AA units that are organized and equipped to land the surface assault elements of the LF in a single lift from amphibious shipping to inland objectives.

The specific tactics and operating doctrine for the EFV are still being developed. However, the current guiding publication is MCRP 3-13.1A, Tactics, Techniques, and Procedures for Expeditionary Fighting Vehicle.

1.4.8.1 Expeditionary Fighting Vehicle – Personnel Variant

The EFV (P)'s C4I system extends beyond the LOS to expand the battlespace of the MAGTF while minimizing vulnerability. With land mobility characteristics comparable to the M1A1, the EFV (P) can achieve speeds of up to 45 miles per hour and can cross the same obstacles and terrain features (i.e., trenches, hills, walls, and soft soils) as the tank. The EFV can achieve a high-water speed of 20 knots in a significant wave height (SWH) of 2 feet and can operate in a SWH of 3 feet. It can also traverse the SZ with 6- to 8-foot plunging surf.

1.4.8.2 Expeditionary Fighting Vehicle – Command and Control Variant

The EFV (C) is designed and well suited for providing the EFV commander with a maneuver capability and the requisite information access to conduct close operations and maintain a high level of SA. It also provides the functional interface via systems applications to support the intelligence, maneuver, and firepower requirements of the commander and his staff, and provides secure voice and data capability between the crew and the embarked commander and staff. This vehicle provides a full range of C4I functionality required to operate as a regimental or battalion tactical echelon CP, and has workstation positions, communications systems, and MAGTF tactical C2 software systems applications for the commander and six staff stations.

1.4.9 Special Purpose Craft

Special purpose craft are employed by LF RECON teams, SEALs, and troop units to conduct advance force operations, clandestine assault support, and raids. They are also used to carry boarding and inspection teams during MIO and VBSS operations. These small rubber or fiberglass boats are significantly impacted by surf conditions. Therefore, when opting to use these craft, operating areas are carefully selected and environmental conditions continuously monitored. Additional information on special purpose craft is found in NWP 3-05, Naval Special Warfare. The special purpose craft used in amphibious operations are described in Paragraphs 1.4.8.1 through 1.4.8.3.

1.4.9.1 Combat Rubber Raiding Craft

Capable of OTH operations, the CRRC is used for clandestine surface insertion and extraction of lightly armed SOF. Although primarily used by Marine Corps boat companies, CRRCs can be launched by various aircraft (airdrop (C-130 and C-141 and larger)/helo-cast), amphibious Mk V SOC, NSW RIBs, amphibious ship well decks, and from surface vessels with appropriate low davits. CRRCs may also be deck launched or recovered from surfaced SSNs or locked in and out from submerged DDS-equipped SSNs.

The CRRC has a low visual electronic signature and can be cached by its crew once ashore. It is 15 feet, 5 inches long, weighs 265 pounds, and uses one 35 to 55-hp engine to attain speeds of up to 18+ knots depending on its load. Limited to operating in 8-foot combined seas, its nominal range is 36 nm, although its endurance is dependent on the size of the motor and amount of fuel carried.

A small CRRC is also sometimes employed. The small CRRC is a 130-pound, motor-driven raft that can carry a maximum payload of 2,000 pounds. It can operate at 15 knots with four combat swimmers and equipment that weighs up to 1,000 pounds. Small CRRCs have not been modified to allow for submerged recovery.

1.4.9.2 Naval Special Warfare Rigid Inflatable Boat

The primary mission of the NSW RIB is to provide a short-range, ship-to-shore insertion and extraction capability for SOF in a low to medium threat environment. Its secondary missions are to resupply SOF along coastal littorals, and conduct maritime surveillance and OTH operations. NSW RIB detachments can be configured for 6-month deployments and are fully compatible with Navy amphibious ships to include launch and recovery at sea. The detachment consists of two NSW RIBs, two trailers, two prime movers, two boat crews, and a DDP. The "A" model NSW RIB can be launched from ramps or amphibious ships.

The NSW RIB is nearly 35 feet 11 inches long and weighs 14,000 pounds. Powered by a 470-hp diesel engine, it can achieve speeds of up to 47 knots with a 200-nm range. It carries a crew of three with up to eight passengers and is seaworthy up to sea state 3.

The NSW inventory also includes 24- and 30-foot RIBs that are high-speed, high-buoyancy, extreme weather craft with the primary mission of tactical insertion and extraction of SEALs. During normal operations, these craft are limited to 10-foot combined seas and winds of 35 knots or less.

1.4.9.3 Mk V Special Operations Craft

While not organic to the AF, the Mk V SOC can operate in support of amphibious operations, including ship-to-shore movement. Its primary mission is medium-range SOF insertion and extraction in a low to medium threat environment. The craft's secondary mission is limited CPI. It is fully interoperable with PCs and NSW RIBs.

Generally operated from an FOB ashore, a Mk V detachment is deployable by land, rail, two USAF C-5 aircraft, or a ship with appropriate flight deck and crane capabilities. A detachment consists of an MST and includes 10 crew and 8 MST personnel. It can be ready to deploy within 48 hours of notification and ready to commence day or night operations in support of amphibious operations within 24 hours of arrival at the FOB.

The Mk V SOC is 82 feet long, weighs 57 tons, and powered by two 2285-hp waterjet engines. It has a maximum range of 600 nm and can achieve speeds in excess of 45 knots for 250 nm in seas state 2, 25 to 40 knots in sea state 3, and can survive through sea state 5.

1.4.10 Miscellaneous Vehicles and Support Components

Miscellaneous craft, boats, and special purpose equipment and systems used to support ship-to-shore movement, establishing the logistics support operations, and the LF CONOPS are described in Paragraphs 1.4.8.1 through 1.4.8.8. Details are delineated in NTRP 3-02.1.2 and NTTP 3-02.3M/MCWP 3-32.

1.4.10.1 Improved Navy Lighterage System

INLSs are composed of 80-foot-long barge sections that lock together like building blocks to create a variety of floating structures. Powered barges will be equipped with 360° rotating water thrusters for propulsion and station keeping. These structures allow Navy ships to transport heavy equipment (such as tanks and trucks, and cargo) to include food, water, and equipment parts, between ships and from ship to shore when moorings have been damaged or are unavailable. The system assembles to form ferries, causeway piers, or ship ramps roll-off discharge platforms, providing the Navy and Marine Corps with a method of discharging cargo from strategic sealift ships and moving cargo to shore in case a port is denied, degraded, or unavailable, in environments greater than sea state 2. Fielding to MPSRONs will begin in fiscal year 2007 with full rate production modules (29 in total) slated for delivery to all three MPF squadrons through 2010.

1.4.10.2 Lighter, Amphibious Resupply, Cargo-5 Ton

LARC Vs are used by the BP for SZ salvage, recovery, dewatering, ship-to-shore movement, MEDEVAC/CASEVAC, C2 roles, ramp checks, and occasionally for transporting personnel and equipment.

1.4.10.3 Causeway Section, Nonpowered

Causeway sections are used to bridge the gap between LCUs and the beach, and facilitate the rapid ship-to-shore movement of wheeled and tracked vehicles and equipment. While CSNPs can be carried in LHA, LHD, LPD, and LSD well decks, they are almost exclusively carried to the AO by MPS ships, where they are assembled by the PHIBCBs into a causeway pier or CF.

1.4.10.4 Causeway Section, Powered

CSPs are causeway sections configured with propulsion and steering equipment. When used in CF operations, a CSP section and CSNP sections are married together and can enter well decks or be loaded by crane while moored alongside to transfer cargo from ships in the transport areas to the beach. CSNPs can be carried in LHA, LHD, LPD**,** and LSD well decks, but are almost exclusively carried to the AO by MPF ships.

1.4.10.5 Causeway Ferry

The CF's general mission is to provide a means for transferring equipment ashore when sufficient sections of causeway are unavailable to construct a shore-fast causeway pier. Additionally, the CF provides a method of transferring equipment and cargo from a cargo ship using ship cranes for offload.

1.4.10.6 Side-Loadable Warping Tug

SLWTs are CSPs specially configured to maneuver, connect, beach, and retrieve other causeway sections; place and retrieve anchors; install offshore bulk fuel systems; act as pushers for causeway ferries; and be salvage craft. LHAs, LHDs, LPDs, or LSDs are capable of transporting SLWTs to the AO in their well decks; however, they are rarely delivered by anything other than MPF ships.

1.4.10.7 Amphibious Bulk Liquid Transfer System

The ABLTS has replaced the AABFS and the AABWS. It is a buoyant system that prior to the availability of docking facilities provides the initial means whereby liquid products (fuel and water) are delivered to the LF ashore from MPF ships or amphibious ships. The ABLTS is assembled by the PHIBCBs.

1.4.10.8 Salvage Craft

As mentioned in Section 1.4.9.1, LARC Vs may be used as a salvage craft and the PWC, or waverunner, is being tested as an alternative to the LARC V. If available, the LCM 8 may also be used as a salvage boat. Additional discussion of salvage operations is contained in Appendix D.

1.4.10.9 Free Boats

Free boats are AAVs or landing craft that are available to carry commanders, C2 groups, or LF personnel ashore. The supported commander establishes free boat requirements. Requirements are weighed against the landing craft and AAV availability, as craft dedicated to this purpose are unavailable for troop lift, except on a second-trip basis. Operation of free boats in the vicinity of the LD and boat lanes prior to the landing of scheduled waves is coordinated with the PCO.

CHAPTER 2

Organization and Command

2.1 PURPOSE

This chapter provides the organizational and command relationships for planning and executing ship-to-shore movement.

2.2 ORGANIZATION

The order initiating the amphibious operation specifies command relationships, available forces, and tasks. Flexibility is essential when developing the operational organization, and special emphasis is given to task grouping for economy of force, unity of command, and supported and supporting relationships between ATF and LF parallel chains of command.

2.2.1 Small-Scale Amphibious Operations

A MEU-sized LF conducts small-scale amphibious operations. The LF is routinely embarked in forward-deployed amphibious shipping (an LHA/LHD, an LPD, and an LSD) referred to as an ESG or an ATF. The ESG has organic air, surface, and subsurface defense provided through the assigned surface combatant ships; expeditionary warfare provided by the amphibious ships and the MEU(SOC). ESGs provide JFCs and component commanders with credible, responsive, and flexible options to shape the battle, respond to crises, and protect U.S. and allied interests. The ESG includes the smallest MAGTF; nevertheless, it comprises a versatile, sea-based, operational force that can be tailored to a variety of missions in support of U.S. policy. These missions include quick-reaction crisis-response options in the maritime, littoral, and inland environs.

2.2.2 Medium- to Large-Scale Amphibious Operations

A MEB- /MEF-sized LF conducts medium- to large-scale amphibious operations. Additional information is as follows:

1. Limited amphibious shipping dictates that the LF be organized into echelons. Only those AE LF units and equipment to execute the initial action are embarked in amphibious shipping.

2. To maintain LF integrity in the execution of the landing plan, a transport group may be formed for each landing area. This group can be subdivided into movement groups or units based on such factors as ports of embarkation, individual ship's speed, mutual protection en route to the AOA, and when they are required in the landing area.

3. The interrelationship of ATF and LF tasks during planning and executing amphibious operations requires the establishment of parallel chains of command with corresponding ATF and LF commanders designated at various echelons for planning and C2. Such parallel chains of command also create special coordination requirements as supporting and supported relationships shift with the mission requirements of the amphibious operation. Therefore, except in emergencies, no significant decision contemplated by a commander in the chain of command that affects the plans, disposition, or intentions of a corresponding commander in another chain of command is made without consultation with the commander concerned. In emergency situations, the commander making an emergency decision will notify the affected commanders of action taken at the earliest practicable time.

JP 3-02 and JP 3-02.2 provide more information on echelons, and transport and movement groups.

2.2.3 Ship-to-Shore Movement Control

The initial stage of a ship-to-shore movement is characterized by decentralized execution of the landing plan by subordinate commanders with the ATF and LF commanders maintaining centralized control through their respective control groups. The ATF commander controls the ship-to-shore movement and exercises control through the Navy control group for the surfaceborne assault and the TACGRU for the airborne assault. Each control group provides the positive control functions and coordination with supporting arms that are necessary to conduct their part of the ship-to-shore movement. A CCO and TACAIR officer plan and conduct the surfaceborne and airborne operations, respectively.

The LF commander establishes two organizations, the TACLOG group and the LFSP, to facilitate and influence the ship-to-shore movement and to keep advised of the progress of the assault.

2.2.3.1 Navy Control Group

The surfaceborne ship-to-shore movement of MEB- /MEF-sized LF may involve multiple landing beaches identified by colors. The organization of the Navy control group is based on the arrangement and number of those beaches. If several beaches are specified in the landing plan, the CCO designates a PCO at each landing beach for coordination and control of the surfaceborne operations.

The airborne ship-to-shore movement from amphibious shipping into LZs is controlled by the TACGRU. The ATF commander controls air operations in the AOA through the TACC afloat. Air traffic control functions for the airborne ship-to-shore movement are delegated from the HCS of the TACC afloat to primary AATCCs, which are the air traffic control agencies for helicopter transport group and helicopter transport unit commanders.

Chapter 4 provides a detailed discussion of the surfaceborne ship-to-shore movement and the Navy control group. Chapter 5 provides a detailed discussion of airborne ship-to-shore movement and the TACGRU.

2.2.3.2 Tactical Air Control Group

The TACGRU, through the ATF TACC, is organized and equipped to control air operations in the AO. Control of aircraft movement from ships in sea echelon areas to LZs is exercised through the TACAIR officer in the ATF TACC. During airborne ship-to-shore movement, the ATF TACC coordinates aircraft movements with supporting arms and other air operations, and maintains a current status of vertical lift aircraft, fixed-wing aircraft, and landing platforms.

The TACAIR officer is assisted by the HCSO when planning an air assault or other air operations. The HCSO is the OIC of the HCS of the ATF TACC. If several LZs are specified in the landing plan, the TACAIR officer designates a primary AATCC to provide air traffic control of the airborne operations into each HLZ.

Chapter 5 provides a detailed discussion of the airborne ship-to-shore movement and the ATF TACC.

2.2.3.3 Landing Force Support Party

The LFSP is the forward echelon of the LCE formed to facilitate ship-to-shore movement. It is a temporary task organization composed of ATF and LF elements established via a formal activation order from the LF commander to provide the LF with initial landing support and CSS during the ship-to-shore movement until relieved by the LCE. The LFSP's mission is to:

1. Facilitate the landing and movement of troops, equipment, and supplies across beaches and into LZs, ports, and airfields.

2. Assist in the evacuation of casualties and EPWs from beaches and LZs.

3. Assist in the beaching, retraction, and if necessary, the salvage of landing craft and amphibious vehicles.

4. Facilitate the establishment of the LCE, ACE, and NBG ashore.

NTRP 3-02.1.2 and Appendix G also discuss the organization and functions of the LFSP.

2.2.3.4 Tactical-Logistical Group

The TACLOG is comprised of representatives designated by LF troop commanders to assist Navy control officers aboard control ships in the ship-to-shore movement of troops, equipment, and supplies IAW the landing plan. TACLOG detachments are collocated with the ATF and LF commanders, the TACAIR officer, CCO, PCOs, and helicopter transport group unit commander(s).

The TACLOG is manned with personnel intimately involved in the development of the landing plan and capable of making rapid, sound decisions concerning ship-to-shore movement based on the LF commander's guidance and intent. Appendix F and JP 3-02.1, Joint Tactics, Techniques, and Procedures for Landing Force Operations, discuss the organization and functions of the TACLOG group.

Figure 2-1 shows the control organization for conducting the ship-to-shore movement.

2.3 AMPHIBIOUS FORCE

As defined in Paragraph 1.1, JP 3-02, and JP 1-02, an AF is an ATF and an LF together with the other forces that are trained, organized, and equipped for amphibious operations.

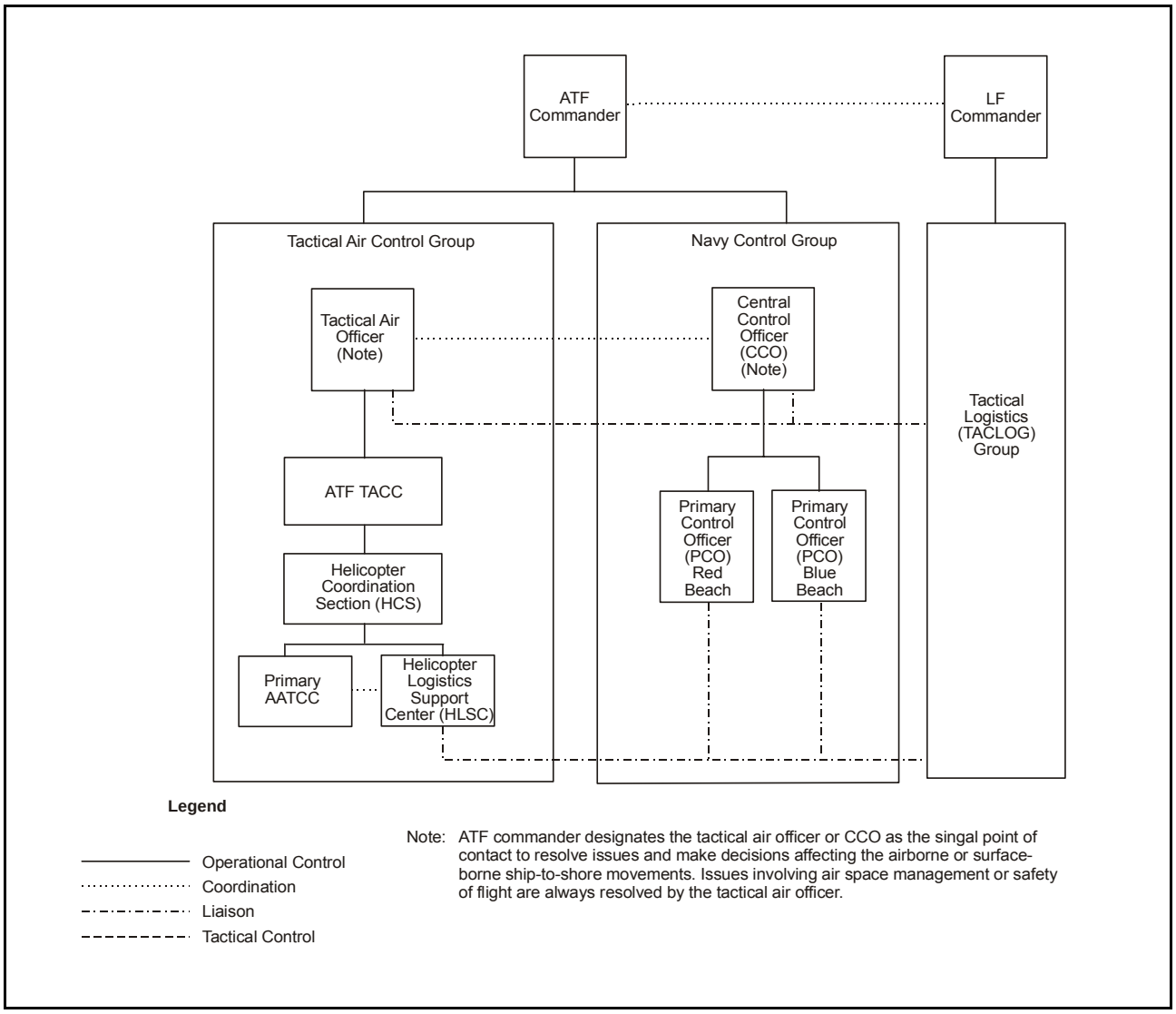


Figure 2-1. Control Organization During the Ship-to-Shore Movement

2.3.1 Amphibious Task Force

Depending on the size of the operation, the LF embarks in amphibious shipping or a combination of amphibious and MSC-chartered shipping for movement to the AOA. Chapter 1 provides the general characteristics of this shipping. The ATF commander is responsible for the task organization of the ships, preparation of the movement plan, selection of sea echelon areas in the vicinity of the landing area, logistics, and FP.

The ATF commander is also responsible for LF debarkation until termination of the amphibious operation. The ATF staff ensures troops, equipment, and supplies are landed at the prescribed times, places, and in the formation required by the LF scheme of maneuver. Elements specific to this staff are described in the sections below.

2.3.1.1 Naval Beach Group

The NBG is a permanently organized Navy command within an AF comprised of a commander and staff, a BMU, a PHIBCB, and displacement/nondisplacement ACUs. Designed to provide an administrative group from which required naval tactical components may be made available to the ATF and LF commanders for any size amphibious operation, the NBG provides Navy elements in support of surfaceborne ship-to-shore movement and LFSP operations. During MPF operations, the NBG is task organized to form the NSE. The CNBG becomes the NSE commander and may also direct the NCHB elements, if assigned.

The NBG and its support elements are discussed briefly in Paragraphs 2.3.1.1.1 through 2.3.1.1.3, and covered in NTRP 3-02.1.2; NTTP 3-02.3M/MCWP 3-32; and JP 4-01.6, Joint Logistics Over-the-Shore.

2.3.1.1.1 Beachmaster Unit

The BMU is a commissioned naval unit of the NBG designed to provide the LFSP with a Navy component known as the BPT. The BMU facilitates the landing and movement of troops, equipment, and supplies across the beach and the evacuation of casualties and EPWs.

2.3.1.1.2 Amphibious Construction Battalion

The PHIBCB is a permanently commissioned naval unit subordinate to the CNBG. The PHIBCBs provide an administrative unit from which personnel and equipment are formed into tactical elements and made available to appropriate commanders to operate pontoon causeways, transfer barges, warping tugs, assault bulk water and fuel systems, perform salvage tasks, and provide camp support and security. The PHIBCBs also have a limited construction capability for beach improvements and egress routes.

2.3.1.1.3 Assault Craft Units

The ACUs are permanently commissioned naval organizations, subordinate to the CNBG, with the landing craft and crews necessary to provide lighterage for an amphibious operation. The ACUs provide displacement LCUs, LCMs, and nondisplacement LCAC in support of ship-to-shore operations, general offload, or MPF operations.

2.3.1.2 Tactical Air Control Squadron

The TACRON is a component of the TACGRU that provides aircraft control and warning facilities afloat. The TACRON operates the ATF TACC, or TADC, to control aircraft in the AOA.

2.3.2 Landing Force

An LF may be composed of Marine Corps and/or Army forces, other forces, and multinational forces. It is the highest troop echelon in the AF and, when Marines are embarked, encompasses the entire MAGTF.

2.3.2.1 Marine Air-Ground Task Force

The MAGTF is the Marine Corps' principal organization for missions across the range of military operations. It is composed of forces task organized under a single commander and can respond rapidly to a contingency anywhere in the world.

MAGTF forces are functionally grouped into four core elements: a CE, a GCE, an ACE, and an LCE. These elements are categories of forces, not formal commands. The basic structure of the MAGTF never varies, but the number, size, and type of Marine Corps units comprising each of the four elements are always mission dependent. MAGTFs are normally sized as a MEU/MEB/MEF, or in some cases, an SPMAGTF. MAGTF elements and organizations are discussed later in this chapter and in JP 3-02.1. Figure 2-2 shows the MAGTF organization.

2.3.2.2 Command Element

The CE is a permanent organization composed of the commander, executive and special staff sections, HQ section, and requisite communications support, intelligence, and RECON forces necessary to accomplish the MAGTF mission. The CE provides C2, intelligence, and other support essential for effective planning and execution of operations by the other MAGTF elements. The CE varies in size and composition, and conducts liaison with higher, adjacent, and supporting commands.

2.3.2.3 Ground Combat Element

The GCE is task organized to conduct ground operations. It is usually constructed around an infantry organization, but can vary in size from a small ground unit of any type to one or more MARDIVs that can independently maneuver. The GCE also includes appropriate CS units. It is generally through the GCE that the MAGTF generates combat power through the use of firepower and mobility. Normally, there is only one GCE in a MAGTF.

2.3.2.3.1 Division

The MARDIV is a balanced force of combat and CS units. It is a major administrative and tactical unit that is organized and equipped to conduct sustained combat operations with or without reinforcement.

2.3.2.3.2 Regimental Landing Team

An RLT is a task organization for landing comprised of an infantry regiment reinforced by those CS organizations such as assault craft (AAV, EFV), LAVs, tanks, and combat engineers that are required for initiating its mission ashore. The RLT may be employed as an integral part of a division or as a semi-independent or independent maneuver unit.

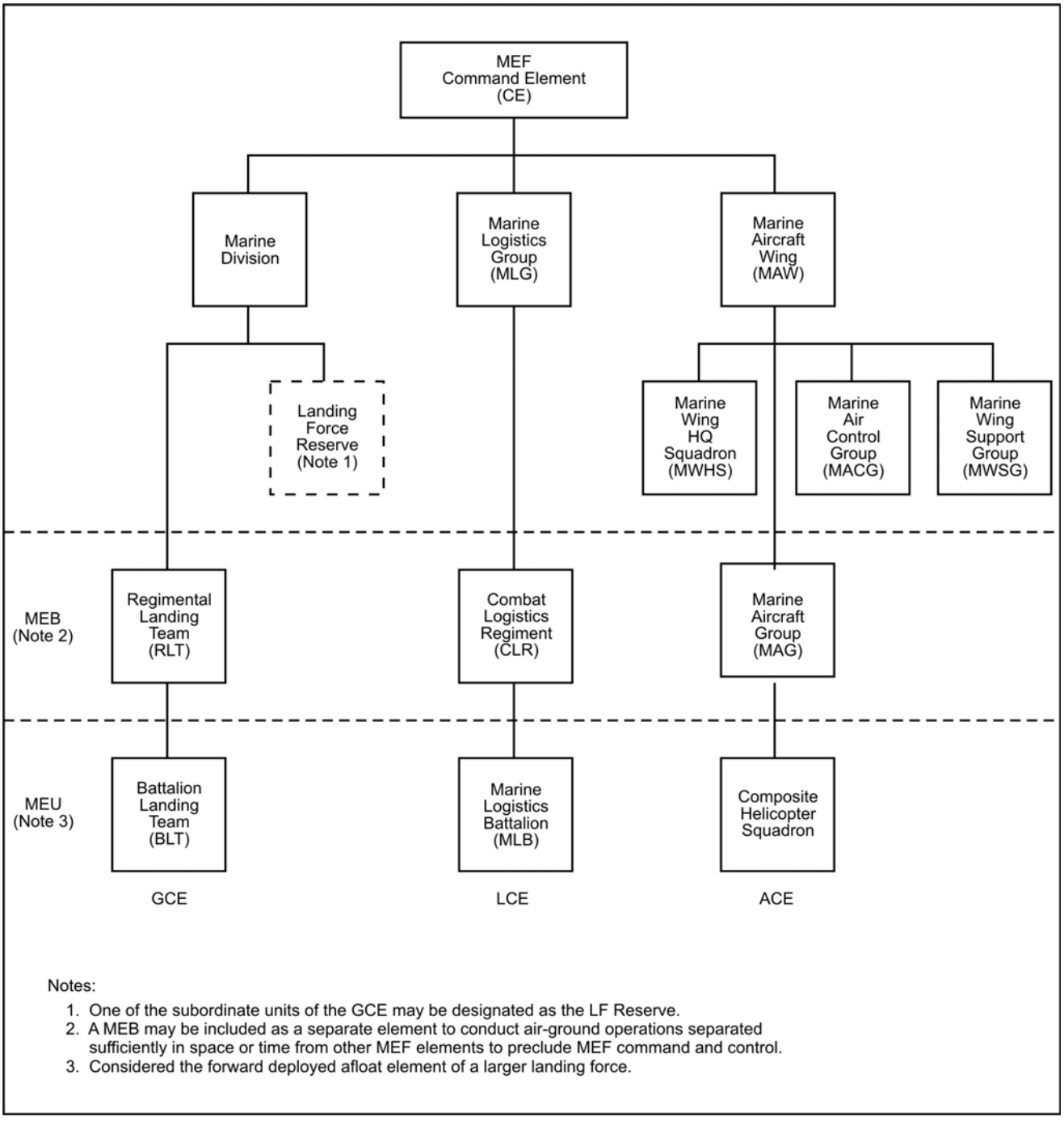


Figure 2-2. Marine Air-Ground Task Force Organization

2.3.2.3.3 Battalion Landing Team

A BLT is an infantry battalion normally reinforced by necessary combat and CS elements. It is the basic unit for planning an assault landing. The BLT may be employed as an integral part of the RLT or as a semi-independent or independent maneuver unit.

2.3.2.4 Aviation Combat Element

The ACE is task organized to conduct aviation operations in direct support of the MAGTF. The ACE executes all or a portion of the six functions of Marine aviation necessary to accomplish the MAGTF's mission. These functions are AAW, OAS, assault support, EW, aerial RECON, and control of aircraft and missiles. The ACE is usually composed of an aviation unit HQ, and elements of MAGs, MACGs, and MWSGs. Depending on the MAGTF mission, the ACE may be reinforced with various other aviation units or their detachments. The ACE can vary in size from a small aviation detachment to a MAW. The ACE includes C2 agencies, combat, CS, and CSS units. Normally, there is only one ACE in a MAGTF.

As the ship-to-shore movement progresses, the ACE commander phases the MACCS ashore.

2.3.2.5 Logistics Combat Element

The LCE is task organized to provide the direct and general support necessary to accomplish the MAGTF mission. The LCE varies in size from a MLB associated with a MEU(SOC) to a CLR associated with a MEB, or a MLG associated with a MEF. It provides supply, maintenance, transportation, general engineering, health services, and a variety of other services to the MAGTF. Normally, there is only one LCE in a MAGTF. The development of combat logistics capability ashore to support the LF during the initial stage of an amphibious operation is accomplished by the LFSP.

2.3.3 Marine Expeditionary Force

A MEF is the largest MAGTF and is the Marine Corps' principal warfighting organization. Normally commanded by a lieutenant general, it is task organized around a permanent CE and normally contains one or more MARDIVs, a MAW, and an MLG. Elements comprising the GCE, ACE, and LCE of MEBs and MEUs are drawn from MEF MSEs, and may be augmented by forces from other MEFs, MARFORES, other services, or coalition military forces. The size and composition of a deployed MEF can vary greatly depending on the requirements of the mission. A MEF is capable of missions across the full range of military operations, including amphibious assault and sustained operations ashore in any environment. It can operate from a sea base, land base, or both, and typically deploys with more than 45,000 personnel with up to 60 days of sustainment.

The MEF is a tailorable and scalable force that can be equipped and task organized to conduct forcible entry operations, operate independently, serve as a JTF HQ, or operate as a JTF. When operating as part of a JTF, the MEF commander can operate as the JFLCC. The MEF is the only means for self-sustainable forcible entry into a region in which U.S. forces have been denied access. It can seize and hold airfields and ports to enable the introduction of follow-on MEF or joint forces. The MEF can then remain in theater to conduct the full range of military operations in support of the joint campaign.

2.3.3.1 Ground Combat Element

The MEF GCE is normally a MARDIV reinforced with appropriate CS units. The GCE may consist of multiple reinforced divisions under a single GCE commander. A MARDIV includes three infantry regiments, and artillery regiment, a tank battalion, AAV/EFV, LAV battalions, a RECON battalion, and a combat engineer battalion.

2.3.3.2 Aviation Combat Element

The MEF ACE is a MAW task organized to perform all six functions of Marine aviation in direct support of the MAGTF commander. A single MAW, which may be augmented from other standing MAWs, MARFORES, other services, or coalition forces, forms the MEF ACE. The MAW is organized to form the ACE and can operate from ships, forward operating sites, permanent airfields, or expeditionary airfields. At the MAW level, when the MEF operates as part of a JTF, the ACE commander can operate as the JFACC.

2.3.3.3 Logistics Combat Element

The MEF LCE is an MLG. It is structured to support a single-division GCE and a single-MAW ACE. The MLG is a task-organized grouping of functional and cross-functional battalions that provide tactical-level ground logistics support to all elements of the MEF. The MLG may be tasked to provide operational-level logistics support in theater as a Marine logistics command. It provides a full range of supply, maintenance, transportation, general engineering, health services, and other services capabilities.

2.3.4 Marine Expeditionary Brigade

A MEB is a mid-sized MAGTF that is smaller than a MEF, but larger than a MEU(SOC). It can deploy in amphibious shipping or be transported by strategic lift into an AO to link up with MPF assets. It is constructed around a reinforced infantry regiment, a composite MAG, and a CLR. Normally commanded by a BGEN or MGEN, and comprised of 8,000 to 18,000 Marines and Sailors, a MEB varies in size and composition and is task organized to meet the requirements of a specific situation. It typically deploys with up to 30 days of sustainment and can conduct combat operations of a limited scope. If the scope of operations expands beyond the capability of the MEB, additional forces can readily deploy to expand to a MEF.

Capable of conducting forcible entry operations and other missions across the full range of military operations, a MEB can function as part of a JTF, as the JFEC, as the lead echelon of a MEF, or alone. A MEB may have other service or foreign military forces assigned or attached. It may be tasked with preparing for the subsequent arrival of the MEF, joint, or international forces, or the conduct of other specified tasks. Key MEB capabilities include:

1. Conduct amphibious operations as discussed in Paragraphs 1.2.2 through 1.2.2.4.

2. Conduct sustained combat operations ashore as part of a larger force to achieve campaign or major operations objectives.

3. Conduct close and deep air support and aerial RECON operations.

4. Coordinate naval support of ground operations.

2.3.4.1 Ground Combat Element

The MEB GCE is tailored to accomplish a specific assigned mission. It is normally an infantry regiment reinforced with selected division units.

2.3.4.2 Aviation Combat Element

The MEB ACE is a composite MAG that is task organized for the assigned mission and may consist of rotary- and fixed-wing aircraft, aviation C2 (including LAAD), and aviation ground support units. It is organized so the ACE commander's MACCS can be phased ashore early in amphibious operations. The MEB ACE is usually organized and equipped for early establishment in the objective area as forward operating sites are established or airfields in the objective area become available.

2.3.4.3 Logistics Combat Element

The MEB LCE, or CLR, is task organized from the organic battalions of the MLG to provide CSS beyond the capability of the supported CE, GCE, and ACE. The CLR is organized to provide maintenance support, transportation support, general engineering support, supply support, disbursing services, legal services, health services, and postal services.

2.3.5 Marine Expeditionary Unit (Special Operations Capable)

A MEU(SOC) is organized as a MAGTF consisting of a CE, a reinforced infantry battalion, a reinforced composite aviation squadron, a task-organized LCE, and an MSOC. The MEU(SOC) is typically associated with an ESG, consisting of approximately 2,200 Marines and Sailors, and is commanded by a COL. It normally fulfills the Marine Corps' forward, sea-based deployment requirements and deploys with up to 15 days of supplies. The forward-deployed MEU(SOC) is uniquely organized, trained, and equipped to provide the RCC or JFC with an expeditionary force. This force is inherently balanced, sustainable, flexible, responsive, expandable, and credible.

The "special operations capable" designation means that the MEU(SOC) is embarked and deployed with an MSOC. The MSOC is normally under the OPCON of theater special operations command commander and under the TACON of the MEU commander. The MSOC is chartered to perform three core special operations missions (SR, DA, and limited FID). The MEU is chartered to perform the following missions:

1. Conduct amphibious raids.
2. Conduct amphibious assault.
3. Conduct security and stability operations.
4. Execute R2P2.
5. Conduct foreign military training.
6. Conduct HA/DR.
7. Conduct NEOs.

8. Conduct TRAP.
9. Conduct limited expeditionary airfield operations.
10. Conduct airfield seizure operations.
11. Conduct joint/combined enabling operations.
12. Develop intelligence.

Should the MSOC be tasked to disembark amphibious shipping for a separate mission, the MAGTF would be referred to as a MEU vice a MEU(SOC).

A MEU(SOC) does not routinely conduct opposed amphibious assault operations and can only conduct amphibious operations of limited duration. Its special operations capability makes it well suited for crisis response, immediate reaction operations such as NEO, limited objective attacks, raids, and for acting as an advance force for a larger follow-on MAGTF. A MEU(SOC) can rapidly deploy and employ via amphibious shipping, by strategic airlift, through marshaling with MPF assets, or any combination thereof.

2.3.5.1 Ground Combat Element

The MEU(SOC) GCE is normally a BLT, which is a reinforced rifle battalion of approximately 1,200 Marines. GCE attachments normally include artillery, engineers, LAR, antiarmor, AA, division RECON units, and a tank platoon.

2.3.5.2 Aviation Combat Element

The MEU(SOC) ACE is normally a reinforced Marine medium helicopter squadron. It does not normally contain the aviation resources to conduct active AD of the landing area. The squadron is normally reinforced with a mix of transport helicopters, light attack helicopters, V/STOL attack aircraft, a detachment from the MACG that includes a LAAD section, and a detachment from the MWSG. A shore-based VMGR detachment is assigned to each MEU(SOC) ACE.

2.3.5.3 Logistics Combat Element

The MEU(SOC) LCE, or MLB, is a task-organized component of the MLG. Relatively small, usually comprised of less than 300 Marines and Sailors, a CLB provides CSS such as supply support, maintenance support, transportation support, general engineering support, landing support, disbursing services, medical and dental services, legal services, and postal services.

2.3.6 Special Purpose Marine Air-Ground Task Force

The SPMAGTF is a MAGTF organized, trained, and equipped with narrowly focused capabilities. It is designed to accomplish a wide variety of expeditionary operations, but its missions are usually limited in scope and duration. It is configured to accomplish specific mission(s) for which a MEF/MEB/MEU(SOC) would be inappropriate or too large. A SPMAGTF may be any size, but is usually a small force, MEU-sized, or smaller. SPMAGTFs are organized, trained, and equipped to conduct a wide variety of expeditionary operations.

The SPMAGTF CE is structured for the conduct of operational functions and is tailored to the mission and task organization of the SPMAGTF.

2.3.6.1 Ground Combat Element

The SPMAGTF GCE is normally at least a platoon-sized element.

2.3.6.2 Aviation Combat Element

The SPMAGTF ACE is a task-organized Marine aviation detachment under an OIC. If required, the full range of Marine aviation capabilities can be included.

2.3.6.3 Logistics Combat Element

The SPMAGTF LCE is typically a CSS detachment that is task organized to meet the specific service support requirements of the SPMAGTF. Depending on the mission of the SPMAGTF, the CSS detachment will be based on the unit upon which the operation is focused. For example, the CSS detachment can provide a landing support detachment if a port or airhead is used, or an engineer detachment if the mission entails construction or bulk fuel movement.

2.3.7 Marine Air-Ground Task Force Movement

Amphibious assault shipping constraints and tactical considerations dictate that MEB- /MEF-sized MAGTFs be echeloned into the AO. For embarkation and movement, MEU(SOC)s are employed as single units. JP 3-02 describes echelons, organization for movement (transport and movement groups), sea routes, sea areas, and regulating points.

2.4 COMMAND RELATIONSHIPS

The command relationships discussed in this section pertain to the planning and execution of ship-to-shore movement.

2.4.1 Amphibious Task Force and Landing Force Commander Command Relationships

In the past, the terms "commander, amphibious task force" and "commander, landing force" have been used doctrinally to signify the two principal commanders assigned to plan, organize, and execute amphibious operations. These terms no longer connote command relationships. They are titles that refer to the senior ATF and LF commanders who are instrumental to the planning and execution of amphibious operations.

The JFC, common superior commander, or establishing authority ensures unity of effort in achieving operational objectives by organizing the forces in a manner that best supports and accomplishes the overall mission. The most effective means are:

1. Establishing a support relationship between functional/service components

2. Delegating OPCON or TACON of AFs to functional/service components.

Command relationship options are discussed in Chapter II of JP 3-02 and in JP 0-2, Unified Action Armed Forces (UNAAF). In amphibious operations, typically a support relationship is established between the ATF and LF commanders based on the complementary, rather than similar nature and capabilities of the maritime and ground forces. Other factors that impact on the type of command relationship chosen are the mission, nature and duration of the operation, force capabilities, C2 capabilities, assigned battlespace (threat), and recommendations from subordinate commanders. However, there is no intent to limit the common superior's authority to establish any appropriate command relationships.

2.4.1.1 Planning Relationships

Regardless of the command relationships**,** when the order initiating planning for an amphibious operation is received, specified relationships are observed during the planning phase. The commanders designated in the order initiating the amphibious operation are coequal in planning matters and decisions. Planning for amphibious operations is discussed in detail in JP 3-02 and Chapter 3 of this publication.

2.4.2 Relationship Between the Ship's Commanding Officer and the Commanding Officer of Troops

The CO of a ship transporting troops exercises command authority over persons embarked as prescribed by NAVREGS. While embarked, troop administration is a function of the COT, but is also subject to ship's regulations and SOP.

2.4.3 Relationship Between the Ship's Commanding Officer and Embarked Aircraft Units

This relationship is discussed in Chapter 5.

2.4.4 Command of the Landing Force Support Party

The LF commander commands the LFSP. The ATF commander directs NBG elements to form the BP and to report to the LF commander for planning. OPCON of the BPT may be passed to the LF commander at this time. Navy BPT commanders, as LFSP subordinates, retain command of Navy units ashore. The LFSP is discussed in detail in NTRP 3-02.1.2 and Appendix G of this publication.

2.4.5 Relationship Between the Amphibious Task Force Surgeon and Landing Force Surgeon

The ATF surgeon is the designated medical officer on the ATF staff and represents the commander in matters pertaining to the HSS for a projected amphibious operation. The LF surgeon is usually identified as the MEU or MEF surgeon, as is appropriate for the MAGTF level. In consultation with the LF surgeon, the ATF surgeon provides the OPLAN/OPORD medical section (Annex Q, Medical Services) to the ATF to ensure mutual support and fulfillment of supporting and supported commands' medical requirements to include patient movement.

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CHAPTER 3

Planning

3.1 PURPOSE

This chapter covers the planning process for ship-to-shore movement and the manner in which the AO and landing area are organized to facilitate amphibious operations. It discusses the planning sequence followed by ATF and LF commanders to land troops, equipment, and supplies at prescribed times and places, and the organization to support the LF CONOPS ashore.

3.2 BACKGROUND

Detailed planning for ship-to-shore movement begins after the ATF and LF commanders approve the LF CONOPS ashore. The ship-to-shore movement plan is the compilation of detailed plans, tables, diagrams, and schedules prepared by the ATF and LF commanders. These documents are issued as an appendix to the amphibious operations annex of the OPORD; message OPORD supplements; or an APP 11, NATO Message Catalogue (NMC), formatted message (the OPTASK AMPHIB). The ship-to-shore movement plan is integrated with other elements of the OPORD and supporting plans (as applicable), and provides for the requisite CSS to the LF during the early stages of the operation. Altogether, the aforementioned documents comprise the landing plan. Many of these documents are designed for MEB- /MEF-sized operations; however, most can be modified for use in ESG/MEU(SOC)-sized missions. Those not included in this chapter are consolidated in Appendix J.

3.3 PLANNING PROCESS

Developing a ship-to-shore movement plan demands a systematic approach to planning and adherence to doctrine. The six-step planning process highlighted in Paragraph 3.3.3 is derived from, and essentially mirrors, the Marine Corps planning process delineated in MCWP 5-1, Marine Corps Planning Process. This planning process involves coordination among all levels of the established parallel chains of command supporting the ATF and LF commanders. Commanders' guidance, based on the LF CONOPS ashore, is provided to subordinate commanders, and the means to land the LF are tabulated and apportioned. Subordinate ATF and LF commanders prepare individual documents, such as those listed in Paragraph 3.2, for approval and consolidation with the naval and LF landing plans.

JP 5-0, Doctrine for Planning Joint Operations; and MCDP 5, Planning, also provide detailed information on the planning process.

3.3.1 Tenets of the Marine Corps Planning Process

The three tenets of the Marine Corps planning process, when adapted for amphibious operations, guide the ATF and LF commanders' use of their staffs and subordinates to execute assigned missions. These three tenets, top-down planning, the single-battle concept, and integrated planning, are discussed in Paragraphs 3.3.1.1 through 3.3.1.3. The overall process is explained in detail in JP 3-02 and MCWP 5-1.

3.3.1.1 Top-Down Planning

Because planning is a fundamental responsibility of command, commanders not only participate in the process, but they drive it. Two-way communication is the key to top-down planning, as commanders convey their intent and guidance to staff and component planners. They use planning as an avenue to gain knowledge and SA to support their decision making process. The commanders' guidance is translated into a CONOPS, and subordinate commanders use their guidance and CONOPS to accomplish assigned missions.

3.3.1.2 Single-Battle Concept

Operations or events in one part of the battle or AO may have profound and often unintended effects on other areas and events. Therefore, commanders view the battlespace or AO as an indivisible entity. In amphibious operations, the single-battle concept allows commanders to effectively focus the efforts of AF elements on mission accomplishment. While the battlespace or AO may be conceptually divided into sectors such as deep, close, and rear to facilitate planning and decentralized execution, the commanders' intent ensures unity of effort by focusing on a single battle or mission.

3.3.1.3 Integrated Planning

Integrated planning is a disciplined approach to planning that is systematic, coordinated, and thorough. It is based on the warfighting functions of C2, maneuver, fires, intelligence, logistics, and FP. Planners use these functions to integrate the planning effort and supervise execution of the plan. Planners use integrated planning to consider all relevant factors, reduce omissions, and share information across the functions. The key to integrated planning is the assignment of appropriate knowledgeable and experienced personnel to represent each functional area.

3.3.2 Planning Methods

Two planning process methods may be applied to any amphibious operation: deliberate planning or rapid planning (also called CAP).

While distinct in concept, in practice, deliberate and rapid planning can form a continuum and complement each other. If appropriate, deliberate planning is conducted early in the planning process. As the time for execution approaches, a transition can be made to rapid planning. Thus, deliberate planning may form the basis for later CAP, and rapid planning may be the revision of earlier deliberate plans.

3.3.2.1 Deliberate Planning

Deliberate planning is the process generally used in MEB- /MEF-level operations for the deployment and employment of apportioned forces and resources. Deliberate planning occurs in response to a hypothetical situation or a situation that develops over an extended period of time. It is performed well in advance of expected execution, often during peacetime or before initiating a deliberate operation. Deliberate planners rely heavily on assumptions regarding the circumstances that will exist when the plan is executed.

3.3.2.2 Rapid Planning or Crisis Action Planning

Rapid planning, or CAP, is the time-sensitive planning for the deployment, employment, and sustainment of assigned and allocated forces and resources that occurs in response to a situation that may result in actual military

operations. Crisis action planners base their plan on the conditions and circumstances that exist at the time the planning occurs. CAP is usually less formal than deliberate planning and more responsive to changing events.

In MEU(SOC) amphibious operations, CAP is also called R2P2. The R2P2 uses a compressed timeline that provides commanders and staffs with an accelerated planning mechanism that facilitates mission execution within 6 hours of WARNORD or alert order receipt.

The goal of rapid planning is to expend less time on planning to provide the executing forces with the maximum time allowable to prepare for the mission. The key to successful rapid planning in amphibious operations is for ATF and LF commanders and their staffs to achieve a balance among sufficient time to develop a feasible COA, sufficient time to coordinate its essential details, and sufficient time to prepare for its execution. This balance cannot be achieved without proper mutual training and the conduct of exercises in preparation for real-world operations. Therefore, dedicated and well-coordinated training for MEU(SOC) and ATF staffs is always structured in detail and included from the very first days of predeployment workups.

In rapid planning, commanders and their staffs shall be thoroughly familiar with potential contingencies or missions, and every individual involved with planning shall be thoroughly familiar with their role(s) in the planning process. A successful R2P2 process is predicated on each unit's early and timely retrieval or receipt of sufficient intelligence and related information; planning experience, foresight to make significant preparations in organizing, training, and equipping; information management; and highly refined, well-rehearsed SOPs.

To best employ R2P2, commanders and their staffs develop capabilities in four areas: planning cells, planning and operations SOPs, intelligence, and information management.

3.3.2.2.1 Planning Cells

To achieve the most effective and efficient rapid planning results, every effort should be made to standardize and maintain the composition and membership of the various planning cells. This is particularly true during the pre-deployment training program and deployment of the MEU(SOC), and its Navy counterpart, the PHIBRON. The planning cells usually employed by these two organizations include the CAT, or central planning cell; the battle staff, if employed; and the mission planning cells. These planning cells participate in frequent planning exercises that involve scenarios similar to those that might be encountered in real-world missions. These exercises serve to refine the LF and ATF staffs' ability to plan rapidly; allow planners to get to know the commanders and each other; and heighten SA regarding likely contingency missions and areas of operations. A comprehensive and challenging training program should result in planning cells understanding where they are to meet, what they will accomplish, and how much time they have to complete their planning efforts. Planning cells shall also be capable of conducting concurrent planning (simultaneous at different echelons of the same command) and parallel planning (between equivalent echelons of different) commands.

3.3.2.2.2 Planning and Operations Standard Operating Procedures

SOPs are essential to rapid planning. ATF and LF planners should be familiar with their respective planning SOPs. Operations SOPs are equally important, because they allow planners to select proven and practiced tasks that provide viable solutions to tactical problems. SOPs also enable ATF and LF MSEs to carry out familiar tasks effectively and efficiently with minimal or no higher level guidance or communication. While it is impossible to develop SOPs for every conceivable mission, those that do exist should include a predesignated task organization;

equipment and ordnance lists; elements of a landing plan; mission execution procedures; and an execution checklist with code words. SOPs should be studied, rehearsed, and executable with little advance notice.

3.3.2.2.3 Intelligence

The LF and ATF commanders and their staffs shall be adept at anticipating possible contingencies based on continual analyses of classified and unclassified intelligence reports. Commanders should ensure their staffs, particularly the planners, are provided the latest intelligence, possible targets, area studies, and other information relevant to the assigned mission or potential missions.

3.3.2.2.4 Information Management

The speed and volume of information flow in most operations, including ship-to-shore movement and related amphibious operations, can be voluminous and therefore overwhelming if not properly managed. Due to the time constraints inherent in R2P2, there is also less time for the commanders and their staffs to analyze information requirements. Therefore, it is critical that participants in the planning process be fully engaged in and realize the importance of their mission area, and that they share their knowledge and information in clear, concise, and simple presentations.

3.3.3 Six-Step Planning Process

The cornerstone of amphibious operations execution is the six-step planning process. Its effectiveness and efficiency are enhanced if the methods and training discussed in Sections 3.3.1 and 3.3.2 are incorporated and implemented. Used in deliberate planning and CAP, and applicable to operations of any size or type, the six-step planning process provides logical procedures to follow from receipt of the order initiating an amphibious operation through the development of OPLANs, OPORDs, or OPTASKs. The process also provides ATF and LF commanders and their staffs with a means to organize planning activities, transmit plans to subordinate commands, and share a common understanding of the mission and the commanders' intent. Interactions among various planning steps allow a concurrent, coordinated effort that maintains flexibility, makes efficient use of time available, and facilitates continuous information sharing. It enhances the commanders' ability to make the primary decisions in most amphibious operations. Figure 3-1 provides a matrix of those decisions.

The six steps of the process are:

1. Mission analysis
2. COA development
3. COA war game
4. COA comparison and decision

Primary Decision	May be Contained in the Order Initiating the Amphibious Operation	Decision	Decision Made Not Later Than Planning Step
Determine AF Mission(s)	X	Mutual	1
Select AF Objectives	X	Mutual	1
Determine COAs for Development	X	Mutual	2
Select COA		Mutual	4
Select Landing Areas		Mutual	4
Select Landing Beaches		Mutual	4
Determine Sea Echelon Plan		ATF Commander	4
Select LF Objectives		LF Commander	4
Select LZs and Drops Zones (DZs)		LF Commander	4
Select Date and Hour of Landing	X	Mutual	4

Figure 3-1. Primary Decisions Responsibilities Matrix

5. Orders development

6. Transition.

Chapter IV of JP 3-02, NTRP 3-02.1.2, MCWP 5-1, and MCDP 5 provide additional detailed information on the six steps of the amphibious planning process.

3.3.3.1 Naval Planning

Naval planning for ship-to-shore movement focuses on developing the unloading, landing control, and patient movement plans/MEDREG. Unloading and landing control are discussed in detail later in this publication. MEDREG is covered in JP 3-02 and JP 3-02.1.

3.3.3.2 Landing Force Planning

LF planning for ship-to-shore movement focuses on developing the sequence and organization for landing. The LF landing plan is the integration of naval and LF surfaceborne, and airborne ship-to-shore movement plans into a single coordinated plan that provides for the rapid projection of combat power or other LF assets ashore with the requisite CSS for sustained operations. To achieve this goal, the LF develops the sequence and organization for landing by:

1. Allocating or specifying landing means to subordinate commanders based on availability and IAW the LF CONOPS ashore

2. Allocating blocks of serial numbers to subordinate commanders

3. Determining the LF's landing priorities

4. Tasking the GCE commander to prepare landing plans based on assigned tasks and priorities

5. Correlating and consolidating subordinate landing plans into the LF landing plan.

3.3.4 Landing Serials

Discussion of the movement categories listed in Paragraph 3.3.5 requires an understanding of landing serials and serial numbers.

3.3.4.1 Serials

A serial is an element or group of elements (e.g., people, vehicles, equipment, supplies) within a series assigned a numerical or alphabetical designation to facilitate planning, scheduling, and control.

3.3.4.2 Serial Numbers

A serial number is a reference number assigned to each serial to identify each element of the LF and those Navy elements to be landed prior to general unloading. Serial numbers are a means of identification, not a statement of priority, and are published in the SAT, which is included in the landing plan. The planned order for landing serials is published in the landing sequence table of the landing plan.

Additional discussion of serial numbers and their allocation and assignment is contained in Paragraphs 3.4.2.5 through 3.4.2.5.2.

3.3.5 Troop and Equipment Movement Categories

LF troops and supplies are arranged in five movement categories for planning ship-to-shore movement:

1. Scheduled waves

2. On-call waves

3. Nonscheduled units

4. Prepositioned emergency supplies

5. Remaining LF supplies.

3.3.5.1 Scheduled Waves

The ATF and LF commanders predetermine the time, place, and formation for landing scheduled waves. Scheduled waves consist of vertical lift aircraft, landing craft, or AAVs carrying serialized troops and their initial CSS ashore.

After surfaceborne waves cross the LD or CDP, and vertical assault waves leave their DPs, the landing of scheduled waves normally proceeds without change to maintain the momentum of the operation and provide for the rapid projection of combat power or troops and equipment ashore. Surfaceborne waves land IAW the assault schedule. Airborne waves proceed IAW the HEALT. The assault schedule and the HEALT are discussed later in this chapter.

3.3.5.2 On-Call Waves

On-call waves consist of elements the LF expects to need ashore early in the operation; however, their time and place of landing cannot be accurately predetermined. They are subject to immediate or emergency call and are positioned to be readily available after the specific hour on D-day at which landing troops land on the beach or LZ (H-hour). On-call waves consist of serialized combat units, CS units, and CSS, and are requested by tactical commanders ashore through the TACLOG group. Airborne on-call waves are positioned aboard ship, while surfaceborne on-call waves remain aboard ship or are positioned at the LD for displacement landing craft or at the CLA for LCAC. Because the units in on-call waves have a high priority for landing, their number should be kept to a minimum consistent with transportation asset availability and expected requirements ashore. The landing of any other elements may be pre-empted to permit the landing of on-call waves. If adequate numbers of landing craft are unavailable, on-call serials may wait aboard ship pending landing craft availability. Surfaceborne on-call waves are listed in the assault schedule, and vertical lift on-call waves are listed in the HEALT.

3.3.5.3 Nonscheduled Units

Nonscheduled units are the remaining serialized units, with their CSS, whose landing is expected before the commencement of general unloading. The need for nonscheduled units ashore is usually not an immediate or emergency requirement. They are requested by tactical commanders through the TACLOG group via the supported unit TACNET or LFSP Control or Command Net, but are not normally landed or boated until requested. When the request is received by the PCO, the Bravo Net is used to direct the ship carrying the serial to prepare it for offload, and the PCO provides appropriate landing craft for moving the serial to the beach.

When the serial is boated, displacement landing craft are dispatched to the PCS. LCAC are dispatched to the PCS or LCAC control ship. When the PCO has verified the LFSP is prepared to receive the serial, landing craft are dispatched to the beach. Once started, landing of nonscheduled units may be interrupted to permit landing of on-call waves, pre-positioned emergency supplies, or other selected supplies or equipment for which there is a greater requirement ashore. Nonscheduled units are generally moved ashore after completing scheduled landings by surface means, not via vertical lift aircraft.

Each supporting LF commander prepares a landing sequence table that prioritizes the landing order for nonscheduled units. If the landing plan calls for landing CSS, Navy, or aviation elements across a beach before the nonscheduled units have finished landing, the LF commander informs the GCE commander of these units and their priority so they can be listed in the GCE commander's landing sequence table and forwarded to the LF commander for consolidation. The LF commander's staff then prepares an LF sequence table that consolidates all nonscheduled units.

3.3.5.4 Prepositioned Emergency Supplies

Prepositioned emergency supplies are designated by the LF commander to meet expected critical needs for CSS replenishment early in the ship-to-shore movement. These serialized supplies, available for immediate delivery ashore, are organized into floating dumps and prestaged airlifted supplies.

3.3.5.4.1 Floating Dumps

Floating dumps are emergency supplies such as ammunition, water, lubricants, etc., preloaded in landing craft or AAVs. Floating dumps minimize the time that critically needed supplies can be transported to the beach. They are located in the vicinity of the appropriate control officer and are landed when requested by a tactical commander through the TACLOG group. The number and types of landing craft or AAVs used as floating dumps are specified in the landing craft employment plan, and landing craft and amphibious vehicle assignment table. Floating dumps report to PCS and are positioned per the assault wave diagram. Due to the limited numbers of landing craft or AAVs, floating dumps are not typically required or available in MEU(SOC)-sized operations.

3.3.5.4.2 Prestaged Airlifted Supplies

Similar in purpose to floating dumps, prestaged airlifted supplies are positioned aboard air-capable ships. These serialized supplies can be delivered to either airborne or surfaceborne units. They are requested by a tactical commander ashore through the TACLOG group, and are listed in the HWSAT. A heliteam is analogous to a boat team.

3.3.5.5 Remaining Landing Force Supplies

Remaining LF supplies are serialized, and consist of replenishment supplies and equipment not included in a unit commander's basic loads, floating dumps, or prestaged airlifted supplies. In MEB- /MEF-sized operations, these supplies are listed in embarkation documents and constitute the major portion of CSS transported into the landing area with the AE or AFOE. Certain supplies are selectively offloaded to maintain adequate replenishment levels ashore; however, the bulk of the remaining LF supplies are landed during general unloading. These supplies are generally moved ashore via surface means, not via air.

3.3.5.6 General Unloading

The general unloading period is that part of the ship-to-shore movement in which unloading is primarily logistic in character, and emphasizes speed and volume of unloading operations. It encompasses the debarkation of LF units, supplies, and equipment from ships as rapidly as facilities on the beach permit. It is usually initiated on the recommendation of the LF commander and proceeds without regard to class, type, or priority of cargo, as permitted by the cargo-handling facilities ashore. General unloading begins when:

1. The progress with the CONOPS ashore is favorable, and permits discontinuing the structurally controlled movement of units and supplies ashore.

2. Sufficient quantities of all classes of supplies are already ashore.

3. BSAs are established.

4. Adequate security exists for supply installations ashore.

3.3.5.6.1 Landing Craft Employment for General Unloading

When general unloading commences, all available landing craft are employed. Landing craft previously ordered to load nonscheduled units complete those assignments, and if so designated, floating dumps are directed to the beach to offload.

3.3.5.6.2 Control

Control during general unloading is delegated to each ship with DIRLAUTH with the PCO for the most efficient use of landing craft. Each ship advises the PCO hourly on offload progress by reporting the percent of PCVT operations (PCVT report).

3.3.5.7 Offloading the Assault Follow-On Echelon

The Navy control group organizes and conducts the offload of AFOE shipping. NTRP 3-02.1.2; NWP 3-02.21; and NTTP 3-02.3, Maritime Prepositioning Ship Operations, contain additional information on AFOE offloading and operations.

3.4 PREPARATION AND PROMULGATION OF KEY DOCUMENTS

In ship-to-shore movement, the landing plan is composed of certain specific documents that present, in detail, the numbers of landing craft, aircraft, and other surface craft available for use and the exact personnel and equipment that will be loaded on each, along with embarkation and landing times. Some of these documents are applicable to amphibious operations of any size. Figure 3-2 lists landing plan documents. Figure 3-3 illustrates the preparation sequence and relationship between each document prepared by the ATF.

While several of the documents listed in Figure 3-2 are more applicable to MEB- /MEF-sized amphibious operations, all are valuable tools for any size mission. The creation of and details in these documents are dependent on the landing plans, CONOPS ashore, and the guidance provided by the ATF and LF commanders.

3.4.1 Documents Prepared by the Navy

Some or all of these documents may be included in the OPORD or OPTASK AMPHIB.

ATF Commander's Responsibility	LF Commander's Responsibility
Naval Landing Plan	Landing Force Landing Plan
Landing Craft Availability Plan	Amphibious Vehicle Availability Table
Landing Craft Employment Plan	Landing Craft and Amphibious Vehicle Assignment Table
Debarkation Schedule	Landing Diagram
Ship's Diagram	Landing Force Serial Assignment Table
Pontoon Causeway Plan	Landing Priority Table
Unloaded Plan	Landing Force Sequence Table
Approach Schedule	Assault Schedule
Assault Wave Diagram	Amphibious Vehicle Employment Plan
Landing Area Diagram	Helicopter Availability Table
Transport Area Diagram	Heliteam Wave and Serial Assignment Table
Beach Approach Diagram	Helicopter Enplaning Schedule
Sea Echelon Area	Helicopter Landing Diagram
Landing Control Plan	Helicopter Employment and Assault Landing Table
Medical Regulating Plan	Ground Combat Element Landing Plan
Amphibious Assault Bulk Fuel System and Offshore Petroleum Discharge System Plan	Consolidated Landing and Approach Plan
	Aviation Combat Element and Landing Force Aviation Landing Plan

Figure 3-2. Landing Plan Documents

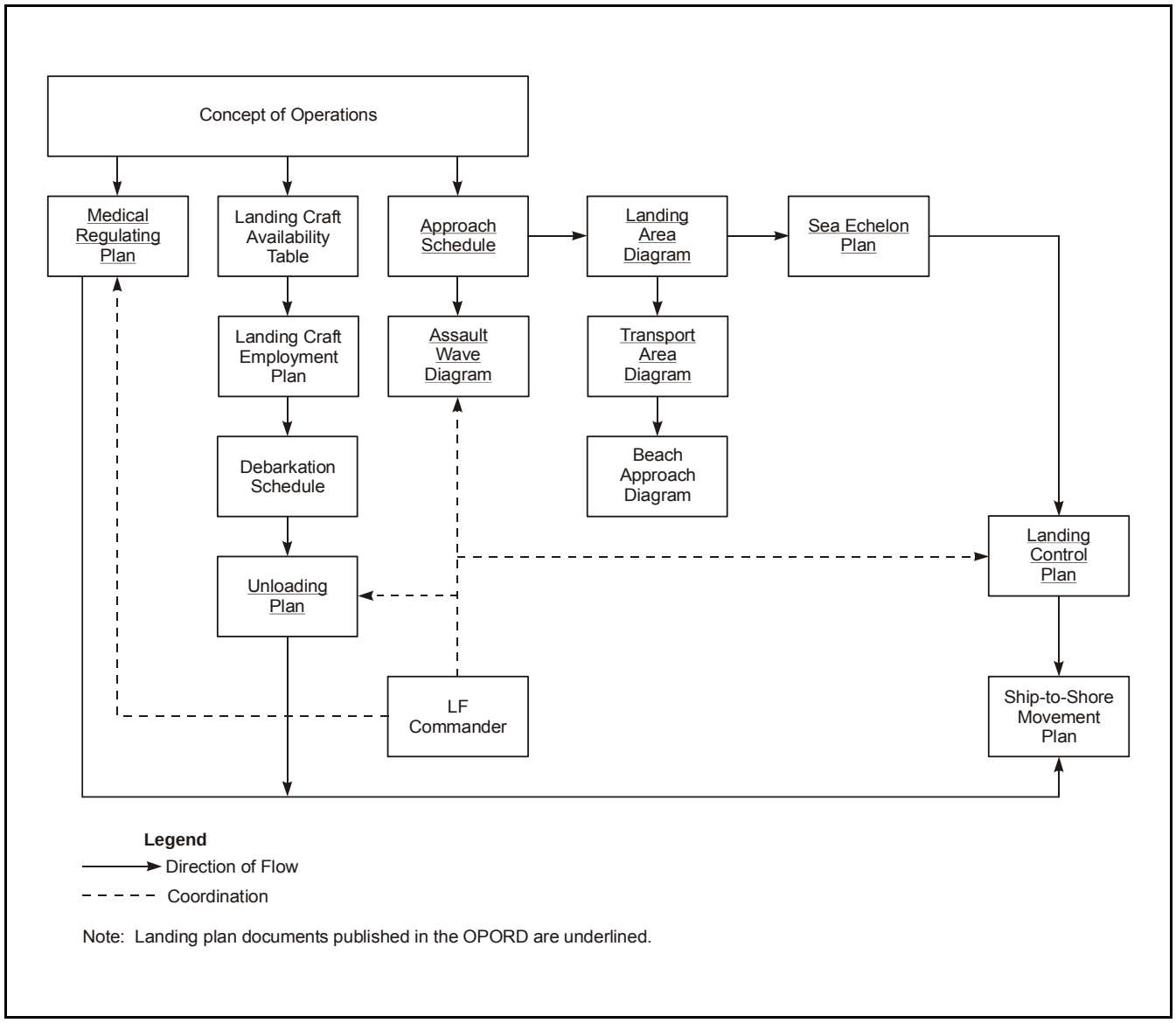


Figure 3-3. Navy Planning Responsibilities and Sequence

3.4.1.1 Naval Landing Plan

The naval landing plan organizes the landing area to facilitate the conduct and control of ship-to-shore movement, offload the LF, and provide for MEDREG. It incorporates the unloading, landing control, MEDREG plans, and other naval documents prepared to support LF planning. Figure 3-4 contains the format for the naval landing plan.

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101200 March 20___

Appendix 3 (Ship-to-Shore Movement) to Annex R (Amphibious Operations) to CTF XX OPORD 1 – { }

Ref: (a) INITIATING DIRECTIVE DTG

(b) CTF XX OPORD 1 – { }

NTTP 3-02.1M/MCWP 3-51.5

Time Zone: H

1. Amphibious operations will be conducted as directed by Ref (a). D-day is 04 June 20XX. H- and L- hours are 040430H June 20XX. The LF landing plan is in Ref (b).

2. Advance force operations will be conducted per Appendix 1. The AF will enter the AOA at 032100H and proceed to the designated transport area.

3. Surfaceborne and airborne assaults will be conducted at Red and Green beaches and LZs Hawk and Falcon. AF movement to the transport area and the conduct of amphibious operations to achieve AF objectives will be per this appendix and Ref (b). Tab A provides detailed instructions for the Navy control and offload plans. Ref (b) is the LF's landing plan, and when combined with these instructions, constitutes the AF landing plan.

4. Force protection measures in the AOA will be established per Appendix 2.

5. Re-embarkation plan is in Appendix 5.

6. Supporting arms will be conducted per Appendix 7.

7. Medical regulating planning will be carried out per JP 3-02 and JP 3-02.1.

David Jones
Captain, USN
Chief of Staff

Tabs:

A. Surfaceborne ship-to-shore movement
B. Airborne ship-to-shore movement

Distribution: Per Annex Z

Page Number **CLASSIFICATION**

Figure 3-4. Example of a Naval Landing Plan Format (Sheet 1 of 2)

Representative Enclosures to Tabs A and B

Tab A

Enclosure (1) Unloading Plan
(2) Approach Schedule
(3) Assault Wave Diagram
(4) Assault Area Diagram
(5) Transport Area Diagram
(6) Beach Approach Diagram
(7) Sea Echelon Plan
(8) Landing Control Plan
(9) Alternate Landing Plan

Tab B

Enclosure (1) Helicopter Landing Diagram
(2) HEALT

Figure 3-4. Example of a Naval Landing Plan Format (Sheet 2 of 2)

3.4.1.2 Landing Craft Availability Table

This table lists the type and number of landing craft available from each amphibious ship, and is categorized by total landing craft for naval and LF use. The landing craft availability table is the basis for landing craft assignment for ship-to-shore movement. It is prepared by the CCO. An example of the landing craft availability table is depicted in Appendix J, Figure J-1.

3.4.1.3 Landing Craft Employment Plan

The landing craft employment plan is prepared by the CCO in conjunction with the NBG support element OIC. It specifies the number and type of landing craft, parent ship, ship to which craft will report, time to report to that ship, and the time period the craft will be attached. It allocates landing craft for pre-H-hour transfers as well as scheduled and on-call waves. Figure J-2 depicts an example of this document.

3.4.1.4 Debarkation Schedule

The debarkation schedule provides for the timely and orderly debarkation of troops and equipment, and emergency supplies for surfaceborne ship-to-shore movement. It is dependent on the designated H-hour, and is approved jointly by the CO of each ship and the senior operational troop commander or COT. It lists the type of landing craft assigned, and the order of troop and equipment debarkation from the ship's well deck. It is supplemented by a ship's diagram. Figure J-3 provides a debarkation schedule example.

3.4.1.5 Unloading Plan

The unloading plan establishes the sequence and designates the means for offloading the LF. It consists of the landing craft availability table and landing craft employment plan. The unloading plan must be closely coordinated with the amphibious vehicle availability table (Paragraph 3.4.2.2), the amphibious vehicle

employment plan (Paragraph 3.4.2.9), the debarkation schedule, and the HWSAT (Paragraph 3.4.2.11) prepared by the LF.

3.4.1.6 Approach Schedule

The approach schedule indicates, for each scheduled wave, the times of arrival/departure from various points, including parent ship, rendezvous area, LD, CHA, CPs, and the landing beach. It provides wave numbers, courses and speeds to be followed by landing craft, and control ships' hull numbers. This schedule is prepared by the PCO, and submitted via the CCO to the ATF and LF commanders for consolidation and coordination with the overall ship-to-shore movement. An example of an approach schedule is depicted in Figure J-4.

3.4.1.7 Assault Wave Diagram

The assault wave diagram displays scheduled waves, landing craft, control ships (if assigned), on-call waves, and floating dumps as they appear at H-hour. This diagram is prepared by the PCO based on wave compositions in the landing diagram (Paragraph 3.4.2.4). The assault wave diagram presents a picture of the boat and LCAC groups, control ships' positions for each landing beach, and CLZs. An example of an assault wave diagram is depicted in Figure J-5.

3.4.1.8 Landing Area Diagram

The landing area diagram provides the overall picture of the seaward approaches in the landing area and overlays an appropriate scale chart. It graphically depicts the landing area's most important details, such as beach designations, boat lanes, the LD, amphibious vehicle and CLAs, transport areas, and FSAs in the immediate vicinity of the boat and LCAC transit lanes. This diagram is prepared by the CCO. Figure J-6 provides an example of a landing area diagram.

3.4.1.9 Transport Area Diagram

The transport area diagram overlays an appropriate scale chart that shows the area from the beach to 1,000 yards seaward of the outermost transport area anchorage or underway sector. The diagram is prepared by the CCO and provides a detailed picture of the transport area. An example of this diagram is depicted in Figure J-7. It includes the following:

1. Transport area(s) and assignment of ships to anchorage or underway sectors
2. HTAs
3. Control ships' positions (if assigned)
4. Boat and approach lanes
5. CLA, CDP, transit lanes, CPs, and CPPs
6. LDs
7. Amphibious vehicle launching area/AAV underway launch track

8. Causeway operating areas (MPF operations)
9. Designation of landing beaches
10. The following distances:
    a. From beach to center of transport area
    b. From beach to LD
    c. From approach lane marker ships to LD
    d. Width of beaches
11. The following courses (true and magnetic):
    a. From LD to beaches
    b. From approach lane marker ships to LD (if/as available or required).

3.4.1.10 Beach Approach Diagram

The beach approach diagram is typically used during MEB- /MEF-sized operations. The beach approach diagram is an overlay for a large-scale chart extending from each colored beach seaward to 500 yards beyond the LD. It can be modified for use in smaller scale operations. This diagram shows the position of control ships and landing craft in the vicinity of the LD after the last scheduled wave has landed, and identifies boat lanes for on-call waves and subsequent serials. An example of a beach approach diagram is shown in Figure J-8. It is prepared by the PCO and includes the following:

1. Designation (color and number) and dimensions of the colored beach
2. LD
3. Distance from beach to LD
4. Position of:
    a. PCS and SCSs
    b. Control boats (if/as available or required)
    c. Medical boats
    d. Salvage boats
    e. Traffic control boats (as available or required)
5. Return boat lanes.

3.4.1.11 Sea Echelon Plan

A sea echelon is a portion of the amphibious shipping that withdraws from, or remains out of, the transport area during a landing or an operation. It operates in designated areas to seaward in an on-call or unscheduled status. The sea echelon plan provides for the dispersion of amphibious shipping and establishes the sea echelon area. It also organizes the landing area to reduce the minehunting and minesweeping effort, and promulgates the priority sequence table (Paragraph 3.4.1.11.2). The sea echelon plan is the responsibility of the ATF commander; however, because of its influence on embarkation and the landing plan, the decision to employ a sea echelon is reached jointly by the ATF and LF commander early in the operation's planning phase. An example of a sea echelon area is depicted in Figure J-9.

3.4.1.11.1 Considerations for Using a Sea Echelon

In arriving at a decision regarding the use of a sea echelon, the commanders balance the desirable aspects of dispersion and mobility against the limitations they may impose on the landing plan. Factors that might drive amphibious ships further offshore and into a more widely dispersed formation include:

1. Range of shore-based weapons or missiles
2. Potential for CBR weapons attack
3. Littoral mining threat
4. Likelihood of attack by swimmers, small boats, or light aircraft
5. Increase in the element of surprise by remaining beyond the adversary's electronic sensor range.

The following factors are balanced against others that dictate moving amphibious ships closer to the beach:

1. Achieving air supremacy, and surface and undersea superiority in the AOA
2. The delay associated with longer transit distances to the beach, thereby, slowing the rapid buildup of combat power, equipment, and supplies ashore
3. The possibility of an insufficient number of naval defensive forces that might dictate concentrating the ships to reduce screen requirements.

3.4.1.11.2 Navy Sea Echelon Responsibilities

Employing a sea echelon does not alter the ATF commander's doctrinal or tactical responsibility to land and support the LF. The sea echelon plan should not be so restrictive that it reduces the LF's tactical effectiveness or jeopardizes mission accomplishment.

Plan development includes:

1. Selecting the sea echelon area
2. Organizing the sea echelon area into operating areas

3. Phasing amphibious shipping into the transport area

4. Providing C4I for ship-to-shore movement

5. Incorporating FP measures

6. Providing an alternate plan to land and reinforce the LF.

A sea echelon plan is developed to regulate and efficiently control the movement of amphibious shipping and minimize connectivity requirements. This plan includes:

1. The composition of the sea echelon

2. A sea echelon overlay

3. A priority sequence table that delineates the order that ships from the sea echelon will enter the transport area to land scheduled waves, and conduct the initial and general unloading phases

4. The procedure for requesting ships out of priority sequence

5. General instructions, such as defining traffic patterns for entering and departing the transport area, unloading instructions, regulating points, etc.

3.4.1.11.3 Landing Force Sea Echelon Responsibilities

The LF commander develops a CONOPS ashore that is compatible with the employment of amphibious ships in the sea echelon. During ship-to-shore movement, the ATF and LF commanders work together; however, the LF commander keeps commanders advised regarding logistic problems associated with the time delay inherent in the sea echelon concept. The time delay is the time involved from the moment a request for a ship out of sequence is received until that ship arrives in the transport area.

3.4.1.11.4 The Sea Echelon Commander

The ATF commander usually delegates control of the sea echelon to a subordinate commander. The sea echelon commander exercises movement control over shipping in the sea echelon area, but coordinates with the CCO and TACLOG group to help ensure ship movements conform to the priority sequence table. This commander also ensures ships in an on-call status or with nonscheduled units embarked can be moved into the transport area in the shortest possible time.

3.4.1.12 Landing Control Plan

The landing control plan organizes the landing area into operating and control areas to:

1. Regulate and deconflict the movements of amphibious shipping

2. Launch landing craft, AAVs, and aircraft

3. Establish control areas, points, and stations for the surfaceborne and airborne ship-to-shore movements

4. Provide operating areas for supporting forces protecting the landing area.

The landing control plan consists of the approach schedule, assault wave diagram, landing area diagram, transport area diagram, beach approach diagram, HEALT, causeway plan (if MPF operations are to be conducted), and sea echelon plan (if applicable). These documents provide ATF commanders and the TACGRU or TACRON with the instructions necessary to control surfaceborne and airborne ship-to-shore movement.

3.4.1.13 Medical Planning

In consultation with the LF surgeon, the ATF surgeon provides the OPLAN/OPORD medical section (Annex Q, Medical Services) to the ATF to ensure mutual support and fulfillment of medical requirements to include patient movement. Concurrent medical planning is essential across all phases of an operation to address HSS considerations in a timely, effective, and coordinated manner, and to ensure adequate and sustainable health care in theater. HSS is discussed further in JP 3-02.

3.4.2 Documents Prepared by the Landing Force

LF-prepared documents are included in the LF commander's OPORD (per MCWP 5-1: Tab C (Landing Plan) to Appendix 14 (Amphibious Operations) to Annex C (Operations)) or issued as supplements to that OPORD. Figures 3-5 and 3-6 depict LF documents for surfaceborne and airborne operations, respectively.

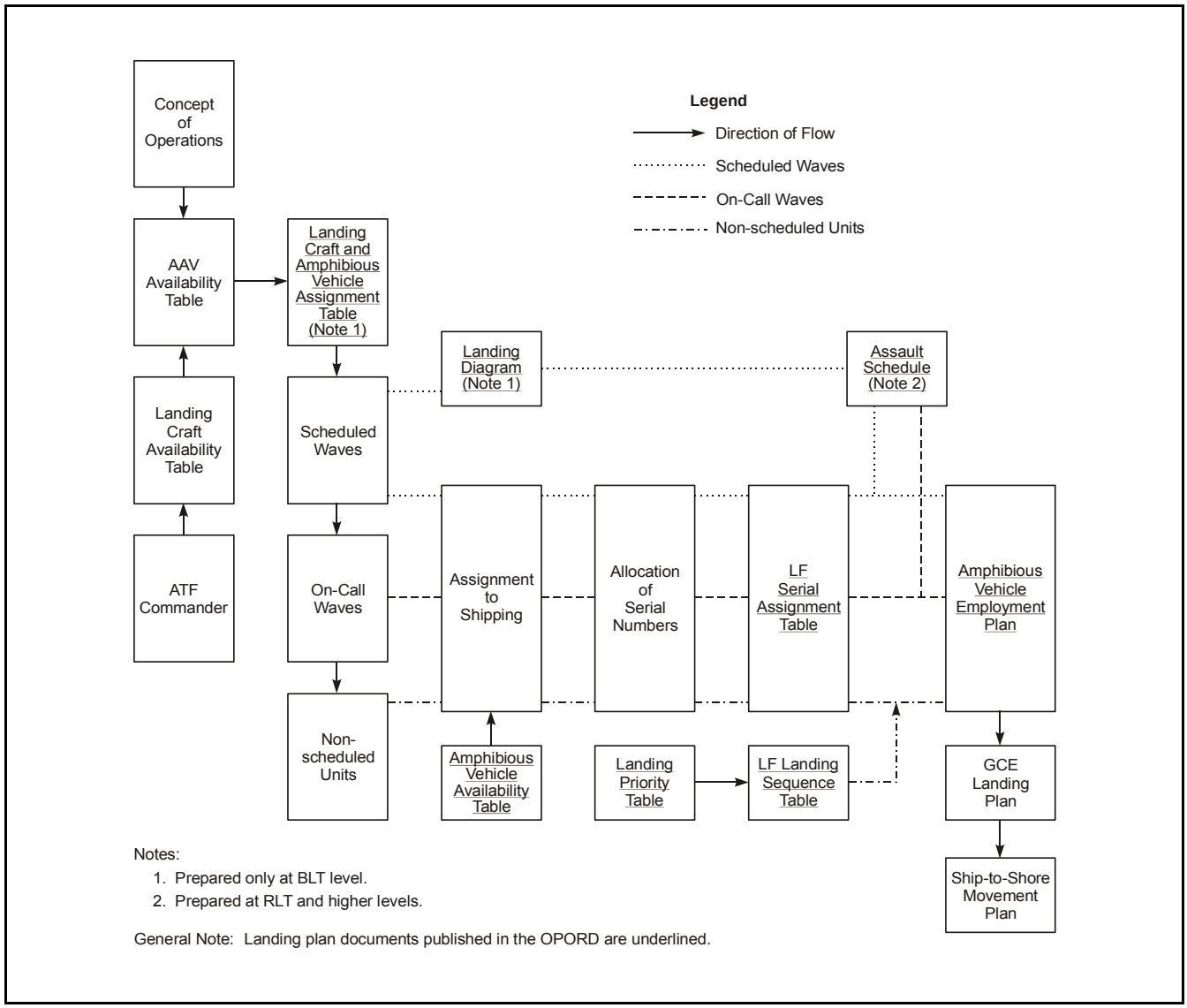


Figure 3-5. Landing Force Planning Responsibilities and Sequence (Surfaceborne)

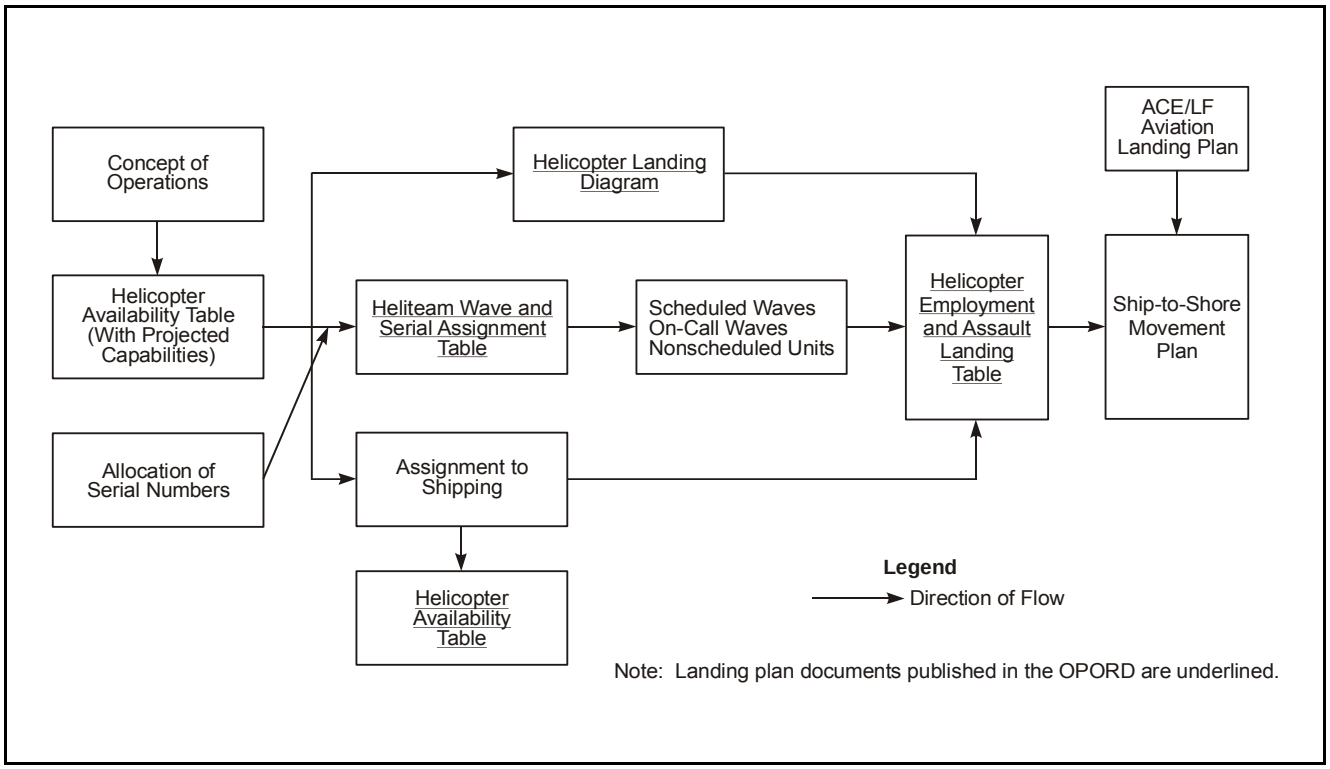


Figure 3-6. Landing Force Planning Responsibilities and Sequence (Airborne)

3.4.2.1 The Landing Force Landing Plan

The LF landing plan is the compilation of detailed plans prepared by the LF. It designates the forces going ashore, and promulgates the means, organization, sequence, and landing priorities. Figure 3-7 shows the format for this plan.

In addition, the LF landing plan:

1. Allocates blocks of serial numbers to subordinate commands
2. Correlates the landing sequence for units not landed with the GCE, but landing prior to general unloading
3. Coordinates GCE landing plans.

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Appendix 3 (Ship-to-Shore Movement) to Annex R (Amphibious Operations) to CTF XX OPORD 1 – { }

Ref: (a) NTTP 3-02.1M/MCWP 3-31.5
(b) CTF XX OPORD 1 – { }
(c) MARFOR Order [ ] (CSS SOP)

Time Zone: H

1. Ship-to-shore movement operations will be conducted IAW Annex C to CTF XX OPORD 1 – { }. Tabs A and B of this appendix, and References (a) through (c) provide detailed procedures or amplifying instructions.
2. Advance force operations are IAW Appendix 1 to Annex R of Reference (b).
3. FP measures in the AO are IAW Appendix 2 to Annex R of Reference (b).
4. Re-embarkation plan is IAW Appendix 5 to Annex R of Reference (b).
5. LF CONOPS ashore is IAW Appendix 6 to Annex R of this OPORD.
6. Supporting arms are IAW Appendix 7 to Annex R of Reference (c).

BY COMMAND OF LTGEN HATFIELD

I. M. MCCOY
Brigadier General
U.S. Marine Corps

Tabs:

A. Surfaceborne ship-to-shore movement

B. Airborne ship-to-shore movement

Distribution: IAW Annex Z to CTF XX OPORD 1 – [ ]

Page Number **CLASSIFICATION**

Figure 3-7. Example of a Landing Force Landing Plan Format (Sheet 1 of 2)

Representative Enclosures to Tabs A and B	
Tab A	
Enclosure	(1) Landing Craft and Amphibious Vehicle Assignment Table
	(2) Landing Diagram
	(3) Amphibious Vehicle Availability Table
	(4) Assault Schedule
	(5) Landing Priority Table
	(6) Landing Sequence Table
	(7) Landing Craft and Amphibious Vehicle Employment Plan
	(8) Division/Regimental/Battalion Landing Plan
	(9) Shore Party Plan
	(10) Alternate Landing Plan
Tab B	
Enclosure	(1) Heliteam Wave and Serial Assignment Table
	(2) Helicopter Availability Table
	(3) Helicopter Landing Diagram
	(4) HEALT
	(5) ACE/LF Aviation Landing Plan

Figure 3-7. Example of a Landing Force Landing Plan Format (Sheet 2 of 2)

3.4.2.2 Amphibious Vehicle Availability Table

The amphibious vehicle availability table is prepared by a GCE representative and lists the number and type of amphibious vehicles available for landing, the LF units embarked in them, and the ships carrying them. An example of this table is depicted in Figure J-10.

3.4.2.3 Landing Craft and Amphibious Vehicle Assignment Table

The landing craft and amphibious vehicle assignment table organizes the LF surfaceborne AE into boat teams; and assigns boat teams to scheduled waves, on-call waves, or nonscheduled units. It also lists the LF units assigned to boat teams, shows the precise position of the boat teams in the assault waves, and if necessary, includes instructions for floating dumps. It is typically used for MEB- /MEF-sized operations, but can be modified for smaller scale operations. This document and the debarkation schedule provide ships' COs with the information needed for debarking troops. The table is prepared by a GCE representative and promulgated concurrently with the landing diagram. An example of this table is depicted in Figure J-11.

3.4.2.3.1 Tactical Integrity

Boat teams are assigned positions in waves to maintain tactical integrity. For example, a rifle squad and its equipment are assigned in the wave formation in proper relation to other squads of the platoon to facilitate unit employment on landing. Nonscheduled units are also boated tactically.

3.4.2.3.2 Guidelines for Assignment to Boat Teams

HQ units and attached or supporting troops (such as FOs, naval gunfire spotters, and communications personnel) are assigned to landing craft or amphibious vehicles carrying the unit to which they are attached or directly support. If these units are assigned to separate craft, the craft is positioned in the wave formation in a manner that facilitates tactical integrity upon landing.

The priority for landing craft and amphibious vehicle assignment in a MEB- /MEF-sized operation is assault units, support units, and reserve units. In MEU(SOC)-sized operations, the priority order is similar; however, a MEU(SOC) does not routinely conduct amphibious assault operations. The initial LF units ashore are those carrying out SOC mission actions.

3.4.2.4 Landing Diagram

The landing diagram depicts the tactical deployment of boat teams in scheduled waves. It provides the wave composition that shows AAVs, landing craft, and boat teams, as well as touchdown times for landing beaches and CLZs. This diagram is prepared by a GCE representative, promulgated concurrently with the landing craft and amphibious vehicle assignment table, and distributed to ATF commanders. An example of a landing diagram is provided in Figure J-12.

3.4.2.5 Landing Force Serial Assignment Table

The LF SAT lists in numerical order the serial numbers of units landed prior to general unloading. It is a ready reference for the composition of each unit assigned a serial number. An example is depicted in Figure J-13.

3.4.2.5.1 Serial Numbers

A serial is a grouping of LF personnel and equipment that originates from the same ship and, for tactical or logistical reasons, will land on a specified beach, CLZ, or LZ at the same time. A reference number is assigned to each serial. A serial number does not preclude the use of code names, designations, or unit titles when expedient, and it does not prescribe any priority in landing. PCS, SCS, LCAC, LCAC control ship, TACLOG detachments, and each AATCC maintain a log that includes serial number, time requested, time dispatched, and time of arrival at the beach or LZ.

3.4.2.5.2 Allocation and Assignment of Serial Numbers

The allocation of block serial numbers to units is based on the administrative organization, and the assignment of individual serial numbers is based on the tactical organization for landing. This allows serial numbers to be issued early in the planning phase before the tactical organization has been determined. Changes can be made at the appropriate level as planning progresses. Allocation begins at the highest echelon. Each unit then allocates a consecutive portion of its block to subordinate units, and allocation continues until each element within the LF has a block of consecutive numbers.

After subordinate units have prepared SATs, they are forwarded to the next higher echelon for consolidation. The serial tables are ultimately forwarded to the LF commander's staff where the LF SAT is prepared.

Information contained in the LF SAT is duplicated in other LF documents that list serials in priority order for landing, rather than in numerical sequence. An example of block serial number allocation is depicted in Figure J-14.

3.4.2.6 Landing Priority Table

The landing priority table is a worksheet used by the LF commander and planners to show the planned buildup of forces ashore. It is based on the LF CONOPS ashore and provides the basis for the phased deployment of LF units ashore. It lists major units to be landed in priority order, the landing day, the landing hour for smaller scale operations, and the designated beaches or LZs, if known. An example of a landing priority table is depicted in Figure J-15.

3.4.2.7 Landing Force Landing Sequence Table

The LF landing sequence table is a complete listing of the estimated landing sequence of nonscheduled units (including CS, CSS, and aviation units). It is the principal document used by control agencies in directing the ship-to-shore movement of these units. The LF commander prepares this table. Subordinate commanders extract pertinent sections of the table for their use. This table is the basis for developing embarkation and loading plans for unscheduled units. An example of an LF landing sequence table is depicted in Figure J-16.

3.4.2.8 Assault and Other Landing Schedule

This schedule provides the formation, composition, and timing of scheduled and on-call waves. When preparing this schedule, the GCE commander considers subordinate commanders' recommendations regarding numbers of waves directed to designated beaches, as well as numbers and types of amphibious vehicles and landing craft in each wave. An example of an assault schedule is depicted in Figure J-17.

3.4.2.9 Amphibious Vehicle Employment Plan

This plan shows the planned employment of AAVs and LARC Vs in the operation, including their employment after arrival at the beach. The GCE commander considers subordinate commanders' recommendations when preparing the plan in addition to information contained in the landing diagram and assault schedule. An example of an amphibious vehicle employment plan is depicted in Figure J-18.

3.4.2.10 Helicopter Availability Table

The helicopter availability table shows the number of helicopters available for the airborne ship-to-shore movement. It lists helicopter units and their call signs, the number of aircraft available for the initial and subsequent trips, aircraft model, parent aircraft transport ship, maximum deck launch spots available on each platform, and tentative load capacity. The table is prepared by an ACE representative and pertains only to D-day operations. An example of this table is depicted in Figure J-19.

3.4.2.11 Heliteam Wave and Serial Assignment Table

The HWSAT specifies the troop units, supplies, and equipment to be loaded into each aircraft. It identifies each heliteam by serial number with the wave number and aircraft position in the wave.

The table contains a weight column to assist planners in ensuring the serial does not exceed the maximum aircraft lift capability. A helicopter boarding schedule is prepared to show each boarding station on the flight deck, the sequence and location for spotting aircraft on the flight deck, and the heliteam serials assigned to that boarding station. This table and schedule is prepared on each helicopter transport ship by the helicopterborne unit commander, is assisted by the AMC or AFL, and is coordinated with the ship's CO and aviation department. An example of this table is depicted in Figure J-20.

3.4.2.12 Helicopter Landing Diagram

The helicopter landing diagram graphically displays routes to and from LZs. It shows the HTA, RPs, DPs, approach and retirement routes, other CPs, LZs, and remarks for clarity. The diagram is prepared by a MAGTF CE representative in coordination with the HCS and the helicopter transport group/unit commander. The diagram is coordinated with Navy planners to ensure deconfliction with the surfaceborne assault and fire support planning. An example of a helicopter landing diagram is shown in Figure J-21.

3.4.2.13 Helicopter Employment and Assault Landing Table

The HEALT is a detailed plan for the movement of helicopterborne troops, equipment, and supplies. It is the landing timetable for the airborne ship-to-shore movement and specifies the assignment of serials to helicopters for scheduled on-call waves.

This document is the basis for preparing flight schedules and is used by the primary AATCC to coordinate helicopter movements. The HEALT is prepared by the MAGTF in coordination with the AATCC and other ATF planners to ensure deconfliction with any planned supporting fires. The LF commander publishes the approved diagram, and subordinate commanders publish extracts for their units. An example of a HEALT is depicted in Figure J-22.

3.4.2.14 Ground Combat Element Landing Plan

The GCE commander and the GCE staff conduct the majority of LF detailed planning for ship-to-shore movement. The ATF and LF commanders provide the following information to subordinate units before planning begins:

1. Combat, CS, CSS, LF aviation, and naval elements to be landed

2. Availability of landing craft, aircraft, and amphibious vehicles.

3.4.2.14.1 Contents of the Plan

The GCE landing plan contains a general description of the forces to be landed, the ship-to-shore control procedures, and the organization of the GCE TACLOG detachment. An example of the format for this document is shown in Figure J-23.

3.4.2.14.2 Assault and Other Landing Units

Pertinent information for landing subordinate units is furnished to those units by the GCE HQ. On the basis of this information, these units submit their landing plans to that HQ, and the GCE landing plan is published with the documents discussed in Paragraphs 3.4.2.5 through 3.4.2.13.

3.4.2.14.3 Landing Force Reserve

Reserve units prepare a landing plan in the same manner as a primary landing unit. However, if the entire LF reserve is a nonscheduled unit, its deployment is prescribed in the LF landing sequence table. SATs are prepared for units landing prior to general unloading.

3.4.2.15 Regimental Landing Plan

The RLT commander considers the tactical recommendations of BLT commanders and submits a recommended landing plan to the GCE commander. After the GCE landing plan is published, the RLT commander extracts pertinent information and publishes it as the RLT landing plan. The documents comprising the RLT landing plan are the same as those in the GCE landing plan.

3.4.2.15.1 Other Regiments

Reserve RLTs prepare landing plans according to the landing category they are assigned. Units to be landed prior to general unloading prepare SATs. Landing of nonscheduled units is prescribed in the LF landing sequence table. Elements of regiments and reserve RLTs to be landed in on-call waves appear in the assault schedule, landing diagram, or HEALT.

3.4.2.16 Battalion Landing Plan

BLT commanders prepare the following documents as appropriate:

1. HEALT.
2. HWSAT.
3. Landing craft and amphibious vehicle assignment table.
4. Landing diagram.
5. Consolidated landing and approach plan prepared jointly by the BLT and PCO in lieu of using separate employment plans and approach schedules. An example of this plan is contained in Figure J-24.
6. Debarkation schedule prepared jointly by the ship's CO and the COT.

3.4.2.16.1 Battalions Other Than Infantry

CS unit commanders prepare the same documents as the BLT for orderly debarkation and movement ashore.

3.4.2.17 Aviation Combat Element/Landing Force Aviation Landing Plan

In most operations, the ACE operates from the sea base comprised of the amphibious ships. The tactical situation or size of the operation may require that ACE units operate simultaneously from the sea base and land-based sites to support the MAGTF. In this case, the ACE/LF aviation landing plan outlines the ACE commander's plans for establishing aviation units ashore by air and surface means. An example of the ACE/LF aviation plan format is depicted in Figure J-26.

3.4.2.17.1 Contents

This plan provides the echelon plan and landing sequence of aviation units established ashore. It contains:

1. Detailed landing documents for air elements that move ashore prior to general unloading

2. Ship-to-shore control provisions

3. Confirmation on fuel handling systems and landing PHIBCB elements to support aviation facilities ashore.

3.4.2.17.2 Composition of Echelons

For assault operations in which the ACE operates from ashore, elements of aviation ground support, TACAIR control support, and fixed-wing V/STOL and rotary-wing units comprise the first echelon, and are landed by vertical lift aircraft and landing craft. The second echelon deployed aboard MPF shipping or the T-AVBs may remain afloat or be transported ashore to support the MAGTF commander's CONOPS.

The initial echelon of the remaining ACE is comprised of personnel and heavy equipment for improving forward operating sites and performing sustained maintenance. This echelon may be surface- or air-lifted into the AOA and landed across beaches. The second echelon, comprising aircraft, pilots, and crews, is flown into the AOA when facilities are ready.

The LF aviation organization for landing may differ greatly from its task organization for combat operations because of the division combat, CS, and CSS into landing elements and the wide variation in time and method of landing these elements. The ACE/LF aviation landing plan provides for regrouping these elements into echelons, if necessary.

3.4.2.17.3 Scheduled, On-Call, and Nonscheduled Elements

The ACE/LF aviation plan lists air elements landed in scheduled and on-call waves, or as nonscheduled units. It also contains landing documents extracted from the GCE landing plans to describe the method and sequence for landing. These documents are:

1. Extracts from assault and other landing schedules, HEALTs, and HWSATs

2. SAT

3. Landing sequence table.

Serials not part of the LFSP that are to be landed in scheduled and on-call waves are submitted to the LF commander for coordination and approval. The GCE commander is then furnished the information to integrate air serials into the assault schedule. Nonscheduled units are serialized and incorporated into the LF landing sequence table.

TACPs, ASLTs, and HSTs are landed in scheduled waves. ASEs are usually landed in on-call waves and are shown in the GCE commander's assault schedule or HEALT. Other air elements landed early in the ship-to-shore movement are serialized and shown in the GCE and LF landing sequence tables as nonscheduled units.

3.4.2.17.4 Airfields, Fuel Handling Systems, and Engineering Operations

The priority of establishment and status of forward operating locations ashore determines when ACE units are scheduled to land. Information on when the facilities are expected to be ready is provided in the ACE/LF aviation landing plan and includes:

1. Time construction/engineering elements will land and commence work on expeditionary airfields, forward operating locations, or forward arming and FARPs

2. Estimated time landing heavy aviation equipment will be completed

3. Estimated time airfields will be operational

4. Estimated time a fuel delivery system from the beach to the aircraft operating sites will be operational.

3.5 ORGANIZATION OF THE SEA OPERATING AREAS

Sea, beach, and inland operating areas in the landing area are selected to meet tactical requirements and facilitate the control of ship-to-shore movement. Sea operating areas are established to minimize the possibility of interference by other amphibious units or support forces.

3.5.1 Ocean Operating Areas

The ATF commander or higher authority establishes ocean operating areas. They are located outside the landing area and may be located inside or outside the AOA. These areas, as depicted in Figure J-27, are:

1. Close support area. Supporting forces such as CSGs, surface combatants assigned to an ESG, and logistics groups operate in this area. As supporting forces, their actions are integrated and coordinated with the landing plan.

2. Distant retirement area. This area is divided into a number of operating areas for amphibious shipping to retire to in the event of heavy weather, hostile fire, or for protective dispersion.

3.5.2 Sea Areas in the Landing Area

The ATF commander establishes sea areas in the landing area and promulgates them in the OPORD or OPTASK AMPHIB.

3.5.2.1 Screening Areas

Supporting aircraft, ships, and submarines provide antiair, antisurface, and undersea warfare (AAW, ASUW, USW) protection for amphibious ships operating in screening areas. In larger scale operations or operations in which there is sufficient time, defensive minefields may also be employed to augment the protective screen.

These areas are not depicted in Figure J-27, because tactical, atmospheric, and oceanographic variables are considered when making station assignments.

3.5.2.2 Landing Area

The landing area is that part of the AOA in which landing operations are conducted. It includes the beach, transport areas, FSAs, the airspace for CAS, shaping operations, AD, assault support, aerial RECON, EW support, and the land included in the advance inland to the initial objective area.

The nature and size of the mission determine the number of landing areas. In larger scale operations, a transport group is normally formed to conduct landing operations at each landing area.

3.5.2.3 Transport Area

The transport area is an area assigned to a transport organization for the purpose of debarking troops and equipment. This area may be divided into an inner and outer area, and includes the HTA. A transport area supports one or more landing beaches or LZs depending on the distance between them.

3.5.2.3.1 Outer Transport Area

The OTA is an area located inside the screening area to which ships proceed upon entering the AOA. It is located sufficiently seaward of landing beaches to be beyond shore battery or antiship missile range. Ships involved in the operation remain underway in this area and may initiate OTH operations or be phased into the ITA for a nearshore assault.

3.5.2.3.2 Inner Transport Area

The ITA is an area located as close to the landing beach as depth of water, navigational hazards, boat traffic, and hostile action permit. Amphibious shipping moves into this area to expedite unloading.

3.5.2.3.3 Helicopter Transport Areas

HTAs for launching and recovering aircraft are located in the transport area. This area should provide ample maneuvering room for ships to maintain relative winds for rotary-wing and tilt-rotor operations.

3.5.2.4 Amphibious Vehicle Launching Areas

Amphibious vehicle launching areas are located as close to the AAV LD as possible to minimize AAV transit times.

3.5.2.5 Landing Craft Air Cushion Launch Area

CLAs are located in the transport area. Transit lanes connect the CLA and CLZ. In an OTH operation, CLAs may be located anywhere in the landing area.

3.5.2.6 Fire Support Areas

An FSA is an appropriate maneuver area assigned to NSFS ships by the naval force commander from which they can deliver gunfire or missile support to the LF in an amphibious operation. FSAs are located to provide optimum fields of fire; to be as nearshore as safe navigation permits; and to remain clear of boat, approach, and transit lanes.

3.5.2.7 Sea Echelon Area

The sea echelon area is an area seaward of a transport area in which amphibious ships operate for dispersion and mobility. A detailed discussion of the sea echelon concept is contained in Paragraphs 3.4.1.11 through 3.4.1.11.4.

3.6 ORGANIZATION OF THE BEACH AND INLAND AREAS

ATF and LF commanders and their planners jointly recommend to the JFC or RCC the battlespace geometry and AOA requirements. They also jointly select the location of beaches and inland operating areas, including the inland routes and CPs for aircraft and LCAC movement.

3.6.1 Combat Service Support Area

The CSSA is an area ashore, inland of the BSA that is organized to contain the necessary supplies, equipment, installations, and elements to provide the LF with CSS throughout the operation.

3.6.2 Beach Support Area

The BSA is the area to the rear of an LF or elements thereof, established and operated by the LFSP, that contains the facilities for unloading troops, and materiel and the support of the forces ashore. It includes facilities for evacuating wounded, EPWs, and captured materiel. BSAs are disestablished when the CSSA becomes operational.

A BMU directs the offload and onload of cargo operations from the 5-fathom curve to the HWM under the direction of the LFSP.

3.6.3 Landing Zone Support Area

An LZSA is established by the LFSP when a planned CSS buildup in an LZ commences. A planned CSS buildup is discussed in Appendix H.

3.6.4 Helicopter Inland Areas

Helicopter inland areas, helicopter routes, and HLZs are discussed in Chapter 5 and depicted in Figure J-21.

3.6.5 Landing Craft Air Cushion Inland Areas

LCAC inland routes and CLZs are discussed in Chapter 4.

3.6.6 Forward Arming and Refueling Point

A FARP is a temporary site organized, equipped, and deployed within the AO by the ACE commander to rapidly generate sorties. The FARP, when employed, reduces transit time and permits programmed sorties to rapidly refuel and rearm.

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CHAPTER 4

Conducting Surfaceborne Ship-to-Shore Movement

4.1 PURPOSE

This chapter discusses the doctrine, methodology, and C2 for LF debarkation and surfaceborne ship-to-shore movement.

4.2 BACKGROUND

This chapter discusses the range of options available to ATF and LF commanders for initiating LF debarkation and subsequent ship-to-shore movement. In a hostile or less than benign environment, debarkation is accomplished underway; however, should conditions in the AO permit, an administrative offload can be effected from ships at anchor or lying to. This chapter also covers planning considerations that should be taken into account in preparation for ship-to-shore operations.

4.2.1 Final Preparations and Approach

Prior to entering the landing area, the ATF and LF commanders and their staffs evaluate the readiness of their units to commence amphibious operations. SEALs or Marine RECON teams provide SUROBS reports and final beach survey reports that are vital in determining whether conditions are suitable for surfaceborne ship-to-shore movement. A key factor in determining the suitability of environmental conditions for surfaceborne ship-to-shore operations being carried out by displacement landing craft is the calculation of the MSI. The Joint Surf Manual, COMNAVSURFLANT/COMNAVSURFPACINST 3840.1 (series) provides the methodology for calculating the MSI and displacement craft operating parameters. Based on this information, the landing plan is confirmed, and the ATF commander initiates the landing by executing the signal "land the landing force." H-hour is the time at which initial LF troops land on the beach or in CLZs, HLZs, or VLZs. The time between "land the landing force" and H-hour depends on several factors, such as size and complexity of the ship-to-shore movement, numbers of pre-H-hour transfers, method of debarkation (underway or static), and distances from launch points to the landing beach.

4.2.1.1 Positioning

When the transport group arrives in the landing area, the ships complete preparatory actions (usually without signal). The landing control plan organizes the landing area and specifies the level of control for ship-to-shore movement. The number of control stations for underway launches during ship-to-shore movement has been reduced, because launch ships' navigation equipment can accurately position landing craft and assault vehicles closer to the LD. This reduces or eliminates requirements for approach ships, and PCSs and SCSs. The installation of GPS navigation equipment in landing craft has further reduced the requirement for PCS and LCAC control ship to control scheduled waves.

4.2.1.2 Landing Sequence

Surfaceborne troops and equipment are arranged in the categories delineated in Paragraph 3.3.5. When the tactical situation ashore permits, the LF commander requests the commencement of general unloading.

4.3 PLANNING CONSIDERATIONS AND DECISIONS

The overall amphibious planning process is covered in Chapter 3 and JP 3-02; however, some additional discussion of the primary considerations and decisions made by ATF and LF commanders and staff planners for surfaceborne ship-to-shore movement is required.

4.3.1 Planning Considerations

The LF CONOPS ashore determines the major aspects of ship-to-shore movement. Specific planning considerations for surfaceborne ship-to-shore movement are:

1. OTH or nearshore launch
2 Surface craft availability and crew status
3. Location of landing beaches and CLZs
4. Only surfaceborne or combined with airborne
5. Amphibious and combatant ship dispersion
6. Composition and timing for displacement and nondisplacement landing craft
7. LF tactical integrity, and achieving the degree of troop and equipment concentration or dispersion
8. Threats to and protection available to the AF
9. MIW threat, and MCM units and time available for clearance
10. Availability and planned utilization of supporting arms
11. Establishment of go/no-go criteria
12. Need for speed and positive centralized control
13. Need to maintain sufficient flexibility to allow exploitation of adversary weaknesses
14. Anticipated weather, sea state, and tidal conditions
15. Adversary disposition and barriers, or other natural or manmade obstructions in the vicinity of the landing beaches and CLZs
16. EMCON requirements.

Selection of these considerations is affected by:

1. Hydrographic features of beach approaches
2. Beach size and trafficability
3. CLZ characteristics.

4.3.2 Planning Decisions

ATF and LF commanders make certain primary decisions during the planning process before an amphibious operation can proceed. Some of these decisions may have been made by the establishing authority and promulgated in the order initiating the amphibious operation. These decisions, as they apply to surfaceborne ship-to-shore movement, are listed below and described in detail in Chapter IV of JP 3-02.

1. Determine AF mission(s)
2. Select AF objectives(s)
3. Determine COAs for development
4. Select COA
5. Select landing areas
6. Select landing beaches
7. Determine sea echelon plan
8. Select LF objectives
9. Select landing date and hour.

4.3.3 Embarkation Flexibility

The organization for embarkation is compatible with the LF landing plan. In MEB- /MEF-sized operations where the landing results in the LF being ready to initiate combat, the LF is organized for the tactical integrity of the participating surfaceborne units. This is accomplished by combat loading troops, equipment, and supplies in amphibious shipping to the greatest extent possible. In any amphibious operation, the main focus of the landing plan and organization for embarkation is to facilitate a smooth flow of units and supplies ashore and inland during ship-to-shore movement. Cross-decking units from one ship to another prior to commencing ship-to-shore movement, referred to as pre-H-hour transfers, can enhance ship-to-shore movement efficiency, particularly when the LF was embarked prior to completing detailed planning. If feasible, cross-decking during the rehearsal phase should be considered; however, caution should be exercised, as extensive cross-decking can have an adverse impact on landing craft availability. When MSC or MSC-chartered ships are involved in the operation, the ability to discharge troops and supplies is also a critical consideration in the organization for embarkation.

The organization for embarkation also provides for maximum flexibility to support alternate plans should they become necessary. For decades, the traditional order in which the phases of an amphibious operation were conducted was PERMA. The LF landing plan and CONOPS ashore were often based on conditions and adversary capabilities that existed in the AO prior to LF embarkation. The extent to which changes in the LF landing plan can be accomplished depends on the organization for embarkation. JP 3-02.2 discusses embarkation in detail.

4.3.4 Oceanographic Considerations

Surfaceborne ship-to-shore movement can be influenced by several oceanographic conditions. Principal oceanographic considerations are:

1. Hydrographic features of offshore areas

2. Extent and feasibility of transiting through mineable waters

3. Capacity of beaches for landing troops, equipment, and supplies

4. Suitability of beaches for beaching landing craft as well as employing causeways (MPF operations) and AAVs under expected weather, tidal, and sea state conditions.

4.3.5 Mine Countermeasures and Mine Warfare Considerations

The capability to counter mines and obstacles is essential to the conduct of amphibious operations, particularly ship-to-shore movement. The integration of MCM operations into the overall strategy of a landing plan is a responsibility shared by the ATF and LF commanders. In short, the ATF commander has primary responsibility for MCM operations in the water, and the LF commander is responsible for MCM operations ashore.

To facilitate planning, participating commanders should provide detailed requirements for amphibious operations to the ATF commander, or MCMC, if assigned, as early as possible. Considerations should include location and size of the AO in comparison to available MCM assets, slow SMCM transit times to the AO, rate of MCM operations to meet established deadlines, and requirements for protecting assets involved in the MCM effort.

NTTP 3-15.3, Mine Countermeasures in Support of Amphibious Operations, a publication under development, will become the source document for MCM in support of amphibious operations by consolidating TTP for current technologies, equipment, concepts, and practices.

Preparations for and the conduct of MCM operations are guided by considering several factors discussed in the following paragraphs.

4.3.5.1 Mined Environment Effects on Amphibious Force

Mining may interfere with the AF's ability to maneuver within the AO, at sea, and on the beach. It may affect advance force operations and ship-to-shore operations through general unloading, and possibly hinder or preclude unloading of the AFOE. Planning for operational timelines, allocation of intelligence collection assets, AF task organization, the rate at which forces are established ashore, and deception operations can be affected by a mine threat. In a mined environment, available organic, national, and theater intelligence assets are obtained and used as soon as possible to gather information. Accomplishing this will enhance the ability of AF units to achieve

surprise and rapidly project combat power or build up troops, equipment, and supplies ashore and into objectives located deeper inland.

4.3.5.2 Mine Threat Intelligence

Because mines continue to proliferate and incorporate new technology, information on potential adversary's mine resources (types, characteristics, numbers, stowage locations, and transportation facilities) and at-sea delivery capabilities are crucial inputs to MCM planning. This information highlights the extent of the mine threat and the MCM assets to conduct responsive clearance operations.

4.3.5.3 Offensive Mine Countermeasures

If the ROE permit, MCM is best accomplished by destroying mines prior to their being deployed. Proactive MCM includes attacks on production, stowage, and transportation facilities, and the forces used to plant mines.

4.3.5.4 Area Reconnaissance and Surveillance

Exploratory searches of areas that may be included in the AO provide early threat assessments to commanders and tactical planners. These search missions, conducted largely by AMCM and SPECWAR forces, should be coordinated with deception or other preparatory operations to better confuse adversarial forces.

4.3.5.5 Area of Operations Characteristics and Mine Environment Conditions

The effort to clear, remove, or sweep mines depends on the types of mines that may be planted and the characteristics of the mined area. Commanders shall make every effort to learn as much as possible about any suspected mined area to include water depth, bottom type, density of mine-like objects on the bottom, tidal variations, and the presence of natural and manmade obstacles.

4.3.5.6 Mine Operating Characteristics

There are a significant number of mine types available worldwide that can devastate an amphibious operation. Their types and capabilities are listed in various classified publications of the NWP 3-15/MCWP 3-31.2, Naval Mine Warfare.

4.3.5.7 Mine Countermeasures Forces and Systems

A variety of forces and systems to counter mines are available. AMCM helicopters and SMCM ships are equipped with minehunting sonars, cutters for severing cables of moored mines, and influence sweep gear.

4.3.5.8 Ship Vulnerabilities

A particular ship's vulnerability to an exploding mine is a function of its size, the mine's warhead, and the ship-mine separation (and relative location) at mine detonation, and the water depth.

4.3.5.9 Coordination of Amphibious Warfare and Mine Countermeasures Operations

The MCM commander or the MCM LNO has the responsibility for working with AF planners to develop an MCM plan that is integrated with and supports the landing plan. The landing plan provides for the protecting

MCM forces and normally involves flexibility at sea and on the beach to reduce the vulnerability of forces. MCM, NSW, Navy EOD, Marine RECON, and combat engineers operations are integrated into the landing plan to optimize the timelines, effectiveness, and responsiveness of the MCM plan. The MCM plan accommodates the following:

1. RECON of the AO
2. Sea echelon area definition
3. OTA MCM
4. HTA MCM
5. Transport lane and ITA MCM
6. Mine clearance from approach and retirement lanes
7. Clearance of the SZ
8. Clearance of the beach zone to CLZs
9. Follow-on clearance operations to support MPF operations and force sustainment (port opening/JLOTS).

4.3.6 Mine Countermeasures Techniques and Equipment Considerations

The equipment, tactics, and systems available to the MCMC or ATF commander can significantly affect the speed of execution in amphibious operations. Because the development of effective MCM tactics and equipment has not kept pace with the amphibious warfighting capabilities available to ATF and LF commanders, commanders are mindful of current shortfalls and consider the following when planning an amphibious operation:

1. MCM SW and VSW hunting or sweeping systems
2. AMCM night capability
3. Accessibility of comprehensive mine RECON systems to commanders on-scene
4. Lack of clandestine autonomous mine RECON, classification, and mine neutralization/destruction capability in MCM forces
5. SZ obstacle reduction system
6. SZ and beach mechanical proofing system
7. Improved night capability for MCM systems
8. Breach lane marking systems (land and sea)
9. Breach lane navigation systems (land and sea) for ingress and egress

10. Man-portable mine detection systems for swimmers

11. An MCM C4I system that can integrated with the ATF commander's C4I system

12. Sufficient number and availability of heavy lift ships to transport SMCM ships in and out of theater to mitigate the slow speed (7.5 knots) inherent in SMCM ship self-deployment.

4.3.7 Supporting Amphibious Operations

Forces other than those assigned to the AF conduct supporting amphibious operations. They are ordered by higher authority, usually the commander who issued the order initiating the amphibious operation, and are normally based on a request from the AF commanders. Such operations may be executed inside or outside the AO and prior to or during the amphibious operations. They may set the conditions for the advance force to move into the AO; however, in all cases, supporting operations are coordinated with appropriate AF commanders. Supporting amphibious operations are covered in detail in Chapter XIII of JP 3-02. Examples of supporting amphibious operations are:

1. Tactical and strategic military deception operations to influence the actions of adversary forces

2. Interdiction operations to isolate the landing area

3. Destruction of specific targets ashore

4. AAW, USW, and SUW operations to gain air superiority and naval supremacy prior to combat operations in the AO

5. Special operations to secure information

6. Psychological or unconventional warfare

7. Minehunting, minesweeping, or minelaying

8. IO.

4.3.8 Advance Force Operations

Advance force operations are conducted to shape the AO and prepare the objective for the main assault or operation. These operations typically include RECON, surveillance, and intelligence collection, seizure of supporting positions, minesweeping and clearing, preassault/preaction preparatory fires, underwater demolition, IO, and deep air support. Advance force operations are covered in greater detail in Chapter XIII of JP 3-02.

4.3.9 Preassault/Preaction Operations

Preassault/preaction operations are conducted by the AF upon its arrival in the AO and prior to H-and/or L-hour. These operations are designed to gain information regarding opposing forces and to prepare the landing area. Pre-assault/preaction operations are discussed in greater detail in Chapter XIII of JP 3-02. Examples are:

1. RECON

2. Hydrographic surveys, SUROBS, and weather reporting
3. Destruction of targets by TACAIR and NSFS
4. Raids
5. EW (including cryptologic support)
6. Military deception operations
7. Minesweeping and/or mine breaching
8. Underwater demolition
9. IO.

4.3.10 Prelanding Operations

The main body of the AF conducts prelanding operations between the commencement of the assault or action phase and commencement of ship-to-shore movement. They encompass preparations similar to those executed by the advance force, but focus on the landing area, concentrating specifically on the landing beaches, CLZs, HLZs, and VLZs to be used by the LF. Prelanding operations also encompass final preparations for ship-to-shore movement. The transition between preassault/preaction and prelanding operations may not be discernible, as there is no break in activity.

4.3.11 In-Stride Operations

In-stride operations are similar to prelanding operations, but are conducted as an integral component of the ship-to-shore movement. They immediately precede the arrival of troops at landing beaches or LZs.

4.3.12 Landing Craft and Amphibious Vehicle Considerations and Support Requirements

The employment of landing craft or AAVs requires that ATF and LF commanders and planners consider a number of factors that are critical to safe and effective operations and support for displacement landing craft, LCAC, and AAVs. These considerations and requirements are delineated in Paragraphs 4.3.12.1 through 4.3.12.3.1.

4.3.12.1 Considerations for Displacement Landing Craft Employment

The principal considerations for employing displacement landing craft are:

1. The role of the surfaceborne operation in the LF CONOPS
2. Beach location, size, hydrographic features, and trafficability
3. Availability of hydrographic surveys for beach approaches
4. Suitable sea state and surf conditions

5. Adversary force capabilities and disposition, especially location, type, and density of antisurface weapons

6. Requirement for supporting arms and CSS

7. Displacement craft and LCAC comprise the principal heavy lift offload capability for LF equipment and supplies.

4.3.12.1.1 Displacement Landing Craft Support Requirements

Amphibious ships have a landing craft refueling capability and the capability to accomplish general maintenance. Well deck ships designated as boat havens can provide dry docking facilities for structural repairs and a safe haven during inclement weather.

The ship-to-shore movement plan should contain provisions for salvage operations. Appendix D and NTRP 3-02.1.2 provide additional information on displacement landing craft salvage operations.

4.3.12.2 Considerations for Landing Craft Air Cushion Employment

The principal considerations for the employment of LCAC are:

1. LCAC can be employed from OTH to achieve tactical surprise.

2. LCAC are not constrained by tidal conditions and most hydrographic features.

3. LCAC complement the vertical assault.

4. LCAC can complement AAV surface assault.

5. Alternate beaches can be included to tactically employ the LCAC's speed and mobility.

6. LCAC operations beyond the HWM could influence the CONOPS ashore.

7. A beach separation of 500 yards from displacement craft is required.

8. Anticipated SWHs, and prevailing windspeed and direction in CCAs, CLAs, DPs, transit lanes, and CHAs exist.

9. LCAC require a minimum 100-yard diameter area in the CLZ in which to discharge their load and return to the ship.

NTRP 3-02.1.2, LCAC SEAOPS, and NWP 3-02.12/MCRP 3-31.1A discuss LCAC operations in greater detail.

4.3.12.2.1 Landing Craft Air Cushion Support Requirements

The LSD class is designed specifically to operate and support LCAC. The LHD, LHA, and LPD class ships have been designed and modified to conduct LCAC operations, provide boat haven facilities, and limited maintenance support, but do not have the organic administrative and logistics support required by LCAC. If sufficient LSD class ships are unavailable, the LHD or LHA can be fitted with MILVANs to increase the level of logistic

support. MILVANs are not movable; therefore, their use is weighed against the reduction in space for LF equipment and supplies.

Provisions for the salvage and land-based recovery of AAVs with the LCAC are discussed in Appendix D and LCAC SEAOPS.

4.3.12.3 Considerations for Amphibious Assault Vehicle Employment

The AAV launch is planned to facilitate the expeditious formation of waves and/or tactical formations to execute the landing plan. Considerations for AAV planning and employment include:

1. Launching ship's AAV launch and recovery safety criteria
2. Ship-to-shore timeline
3. Serial numbers
4. Unit
5. Relationship of AAV launch area to LD
6. Launching ship's speed (if launched underway)
7. Availability of well deck lighting
8. Staging of AAVs for launch
9. Communications and signals
10. Launch intervals
11. Location, disposition, and mission of other amphibious ships in the AAV launch area, transport area, LD, or boat lane
12. Safety boat location
13. Coordination of safety procedures for AAV emergencies.

MCWP 3-13, Employment of Amphibious Assault Vehicles (AAVs), contains a more detailed discussion of AAV procedures, planning, and employment.

4.3.12.3.1 Amphibious Assault Vehicle Support Requirements

AAVs are normally embarked and transported to the AOA in LPD or LSD class ships, but may embark in LHAs or LHDs. Employment of these ships allows for rapid embarkation and debarkation, ease of maintenance and preparation for assault, and the high-speed underway launch on or near the LD.

Ships supporting embarked AAVs are prepared to:

1. Accommodate AAV command, maintenance, and communication personnel; equipment; and wheeled vehicles to support AAV operations.

2. Load supplies, POL; repair parts or equipment to support embarked vehicles.

3. Staff and equip ships designated as AAV repair ships.

4. Preload the supported unit's equipment and cargo as required.

Provisions for the salvage and land-based recovery of AAVs with the AAVR7A1 are discussed in Chapter 1, Appendix D, and MCWP 3-13.

4.3.12.4 Considerations for Expeditionary Fighting Vehicle Employment

The considerations for EFV employment are much the same as those delineated in Paragraph 4.3.12.3 for AAVs. However, the considerations listed below are more specific to the EFV because of its significantly increased capabilities. EFVs have the following characteristics:

1. Can perform at slower water speeds to facilitate operations with AAVs or move rapidly up on plane to execute high-speed and/or long-distance operations with or without LCAC.

2. Can accurately engage targets day or night from the water onto the beach up to 2,000 meters using its stabilized 30-mm cannon. This capability can have a significant impact on the CONOPS during ship-to-shore movement.

3. If operating with LCAC, must share tactical information to avoid high-speed collisions and coordinate the action. Common tactical pictures and GPS allow for precision navigation and precise timing of touchdowns ashore while only under advisory control from the PCS. EFVs use operational graphics and forms of control similar to those used by LCAC.

4. Require at least 15 feet of water depth to come off plane and retract planes before moving into shallower water and proceeding to land. EFVs must slow down to perform this procedure.

5. Require time to allow the download of operational information onto the vehicles' software when in the well deck. AAVs do not have this capability. Once launched, EFVs can receive wireless operational information updates.

6. Have the range, speed, and tactical flexibility to launch beyond a 25-nm range from shore.

7. Require a well-coordinated and comprehensive SAR plan when operating at significant distances from shore.

While the EFV is taking the place of the AAV, in many ways, it will operate in a manner similar to LCAC. LCAC SEAOPS provides comprehensive guidance on LCAC operations and tactics. Operating doctrine for the EFV continues to evolve as the program matures. Some of the issues that will require careful coordination between Navy and Marine Corps planners and doctrine developers are:

1. EFV control procedures (who and by what method(s)).

2. EFV control areas.

3. EFV wave composition and formation(s).

4. EFV and AAV interoperability. Both vehicles will almost certainly operate together until all AAVs are phased out of the Marine Corps inventory.

4.3.12.4.1 Expeditionary Fighting Vehicle Support Requirements

EFVs are normally embarked and transported in LPD or LSD class ships, but may embark in LHAs or LHDs. Employment on these ships allows for rapid embarkation and debarkation, ease of maintenance and preparation for assault, and the high-speed underway launch on or near the LD. Ships supporting embarked EFVs must be prepared to:

1. Accommodate EFV command, maintenance, and communication personnel, equipment, and wheeled vehicles to support EFV operations.

2. Load supplies and POL, and repair parts or equipment to support embarked vehicles.

3. Staff and equip ships designated as EFV repair ships.

4. Preload the supported unit's equipment and cargo as required.

Provisions for the salvage and recovery of EFVs are not fully developed at this time.

4.4 EXECUTION

This section covers the period of time between the execution signal to "land the landing force" and the completion of the offload of amphibious shipping.

4.4.1 Pre-H-Hour Transfers

If pre-H-hour transfers are necessary, they should be carried out by the most efficient means available: vertical lift, boat, or landing craft. If transfers are required, accountability and the location of personnel, equipment, and supplies transferred are maintained. The LF commander and planners provide the following information for the landing craft employment plan:

1. Boat team or serial number

2. Ship on which the boat team or serial is embarked

3. Ship to which the boat team or serial will be delivered

4. Number of personnel, equipment, and supplies to be moved.

4.4.2 Surfaceborne Ship-to-Shore Movement Control Organization

The surfaceborne ship-to-shore movement control organization is the Navy control group. This group consists of personnel, ships, boats, and landing craft designated to plan and control the ship-to-shore movement. Figure 4-1 depicts the organization of the Navy control group for surfaceborne ship-to-shore movement. The figure shows only the organization for one landing beach. A similar organization would be established for other beaches.

This organization supports large-scale amphibious operations, but ships, landing craft, and personnel can be scaled back for smaller scale operations.

4.4.2.1 Central Control Officer

The CCO is designated by the ATF commander and is normally embarked in a CCS. The CCO is responsible for:

1. Planning and supervising the surfaceborne ship-to-shore movement
2. Organizing the Navy control group to best support the landing plan
3. Maintaining liaison with the TAO
4. Maintaining liaison with the TACLOG group (discussed in Appendix F).

4.4.2.2 Assistant Central Control Officer

An ACCO may be designated when two or more landing areas are used or when the landing area has beaches located where centralized control is impractical.

4.4.2.3 Primary Control Officer

The PCO is embarked in a PCS assigned to control the movement of the landing craft and amphibious vehicles. A PCO is assigned for each landing beach and is responsible for:

1. Providing detailed plans, or PCO instructions, for conducting ship-to-shore movement across a landing beach
2. Maintaining the current location and status of ships, landing craft, and boats assigned to conduct a landing on a specific beach
3. Monitoring surf and weather conditions, and recommending the termination of boating should conditions warrant

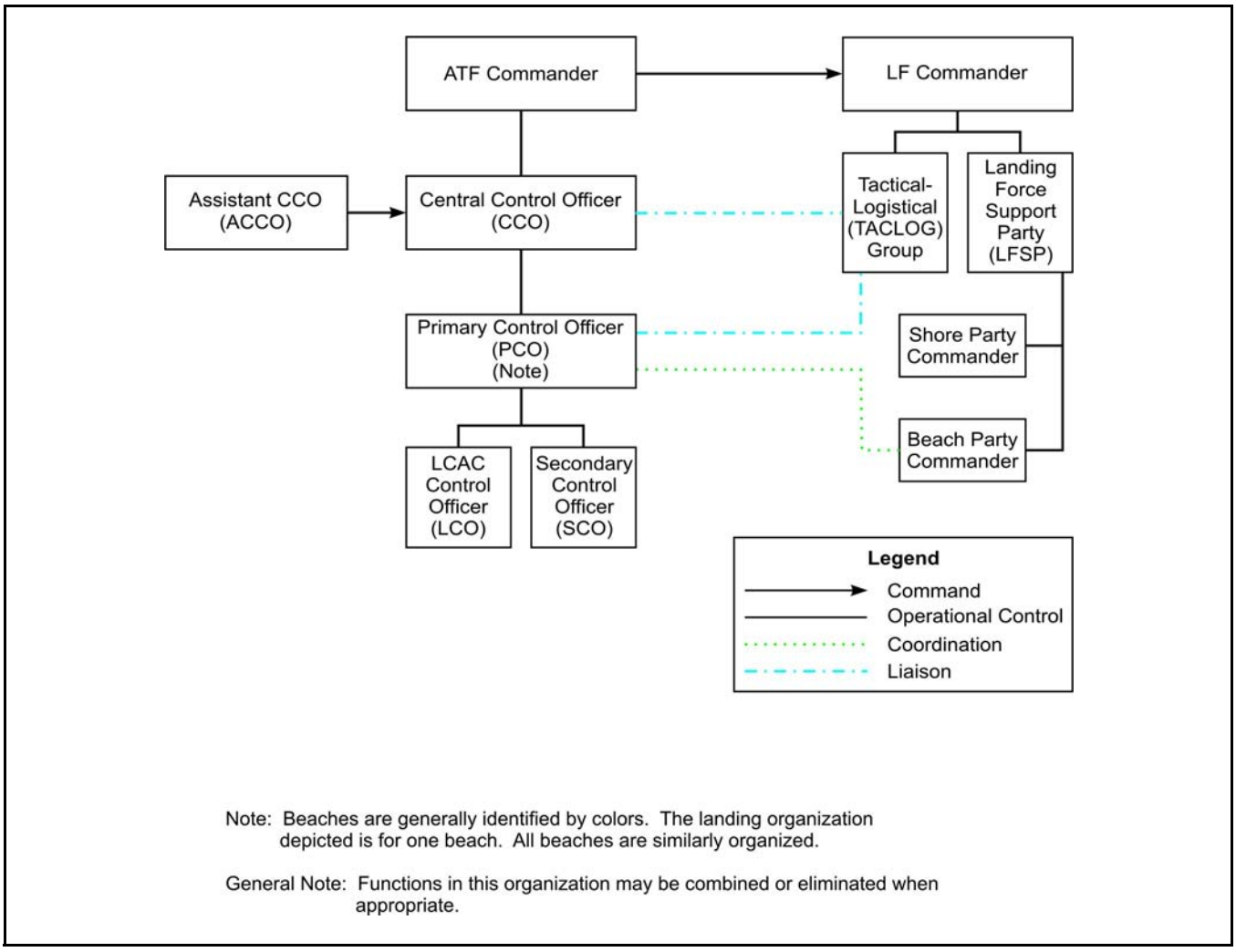


Figure 4-1. Control Organization During Surfaceborne Ship-to-Shore Movement

4. Maintaining the status of debarkation or embarkation
5. Landing scheduled waves at the correct beach at the specified time
6. Arranging for fueling boats, and providing rest and food for boat crews
7. Providing liaison to the surfaceborne TACLOG detachment
8. Conducting landing craft or amphibious vehicle salvage operations
9. Coordinating landing craft operations with a designated area of responsibility following the initial action.

4.4.2.4 Primary Control Ship

The PCS is an amphibious ship designated to provide support for the PCO and a CIC control team to track and control landing craft and AAVs for a landing beach. PCS responsibilities include:

1. Maintaining a plot of ships, AAVs, and landing craft within the PCO control area
2. Coordinating the movements of on-call waves and nonscheduled units
3. Coordinating landing craft, amphibious vehicle, and boat traffic movements during general offload or LF backload.

The PCS may be assigned a point station or an underway sector in the vicinity of the LD.

4.4.2.4.1 Boat Control Team

The boat control team plots, tracks, and controls the movements of scheduled waves from the PCS's CIC. Team members are:

1. Supervisor
2. Wave controller
3. Grid plotter
4. Radio net operators (Alfa and Bravo nets)
5. Radar operator
6. Visual bearing taker
7. Signalman for visual communications.

Appendixes B and C contain signals and wave control procedures used by the boat control team.

4.4.2.5 Secondary Control Officer and Secondary Control Ship

When designated, the SCO embarks in the SCS and is the principal assistant to the PCO. The SCS**,** when designated, is assigned a point station on the LD or an underway sector in the vicinity of the PCS. SCO/SCS duties include:

1. Maintaining duplicate control records and plots required of the PCO and PCS
2. Monitoring PCO radio circuits
3. Controlling surfaceborne ship-to-shore movement over a landing beach when two or more landing beaches are designated

4. Assuming PCO and PCS duties in an emergency.

In a large-scale operation when two or more landing beaches are designated, an additional SCO and SCS may be assigned as a backup or to control the third or additional landing beaches.

4.4.2.6 Landing Craft Air Cushion Control Officer

The LCO embarks in the LCAC control ship and assists the PCO by providing detailed plans for the ship-to-shore movement of LCAC. LCAC ship-to-shore movement is planned and conducted under the centralized control of the Navy control group organization. The PCO delegates various planning aspects and control from the CDP to the LCC to the LCO. The LCO reports to the PCO and is responsible for:

1. Providing detailed plans, called LCO instructions, to conduct LCAC ship-to-shore movement
2. Controlling LCAC from the CLA to the CPP
3. Maintaining the current location and status of LCAC involved in the landing
4. Monitoring surf and weather conditions, and recommending to the PCO the termination of LCAC operations when conditions warrant
5. Arranging for LCAC fueling and providing replacement crews or crew rest and food
6. Coordinating LCAC salvage operations.

4.4.2.7 Landing Craft Air Cushion Control Ship

The LCAC control ship supports the embarked LCO and CCTs to control LCAC groups from the CLAs to the CPPs.

4.4.2.7.1 Landing Craft Air Cushion Control Team

Analogous to the boat control team, the LCAC control team plots, tracks, and provides advisory or positive control to LCAC groups en route to the CPP. During general offload, the team controls or monitors LCAC movements as directed by the LCO, or the PCO when a separate LCO is not assigned. When the PCS is controlling LCAC and displacement craft concurrently, LCAC and boat control team functions can be combined where feasible to avoid redundancy. LCAC control team members include:

1. A supervisor/evaluator
2. Wave controllers
3. A plotter/computer workstation operator for AN/KSQ-1
4. Radio net operators
5. A radar operator.

4.4.2.8 Landing Craft Air Cushion Wave

An LCAC wave consists of two or more craft comprising a scheduled wave. LCAC do not display any unique visual signals or insignia because of the FOD such devices could cause. Appendix B and NWP 3-02.12/MCRP 3-31.1A depict LCAC beach markings and maneuvering hand signals.

4.4.2.9 Landing Craft Air Cushion Craftmaster

As the person in charge of an individual LCAC, the craftmaster is responsible for safety and well-being of embarked personnel and equipment, and the efficient and safe operation of the craft.

4.4.3 Control Areas

Displacement craft and LCAC control areas are established in the landing area to deconflict tactical operations, define transit lanes/routes for LCAC waves, and identify geographic positions for timing of LCAC ship-to-shore movement. Displacement craft control areas are the approach lane, LD, and boat lane. LCAC control areas include CLAs, CCAs, CDPs, LCAC transit lanes, CCPs, CPPs, ingress and egress routes, CLZs, CLSs, and CHAs. Figure 4-2 depicts examples of displacement craft and LCAC control areas. EFVs use the same type of control areas as LCAC.

4.4.3.1 Displacement Craft Control Areas

AAVs and displacement landing craft begin their timed transit to the beach from the LD. The timing for each scheduled wave is established in the approach schedule.

4.4.3.1.1 Approach Lane

The approach lane is an extension of a boat lane from the LD toward the transport area. It indicates the exact route displacement landing craft use to approach the LD from the transport area in a static launch from amphibious ships. If necessary, a boat or buoys may mark it; however, displacement landing craft are more often vectored to the beach by the PCS boat control team. Adjacent approach lanes may be parallel or diverge seaward to provide for early wave dispersion.

4.4.3.1.2 Line of Departure

The LD is a suitably marked offshore coordinating line to assist landing craft in landing at designated beaches at scheduled times. It marks the seaward end of the boat lane. If multiple beaches are being used in the operation, each landing beach has an LD, and topographic, hydrographic, and tactical considerations determine the specific location. If necessary, it may be marked by PCS/LCAC control ship, boats, or buoys, but when scheduled waves are launched underway, the LD may be unmarked. Displacement craft waves are dispatched to the beach from this line. A separate LD may be provided for AAVs to reduce surfaceborne transit times.

4.4.3.1.3 Boat Lane

A boat lane is a path for landing craft that extends seaward from the landing beaches to the LD. The length of the corresponding landing beach determines the width of the boat lane. A flank of the boat lane is designated as the return boat lane to deconflict traffic patterns of displacement landing craft returning to the beach via the LD.

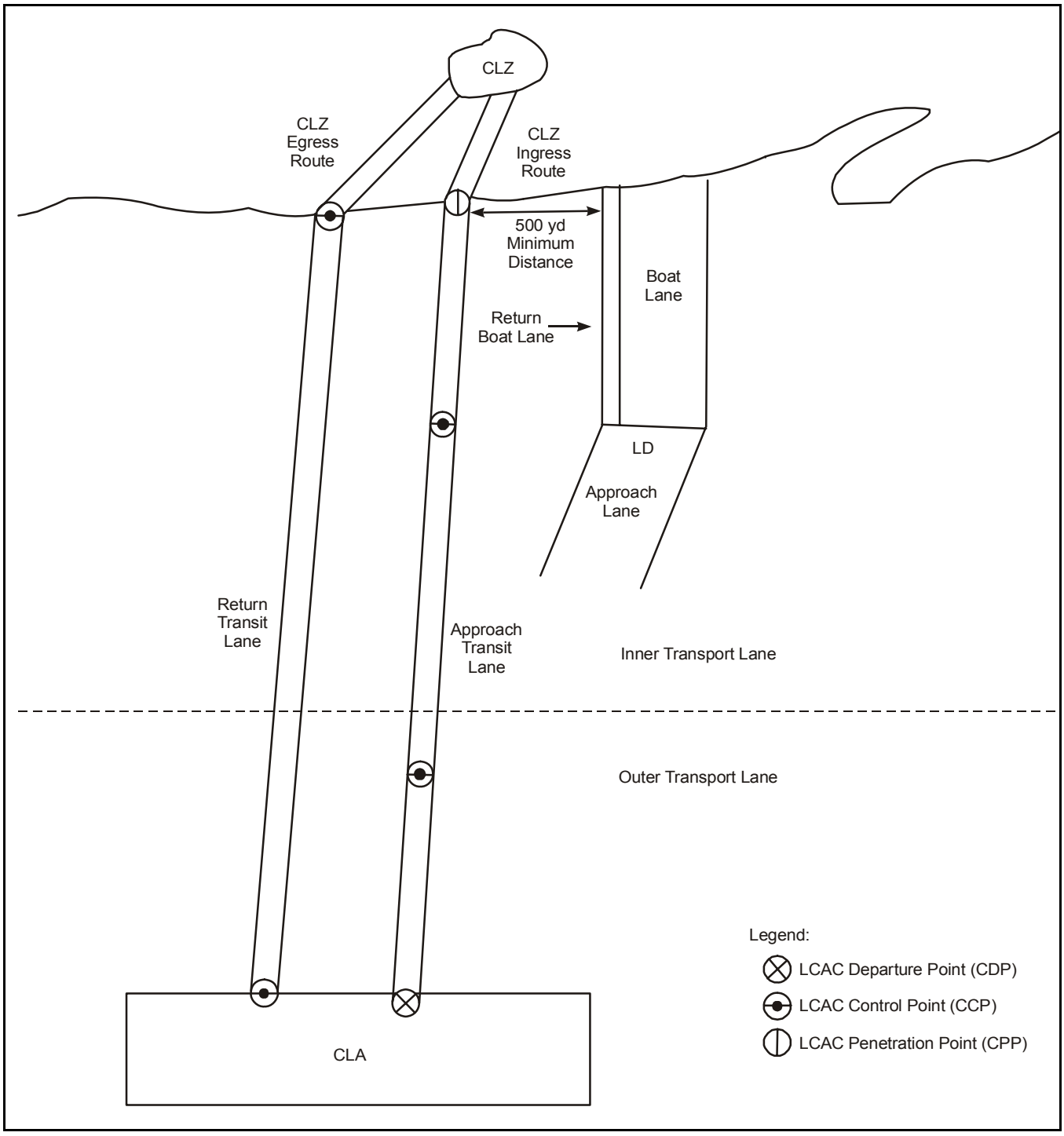


Figure 4-2. Example of Displacement Craft and Landing Craft Air Cushion Control Areas

4.4.3.2 Landing Craft Air Cushion Control Areas

LCAC waves begin their timed transit to the beach from the CDP. The timing for each scheduled wave is established in the approach schedule. When a separate LCO/LCAC control ship is not assigned, the PCO/PCS designates or selects the area.

4.4.3.2.1 Landing Craft Air Cushion Launch Area

The CLA is designated by the ATF commander and is the area in which LCAC are launched. The LCO, using the CCT in the LCAC control ship, assumes OPCON of LCAC in the CLA upon departure from the launching ships. The LCO either dispatches LCAC waves to the CDP IAW the approach schedule or monitors their independent transit to the CCA. Depending on the scope of the operation, the CLA may be located several thousand yards offshore or OTH.

4.4.3.2.2 Landing Craft Air Cushion Collection Area

The CCA is designated by the LCO, and in large-scale operations in which multiple craft are coming from multiple CLAs, it is where LCAC waves are formed. The LCO dispatches LCAC waves to the CDP IAW the approach schedule. The positioning of the CCA relative to the CDP is critical in LCAC getting up to speed and over the hump prior to passing through the CDP. Landing area planning is flexible enough to facilitate CCA positioning on any axis from the CDP just prior to LCAC launch to ensure craft obtain "hump speed" before passing through the CDP.

As the LCAC hovers over the water, it creates a depression in the water surface directly beneath itself, equal in water volume to its weight. As the LCAC begins to move, the depression moves with it and creates a substantial wake. At certain speeds, craft resistance becomes relatively high. These critical speeds are referred to as "hump speeds" due to their "hump"-like appearance on the graph of speed versus resistance. Additional information on "hump speed" and hump transition is found in LCAC SEAOPS and NWP 3-02.12/MCRP 3-31.1A.

4.4.3.2.3 Landing Craft Air Cushion Departure Point

The CDP is the geographic position designated by the LCO that marks the seaward end of the LCAC approach lane. It is analogous to the LD for displacement landing craft and AAVs.

CDPs are not loiter or RPs. Launch and CLA/CCA dispatch timing should permit LCAC to proceed directly to and through the CDP to the next CP on schedule, with the craft already at hump speed and at intended track speed.

4.4.3.2.4 Landing Craft Air Cushion Transit Lanes

LCAC transit lanes are the sea routes designated by the LCO to connect the CPs through which LCAC pass. Approach transit lanes connect CDPs and CPPs, and may contain more than one CCP. Separation between LCAC transit lanes and displacement craft approach and boat lanes is 500 yards or greater. Separate approach transit lanes for each LCAC wave or a single transit lane for all waves may be designated by the LCO based upon dispersion of amphibious shipping. Separate return transit lanes may also be established to deconflict inbound and outbound LCAC flow.

4.4.3.2.5 Cushion Landing Craft Air Cushion Control Point

A CCP is a geographical position established by the LCO along the LCAC transit lane to control the ship-to-shore movement of LCAC waves. Unlike displacement craft control procedures that generally involve minute-by-minute position updates, LCAC waves proceed to the beach independently or receive only periodic position reports relative to CCPs from the LCAC control ship. The position of LCAC waves in the transit lane relative to CCPs can be used by the CCT to verify the waves are on schedule or to adjust craft speed to regain the approach schedule.

A DP is a special type of CCP that marks a split in the LCAC approach transit lane providing alternate routes to shore. The use of a DP may be ordered by the ATF commander to allow the tactical option of changing the CPP and CLZ in response to a changing tactical situation ashore.

4.4.3.2.6 Cushion Landing Craft Air Cushion Penetration Point

A CPP is the geographic position selected by the ATF commander where the LCAC wave crosses the high water mark. It is the point at which OPCON of the LCAC waves shifts from the LCO to the CLZ CCT or RECON/SEAL teams providing ITG into the CLZ. The selection of CPPs is influenced by several factors, including ease of identification, tactical CONOPS ashore, local topography, and suitability for nearby CLZs while avoiding nonnegotiable LCAC features such as dunes with sharp dropoffs or excessively wide trenches.

For night operations, the centers and flanks of CCPs may be marked initially by lights or strobes placed during preassault operations. Marker arrangements and characteristics are shown in Appendix B.

4.4.3.2.7 Cushion Landing Zone

Selected by the ATF commander, a CLZ is the beach or inland area where LCAC waves come off cushion to offload or on load personnel, equipment, and cargo. CLZs are selected based on a variety of factors supporting the LF CONOPS ashore, including the number of LCAC simultaneously in the CLZ, type of cargo discharged, proximity to LZs or landing beaches when troop and equipment assembly is required, general terrain features, trafficability, and avoidance of bottlenecks leading into and out of CLZs. Ideally, CLZs are surveyed to verify suitability and determine the best route from the CPP to the CLZ.

4.4.3.2.8 Landing Craft Air Cushion Control Team and Cushion Landing Zone Support Team

The CCT provides LCAC terminal guidance from seaward of a CPP to a CLZ for unloading or loading, and directs the movement of personnel, equipment and supplies out of the CLZ and back to amphibious shipping. A CCT consists of personnel from the BMU and is moved ashore as soon as practicable after the initial surfaceborne action. CCT personnel provide ITG into the LF CLZ. RECON or SEAL personnel may provide ITG when the beach is less than benign or hostile. Appendix G provides details on CCT and CLZ support team performance during surfaceborne ship-to-shore movement.

The CCT is subordinate to the BP component of the LFSP and is analogous to a BPT within the BP organization. Its counterpart in the CLZ for the unloading and movement of personnel, equipment, and supplies from LCAC is the CLZ support team, a subordinate of the shore party component of the LFSP.

4.4.3.2.9 Cushion (Landing Craft Air Cushion) Landing Site

The CLS is an individual landing spot within the CLZ. The CCT controls LCAC wave movement into and out of CLZs and positions LCAC in an available CLS upon its arrival for an offload or onload by the CLZ support team. As a planning estimate, a 100-yard diameter area is required for each LCAC. The size of the area is increased for sloping terrain or other natural or manmade features which dictate increased LCAC maneuvering room.

4.4.3.2.10 Landing Craft Air Cushion Holding Area

The CHA is the area designated by the ATF commander to which LCAC waves may proceed upon return from the beach to await further assignment by the PCO. An SRP may be located in the CHA where individual LCAC rendezvous with and are recovered by amphibious shipping.

4.4.3.3 Expeditionary Fighting Vehicle Control Areas

During ship-to-shore movement, the EFV formation will proceed from one CCP to successive CCPs along preplanned routes. Given the necessary dispersion of the formation, only selected vehicles will proceed along the precisely specified route (given the inherent precision of navigation aids), while other vehicles will guide off the movements of these vehicles based on the formation desired. As the wave comes off plane at the release point, EFV sections will proceed to their designated position in the formation used to assault the shoreline.

4.4.4 Debarkation

Debarkation is the unloading of troops, equipment, and supplies from a ship or airfield. It can be accomplished from a ship that is underway, lying to, or at anchor.

Visual signals may be used to call displacement craft into well decks to embark troops or cargo for subsequent debarkation on the beach. To visually call LCUs or AAVs into the well deck, the receiving ship hoists the appropriate flag from Figure 4-3 over the WHISKEY flag from either yardarm. For an LHA well deck, signal flags are hoisted on the port or starboard yardarm to indicate which side of the split well deck the displacement landing craft is to make. A light box with three vertical lights is used to call displacement landing craft or AAVs into a ship's well deck during night operations. The top light is white. The bottom light is left blank when calling single landing craft. For LHAs, a steady top white light indicates a displacement landing craft is to make the starboard side of the split well deck. A flashing white top light indicates the port side of the split well deck.

LCAC physical characteristics preclude maintaining an effective visual signal watch. Therefore, directions for LCAC to enter a well are made over the MOM voice radio system or the LCAC Control Net.

4.4.4.1 Debarkation Schedule

The sequence and means for LF surfaceborne debarkation is established in the landing diagram, landing craft and amphibious vehicle assignment table, and LF sequence table.

Information to Signal		Day Signal Flags	Night Signal Flags
Sides	Starboard Port	Starboard yardarm Port yardarm	Top green light Top red light
Boats and Displacement Landing Craft	LCU AAV	U Flag T Flag	Middle white light Middle amber light, flashing
Well Deck	Note 1	WHISKEY Flag	Top white light Middle (type of landing craft to enter well deck is indicated by middle light) Bottom white light Note 2

Notes:
1. Paragraph 4.4.4 contains special signals for the LHA well deck.
2. Turned off for calling single landing craft into well deck.

Figure 4-3. Day and Night Signals for Calling Displacement Landing Craft to Well Decks

4.4.4.2 Static Debarkation Sequence

This doctrine pertains primarily to larger scale operations in which significant numbers of displacement landing craft are employed. It establishes the methodology for displacement landing craft to load, form into waves, and transit to the beach in a well-defined and controlled process from ships lying to or at anchor.

Some of the doctrine herein is adaptable to underway debarkation discussed in Paragraph 4.4.4.4; however, static debarkation is generally administrative in nature, and is therefore conducted in benign or less than hostile environments.

4.4.4.2.1 Rendezvous Area

The rendezvous area is seaward of the LD. It is the area in which landing craft waves form into proper wave sequence. The PCS then dispatches the landing craft to the beach, crossing the LD at the time specified in the approach schedule.

If designated and employed, floating dumps generally remain in the rendezvous area.

4.4.4.3 Amphibious Assault Vehicle and Expeditionary Fighting Vehicle Debarkation

Boat teams are loaded into AAVs and EFVs in the well deck as per the debarkation schedule. They may be launched while the ship is underway, anchored, or lying to, and proceed directly to the beach to minimize transit time and troop fatigue. A more detailed discussion of AAV procedures can be found in MCWP 3-13.

4.4.4.4 Underway Launch of Amphibious Assault Vehicles, Expeditionary Fighting Vehicles, and Landing Craft

AAVs are normally launched underway, while preloaded displacement landing craft may be launched underway, at anchor, or lying to. However, launching while underway minimizes the time the launch ship must be close to the beach, its vulnerability to hostile action, and maximizes the element of surprise that is often integral to the CONOPS ashore. AAVs are launched close to the AAV LD on a timeline so each wave can proceed directly to the beach. The AAV wave commander, the senior LF officer or NCO in each wave, directs each AAV wave.

Depending on the control plan, displacement landing craft are launched in the ITA or close to the LD. Within this area, they are launched in a displacement craft launching area designated by the PCO, and the launch ships retire to the OTA. The procedures for conducting displacement craft underway launches are found in COMNAVSURFLANTINST/COMNAVSURFPACINST 3340.3.

Figure 4-4 depicts an example of an underway launch scenario. Appendixes A and B depict landing craft, AAV, and EFV formations and control signals.

4.4.5 Dispatching Scheduled Waves to the Beach

AAVs and displacement landing craft begin their timed transit to the beach from the LD. The timing for each scheduled wave is established in the approach schedule. Appendixes A and B discuss landing craft and AAV formations and control signals.

4.4.5.1 Amphibious Assault Vehicle Waves

AAVs are normally launched underway close to the LD, but may be launched while the ship is anchored or lying to in the amphibious vehicle launching area.

AAVs are usually launched in a column. Once all AAVs are launched, a flanking maneuver is executed toward the beach, and the AAVs proceed down the boat lane in one of the tactical formations (as outlined in MCWP 3-13) to touchdown at the prescribed time. AAV standard distance is 50 meters/yards or as dictated by the tactical situation, sea state, and visibility constraints. Battle Speed is ordered when approximately 1,000 yards from the beach, unless it would result in an early landing of the first wave.

AAVs normally do not stop at the beach, but continue inland with embarked troops as per the CONOPS ashore. Additional information on AAVs is found in Paragraph 1.4.7 and MCWP 3-13.

4.4.5.2 Expeditionary Fighting Vehicle Waves

EFV formations, also known as waves, use the platoon as its base maneuver unit during OTH ship-to-shore movements. An EFV platoon maneuvering from ship-to-shore will normally be designated as a single wave. An infantry battalion, embarked aboard an EFV company, could be comprised of several waves. The senior EFV officer/SNCO within each wave is normally designated as the WGO. Employing the appropriate C2 network, the PCO provides guidance to the surface element waves based on the chosen method of control to enable the execution of the landing plan.

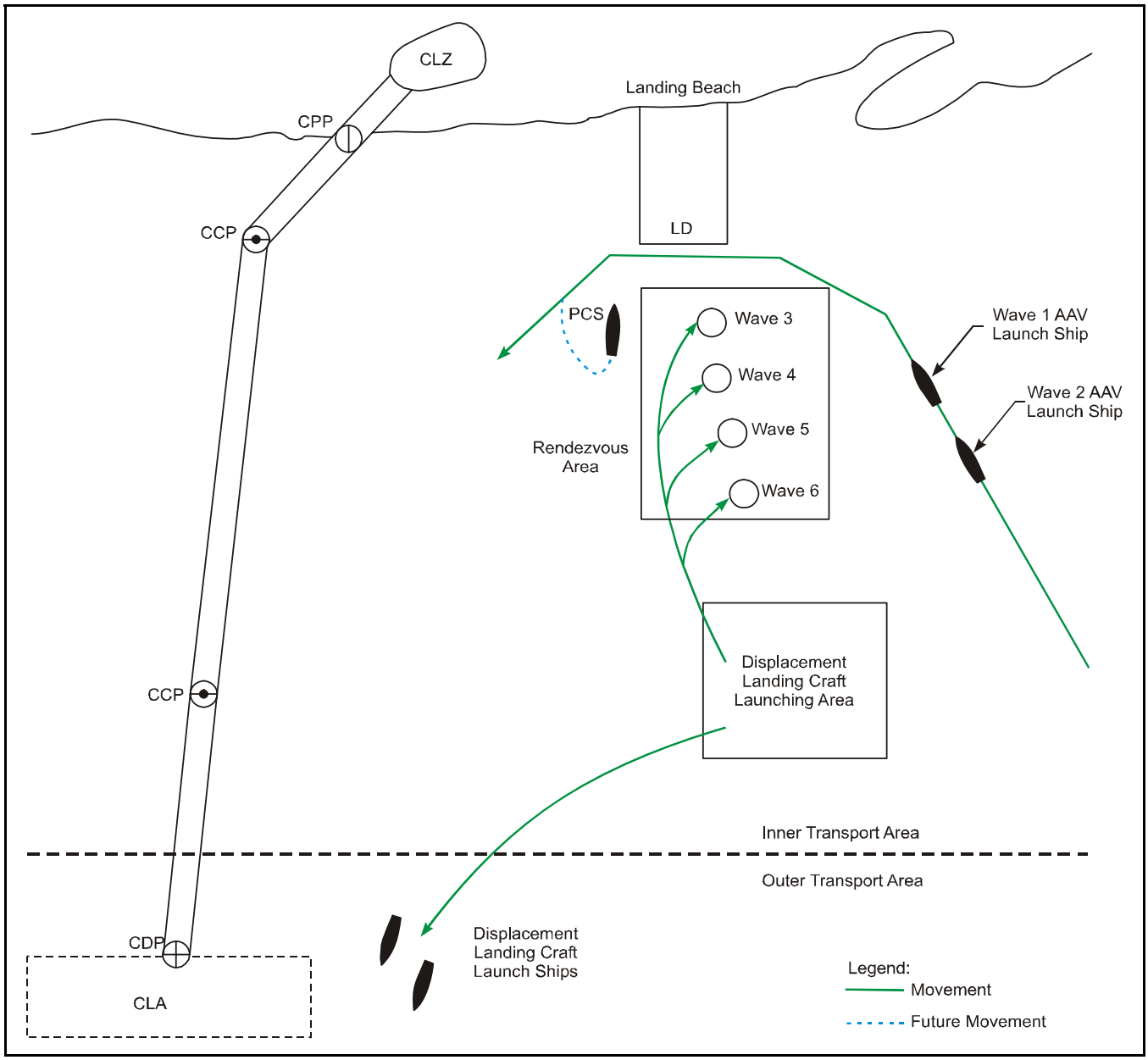


Figure 4-4. Underway Launch Scenario

4.4.5.3 Displacement Landing Craft Waves

When displacement landing craft are launched, they proceed to a rendezvous area or directly to the LD. Depending on the approach schedule, displacement landing craft may assume a line abreast or a column formation after crossing the LD.

Standard distance between displacement landing craft (LCUs) is 50 yards in daylight, at night, or during low visibility. Battle Speed may be ordered when 1,000 yards from the beach, unless it would result in a landing earlier than prescribed in the timeline.

Medical and designated salvage craft are assigned to specific waves.

4.4.5.4 Landing Craft Air Cushion Waves

Although a single LCAC can be considered a wave, LCAC usually proceed to the beach in waves of two or more craft. Also, LCAC movements can take place simultaneously with those of displacement craft. An LWC directs each wave. LCAC are launched in the CLA to pass through the CDP at the time prescribed in the approach schedule. The CDP is analogous to the LD. LCAC proceed down transit lanes with adjustments to course and speed directed by the LCAC control ship according to the method used.

Depending on the approach schedule and the tactical situation, LCAC proceed to the beach in either a line of bearing or line abreast to starboard, a wedge, a column, or a deceptive (from different directions) formation. Standard distance between LCAC is 100 yards. A minimum of five standard distances should be used when in column formation. The LWC in the lead craft specifies the distance between LCAC after considering the type of formation, speed, sea state, and visibility.

4.4.5.4.1 Landing Craft Air Cushion Navigation

LCAC navigation is achieved through a combination of electronic and visual means or through external information received from the Navy control organization. A more detailed discussion of LCAC navigation systems and procedures is contained in LCAC SEAOPS and NWP 3-02.12/MCRP 3-31.1A.

4.4.6 Surfaceborne Ship-to-Shore Movement Control

The Navy control group depicted in Figure 4-1 organizes and controls ship-to-shore movement. Paragraph 2.2.3 discusses the concept of decentralized execution of surfaceborne ship-to-shore movement through that control group. Control of ship-to-shore movement is also discussed in Chapter XV of JP 3-02 and in JP 3-02.1.

4.4.6.1 Expeditionary Fighting Vehicle Control

The EFV OTH ship-to-shore movement is characterized by high-speed maneuver commencing beyond visual and radar range of the shoreline, the potential use of one or more penetration points, and varying methods of control during the movement. The method of control employed is dependent on several factors, including the mission and tactical situation, weather and visibility, volume of surface traffic (military and civilian) and other hazards to navigation, and the communications capabilities of the controlling platform. Normally, this control function is provided by the PCO, located aboard the PCS. However, the operator performing the control function may be aboard an amphibious ship, another landing craft, or airborne platform.

4.4.6.2 Displacement Craft Control

The CCO plans and executes the surfaceborne ship-to-shore movement, including LCAC operations. The PCO is responsible for LCAC operations at a landing beach and delegates LCAC control functions to the LCO. However, the manner in which the LCAC ship-to-shore control organization exercises control depends on the mission, tactical situation, and the PCS C2 capabilities. The three types of LCAC control procedures for exercising control from OTH are discussed in the following paragraphs.

4.4.6.3 Landing Craft Air Cushion Control

The CCO plans and executes the surfaceborne ship-to-shore movement, including LCAC operations. The PCO is responsible for LCAC operations at a landing beach and delegates LCAC control functions to the LCO. However, the manner in which the LCAC ship-to-shore control organization exercises control depends on the mission, tactical situation, and the PCS C2 capabilities. The three types of LCAC control, procedures for exercising control from OTH, and the AN/KSQ-1 amphibious assault direction system are discussed in the following paragraphs.

4.4.6.3.1 Advisory Control

Advisory control involves minimal communications between the CCT and the LWC or craftmasters. It is the preferred type of LCAC control. The control team and the LWC or craftmaster interact and exchange positional information, course, speed, and current time late/early for landings as follows:

1. Two-way voice communications are maintained on the LCAC Control Net between the CCT and the LWC or craftmaster.

2. The launching ship provides LCAC with launch positions and a vector to the CDP or CCA.

3. The LWC or craftmaster uses the MOM voice radio system or another designated voice net to take TACON of the wave or individual LCAC en route to the CDP or in the CCA.

4. The CCT tracks the LCAC wave or individual units and periodically gives the LWC or craftmaster a current position and time early or late based on the approach schedule.

5. The LWC or craftmaster adjusts the wave or individual craft's course and speed to pass through the next point on schedule.

6. If required, the LWC or craftmaster may mark passage through CPs, course and speed changes, and the sighting of navigational aids over the LCAC Control Net.

7. More frequent positioning information may be needed from the control team while the LCAC are in the transit lane.

8. LCAC will remain under advisory control until they pass beyond the radar horizon of the controlling ships.

4.4.6.3.2 Independent Control

Independent control involves emergency-only use of voice communications between the CCT and the LWC or LCAC craftmaster. It is used when operations are conducted under EMCON conditions or OTH transit. With GPS

installed, LCAC are capable of independent movement from OTH to an unmarked CPP. Without GPS, the LWC or craftmaster relies upon DR from a known point at sea to a visually prominent or marked CPP. Transit from OTH under independent control by DR may result in navigator errors and negatively impact arrival time.

4.4.6.3.3 Positive Control

Positive control involves a continuous flow of communications from the CCT to the LWC or craftmaster via electronic means, voice, or data link. Communications are used to provide craft positions, vectors, courses, and speeds, along with wave commands and time and distance to turn.

Radar is the primary sensor for determining an LCAC's position relative to the LCAC control ship or PCS. LCAC OTH operations may preclude positive control due to loss of radar coverage. Voice communications over the LCAC Control Net are used to provide positions and vectors to the LWC or craftmaster. The following procedures are employed in positive control:

1. An LCAC or an LCAC wave is launched and passes through the CDP at a designated time. The craft may maneuver on a zigzag course, execute a 360° turn, or adjust speed to arrive at the CDP on time.

2. Position information is provided every minute by the CCT over the LCAC Control Net using established grid reference procedures.

3. When running multiple legs to the beach in a high sea state, it may be necessary to establish different notional speeds on each leg. Otherwise, LCAC may be "time late or early" on initial legs.

4.4.6.4 Procedures for Landing Craft Air Cushion Over-the-Horizon Control

When LCAC are launched from OTH, arrival at the CLZ or designated rendezvous position requires detailed planning, coordination by the PCO and LCO, and precise navigation by LCAC crews. Radar is the primary sensor for determining landing craft position relative to the controlling station. Voice or other communication means may be used to provide position and vector information to the craft. In many OTH operations, a control ship cannot continuously maintain radar contact; therefore, procedures for LCAC OTH control include independent transit, aircraft relay, and airborne control.

4.4.6.4.1 Independent Transit

LCAC independent transit is conducted under independent control and is the most frequently used method for transiting from OTH. LCAC equipped with GPS are capable of independent transit from OTH at high speed, passing through CCPs to the CPP and arriving at the CLZ on time. Unless LCAC OTH operations are conducted under restrictive EMCON, the craft will remain under advisory control until passing beyond the radar horizon of the controlling ship. Considerations in LCAC independent transit planning include:

1. Navigation skills of LCAC crews

2. Availability and accuracy of GPS position updates in the AO

3. Advance coordination and detailed briefing of participants

4. Shipping in or near LCAC transit lanes

5. Weather, visibility, current, and projected sea state

6. Radar navigation features in the vicinity of the CPPs

7. Criteria for operating LCAC radar in the event of GPS failure during restrictive EMCON transits

8. RTF procedures.

4.4.6.4.2 Aircraft Relay

An ARP can be used to maintain a track on the progress of LCAC or individual LCAC during transit to the beach. The ARP relays LCAC position to the LCAC control team via voice or data link. An ARP can also be stationed "on top" of LCAC, where they can be tracked on radar by the LCAC control team. The aircraft can also act as a radio relay.

4.4.6.4.3 Airborne Control

When airborne control is employed, each landing craft transit lane should be controlled by one aircraft. The airborne controller directs LF units, usually through secure UHF communications.

4.4.6.5 AN/KSQ-1 Amphibious Assault Direction System

This system enables ATF and LF commanders' control teams with a capability to identify, track, and communicate with landing craft in the landing area during ship-to-shore movement. It combines hardware and software in a unified build format with the Marine Corps AN/USQ-90 PLRS data from the GPS, and existing ship and landing craft equipment to provide C2 of surface ships and landing craft. When an aircraft relay is available, it operates without restriction or reduction in capability when unit separation does not exceed 100 nm. The AN/KSQ-1 (V) provides:

1. Accurate, real-time position and identification information

2. AN/KSQ-1 (V) and PLRS-equipped units' identification and position

3. OTH operating ranges via airborne relay

4. Limited text message exchange among network members via digital data link

5. Marine Corps PLRS MS interoperability

6. Cryptographic security

7. Antijam and low probability of intercept operation.

4.4.7 Beaching, Retraction, and Return

Upon touching down on the beach displacement craft, report on the Beach Boat Control circuit (Alfa Net) and then shift to the Beach Boat Operations circuit (Bravo Net). AAV and EFVs shift to the appropriate LF TACNET.

LCAC report on the LCAC Control Net and then shift to the CLZ Control Net.

4.4.7.1 Displacement Landing Craft

Displacement landing craft operations at the beach are controlled by the BP, which is landed as a scheduled or on-call wave. Displacement landing craft, beach, lower the ramp, offload troops and equipment, raise the ramp, and retract off the beach on signal from a BP traffic control person. The BP keeps the PCS advised of the status of debarkation or embarkation via the Bravo Net. Displacement craft back clear of the SZ and return via the return boat lane. Additional information on displacement landing craft operations is contained in Paragraph 1.4.6 and NTRP 3-02.1.2.

4.4.7.2 Landing Craft Air Cushion

LCAC in scheduled and on-call waves are controlled to the CPP by the CCT. The CPP is separated from the boat lane flank by at least 500 yards to deconflict LCAC high-speed approaches from relatively slower displacement landing craft and amphibious vehicles. A CCT is landed as early as possible to provide terminal control for the LCAC from seaward of the CPP to the CLZ using the CLZ Control Net. SEALs or Marine force RECON teams may provide LCAC control and coordination until the CCT is established. The tactical situation, or in some cases, the topography of the beach, governs how far inland CLZs are established. LCAC do not transition to LF control like AAVs and EFVs. They usually land in groups of two or more and proceed to the CLZ by an ingress route. They discharge troops, equipment, and cargo and, when directed by the CCT, return to the beach by a separate egress route. For general unloading, LCAC report directly to the PCS. Additional information on LCAC operations is found in Paragraph 1.4.6 and NTRP 3-02.1.2.

4.5 COMMUNICATIONS

Communications to control scheduled waves and coordinate initial and general offloading at a landing area are shown in Figure 4-5. These communication nets are explained in the following paragraphs.

4.5.1 Control Ship Coordination Net

The Control Ship Coordination Net is a directed net used for overall coordination of surfaceborne ship-to-shore movement.

4.5.2 Primary Control Net

The Primary Control Net is a directed net used to exchange C2 information and for coordinating ship-to-shore movement at a landing beach.

4.5.3 Beach Boat Control (Alfa Net)

The Alfa Net is a directed net used during near ship-to-shore movement to provide displacement landing craft, AAVs, and EFVs control of scheduled waves for a landing beach. Upon reaching the beach, AAVs and EFVs shift to LF tactical, and displacement landing craft shift to Beach Boat Operations (Bravo Net) circuits.

Subscribers	Nets									
	Control Ship Coordination	Primary Control	Beach Boat Control (Alpha Net)	Beach Boat Operations (Bravo Net)	LCAC Control	CLZ Control	LFSP Command	LFSP Control	LF CSS	MEDREG
CCO	C									
PCO	X	C	C	X						
SCO	X	X	X	X						
LCO		X		X	C					
LCAC Control Team		X			X					
LCAC Group				W	X	X				
CLZ Control Team					X	C				
LCAC Airborne Control					W					
Beach Party		X		X		W				
LCU		X	X	X						
AAV			X	X						
PHIBCB				X						
Debark Control				X						
LF TACLOG Det							X	X	C	
GCE TACLOG Det							X	X	X	
Surf TACLOG Det							X	X	X	
LCE							W	W	C/S	
LFSP							C	X	X	
Shore Party							X	C	X	
MRCO										C
MEDREG Teams										X

Legend:

C – NECOS
X – Guard
W – When Directed
C/S – NECOS When Established Ashore

Figure 4-5. Surfaceborne Ship-to-Shore Movement Communications Matrix

4.5.4 Beach Boat Operations (Bravo Net)

The Bravo Net is a free net used to coordinate the launch of displacement landing craft, AAVs, EFVs, and the initial and general offload at a landing beach.



If no more than two waves are being launched, only one net (Alfa or Bravo) is necessary for control of landing craft or AAVs. The other net is activated, but used only in the event of a loss of communications on the active net.

4.5.5 Landing Craft Air Cushion Communications

LCAC are capable of encrypted and unencrypted voice communications and limited visual communications.

Voice communication capabilities include HF, UHF, and VHF radios that may be channelized and quickly shifted. The LCAC crew and the LWC, if embarked, are connected on a craft IVCN that also permits access to any of the external radios. The IVCN can also access the MOMS voice radio system, a short-range UHF walkie-talkie for communicating with well deck and CLZ personnel and between LCAC in formation.

LCAC visual communications are limited to the use of hand signals while maneuvering in well decks and ashore.

4.5.5.1 Landing Craft Air Cushion Operations Net

The LCAC Operations Net is a free net used by launching ships to control assigned LCAC until the PCO assumes OPCON on the LCAC Control Net. Multiple LCAC Operations nets may be employed to provide individual or groups of LCAC-capable ships discrete frequencies.

4.5.5.2 Landing Craft Air Cushion Control Net

The LCAC Control Net is a directed HF or UHF net or data link used to control LCAC from launch to CPPs and during return to amphibious shipping.

4.5.5.3 Cushion Landing Zone Control Net

The CLZ Control Net is a directed net used to control LCAC from the CPP into and out of the CLZ at a landing beach. If AN/KSQ-1 is installed, the AN/KSQ-1 master station to LCAC user link may perform this function.

4.5.6 Landing Force Support Party Command Net

The LFSP Command Net is used by the LFSP to direct CSS operations with TACLOG detachments, the shore party, and the HST.

4.5.7 Landing Force Support Party Control Net

The LFSP Control Net is used by the shore party to coordinate logistics requests from surfaceborne units and the surfaceborne TACLOG detachment.

4.5.8 Landing Force Combat Service Support Net

The LF CSS Net provides CSS coordination with TACLOG detachments when the appropriate LCE is established ashore.

4.5.9 Medical Regulating Net

A dedicated MEDREG Net is established to pass patient movement and information pertaining to medical requirements and capabilities among ATF and LF units. Depending on the phase of the operation, MRCO afloat or PEO ashore controls the MEDREG system. Units designated to guard the MEDREG Net should be designated in Annex K (C4 systems) of the OPLAN.

CHAPTER 5

Conducting Airborne Ship-to-Shore Movement

5.1 PURPOSE

This chapter discusses the doctrine, command relationships, delegation of authority, and the C2 organization for conducting LF debarkation and airborne ship-to-shore movement. It also briefly explains the flexibility added to an amphibious operation when vertical lift aircraft are employed. It stresses the importance of coordinating with surfaceborne assault or action forces, explains the manner in which helicopters and tilt-rotor aircraft may be employed, and the organizations used to control vertical assault operations during ship-to-shore movement.

5.2 BACKGROUND

Vertical lift aircraft add significant flexibility to amphibious operations. Helicopters can operate from all amphibious ships and land in almost any cleared site within the landing area. Helicopters and tilt-rotor aircraft can rapidly inject the LF and other required assets into HLZs and VLZs to achieve surprise, avoid obstacles or defenses, and provide depth in the operation. They can also respond quickly should their capabilities be required in a tactical role. Additional information on helicopter operations is found in NAVAIR 00-80T-122, Helicopter Operations for Air-Capable Ships; and JP 3-04.1, Joint Tactics, Techniques, and Procedures for Shipboard Helicopter Operations.

5.2.1 Helicopter Employment

In ship-to-shore movement, vertical assault aircraft are used primarily for troop transport and logistics resupply. The use of airspace within an AO by aircraft and supporting fires is closely coordinated. During ship-to-shore movement, this coordination is provided through the Navy TACAIR officer and an HCS in the ATF TACC.

5.2.2 Helicopter Capabilities and Limitations

Section 1.4.5 provides a brief description of the types of helicopters routinely employed during amphibious operations as well as helicopter tactical advantages and disadvantages. However, the capabilities and limitations of helicopters must be examined when considering their employment as a means of transporting the LF during ship-to-shore movement or using them in support of the CONOPS ashore.

Helicopter capabilities include the ability to:

1. Ascend and descend vertically into small unprepared areas inaccessible by other means to load and unload troops, equipment, and supplies.

2. Fly safely at low altitudes and operate under conditions of low ceilings and reduced visibility, and during periods when surf conditions preclude the use of landing craft and amphibious vehicles.

3. If required by the mission, provide rotary-wing CAS and emergency support within the AO, including resupply, reinforcement, and withdrawal.

4. Provide fast and flexible airborne platforms that can deploy from OTH and perform numerous missions within the AO, such as C2, observation, liaison shuttling, RECON, SAR, MEDEVAC, and CASEVAC. In addition, missions such as TRAP, illumination roles in support of ground action, heavy lift, transport of artillery and LAVs, armed escort, and anti-armor CAS may be performed.

Helicopter limitations are:

1. Helicopters require more fuel than surface vehicles performing similar tasks. Additionally, greater maintenance effort is required for helicopters than other modes of transportation.

2. In certain operations, secrecy may be compromised by helicopter engine and rotor noise and/or dust in the LZ.

3. Weather can significantly affect helicopter employment. Helicopter operations are severely limited when icing conditions prevail or visibility is significantly restricted, and their lift capability is affected by changes in atmospheric conditions, such as altitude, wind, humidity, and temperature. Additionally, although helicopters may be able to fly under cloud decks, low ceilings may reduce the effectiveness of, or preclude the use of, fixed-wing aircraft that may be counted on to provide escort, LZ preparation, or CAS for the helicopter force.

4. The CH-53E has the best heavy lift capability, up to 36,000 pounds on a sea level, standard day. However, even this aircraft may not provide adequate lift capability for all types of MAGTF equipment under all conditions.

5.2.3 Tactical Organization

When vertical lift aircraft are employed in support of the landing plan, close coordination is required between the ATF and LF parallel chains of command.

The TACRON detachment, manning the ATF TACC aboard the LHA or LHD, is the NSE responsible for organizing all aspects of the air operation. The ATF TACC coordinates between representatives from the helicopterborne unit commander, ACE, ADC, SAR detachment, CCO, LFOC, and the helicopter transport group or unit commander. This coordination is essential to the development of the ATO at the daily ATO meeting, APB, or air planning cell meeting.

The MAGTF prepares the HEALT in close coordination with the ACE, GCE, and the TACRON. (See Paragraph 3.4.2.13.) This document is the principal component of the LF OPLAN that guides vertical assault employment and is incorporated into the theater ATO.

5.3 PLANNING CONSIDERATIONS

Objectives and missions that cannot be carried out by surfaceborne amphibious operations often may be achieved by vertical assault operations.

5.3.1 Helicopter Employment Considerations

The principal considerations for the employment of helicopters are:

1. The role of helicopters in the LF CONOPS.
2. Quantity and types of helicopters available.
3. Numbers of helicopter-capable ships available that can operate and maintain helicopters, and those that can only operate helicopters.
4. Location, nature, number, and size of HLZs and their approach and retirement lanes.
5. Capabilities and dispositions of any opposing forces, especially location, type, and density of any anti-aircraft weapons.
6. Requirements for supporting arms, linkup, and CSS.
7. Oceanographic or weather influences such as:
    a. Suitability of sea state for launch and recovery.
    b. Weather conditions to be encountered at launch and recovery, en route and at the HLZ. This includes ceiling, visibility, icing, and turbulence.
8. Availability of alternate plans for landing serials scheduled for helicopterborne waves aborted during the landing.

5.3.2 Helicopter Support Requirements

When helicopters are included in the landing plan, support requirements to consider are:

1. Ships having helicopter refueling and rearming capabilities should be considered when establishing refueling or rearming cycles.
2. When practicable, decks should have no less than two spots to maintain flight section integrity.
3. As the assault or action progresses, FARPs may be established ashore by the LF.
4. Provisions for downed aircrew and aircraft should be included in the LF OPORD.
5. Requirements for deploying HSTs in LZs.

5.3.2.1 Helicopter Support Team

An HST is a task organization formed and equipped for employment in an LZ to facilitate the landing and movement of helicopterborne troops, equipment, and supplies, and to evacuate selected casualties and EPWs. Additional details regarding the HST are contained in Appendix H.

5.4 AVIATION COMMAND AND CONTROL DURING SHIP-TO-SHORE MOVEMENT

Helicopterborne ship-to-shore movement is coordinated between the ATF and LF commanders and planners. During amphibious operations, particularly ship-to-shore movement, the NTACS and MACCS, two distinct but complementary aviation control and aviation C2 systems, integrate to form one ATACS that executes the helicopter landing plan and phases MAGTF aviation C2 ashore. Additional information on the ATACS is contained in JP 3-09.3, Joint Tactics, Techniques, and Procedures for Close Air Support (CAS).

The ATF commander coordinates vertical assault operations through the TACGRU or TACRON. The LF commander employs the TACLOG group to assist Navy control officers. The ACE commander exercises LF aviation C2 through the ATF's shipboard facilities and through airborne nodes as required while phasing C2 ashore.

Execution of the operation is decentralized to helicopter transport groups and unit commanders by delegating air traffic control of vertical assault aircraft to an ATF TACC or TADC, and specific authority to change certain aspects of the operation is delegated to subordinate LF commanders. The control organization for a single LZ is depicted in Figure 5-1. When multiple LZs are employed in the landing area, a helicopter transport unit commander with a similar air traffic control organization may be designated for each LZ.

5.4.1 Amphibious Task Force Air Control

The paragraphs below discuss the organizations and personnel who execute air control responsibilities within the ATF.

5.4.1.1 Tactical Air Control Group

Previously discussed in Paragraph 2.2.3.2, the TACGRU is a tactical component of the AF that provides aircraft control and warning facilities afloat for offensive and defensive missions within the AOA.

5.4.1.2 Tactical Air Control Squadron

The TACRON is an administrative and tactical component of the TACGRU, and is organized and trained to man the ATF TACC and TADC. Within an ATF, the TACRON performs the same functions as the TACGRU in a larger AF. It is commanded by an aviator and serves as the air planning staff for the ATF commander. Each LHA or LHD embarks a TACRON detachment, which reports to the PHIBRON or ESG commander. One LHA or LHD TACRON detachment will be designated the primary ATF TACC. The others are designated as TADCs, subordinate to the primary ATF TACC.

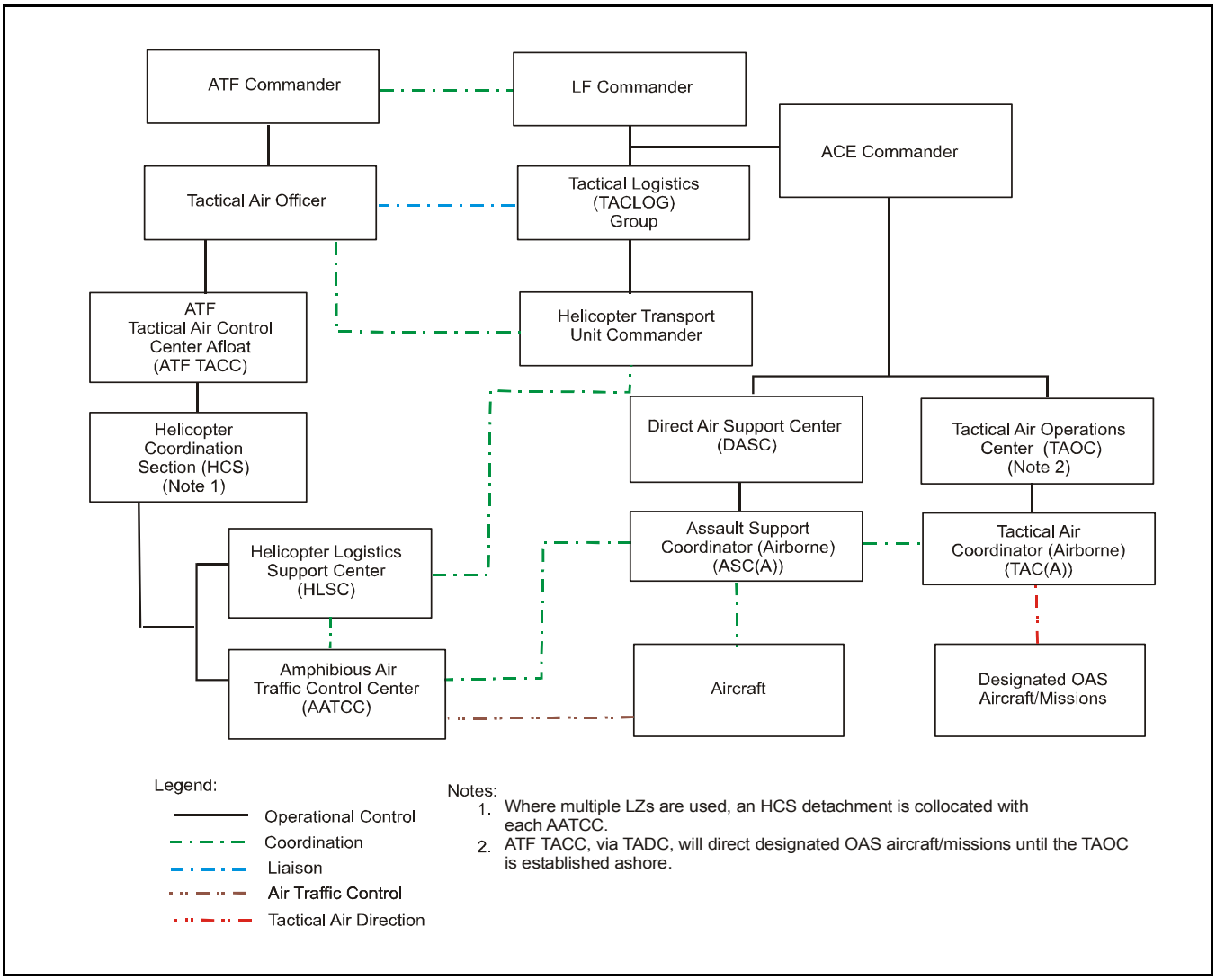


Figure 5-1. Control Organization for Airborne Ship-to-Shore Movement

A TACRON is also capable of splitting its organization in a manner that allows the establishment of a temporary TADC on other platforms, such as LPDs.

5.4.2 Tactical Air Officer

The TACAIR officer is a naval aviator or NFO under the ATF commander who coordinates planning of phases of air participation in the amphibious operation and the air operations of supporting forces en route to and in the objective area. The TACIAR officer is usually the senior TACRON officer embarked during MEU(SOC)-level operations and the TACGRU or TACRON CO when operations require a MEB- /MEF-level LF. The TACAIR officer does not exercise command authority over aviation resources directly, as that authority is vested in the

MAGTF commander and subordinate element commanders within the LF. Until control is passed ashore, this officer exercises control over operations of the ATF TACC and is charged with the following:

1. Control of aircraft in the ATF AO assigned for TACAIR operations

2. Control of other aircraft entering or passing through the objective area

3. Working closely with the JFACC's ACA and AF warfare commanders (particularly the ADC) to ensure a coherent air picture is maintained throughout the ATF AO.

For vertical assault ship-to-shore movement, the TACAIR officer is responsible for:

1. Coordinating and supervising the operation

2. Organizing the TACGRU or TACRON in support of the landing plan

3. Coordinating with the CCO

4. Maintaining liaison with the TACLOG group.

5.4.3 Tactical Air Control Center Watch Officer

The TACCWO is the senior watch officer for a TACC duty section who is in charge of all operations in the ATF TACC. This officer is responsible to the TACAIR officer for control of aircraft and air warning facilities within the AOA.

5.4.4 Amphibious Task Force Tactical Air Control Center

As a critical node in the NC2AO concept, the ATF TACC is the primary shipboard air control agency within an AF and controls air operations supporting the landing plan and the LF CONOPS. Tactical aircraft and helicopter operations are coordinated with supporting arms and other air operations through this center. The ATF TACC is normally located with the ATF commander and staff aboard an LHA or LHD. If the ATF consists of multiple LHAs or LHDs, one of the TADCs is designated as the primary backup in the event the primary ATF TACC suffers a casualty or is otherwise unable to function.

ATF TACC functions include:

1. Acting as the AF's AREC, charged with managing the effective employment of aircraft assigned to support the AF per the ATO

2. Merging the ATOs within the AF into the AF and theater ATO

3. Coordinating with the ADC, the USWC, and the SUWC to ensure airspace deconfliction and an integrated defense for the AF

4. Coordinating aircraft hand-over and integrating TMD with the JAOC

5. Directing CAS and other air support as requested by the LF, providing aircrews with complete and timely briefings on targets to be attacked, and expected conditions in the air or on the ground that might be encountered in the AOA

6. Consolidating and coordinating air support requirements with supporting aviation, and coordinating to ensure sea- and land-based air forces scheduled to execute support missions receive the latest intelligence

7. Monitoring and coordinating aircraft operations to include those conducted by LF assets and supporting other warfare commanders in the SUW and USW efforts

8. Providing TACAIR control and flight following vertical assault waves, unless delegated to a TADC or another air control unit

9. Coordinating and/or controlling air traffic in the AOA to protect friendly aircraft from collision, provide air navigational assistance, provide assistance to aircraft in emergency situations, and assist in early identification of opposition or unidentified aircraft in the area

10. Providing an ASC to advise the SAC and FSC regarding available CAS assets, including weapons load, fuel status, and other pertinent data

11. Representing the AF commander at the JETB

12. Chairing the AF's APB to ensure smooth integration of AF air missions into the theater ATO

13. Controlling MEDEVAC helicopter movements to CRTSs based on the advice of the MRCC

14. Acting on requests for helicopter employment from the HCS, helicopter support transport unit commander, and the helicopterborne unit commander

15. Maintaining a continuous radar plot of assigned transport helicopters and escorts

16. Monitoring ASC(A) operations.

The ATF TACC and its subordinate TADCs conduct their part of the aircraft ship-to-shore movement by combining positive radar control with procedural control and providing air traffic control for aircraft from an RP to an LZ and from an LZ to a breakup point.

ATF TACCs and TADCs are collocated or integrated with the TACLOG, HLSC, and MRCC. During amphibious operations, the TACLOG monitors requests from assault or action units or their CSS elements ashore. Requests for delivery by helicopter are forwarded through the HLSC, which coordinated the debarkation of serials as delineated in the landing plan. The requests are then forwarded to the ATF TACC or TADC for execution.

In large-scale operations when the DASC is established ashore, it assumes control of MAGTF direct support aviation. Complete coordination authority, however, is seldom passed all at one time. The LF assumes responsibility for various fire support functions as the necessity for those functions is established ashore. The DASC is usually the first principal MACCS agency ashore and, once established, normally assumes control of DAS aviation assets overland while the ATF TACC retains control of aircraft moving from ship to shore.

Additional information on the ATF TACC is found in NTTP 3-02.1.3, Amphibious/Expeditionary Operations Air Control, and JP 3-09.3.

5.4.4.1 Tactical Air Direction Center

The TADC is an air operations installation under control of the ATF TACC or the ACE TACC, from which aircraft and air warning service functions of TACAIR operations in support of amphibious operations are directed. A TADC may also be established in support of a subordinate Navy commander charged with conducting other operations, such as a simultaneous or nearly simultaneous operation at a landing area widely separated from the main AOA or supporting the advanced force commander conducting preassault operations. As per Paragraph 5.4.1.2, a TADC may also be established on an alternate flagship.

5.4.5 Helicopter Coordination Section

The HCS is an integral part of the ATF TACC. It is the section that coordinates vertical assault operations decentralized under the control of subordinate air control units. It is normally augmented with personnel from the LF's ACE during amphibious operations. The ACE also provides advice to the HCS on the employment of and availability of its vertical lift aircraft and crews. Close coordination with the MAGTF air officer and the ACE staff is essential to support the priority of effort delineated in the landing plan.

Within the HCS, the HCO supervises the HCS and supporting air control units, advises the TACAIR officer on vertical lift employment, and coordinates the planning and execution of airborne ship-to-shore movement.

The HCS is organized into two subdivisions: an HCU and an HAU. It can also provide a detachment to carry out HCS duties on an alternate platform, such as an LPD.

5.4.5.1 Helicopter Coordination Unit

The HCU is responsible for the coordination and direction of vertical lift aircraft.

5.4.5.2 Helicopter Advisory Unit

The HAU maintains current data on the status, availability, locations, fuel requirements, available deck space, armament, and the progress of the helicopterborne operation. Such information is passed to the HCU to assist in vertical lift employment decisions and actions.

5.4.5.3 Helicopter Coordination Section Functions and Mission

The primary functions and missions of the HCS are to:

1. Coordinate vertical lift movements with supporting arms and other AOA air traffic.

2. Coordinate vertical lift operations conducted by subordinate control agencies and control specific vertical assault missions when required.

3. Assign sectors, routes, and CPs to each primary TADC when not covered in the OPORD or when changes occur.

4. Monitor vertical lift operations conducted by primary TADCs, including CASEVAC or MEDEVAC missions.

5. Maintain readiness data on vertical lift aircraft and vertical lift aircraft-capable ships.

6. Coordinate changes to the HEALT with the HLSC.

7. Act on requests for additional vertical lift support.

8. Coordinate emerging requirements and redirect vertical lift when required.

9. Monitor SAR operations.

10. Prepare the daily vertical assault aircraft air plan and provide input to the ATO.

5.4.5.4 Helicopter Coordination Section Communications

The HCS is the NECOS for the HR, HA, and HC nets. These nets are described in Section 5.8.

5.4.5.5 Tactical Air Direction Center Helicopter Control Subsection

When multiple LZs are designated, each TADC acting as an assault helicopter ACU maintains a collocated HCS detachment. The HC subsection performs functions similar to those of the HCS within its sector of responsibility, thus, allowing the primary TADC to focus on air traffic control. The functions of the HC subsection are to:

1. Advise the helicopter transport unit commander on the most efficient employment of vertical lift.

2. Coordinate vertical lift employment with the HCS when required.

3. Coordinate changes to the HEALT with the helicopterborne TACLOG detachment, HLSC, and the AATCC.

4. Issue instructions to air-capable ships to open flight decks and launch or recover aircraft.

5. Maintain liaison with the MRT to direct CASEVAC or MEDEVAC aircraft to the appropriate CRTS.

6. Guard the HC, HR, and HA nets.

5.4.5.6 Medical Regulating Control Center

Depending on the phase of the operation, the MRCC afloat is normally collocated with AATCC when command and/or net control is sea based, or with the DASC (LF MRCC) when command and/or net control has been passed ashore. Both MRCCs:

1. Maintain the status of current requirements and capabilities of HSS facilities in the AOA.

2. Designate which HSS facilities are to receive and treat casualties.

3. Maintain liaison with MRT members on the MEDREG Net.

The supported and supporting roles and responsibilities of the MRCCs are decided prior to deployment and written into the Annex Q of the OPLAN or OPORD.

5.4.6 Helicopter Transport Group/Unit Commander

In large-scale operations with multiple LZs, a helicopter transport group/unit commander is designated for each LZ and provides air traffic control through its embarked TADC and facilitates the movement of serials ashore by vertical lift. The transport group/unit commander's role is analogous to that of the PCO. This individual is responsible for matters related to the flight control of vertical lift, and to best accomplish all assigned missions, is collocated with the AATCC and HLSC. The helicopter group/unit commander is also responsible for:

1. Assuming TACON of assigned landing craft

2. Launching scheduled waves IAW the HEALT and helicopter landing diagram

3. Maintaining the current location and status of ships and aircraft assigned to conduct the landing

4. Monitoring weather conditions and recommending termination of flight operations should conditions warrant

5. Maintaining the debarkation and embarkation status

6. Maintaining liaison with the helicopterborne TACLOG detachment.

5.4.7 Amphibious Air Tactical Control Center

Each LHA and LHD has an AATCC that is responsible for IMC approach and departure control for its respective ship. The AATCC may be employed in one of two ways to provide decentralized TACON of helicopters during an amphibious operation:

1. AATCC personnel augment the ATF TACC and/or the TADC's HCS.

2. The AATCC reports directly to the ATF TACC's HCS as a subordinate air control unit.

5.4.7.1 Amphibious Air Tactical Control Center Functions and Mission

The functions and mission of the AATCC are to:

1. Control the movement of vertical assault aircraft operating in its assigned control zone to conduct precision approaches to the LHA or LHD.

2. Control the rendezvous of assault and escort aircraft launching within its assigned control zone prior to aircraft being handed over to the ATF TACC for en route control.

3. Maintain and report to the HCS and ATF TACC the status and location of aircraft scheduled for launch.

4. Advise the HCS or ATF TACC on all matters pertaining to vertical lift movement within its control zone that may require coordination with supporting arms.

5.4.8 Helicopter Logistic Support Center

The HLSC is manned by LHD or LHA ship's company personnel and is located in close proximity to the HCS detachment and the TACLOG detachment. The OIC of the HLSC is the HLC. This individual is a Navy officer, comparable to the PCO in surfaceborne ship-to-shore movement.

The HLSC coordinates the debarkation of air serials IAW the landing plan, the HEALT, and the LF landing sequence table. Debarkation of on-call waves and nonscheduled units, including deviations from the landing plan, is coordinated with applicable ships by the HLSC using the Helicopter Logistics Command Net.

The LF commanders ashore request on-call, nonscheduled serials and emergency resupply based on priorities. The LFSP and HSTs may act to facilitate requests. The HLSC processes air requests for serials through the HCS and, once approval is given, notifies the debarkation control officer on the applicable ship of the nature of the mission and instructs that unit to prepare for vertical assault operations. The HLSC then maintains the status of the serials to include the time of the request, verification of the request, the time the serial departs the delivery deck, and identification of the aircraft carrying the serial.

The coordination the HLSC performs is dependent on the logistics communication nets available. The TACLOG, located aboard the CCP, monitors such requests and assists as required. The TACLOG informs the requesting LF commander or the supporting CSS element when the mission is approved, the type and number of aircraft, the expected time of arrival at the LF commander's position, and if applicable, the route. This information is required by the ground unit's FSCC for fire support coordination. Once DAS control is passed ashore, the DASC provides this information.

An alternate HLSC is designated in the same ship as the alternate primary AATCC and is activated concurrently with the HLSC.

5.4.9 Tactical-Logistical Group

The TACLOG group consists of representatives designated by LF commanders to assist Navy control officers aboard control ships in the ship-to-shore movement of troops, equipment, and supplies. The TACLOG group is discussed in detail in Appendix F.

5.4.10 Transitioning Marine Air-Ground Task Force Aviation Command and Control Ashore

The ACE commander's C2 network is transitioned ashore in a manner that, based on the tactical situation, best supports the MAGTF commander's CONOPS. In the initial phases of the amphibious operation, the MAGTF deploys terminal controllers and DAS agencies ashore to control, coordinate, and integrate air operations and fires. As operations progress, the ACE may transition all, or part of, LF aviation C2 ashore depending on an operation's scope, complexity, and duration. The paragraphs below represent the typical sequence of aviation C2 movement ashore in support of the MAGTF. The MCWP 3-25 series contains additional information on the MACCS.

5.4.10.1 Landing Force Terminal Control

Terminal controllers deploy with the initial assault elements ashore. The principal terminal controller is the FAC, who leads a TACP.

5.4.10.1.1 Forward Air Controller

The FAC is a Marine aviator trained, qualified, and designated to control aircraft in support of ground troops from a forward ground position. The FAC aids in target identification and reduces the potential for fratricide.

The FAC is responsible for:

1. Knowing the adversary's situation, selected targets, and the location of friendly units
2. Knowing the supported units' plans, positions, and requirements
3. Locating targets of opportunity
4. Advising the supported ground commander on proper employment of air assets
5. Requesting and controlling CAS
6. Performing BDA.

5.4.10.1.2 Tactical Air Control Party

Administratively, the TACP is not part of the MACG, but functions as an important aviation C2 node. It is located within the GCE, provides an avenue for GCE commanders to request and direct CAS and assault support, and provides aviation advisory personnel and the means to integrate aviation fires and maneuver into combined arms operations.

TACPs are part of the MARDIV and are assigned to each infantry battalion, LAR battalion, and tank battalion. In an infantry battalion, TACPs are typically forward deployed as part of a maneuver company operating in support of an infantry battalion and are usually the first aviation C2 elements ashore.

Depending on the MAGTF's organization for combat, the TACP's principal conduit to coordinate and integrate aviation fires and assault support is through the DASC or a ground-based extension of the DASC, an ASLT, or an ASE. JP 3-09.3 contains additional information on TACPs.

5.4.10.1.3 Air Officer

An air officer at the battalion, regiment, and division monitors TACP operations, and an air officer is also assigned to the MAGTF HQ. Air officers at every level of the GCE and MAGTF ensure thorough coordination and integration of the six functions of Marine aviation discussed in Paragraph 2.3.3.

5.4.10.1.4 Supporting Arms Liaison Team and Firepower Control Team

At the battalion level, the SALT is led by a Marine aviator designated a FAC, who coordinates the activities of subordinate FCTs operating in support of maneuver companies. FCTs, closely followed by SALTS, may be among the first elements of aviation C2 ashore. An ANGLICO air officer at the Army, allied, or coalition brigade and division HQ monitors SALTs and FCTs. The DASC, or a ground-based extension of the DASC, provides the principal conduit for SALTs and FCTs to coordinate and integrate aviation fires and assault support for Army, allied, or coalition forces as part of, or in conjunction with, a MAGTF.

The ANGLICO, organic to the MEF, provides the MAGTF commander a liaison capability with foreign area expertise to plan, coordinate, employ, and conduct terminal control of fires in support of joint, allied, and coalition forces. ANGLICO elements are designed to attach and integrate into units not typically found in a MAGTF at the company, battalion, brigade, and division level.

MCWP 3-25.3, Marine Air Command and Control System Handbook, contains additional information on the ANGLICO.

5.4.11 Direct Air Support Center

The DASC is the ACE's primary direct support air control center. To facilitate DAS operations, ground-based and airborne extensions of the DASC may be employed based on the tactical situation.

5.4.11.1 Air Support Landing Team

As required by the tactical situation, the MASS commander, primarily charged with coordinating and controlling MAGTF DAS, may task organize and deploy ASLTs subordinate to the DASC to maintain liaison and coordination between the DASC and the FSCC, the SACC, or the ATF TACC in an assigned area or for a designated unit. This coordination is vital to the integration of DAS missions with other supporting arms. When employed, ASLTs may be among the first elements of the MASS ashore during an amphibious operation. The ASLT can provide a vital link between the TACPs and the SACC or FSCC, or between the DASC and ATF TACC in an assigned area. The number, size, and composition of ASLTs are determined by the MASS CO based on the experience of FSCC personnel, terrain, OPTEMPO, and the GCE's mobility. An ASLT typically varies in size from a single Marine with a man-portable radio or field telephone to several MAGTF personnel operating from MRCs vehicles.

MCWP 3-25.5, Direct Air Support Center Handbook, contains additional information on ASLTs.

5.4.11.2 Air Support Element

An ASE is a task-organized element that may be employed by the MASS to perform various air support and air control functions. Employment options can range from MEU-level operations with limited assets and endurance, to a multidivision operation where the ASE controls and manages a division's airspace while remaining subordinate to the DASC. The ASE can function as an extension of the ATF TACC or AATCC in conjunction with the battalion TACP. In support of a large-scale operation involving several major subordinate commands, the ASE may be augmented with assets and personnel to support continuous control of DAS over an extended period. In this situation, the DASC is centrally located to facilitate supporting multiple MARDIVs and ASEs. When the

tactical situation dictates its employment, the ASE may be the first DAS element to arrive ashore. ASEs may be equipped with a CASC, MRC vehicles, or man-portable radios.

Additional information on ASEs is contained in MCWP 3-25.5.

5.4.11.3 Direct Air Support Center

Depending on the tactical situation, the MASS commander may employ an entire DASC, DASC(A), ASE, or an ASLT to support the MAGTF CONOPS. Regardless of the elements employed, MASS elements remain subordinate to the DASC. The DASC is the ASE commander's principal air C2 element responsible for the direction and control of CAS or DAS.

The DASC responds to MAGTF major subordinate commanders' requirements for DAS by processing immediate air support requests, coordinating aircraft employment with other combined arms through the GCE's senior FSCC, and directing designated air operations. The DASC's role in DAS provides a vital link among the GCE, LCE, and the ACE. The DASC is subordinate to the ACE TACC, ACE TADC, and the ATF TACC depending on established command relationships and the phase in which MAGTF aviation C2 is passed ashore. The ACE TACC exercises centralized command, and the ATF TACC provides centralized control of air operations within a designated AOA, while the DASC provides centralized control of OAS, assault support, EW, and air RECON missions within its designated area.

MCWP 3-25.5 contains additional information on the DASC.

5.4.11.3.1 Direct Air Support Center Mission

The DASC provides the MAGTF with a land-based agency for the coordination of DAS missions. It functions in a decentralized mode of operation, subordinate to the established TACC or TADC. It is normally the first major air control element established ashore for large-scale amphibious operations. As soon as practicable in the operation, the DASC is established ashore to provide centralized coordination of DAS. The DASC coordinates CAS strikes controlled by the FAC/FAC(A) and TAC(A) with other fire support means to include naval surface fire and artillery.

In addition to TACAIR support operations, DASC controls vertical assault operations when air control is passed from the ATF TACC to the DASC. After control has been passed ashore to the DASC, the AATCC remains prepared to reassume control, if required. Additionally, an alternate AATCC is designated by the ATF TACC and assumes control should the AATCC (or the DASC) suffer a casualty or be otherwise unable to carry out its responsibilities.

The LF MRCC/PET is collocated with the DASC for advice on casualty movement and other medical issues. As long as LF vertical lift aircraft are based afloat, the AATCC remains in operation to provide air traffic control for aircraft operating between ships and the beach.

5.4.11.3.2 Direct Air Support Center Functions

The DASC:

1. Receives the ATO from the ATF TACC or ACE TACC and coordinates preplanned DAS.

2. Receives, processes, and coordinates FRAGORDs and requests for immediate DAS.

3. When delegated authority by the ACE commander, adjusts preplanned schedules, diverts airborne assets, and launches aircraft as necessary.

4. Coordinates with the MAGTF FFCC or GCE senior FSCC.

5. Coordinates the execution of DAS missions with other combined or supporting arms through the appropriate FFCC or FSCC, and as required, with the appropriate ACE C2 agencies.

6. Receives and disseminates pertinent tactical information reported by aircraft conducting DAS missions.

7. Provides aircraft and air control agencies with advisory and threat information to assist in the safe conduct of flight. Such information includes artillery, naval surface fire, TACTOM, air strikes, opposition antiaircraft activity, and restrictive fire plans.

8. Monitors, records, and displays information on DAS missions.

9. Maintains friendly and hostile ground situation displays necessary to coordinate DAS operations, and provides the ATF TACC or ACE TACC with timely SA of friendly and hostile force locations and BDAs.

10. Receives and processes JTARs and ASRs called in over the TAR/HR Net.

5.4.12 Airborne Terminal Control Agencies

Depending on the size and scope of an amphibious operation and the level of C2 required, the ACE commander may direct the employment of airborne coordinators to facilitate operations. Marine aviators organic to the MAG at the squadron level and designated as airborne terminal controllers act as airborne C2 extensions of the DASC or the TAOC.

JP 3-0.3, MCWP 3-25.3, and MCWP 3-25.5 contain additional information on airborne terminal control agencies.

5.4.12.1 Assault Support Coordinator (Airborne)

The ASC(A) is a Marine aviator who, from an aircraft, coordinates aviation asset movements during assault support operations. This officer shall have an extensive knowledge of the MACCS, assault support, airspace management, fire support coordination, fixed- and rotary-wing capabilities, and it is essential that the ASC(A) be thoroughly knowledgeable in every aspect of the amphibious operation.

The ASC(A) performs the following functions:

1. Provides SA to the assault support force regarding routing, weather, delays, and threat status

2. Relays immediate assault support requests to assist the DASC or the ATF TACC

3. Serves as a conduit to the C3 system by relaying information to and from the assault support force

4. Exercises launch authority for immediate and on-call missions when authorized by the ACE, or as directed by the DASC or the ATF's AATCC

5. Coordinates with the TAC(A) for combined arms support of the AMC

6. Provides recommendations to the AMC regarding deviations in assault support asset routing during ingress and egress

7. Establishes connectivity and the means for coordination between the TAC(A) and the ACE commander

8. Collocates with the helicopter unit commander in a single aircraft when feasible to facilitate timely and coordinated decisions

9. Coordinates with TAC(A)s to ensure fixed-wing preparation strikes controlled by the TAC(A)s are conducted in an accurate and timely manner

10. Advises the AATCC on the status of the landing, including changes made in LZ selection.

5.4.12.2 Tactical Air Coordinator (Airborne)

The TAC(A) is a naval aviator trained and designated to coordinate, from an aircraft, the action of aircraft engaged in armed RECON and CAS of ground or sea forces. A TAC(A), when employed, is an onsite airborne extension of the DASC, ATF TACC, or TADC, and is normally the senior coordinator with authority over aircraft operating within an assigned airspace. The TAC(A) may or may not be assigned depending on mission requirements and aircraft availability.

Depending on the phase of the operation, the DASC, ACE TACC, ATF TACC, or TADC defines the TAC(A)'s authority over aircraft operating in the assigned area. During helicopter operations in which a TAC(A) is employed in conjunction with an ASC(A) the ACE commander or a designated representative establishes the relationship between the two. The TAC(A)'s principal responsibilities are to deconflict aircraft and coordinate employment of supporting aircraft with other combined or supporting arms. In carrying out these responsibilities, the TAC(A) coordinates, as necessary, with TACPs, FSCCs, FAC(A)s, ASC(A)s, and the FDCs for artillery and NSFS.

5.4.12.3 Forward Air Controller (Airborne)

A FAC(A) is normally an appropriately trained and qualified naval aviator or NFO operating from an aircraft and designated to perform air RECON and surveillance, conduct terminal control of aircraft engaged in OAS, control artillery and NSFS missions, act as a radio relay as required by ground FACs, and control LZ preparations, including marking those LZs. The FAC(A)'s mission is separate and distinct from that of a TAC(A); the FAC(A) controls aircraft, and the TAC(A) coordinates.

5.4.13 Landing Zone Guidance and Control

TACAIR traffic control functions may be performed in an LZ by a qualified FAC, while guidance and coordination functions may be performed by specialized teams. HSTs and ITG teams provide advisory services to aircraft operating in and around LZs and serve to locate, identify, and mark VLZs. HSTs and ITGs work

independent of a FAC, but when required a FAC may act to control and deconflict LZ operations where the HST or ITG is established.

5.4.13.1 Initial Terminal Guidance Team

ITG is a function that is normally performed by teams from the MEF's force RECON company, the MARDIV's RECON battalion, or ANGLICO. ITG teams have the inherent capability to provide terminal guidance for initial helicopterborne waves in the LZs. The teams are composed of personnel inserted in the LZ in advance of the LZ control team. They execute prelanding RECON tasks, and establish and operate signal devices for guiding the helicopterborne waves from the IP to the LZ.

The ITG team's responsibilities are:

1. Locating and identifying VLZs
2. Marking VLZs, landing sites, and landing points based on the CONOPS and threat
3. Emplacing and operating homing beacons and other guidance devices
4. Locating, identifying, and providing warning if there are obstructions in the VLZ, including CBR hazards
5. Removing VLZ hazards, when feasible
6. Locating, identifying, and providing warning if there are obstacles in aircraft approach and egress routes
7. Removing approach and egress obstacles, when feasible
8. Determining and communicating wind direction and ground surface conditions
9. Providing advance notice of opposing force positions
10. Recommending action by follow-on helicopter waves
11. Reporting opposition force activity that may hinder the landing.

ITG command relationships are described in detail in Appendix H.

5.4.13.2 Helicopter Support Team

An HST is a task organization formed and equipped for employment in an LZ to facilitate the landing and movement of helicopterborne troops, equipment, and supplies, and to evacuate selected casualties and EPWs. Within the Marine Corps, HSTs are organic to the MLG, specifically, the landing support company of the support battalion.

5.4.13.3 Tactical Air Operations Center

The TAOC is the ACE commander's principal AD agency for conducting surveillance, airspace control, TACAIR battle management, and AAW. Through radar inputs from organic sensors and data link information from other

MRUs, the TAOC provides real-time surveillance of assigned airspace in addition to air direction, positive aircraft control, and navigational assistance to friendly aircraft. Its primary function, to conduct and coordinate AAW, is accomplished through the employment of various AD weapons systems, including interceptor aircraft and GBAD weapons. By collecting and displaying information from its own sensors, other Marine Corps sources, and external sources, the TAOC controls assigned airspace, and directs and controls the fires of assigned AD assets. It can be used to enhance the ability of the ACE TACC to prosecute the ACE's support of deep operations. The TAOC functions in a subordinate role to the ACE TACC and is a unit adjacent to the DASC. The MACG's MACS furnishes the equipment and personnel to operate the TAOC and its subordinate facilities and agencies.

The TAOC:

1. Provides airspace control, management, and surveillance for its designated sector or AOI
2. Provides navigational assistance, including itinerant air traffic control, to friendly aircraft
3. Detects, identifies, and controls hostile aircraft and missile intercepts
4. Deploys EW/C sites to supplement or enhance the TAOC's radar coverage
5. Assumes agency coordination functions of the Alt ACE TACC or Alt ACE TADC for limited or specified periods
6. Recommends employment of assigned weapons and surveillance means
7. Recommends AD sectors, subsectors, and WEZs for itself and component elements
8. Deploys sensors and communications systems for air surveillance
9. Detects, identifies, and classifies aircraft and missiles within its assigned sector
10. Displays and disseminates appropriate air and ground information to adjacent, higher, and subordinate agencies, such as the ACE TACC, another TAOC, the DASC, GBAD units, aircraft, and MATCDs
11. Selects and assigns appropriate weapons to engage and destroy opposing force air threats
12. Controls subordinate AD element fires
13. Functions as an Alt TACC/Alt TADC when directed for limited or designated periods
14. Interfaces with adjacent and higher AD agencies
15. Manages AD resources
16. Coordinates and executes EMCON conditions in its assigned sector
17. Conducts itinerant air traffic control and provides navigational assistance to friendly aircraft
18. Provides TBM warning to the MAGTF.

During the initial phases of an amphibious operation, ATF aircraft operating from supporting aircraft carriers, LHAs, and LHDs provide airborne AD ashore. Stinger teams, initially in direct support of the GCE, represent the first dedicated, operational shore-based AD capability responsible for low-altitude threats. Additional AD assets will typically deploy with follow-on elements of the ACE, GCE, CE, and LCE. During the buildup of MAGTF AD ashore, MWCS detachments, MWSS detachments, and MATCDs establish and operate FOBs. FOBs allow MAGTF aircraft, including AAW-capable platforms, to establish forward bases ashore. As LF aircraft begin using FOBs, GBAD assets provide AD. The early introduction of Stinger teams and EW/C radar/control elements ashore extends shipboard weapons employment, radar surveillance, identification, and coordination and control capabilities. EW/C sites provide initial EW, cueing, and surveillance capabilities against opposition air and missile threat, including TBMs. As additional Stinger assets move ashore, the remainder of the TAOC's equipment and personnel also phases ashore. Liaison is established with the landward SAAWC to coordinate MAGTF AAW operations. Once the TAOC and other GBAD assets are operational, they establish and maintain the required voice and digital information links with the landward sector SAAWC.

The ACE commander establishes the TAOC's air control agencies and facilities ashore as soon as possible. These agencies and facilities provide increased surveillance and quicker response, and extend the ATF's weapons control capabilities. Initially air control agencies ashore operate as an adjunct to agencies afloat. The TAOC and/or the EW/C agencies ashore assist as needed and monitor air control aspects, including communications circuits, directly related to their tasking. As the ACE establishes C2 ashore, TACON of various portions of air operations may be transferred ashore. Once the TAOC is established and operational ashore, TACON of landward sector AD, including TBM warning, may be phased ashore. The ATF commander decides when to pass authority from agencies afloat to those established ashore. Control agencies afloat continue to monitor communications and serve as backups to shore-based C2 agencies should they become casualties.

MCWP 3-25.7, Tactical Air Operations Center Handbook, and MCWP 3-25.3 contain additional information on the TAOC.

5.4.13.4 Aviation Combat Element Tactical Air Control Center

When air control is passed ashore, the LF commander exercises control of air operations through the ACE commander in the ACE TACC. When the ACE TACC accepts responsibility for aircraft and air warning functions of TACAIR operations in the AOA, the ATF TACC becomes a TADC. The ACE TACC is generally established for large-scale operations. A MEU(SOC) neither has the assets to operate a TACC nor does the scope of its assigned missions warrant the establishment of such an organization.

The ACE TACC is supported by the ACE commander's staff, personnel, and equipment sourced from the MACG's MTACS and the MWHS. The ACE TACC provides the command post for the MAGTF ACE commander. The ACE TACC may also function as a TADC prior to the transfer of air operations control ashore. The ACE's TADC is subordinate to the ATF TACC prior to control passing ashore.

The unique capabilities of the ACE commander's C2 structure enable the ACE to C2 theater-wide air operations. The ACE commander is normally designated as the TAC for the MAGTF and may also be designated as a JFACC by a JTF commander.

As the focal point for planning and executing MAGTF air operations, the ACE TACC:

1. Manages the execution of the current ACE OPORD or FRAGORD and theater ATO

2. Manages aircraft and surface-to-air weapons in the MAGTF's AO
3. Monitors the status of MAGTF air assets and air operations
4. Serves as the operational POC between the MAGTF and external air control agencies
5. Coordinates the operations of subordinate C2 agencies
6. Develops and issues NAGTF air operations plans and orders, including MAGTF ATOs
7. Recommends apportionment of MAGTF air assets to the MAGTF commander
8. Provides centralized direction for the allocation and tasking of MAGTF air assets
9. Directs execution of MAGTF air operations
10. Coordinates MAGTF air operations with other theater air operations
11. Establishes AD control measures in the MAGTF's AO, including AD warning conditions and weapons control status
12. Evaluates the results of MAGTF air operations.

MCWP 3-25.3 and MCWP 3-25.4, Marine Tactical Air Command Center Handbook, contain additional information on the TACC.

5.5 COMMAND RELATIONSHIPS

As discussed in Chapter II of JP 3-02, the type of relationship chosen for the AF by the common superior commander, or establishing authority, should be based on the mission, nature, and duration of the operation, force capabilities, C2 capabilities, battlespace assigned, and recommendations from subordinate commanders. Command relationship options include either an OPCON, TACON, or support relationship, as described in JP 0-2. As discussed in Paragraph 2.4.1 and JP 3-02, typically, a support relationship is established between the Navy and LF commanders that is based on their complementary rather than similar nature and capabilities.

5.5.1 Relationship Between the Ship's Commanding Officer and an Embarked Helicopter Unit

In helicopterborne operations, helicopters and helicopter units are component parts of the LF under the MAGTF commander's OPCON. During amphibious operations, the helicopter units remain under the command of the MAGTF commander and are not under the OPCON of the ship's CO. However, the ship's CO does retain certain authority over embarked helicopter units, including those delineated in Paragraph 5.5.1.1.

5.5.1.1 Authority of the Ship's Commanding Officer

U.S. NAVREGS establish the authority of the ship's CO with respect to aircraft embarked in or operating from their ships. This includes, but is not limited to:

1. Indoctrinating pilots and crews in safety-of-flight operations related to shipboard operations

2. Requiring all-weather day and night pilot qualifications for shipboard land and launch operations
3. Providing landing, launching, and air traffic control in the vicinity of the ship
4. Controlling flight deck and hangar deck operations and spotting
5. Providing and operating flight deck handling and servicing equipment
6. Providing intermediate aircraft maintenance facilities for embarked squadrons
7. Loading and accurately manifesting personnel, cargo, ammunition, and equipment in aircraft
8. Providing heavy weather protection
9. Knowing aircraft limitations
10. Informing pilots of the ship's aviation fuel status and capability, and providing a fuel sample prior to any aircraft fueling/refueling operation.

5.5.1.2 Coordination Between a Helicopter Unit Commander and the Ship's Commanding Officer

The following matters relate to operating LF vertical lift aircraft from a ship and are coordinated between the helicopter unit commander and the ship's CO:

1. Pilot qualifications and limitations
2. Aircraft limitations
3. Scheduling of aircraft, pilots, and crewmen
4. Pilot briefings
5. Arrival, departure, and en route position reports
6. Fuel, maintenance, and ordnance status reports.

5.6 EXECUTION

The helicopterborne ship-to-shore movement of scheduled waves to LZs is conducted per the HEALT and the vertical assault landing diagram. After launching, vertical assault aircraft proceed to the VLZ via specific CPs and routes, as described in Paragraph 5.6.4.2. After discharging their loads, vertical assault waves rendezvous and proceed via specific CPs and routes to a breakup point. At this point, individual flights return to their respective ships or proceed as directed by the primary ATF TACC.

5.6.1 Embarking in Aircraft

Shipboard debarkation control personnel assisted by LF personnel conduct LF embarkation into aircraft. Troops in the helicopterborne ship-to-shore movement are organized into heliteams IAW the HWSAT, and strict accountability is maintained via passenger manifests.

5.6.2 Troop and Equipment Categories

Helicopterborne troops and supplies are arranged in the same categories as for surfaceborne ship-to-shore movement. These categories are discussed in Paragraph 3.3.5.

5.6.3 Operating Areas, Routes, and Points

The following areas, routes, and points are used to direct and control helicopters during ship-to-shore movement. Airspace requests are coordinated through the ATF TACC using the appropriate ACMREQ form or message.

5.6.3.1 Landing Zone

An LZ is any specified zone used for landing aircraft. They are usually designated by code names (i.e., Hawk, Falcon, other birds).

5.6.3.2 Vertical Landing Zones and Helicopter Landing Zones

A VLZ is a specified ground area for landing VTOL aircraft, and an HLZ is a similar specified area for assault helicopters to embark or disembark troops and/or cargo. Both LZs may contain more than one landing site and are designated by a code name (birds). The operation and organization of HLZs is discussed in Appendix H.

5.6.3.3 Landing Site

A landing site is a designated subdivision of an LZ in which a single flight or wave of vertical assault aircraft land to embark or disembark troops and/or cargo. Landing sites do not have to be geographically contiguous and are designated by color.

5.6.3.4 Landing Point

A landing point is a point within a landing site where one helicopter or V/STOL aircraft can land.

5.6.3.5 Helicopterborne Approach and Retirement Route

Helicopter approach and retirement routes are a track or series of tracks along which vertical assault aircraft move to and from a DP to a specified LZ. Planned routes facilitate coordinating aircraft movement with fire support plans. They are designated by the names of states.

5.6.3.6 Penetration Control Point

The PCP is the point along a helicopterborne approach route at which airborne waves penetrate a coastline. PCPs are one type of air CP and are designated by names of cities.

5.6.3.7 Initial Point

An IP for vertical assault aircraft is an air CP in the vicinity of the LZ from which individual flights of vertical assault aircraft are directed to their prescribed landing sites. To distinguish them from other air CPs, IPs are designated by the names of automobiles.

5.6.3.8 Rendezvous Point

The RP is prescribed at a given altitude and position relative to the DP for assembling vertical assault aircraft. RPs are designated by the names of cities.

5.6.3.9 Departure Point

A DP is an air CP at the seaward end of the helicopter approach lane system from which helicopter waves are dispatched along the selected helicopter approach lane to the IP. DPs are designated by names of cities.

5.6.3.10 Control Point

A CP is a position marked by a buoy, boat, aircraft, electronic device, conspicuous terrain feature, or other identifiable object that is given a name or number and used as an aid to navigation or control of ships, boats, or aircraft. CPs are designated by names of cities.

5.6.3.11 Breakup Point

The breakup point is where aircraft returning from an LZ break formation and are released to return to individual ships or dispatched for other employment. The breakup point may be the same as the DP.

5.6.4 Helicopterborne Operations With Control Afloat

Assault support aircraft are launched at the times and in the order prescribed in the HEALT. Control procedures and requests for on-call waves and nonscheduled units are described in Paragraphs 5.6.4.1 through 5.6.4.6.

5.6.4.1 Helicopterborne Platform Landing/Launch Control

Air traffic control is maintained by PRIFLY control on the Land/Launch Net for takeoff, landing, and operations in the ship's control area (normally, a 5-nm radius). Under IFR, the LHD or LHA's AATCC controls approaches and departures within the ship's control area. Under VFR, air traffic control of aircraft departing the ship's control area is passed from PRIFLY to the AATCC. Air traffic control of returning aircraft is executed in the reverse order. The shift of air traffic control is coordinated between AATCC and PRIFLY.

5.6.4.1.1 LHD/LHA Air Traffic Control Procedures

Air traffic control procedures are contained in the LHA/LHD NATOPS Manual.

5.6.4.1.2 Rescue Helicopter Procedures

Rescue helicopter procedures are contained in NWP 3-04.1M/MCWP 3-24.1 and the LHA/LHD NATOPS Manual.

5.6.4.2 En Route Control

The ATF TACC or a TADC takes control of flights on the Helicopter Direction Net before they arrive at the RP. The ATF TACC or TADC controls the flight to the RP where air traffic control is shifted to an overland air control agency, if available. The HSC detachment reports the status of assault waves to the HCS on the HC Net as they proceed inbound on approach routes. The flight leader checks in at the PCP with the ASC(A), if employed or required. The flight then proceeds to the assigned LZ.

Upon leaving the LZ, the flight leader checks in with the primary ATF TACC or ASC(A) for air traffic control to the breakup point, where flights are cleared for individual ship's control for landing and reloading.

Shipboard and airborne (if available) radar is used to maintain continuous radar surveillance of flights and waves, particularly at night or during marginal weather.

5.6.4.3 Terminal Information

When vertical assault aircraft check in with the ASC(A), they are briefed on any changes to the prebriefed VLZ situation, including:

1. Wind direction and velocity
2. Physical obstructions in the LZ
3. Friendly and enemy positions
4. How the LZ is marked.

Prior to landing, the pilot informs the heliteam leader of the direction in which the aircraft will be heading after landing.

5.6.4.4 Requests for On-Call Waves

Requests for on-call waves are normally made by tactical commanders on the Helicopterborne Unit Command Net. The requests are made to the HST and then transmitted to the helicopterborne TACLOG detachment on the HST Control Net. The TACLOG detachment monitors the supported unit TACNET to anticipate tactical and logistical requirements.

Upon receipt of a request for on-call waves, the helicopterborne TACLOG detachment notifies the HCS detachment and HLSC. The HSC detachment retasks vertical assault aircraft for the requested lift, and the primary TACC directs their movement to the appropriate ship and LZ. The HLSC issues instructions to the ships concerned to prepare troops, supplies, or equipment for debarkation. The HCS and MAGTF CE monitor these requests, and by silence, indicate concurrence.

5.6.4.5 Requests for Nonscheduled Units

After scheduled waves are landed, or sooner if requested by tactical commanders ashore, nonscheduled units are landed IAW the LF sequence table. Requesting nonscheduled units early or out of sequence is accomplished in a manner identical to that for on-call waves.

5.6.4.6 Changes in Landing Sequence

Helicopterborne requests for serials out of sequence are the same as for on-call waves. The helicopter transport unit commander first determines that ship loading can support the request. Once concurrence is obtained, the HLSC directs the appropriate ship to prepare the serials for debarkation. The HCS detachment diverts returning vertical assault aircraft or directs aircraft to launch in support of the mission. The HCS and LF TACLOG detachment monitor these requests, and by silence indicate concurrence.

5.6.4.7 Medical Evacuation and Casualty Evacuation

During the initial phase of helicopterborne ship-to-shore movement, empty aircraft transport casualties to CRTSs on return flights. As the operation progresses, the air plan designates MEDEVAC and CASEVAC aircraft, and casualties are transported to specific CRTSs based on the advice of the MRCO collocated with the HCS. Appendix H describes these procedures and the HST in detail.

5.6.5 Downed Aircrew and Aircraft Recovery Operations

The OPORD specifies plans for recovering downed aircrew and aircraft, and includes at a minimum:

1. Organization of aircraft maintenance recovery teams by type of aircraft
2. Security elements available for employment
3. Designated equipment requirements by type aircraft
4. Designated recovery vehicles, call signs, and frequencies
5. TRAP criteria.

5.6.6 Sequence of Events for Helicopter Tactical or Combat Service Support Request From a Helicopterborne Unit

Paragraphs 5.6.6.1 and 5.6.6.2, and Figure 5-2 detail the sequence of events.

5.6.6.1 Tactical Support Request

Requests for tactical support are made as follows:

1. Units ashore request assault support by transmitting an ASR on the HR Net or by messenger to an air officer.
2. The MAGTF CE monitors the HR Net and negates or provides consent by silence.
3. The HCS detachment receives the request on the HR Net and takes action if aircraft are available. If aircraft are unavailable or the request generates a conflict with present requirements, the HCS detachment refers the request to the HCS on the HC Net.

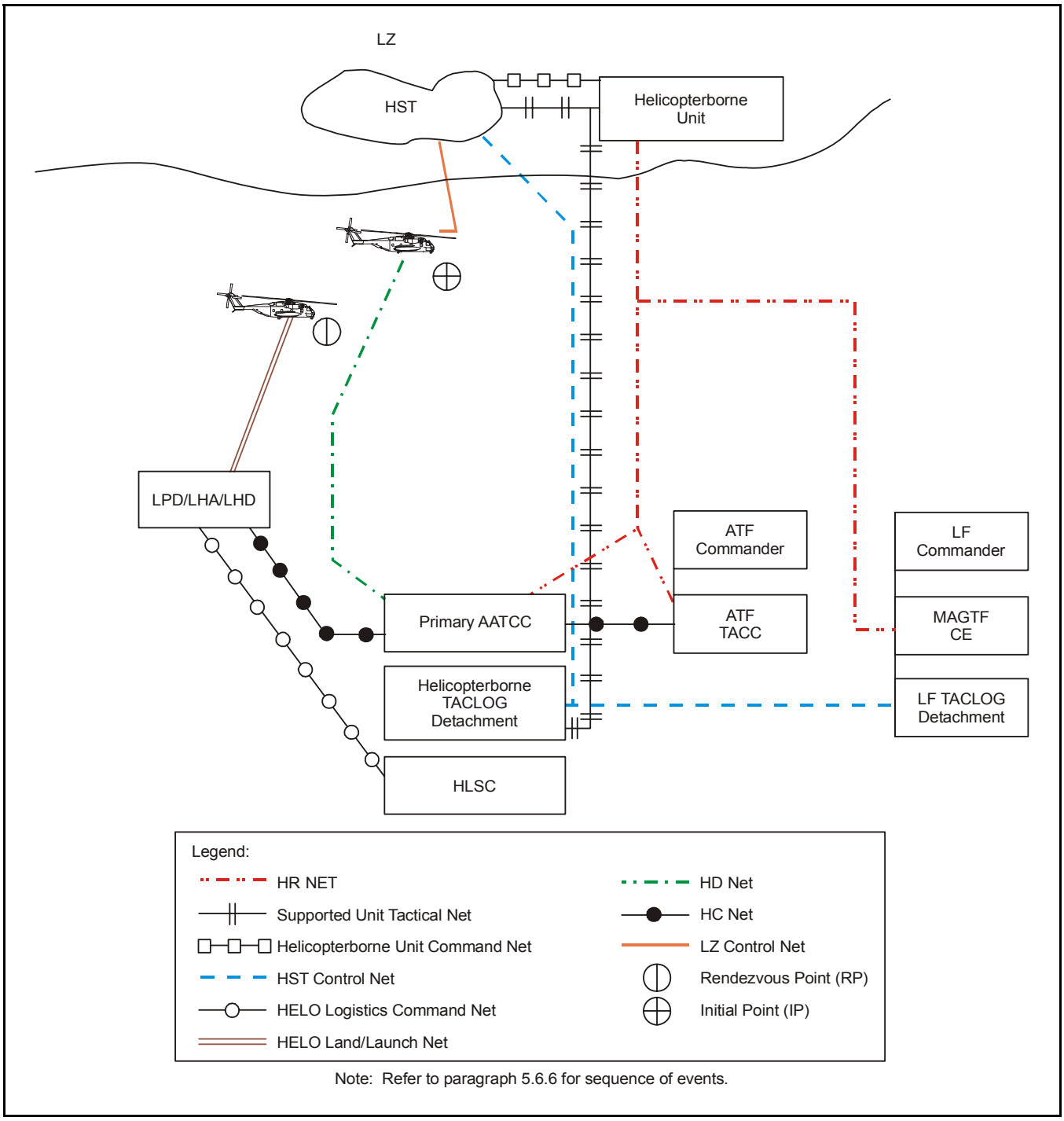


Figure 5-2. Helicopter Tactical or Combat Service Support Requests From a Helicopter Unit

4. The HSC detachment directs that aircraft be launched on the HC Net and provides routing, if required, to the HCS for coordination with supporting arms.

5. The assigned aircraft platform controls helicopters from the ship to the RP on the land/launch or an HD Net. At the RP, air traffic control is turned over to the primary AATCC.

6. The primary AATCC or TADC controls vertical assault aircraft from the RP to the PCP on an HD Net.

7. The ASC(A), if employed and the tactical situation permits, controls aircraft from the PCP to IP on the HD Net.

5.6.6.2 Combat Service Support Request

Requests for CSS are made as follows:

1. A unit ashore requests CSS from the HST on the Helicopterborne Unit Command Net.

2. If the HST finds that supplies are unavailable ashore, supplies are requested from the helicopterborne TACLOG detachment on the HST Control Net.

3. The helicopterborne TACLOG detachment coordinates delivery with the HLSC and HCS detachment.

4. The HLSC locates supplies afloat and via the Helicopter Logistics Command Net, and directs appropriate ships to prepare supplies for helicopter lift.

5. On the HC Net, the HCS detachment directs that aircraft be launched and provides routing, if required, to the HCS for coordination with combined arms.

6. The primary TACC or TADC provides air traffic control en route to the LZ, as described in Paragraph 5.6.6.1.

5.6.7 Sequence of Events for Helicopter Tactical or Combat Service Support Request From a Surfaceborne Unit

Paragraphs 5.6.7.1 and 5.6.7.2, and Figure 5-3 detail the sequence of events.

5.6.7.1 Tactical Support Request

Requests for tactical support are made as follows:

1. Surfaceborne unit requests tactical support from the LFSP on the supported unit TACNET.

2. The LFSP relays the request to the HCS detachment on the HR Net.

3. The MAGTF CE monitors the HR Net and negates or gives consent by silence.

4. The HDC detachment, on the HC Net, directs that aircraft be launched and provides routing, if required, to the HCS for coordination with combined and supporting arms.

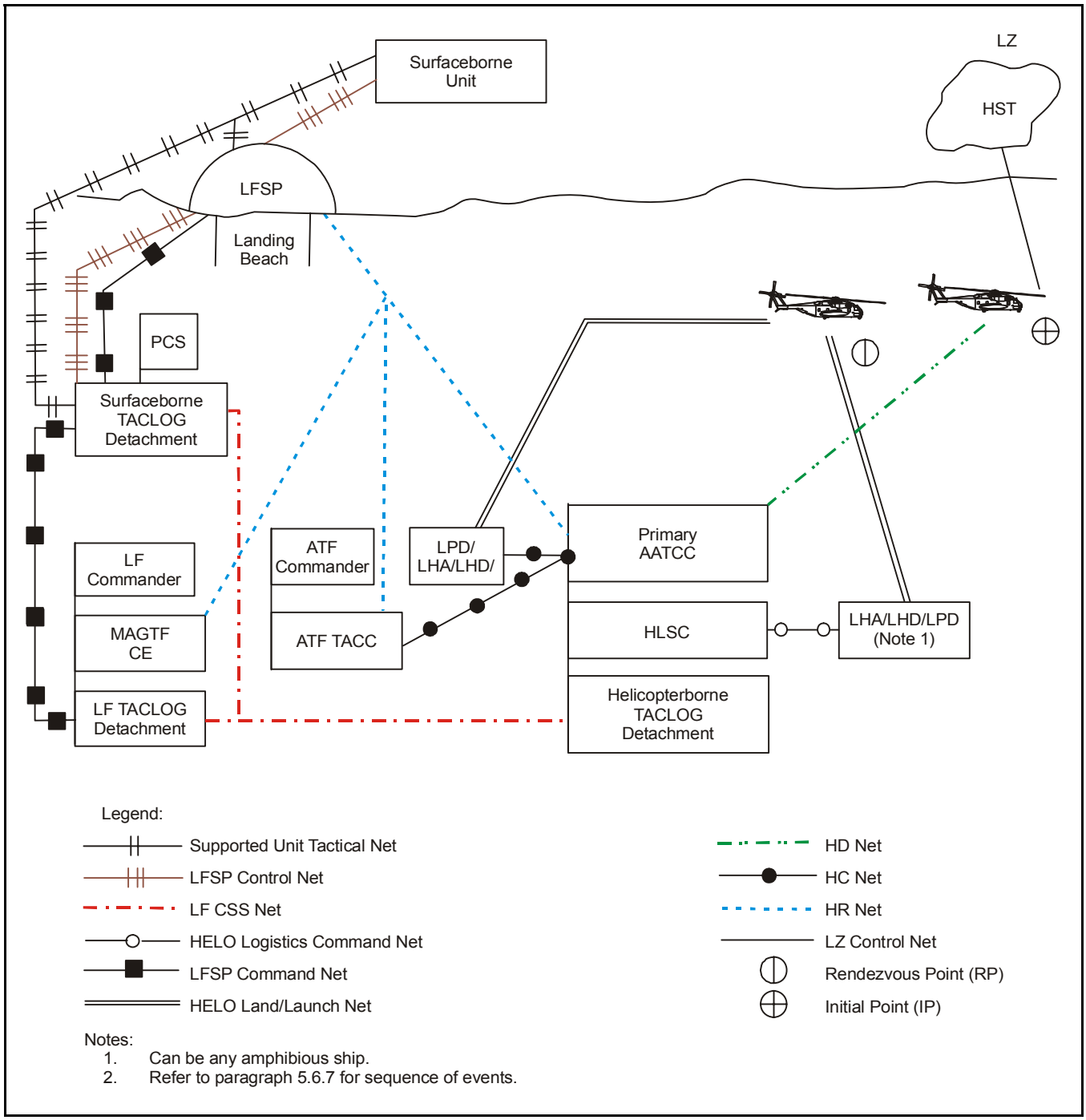


Figure 5-3. Helicopter Tactical or Combat Service Support Requests From a Surfaceborne Unit

5. The assigned helicopterborne platform controls aircraft from the ship to the RP on the land/launch or an HD Net. At the RP, flight control is turned over to the primary ATF TACC.

6. The primary ATF TACC or TADC provides air traffic control en route to the LZ, as described in Paragraph 5.6.4.2.

5.6.7.2 Combat Service Support Request

Requests for CSS are made as follows:

1. The surfaceborne unit makes a CSS request for serials to the LFSP on supported unit TACNET or the LFSP Control Net.

2. If serials are unavailable, the LFSP requests, via the LF CSS or LFSP Command Net, that the surfaceborne TACLOG detachment located on the PCS provide the serials. The surfaceborne TACLOG detachment locates the serials on ships in the transport group and determines whether surface means or helicopterborne delivery is the most expedient method.

3. The surfaceborne TACLOG detachment passes the request to the LF TACLOG detachment to coordinate vertical assault aircraft employment with the MAGTF CE. With MAGTF CE concurrence, the LF TACLOG detachment notifies the helicopterborne TACLOG detachment of the CSS requirement on the LF CSS Net, and this TACLOG detachment coordinates with the HSC detachment and HLSC for delivery of the serials.

4. The HLSC directs the appropriate ships to prepare the serials for vertical lift on the Helicopter Logistics Command Net.

5. On the HC Net, the HEC detachment directs that aircraft be launched and provides routing, if required, to the HCS for coordination with supporting arms.

6. The launch helicopterborne platform shifts aircraft to the Land/Launch Net of the helicopterborne platform with the serial embarked.

7. Aircraft load serials and launch, and are directed to the RP on the land/launch or an HD Net. At the RP, aircraft are directed to shift to the primary ATF TACC or TADC for air traffic control.

8. The AATCC provides air traffic control en route to the LZ, as described in Paragraph 5.6.4.2.

5.6.8 Helicopterborne Operations With Control Ashore

Phasing any portion of air control ashore is formally executed between the ATF and LF commanders. Transfer of air control functions ashore generally occurs when the facilities for coordinating supporting fire and air control are established. To control DAS and integrate with combined arms, the LF establishes the DASC and an FSCC ashore.

As long as vertical assault aircraft are based afloat, primary TADCs continue to function under the direction of the DASC and remain prepared to resume air traffic control in the event of a casualty to the DASC.

The DASC or a subordinate ASE or ASLT controls preplanned aircraft employment. Emergency requests for on-call missions from surfaceborne units are made on the TAR/HR Net. DASC assigns on-call aircraft ashore on an HD (or mission common) Net, and if necessary, coordinates the mission with supporting fire through the FSCC and HCS.

If aircraft ashore are unavailable, the DASC uses the HC or HR Net to request support through the HCS. The HCS issues orders to launch aircraft for missions assigned by the DASC, and the AATCC controls them from the RP to the CP specified by the DASC. At the CP, the DASC assumes flight control of the helicopters and directs the mission.

The MCWP 3-25 (series) publications delineate the procedures for control of LF shore-based aircraft.

5.7 DELEGATION OF AUTHORITY

To maintain maximum flexibility during amphibious operations, commanders delegate some authority to make changes in coordinating some actions. For helicopterborne ship-to-shore movement, the delegation of authority to make changes includes, but is not limited to:

1. Airborne control of aircraft
2. Changing from primary to alternate LZs
3. Changing approach and retirement routes
4. Changing the landing sequence.

Examples of coordination authority that may be delegated are:

1. Coordination of flights with other aircraft and supporting fires
2. Coordination of maneuvers between adjacent troop units
3. Coordination of supporting fires with troop maneuvers.

The authority delegated is clearly delineated in the OPORD. OPORDs establish:

1. The conditions under which these changes are to be made
2. The extent that changes may be made without approval from higher authority
3. The details of reporting changes
4. Other actions to ensure coordination of aviation operations with combined arms or other air operations.

5.7.1 Airborne Control of Vertical Assault Aircraft

The airborne control of vertical assault aircraft in the AOA during multideck operations may be delegated to an ASC(A). Section 5.4 provides additional information on airborne coordination.

5.7.2 Shifting From Primary to Alternate Landing Zones

The LF commander selects primary and alternate LZs for each LF objective based on recommendations from the helicopterborne unit commanders and ACE commander. When the use of any LZ does not affect the CONOPS, plan of supporting fires, or adjacent or higher troop units, the helicopterborne unit commander, in coordination with the ASC(A) and the helicopter unit commander, may be delegated the authority to change from the primary to alternate LZ to exploit a tactical advantage or improve a ground situation. If the use of an LZ can have a negative impact on adjacent or higher level troop units, this authority cannot be delegated below the highest troop unit affected or the SACC.

5.7.3 Changing Approach and Retirement Routes

The HCS, the helicopter transport group/unit commander, and representatives from the ACE coordinate to select primary and alternate approach and retirement routes between VLZs and the DP.

When supporting fires are included in the landing plan, and the supporting fires plan permits both the alternate and primary routes to be designated as an ACA, even when not in use, the authority to shift from primary to alternate routes may be delegated to the helicopter director, the ASC(A) (if employed), the AMC, or AFL.

When the use of other than preselected routes will not affect the CONOPS ashore, the supporting fires plan, or adjacent or higher troop units, the ASC(A) may select routes to accomplish the mission, as required. If the use of any route will affect such units, this authority cannot be delegated below the highest troop unit level or the SACC.

If the decision to use a different route requires establishing an ACA, the authority to change approach and retirement routes cannot be delegated below the ATF TACC or ACE TACC when air control functions have been phased ashore.

5.7.3.1 Airspace Coordination Area

An ACA is a three-dimensional block of airspace in a target area, established by the appropriate ground commander, in which friendly aircraft are reasonably safe from friendly surface fires. The ACA may be formal or informal.

5.7.4 Changing the Landing Sequence

The helicopterborne unit commander designates the landing order in the HEALT. The authority to direct a change in the landing sequence is normally delegated to the helicopterborne unit commander.

Transport group/unit loading is considered when changing the landing sequence of helicopterborne units. Without the concurrence of the helicopter group/unit commander, serials not readily available aboard ship will not be changed to an earlier loading time.

5.8 AIRCRAFT CONTROL COMMUNICATIONS

In helicopterborne operations, flight control is exercised primarily through voice radio. Air control nets are listed in the communications plan. A separate helicopter communications plan is not published. Figure 5-4 provides a communications matrix for helicopterborne ship-to-shore movement.

SUBSCRIBERS	NETS												
	COMMAND, CONTROL, AND COORDINATION					HST				LFSP			
	HELO COMMAND	HELO ADMINISTRATIVE	HELO REQUEST	HELO DIRECTION	LAND/LAUNCH	HST CONTROL	LZ CONTROL	HCE LOCAL (NOTE 1)	HELO/LZ LOCAL	LFSP COMMAND	LF CSS	HELO LOGISTICS COMMAND (NOTE 1)	MEDREG
ATF TACC (HCS)	C	C	C	C									
TADC	P	P	P	P									
AATCC	P	P	P	P									
ASC(A)	X			W	W		W						
TAC(A)				W			W						
SHIPS	X	X		X	C							X	
ESCORT HELOS				W	W								
TRANSPORT HELOS				W	W		W	W					
LF TACLOG DET										X	C		
GCE TACLOG SET						X				X	X		
SURF RLT TACLOG DET						P				X	X		
HELO RLT TACLOG DET						X				X	X		
LFSP			X			X	W			C	X		
SHORE PARTY										P	X		
HST			X			C			C				
HCE							W	X					
LZ CONTROL TEAM				X			C	C	X				
ACE	W	W	W	W		X			W				
LCE									W	W	C/S		
TACP (NOTE 2)			X										
TACP (NOTE 3)			W										
DASC			C/A	C/A									
FSCC			X										
HLSC												C	
MRCO													C
MEDREG TEAMS													X

NOTES:

1. ACTIVATED ON "AS REQUIRED" BASIS ONLY.
2. HELICOPTER-LIFTED LF UNITS.
3. SURFACE-LIFTED LF UNITS.

GENERAL NOTES:

1. NETS MAY BE COMBINED WHERE PRACTICABLE.
2. REFER TO NTTP 3-02.2 FOR SUPPORTING ARMS COMMUNICATION NETS.

LEGEND:

C – NECOS
P – BE PREPARED TO ASSUME NECOS
X – GUARD
C/A – NECOS WHEN AIR CONTROL IS PHASED ASHORE
W – GUARD WHEN DIRECTED
C/S – NECOS WHEN LCE IS PHASED ASHORE

Figure 5-4. Airborne Ship-to-Shore Movement Communications Matrix

5.8.1 Communications Planning

Representatives from the HCS, primary TACCs afloat, TADCs, Navy and LF MRCOs, and helicopterborne and vertical assault units are involved in communications planning.

5.8.2 Helicopterborne Command, Control, and Coordination Nets

Most nets for helicopterborne command, control, and coordination are in use during the movement phase of an amphibious operation. These nets are discussed in Paragraphs 5.8.2.1 through 5.8.2.5.

5.8.2.1 Helicopter Command Net

The HC Net is used by the HCS, HCS detachment, TACCs afloat or TADCs, and ASC(A)s to coordinate and direct helicopterborne employment. It is used by HCS detachments to direct launches from air-capable ships for specific missions.

5.8.2.2 Helicopter Administrative Net

The HA Net is used by the HCS, HCS detachment, and air-capable ships for administrative and logistics matters pertaining to helicopter operations. This net may be combined with the HC Net.

5.8.2.3 Helicopter Request Net

The HR Net is used by the HCS, HCS detachment, TACP, HST, and the LFSP to request immediate assault support. The LFSP guards this net for surfaceborne units.

5.8.2.4 Helicopter Direction Net

The primary ATF TACC or TADCs use the HD Net for positive or advisory air traffic control of vertical assault aircraft. The helicopter air controller uses the HD Net to direct the flight course and altitude of vertical assault aircraft. The ASC(A) and TAC(A) use the HD Net for air traffic control in the objective area.

5.8.2.5 Land/Launch Net

The Land/Launch Net is used by each PRIFLY for helicopter air traffic control within its control area under VMC.

5.8.3 Helicopter Support Team Nets

The HST Nets are used by the HST and are described in Paragraphs 5.8.3.1 through 5.8.3.4.

5.8.3.1 Helicopter Support Team Control Net

The HST Control Net provides for the exchange of logistics information among the HST, TACLOG detachments, and the LFSP. Multiple HST Control Nets may be required depending on the scope of helicopter operations.

5.8.3.2 Landing Zone Control Net

The LZ Control Net is used by the HCE for aircraft guidance from the IP to the LZ. It also provides communications with the ASC(A).

5.8.3.3 Helicopter Control Element Local Net

The HCE commander uses the HCE Local Net to contact landing sites within an LZ.

5.8.3.4 Helicopter/Vertical Landing Zone Local Net

The HST uses the Helicopter/VLZ Local Net to control supply dumps, maintenance sites, evacuation points, etc., within an HLZ or VLZ.

5.8.4 Landing Force Support Party Nets

The LFSP Command Net and the LF CSS Net are described in Paragraphs 4.5.6 and 4.5.8, respectively.

5.8.4.1 Helicopter Logistic Command Net

The Helicopter Logistic Command Net is used by the HLSC for directing ships to prepare serials for offload by helicopters.

APPENDIX A

Landing Craft and Amphibious Vehicle Formations and Control Signals

A.1 LANDING CRAFT AND AMPHIBIOUS VEHICLE FORMATIONS

Figure A-1 shows the various types of formations for landing craft and amphibious vehicles.

A.1.1 Order of Landing Craft in Formation

Assault craft form in numerical order as follows:

1. In column or echelon formations, landing craft and amphibious vehicles form in numerical order from the van to the rear.

2. AAVs in a line-abreast formation are numbered from the left flank to the right flank. They are launched in boat team number sequence (1-1, 1-2, 1-3, etc.) in a column to facilitate a subsequent flanking movement to a subsequent tactical formation.

3. Landing craft in line abreast, wedge, and "vee" formations station the number one craft in the center of the wave with odd-numbered craft to starboard and even-numbered craft to port.

A.2 CONTROL SIGNALS

Improved communications have significantly decreased the use of arm and hand signals for controlling displacement landing craft and amphibious vehicles. However, they may still be used if communications should fail or during NATO or coalition operations, and are therefore depicted in Figure A-2.

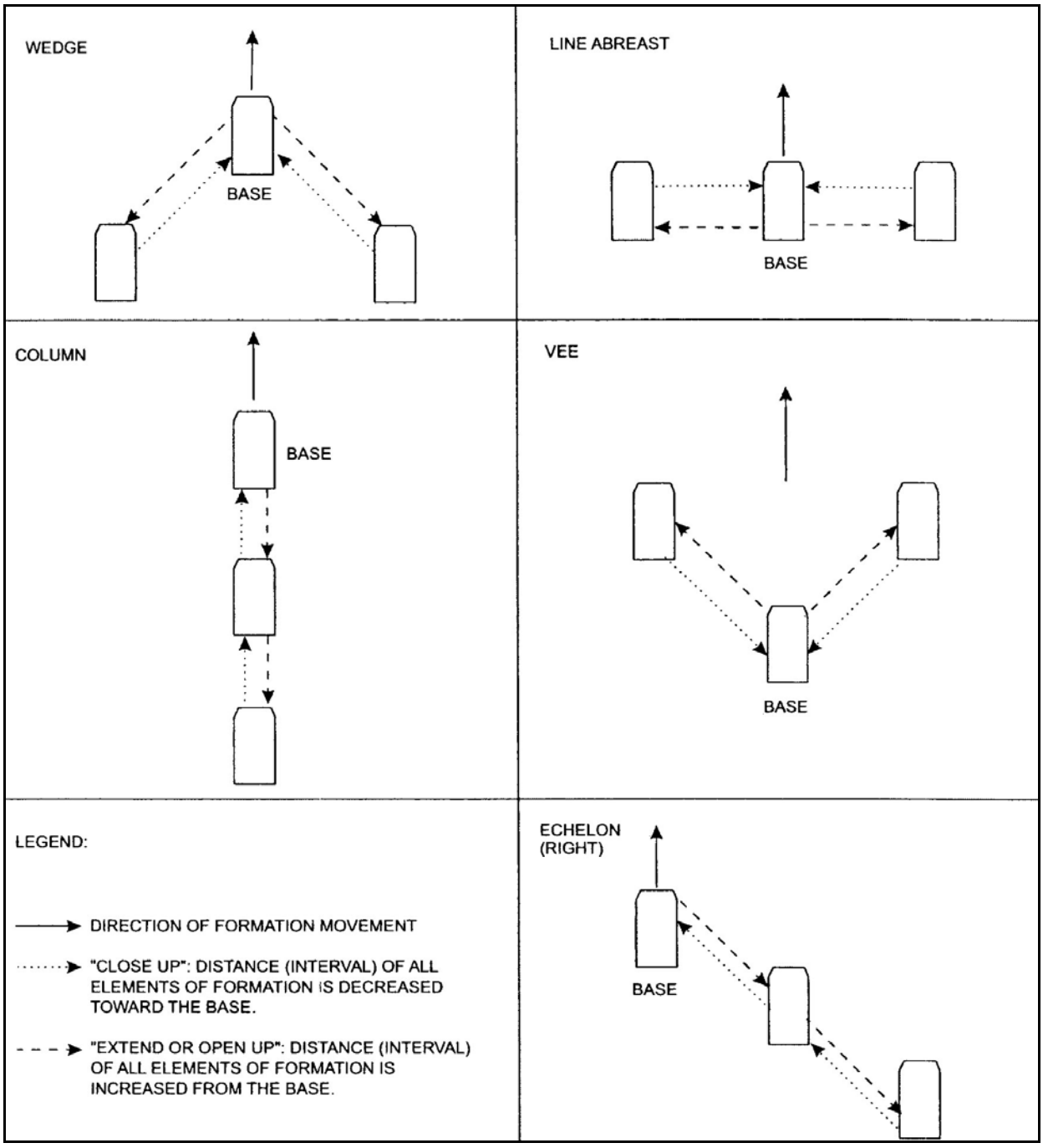


Figure A-1. Landing Craft and Amphibious Vehicle Formations

Assemble or Pass Towline

Day



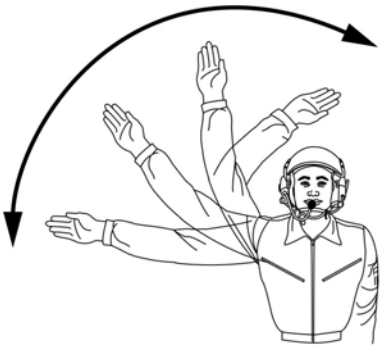
Small circle, palm out.

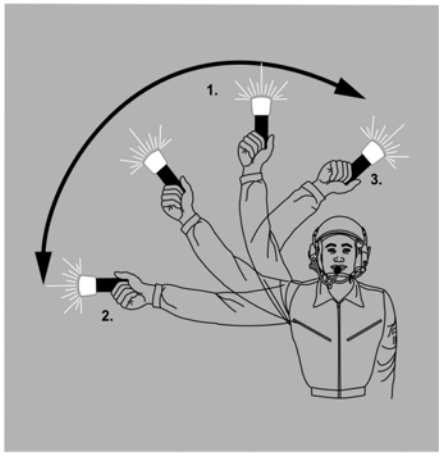
Night



Turn light on when right arm is extended overhead; execute large horizontal circle. Turn light off before lowering arm. Repeat as necessary.

Attention





Turn light on when arm is in the starting position. Turn light off when signal is completed. Repeat as necessary.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Vehicles (Sheet 1 of 20)

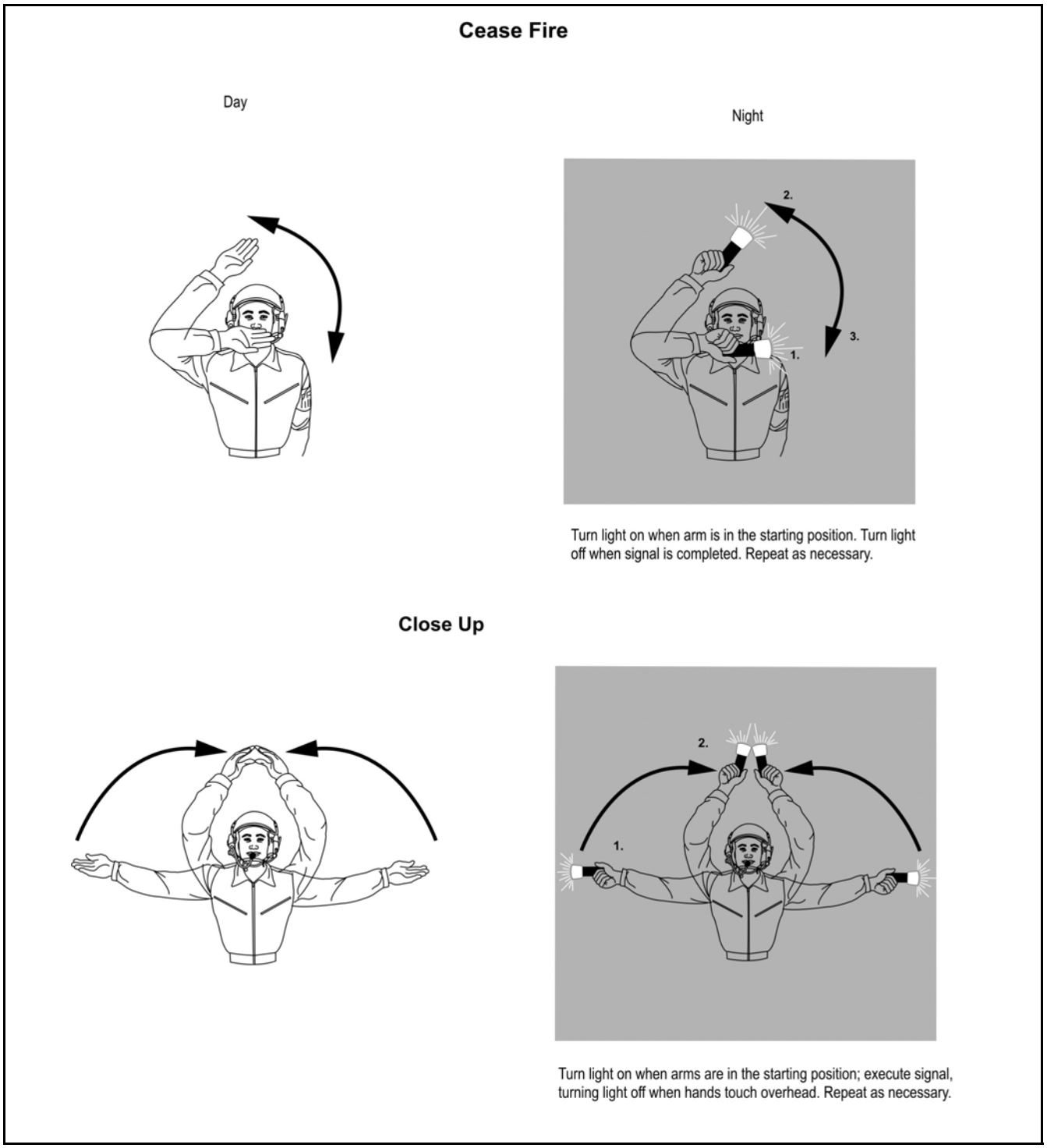


Figure A-2. Arms and Hand Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 2 of 20)

I Do Not Understand

Day

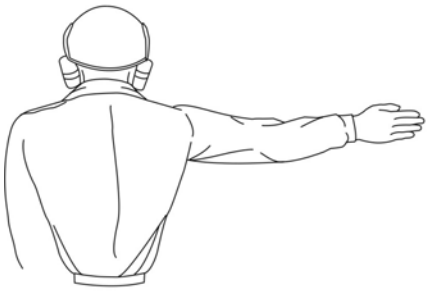


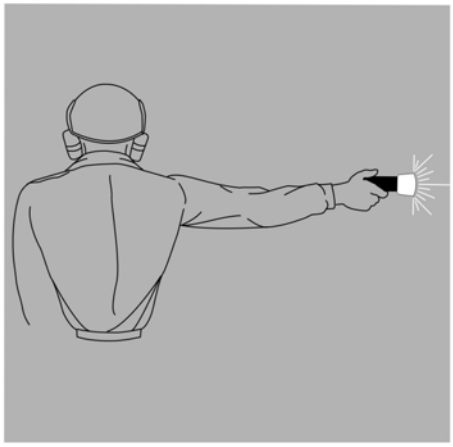
Night



Turn lights on as hands are brought down across the face; hold in position, parallel, horizontal, until acknowledged. Turn lights off while they are still in front of the face.

Column Right (Left)





Turn light on as arm is extended; hold in position until understood or acknowledged or the maneuver is executed. Turn light off while arm is still extended. Repeat as necessary.

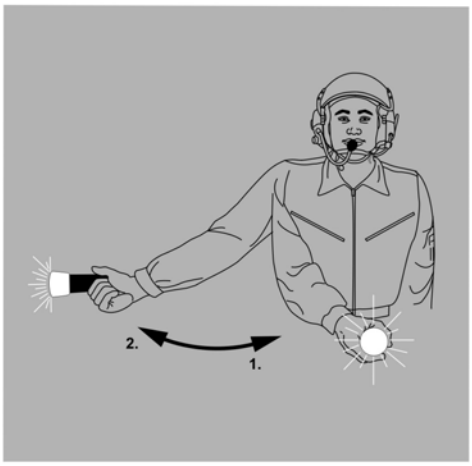
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 3 of 20)

Commence Firing

Day

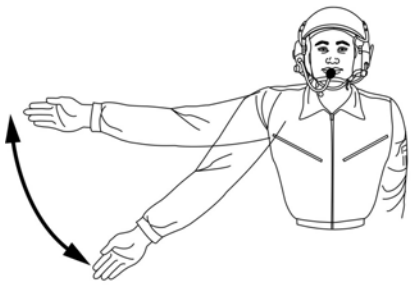


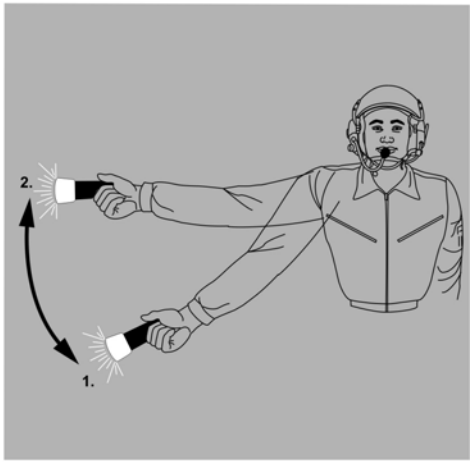
Night



Turn light on when arm is in the starting position. Turn light off when signal is completed. Repeat as necessary.

Decrease Speed (Vehicles)
Quick Time (Dismounted



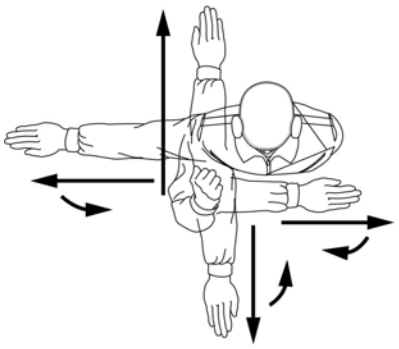


Turn light on when arm is in the starting position. Turn light off when signal is completed. Arm does not move above the horizontal position. Repeat as necessary.

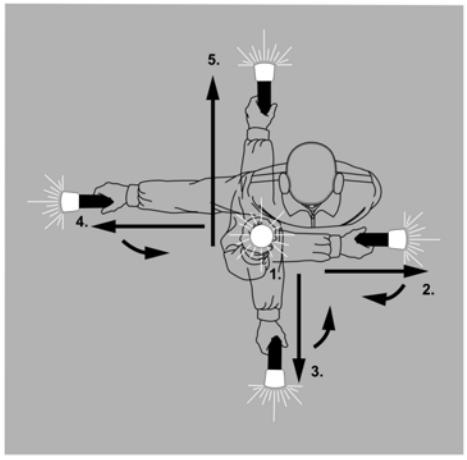
Figure A-2. Arm and Hand Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 4 of 20)

Disperse

Day



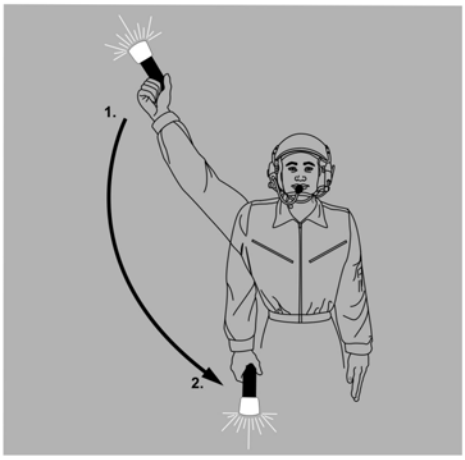
Night



Turn light on when arm is in starting position. Return arm to starting position after each movement in a given direction. Turn light off after arm has been moved to the rear.

Dismount, Down, Take Cover





Turn light on when arm is in the starting position. Turn light off when arm is down at the side. Repeat as necessary.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 5 of 20)

Echelon Right (Left)

Day

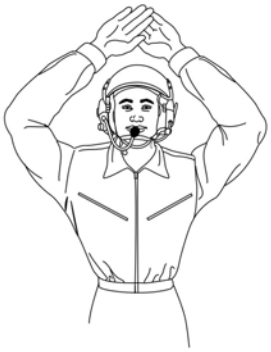


Night



Turn lights on when arms are in correct positions. Turn lights off before taking arms from the signal position. Repeat as necessary.

Disregard Previous Command - As You Were



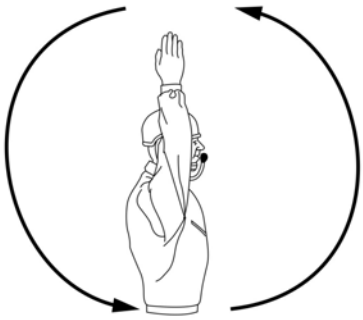


Wands are crossed instead of the hands. Turn lights on when wands are in position overhead. Turn lights off when signal is understood or acknowledged.

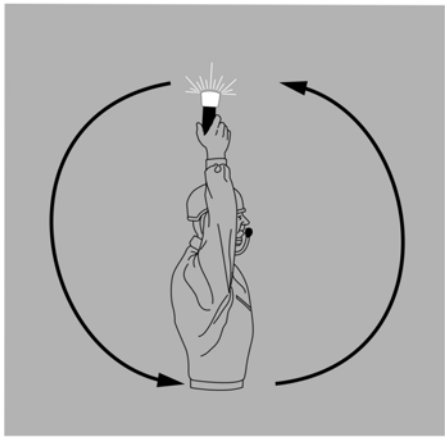
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 6 of 20)

Form Column

Day



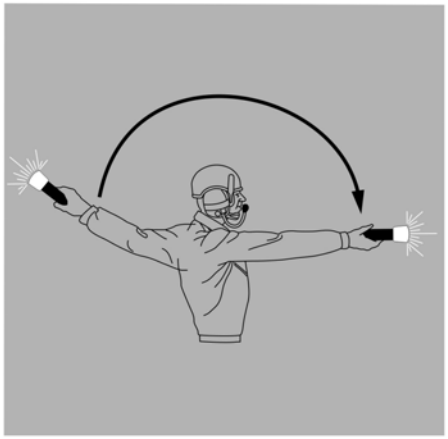
Night



Turn light on when arm movement for signal is started. Turn light off when completed. Repeat as necessary.

Advance or Move Out





Face the desired direction of movement; turn light on when arm is extended to the rear; swing arm overhead and forward in the direction of desired movement. Turn light off when arm is in horizontal position. Repeat as necessary.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 7 of 20)

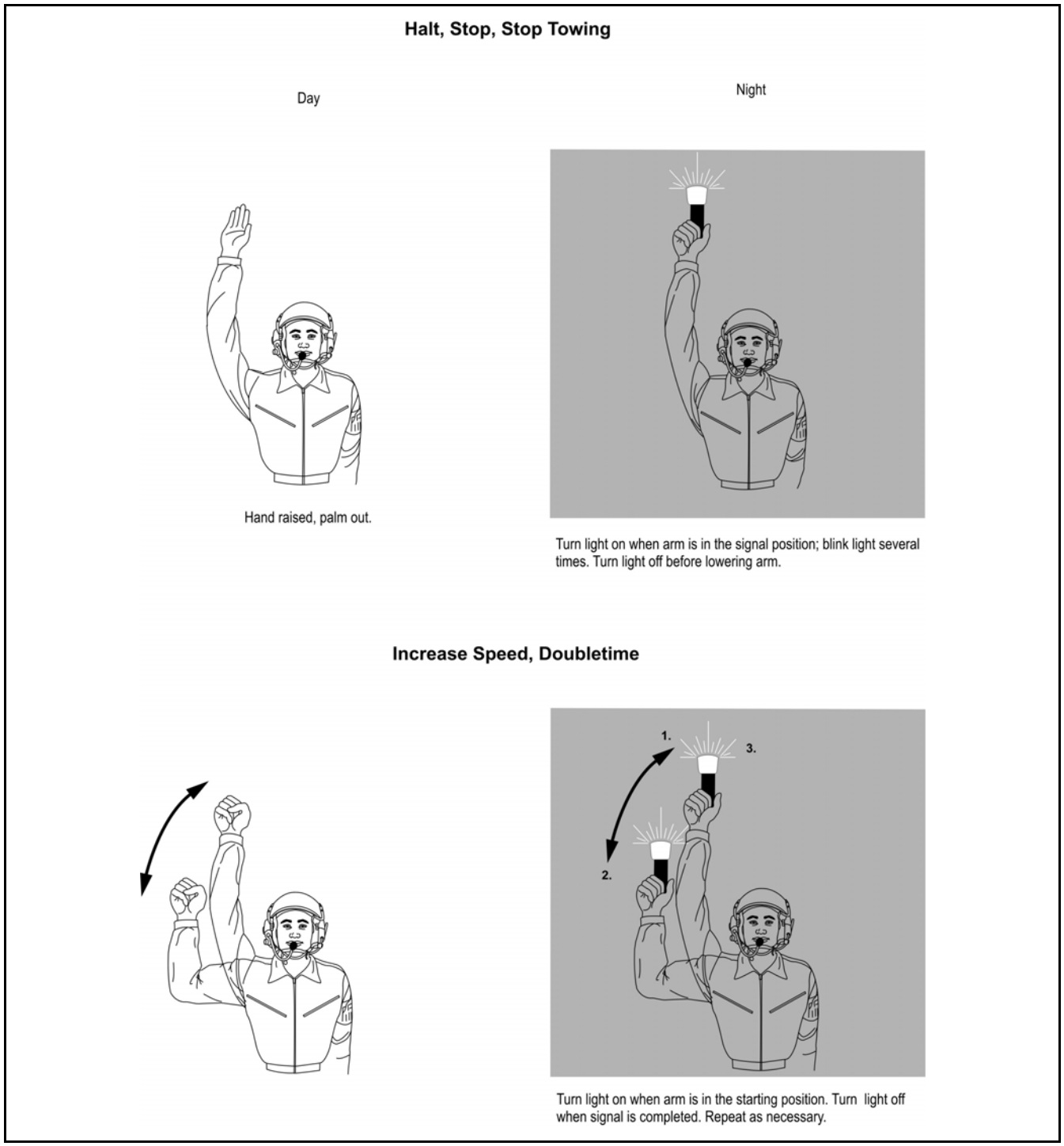


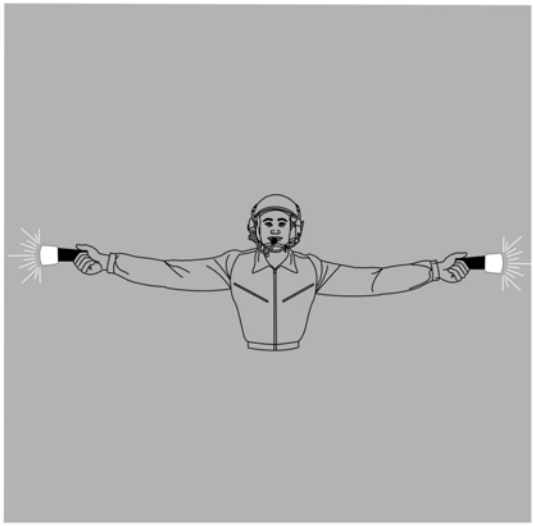
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 8 of 20)

Line Formation, Deploy into Line Abreast, as Skirmishers

Day



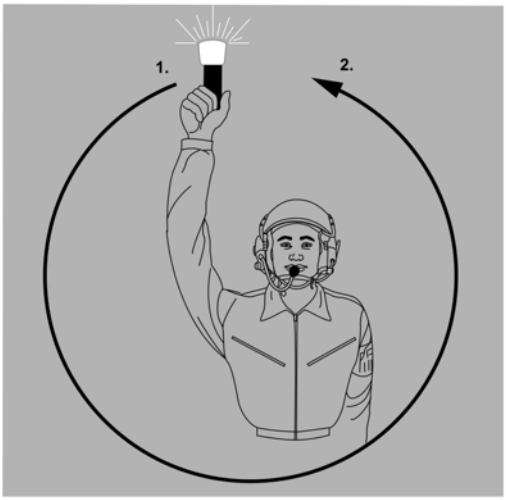
Night



Turn lights on as arms are extended; hold in position until signal is understood or acknowledged or the maneuver is executed. Turn lights off when arms are still in signal position. Repeat as necessary.

Man Overboard





Turn light on when arm is in vertical position; execute complete circle while blinking the light. Turn light off when arm is returned to the vertical position. Repeat as necessary.

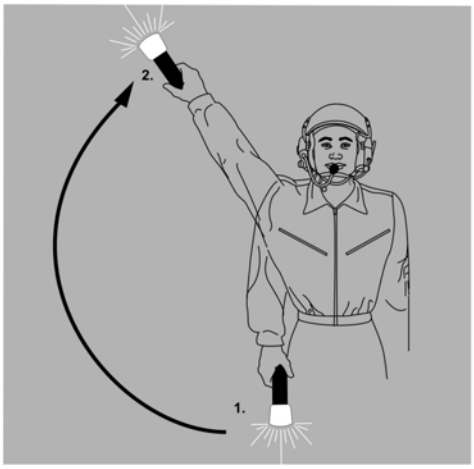
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 9 of 20)

Mount

Day



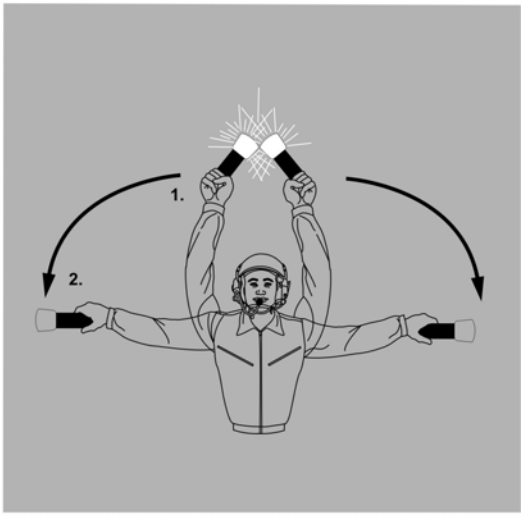
Night



Turn light on in starting position. Turn light off when arm is 45 degrees above horizontal position. Repeat as necessary.

Open Up, Extend



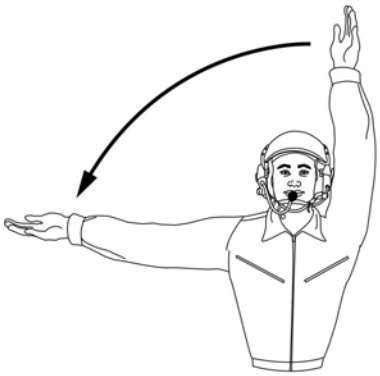


Turn lights on when arms are in starting position. Turn lights off when arms are in in horizontal position. Repeat as necessary.

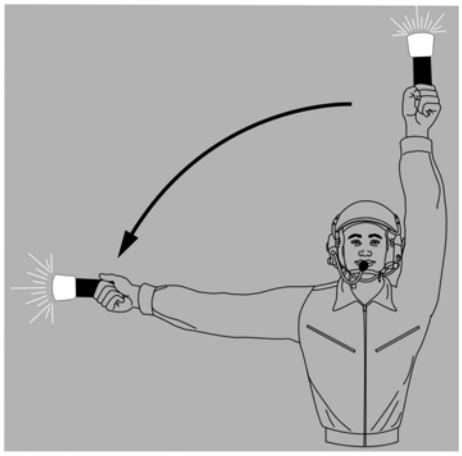
Figure A-2. Arm and Hand Signal Controls – Displacement Landing Craft and Amphibious Vehicles (Sheet 10 of 20)

By the Right (Left) Flank

Day

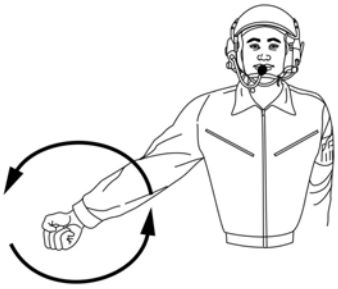


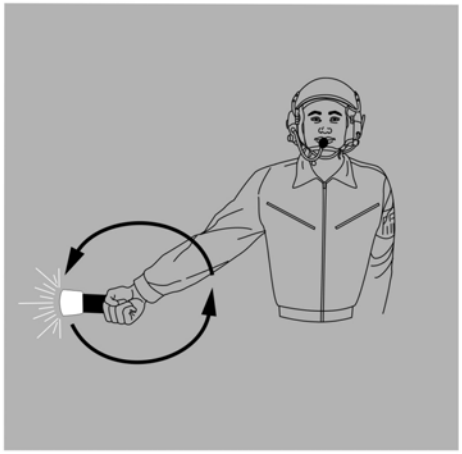
Night



Extend the right (left) arm parallel to the deck and hold the left (right) arm above the head so that both arms are at a 90-degree angle to each other. At the point of execution, move the left (right) arm across the body as if to clap hands together.

Start Engines, Prepare to Move



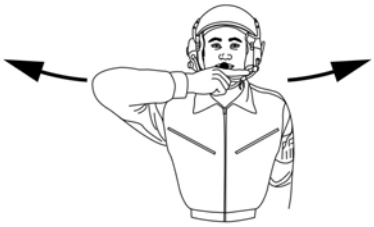


Turn light on when arm is in starting position. Turn light off when signal is completed.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 11 of 20)

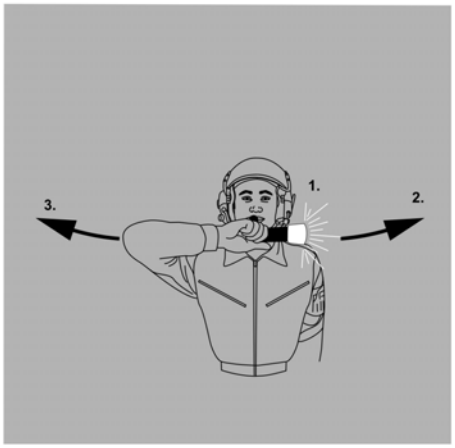
Stop Engine, Cut Engine

Day



CAUTION: This signal should not be used for waterborne AAVs.

Night



Turn light on when arm is in starting position. Turn light off when signal is completed.

V Formation



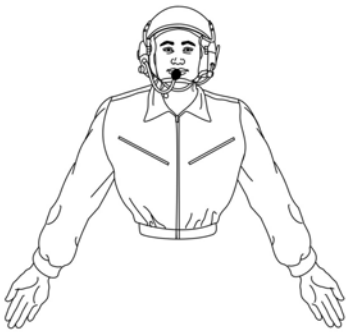


Turn lights on as arms are extended; hold in position until signal is understood or acknowledged or the maneuver is executed. Turn lights off while arms are still in signal positions. Repeat as necessary.

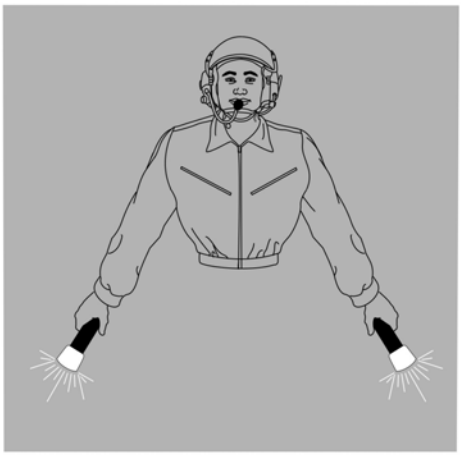
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 12 of 20)

Wedge Formation

Day

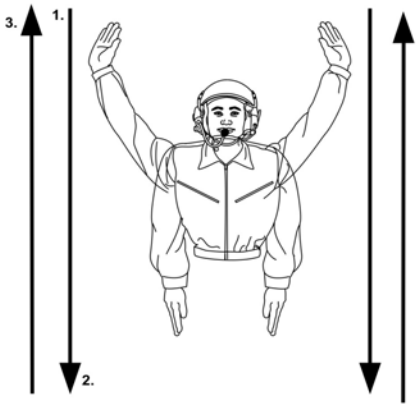


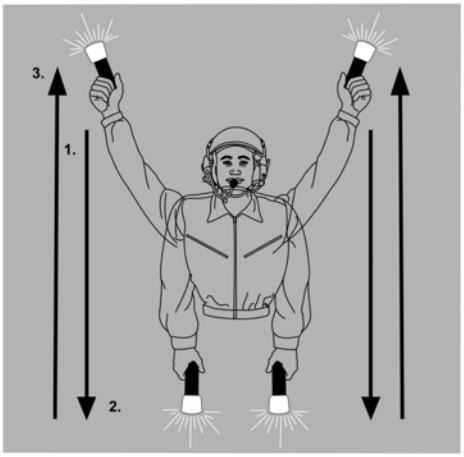
Night



Turn lights on as arms are extended; hold in position until signal is understood or acknowledged or the maneuver is executed. Turn lights off while arms are still in signal position. Repeat as necessary.

Breakdown





Turn lights on with arms extended overhead. Swing arms forward and down to knees. Swing arms forward and upward from knees to overhead. Continue motion until signal is understood.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 13 of 20)

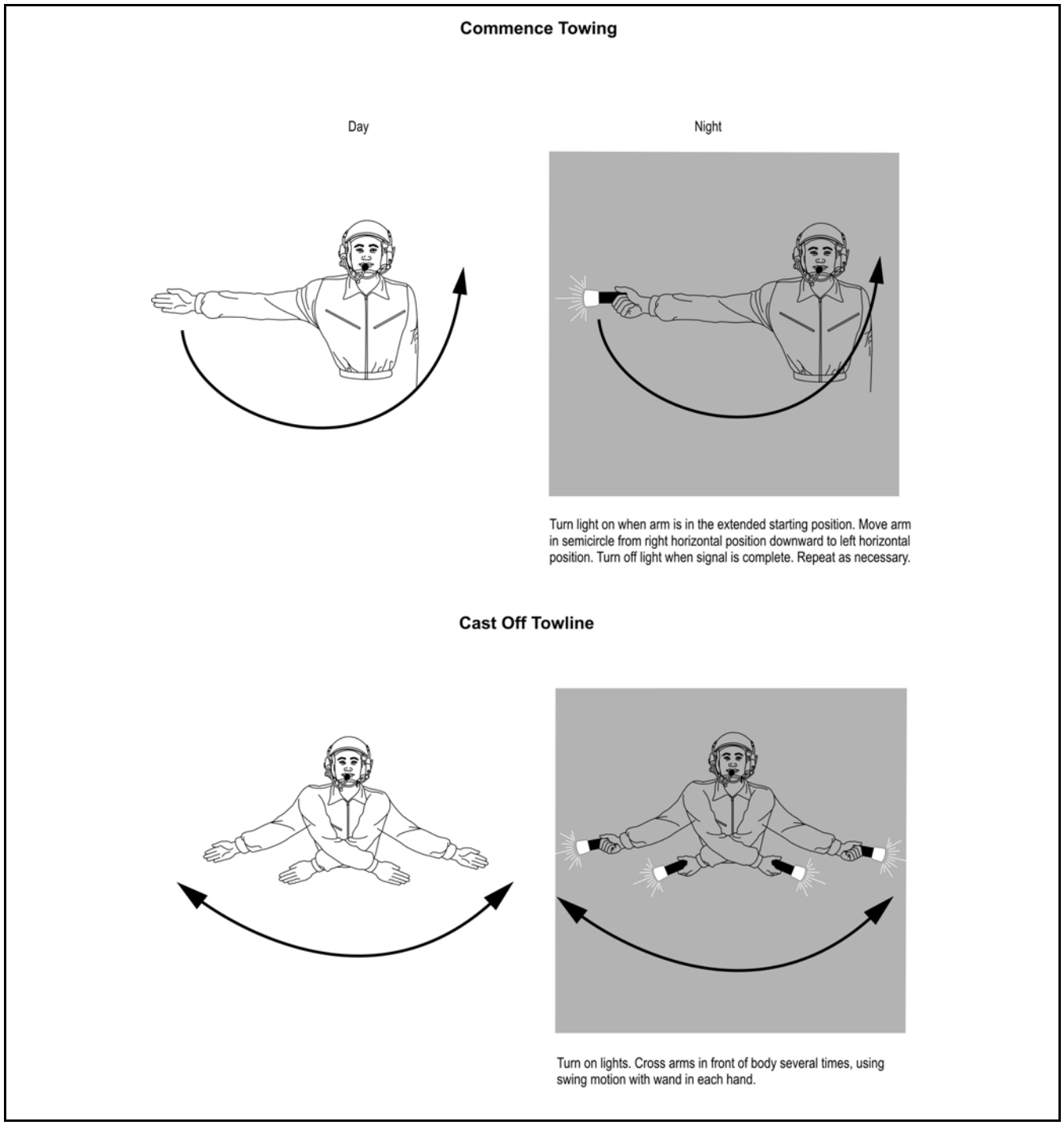


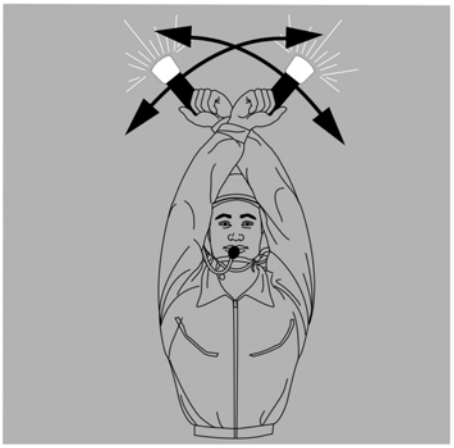
Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 14 of 20)

Air Attack

Day



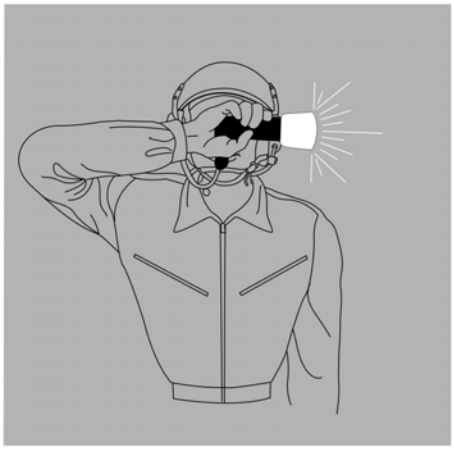
Night



Turn on lights. Rapidly cross and uncross arms fully extended above the head with wand in each hand.

Nuclear Warning





Turn on light. Cover both eyes with wand held in right hand to warn exposed troops to take cover before the detonation of nuclear weapon.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 15 of 20)

Ramp Up and Dogged

Day



Arms held out parallel to deck with hands held open and pointed up. (Coxswains acknowledge with same signal to inform traffic controlman that the ramp is up and dogged.)

Night



Turn on lights. Arms held out parallel to deck with wands pointed straight up. (Coxswains acknowledge with same signal to inform traffic controlman ramp is up and dogged.)

Ramp Down



Arms held out parallel to deck with hands held open and pointed down. (Coxswains acknowledge with same signal.)

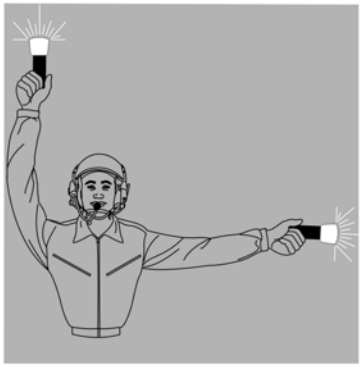


Turn on lights. Arms held out parallel to deck with wands pointed straight down. (Coxswains acknowledge with same signal.)

Vector Left (Right)



Signalman faces beach with one arm straight out to side pointing in direction of the turn, other arm raised straight up, palm forward.



Turn on lights. Light wand in each hand; one straight out to side pointing in direction of the turn, other straight up.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 16 of 20)

Pivot Right

Day



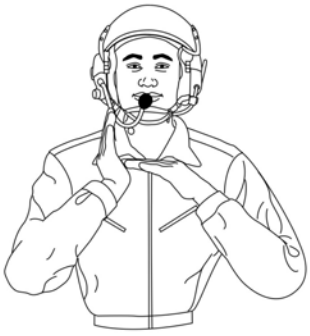
Hold one hand, fingers together, palm facing inboard in front of the chest in vertical position. Hold other hand with fingers joined, palm facing down, in horizontal position against vertical hand, in direction vehicle is to travel.

Night



Same as day, except with use of light wands.

Pivot Left



One hand is in front of chest in a vertical position, fingers joined, palm facing perpendicular to the chest. Hold other hand with fingers joined, palm facing down, in horizontal position against vertical hand, in direction vehicle is to travel.



Same as day, except with use of light wands.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 17 of 20)

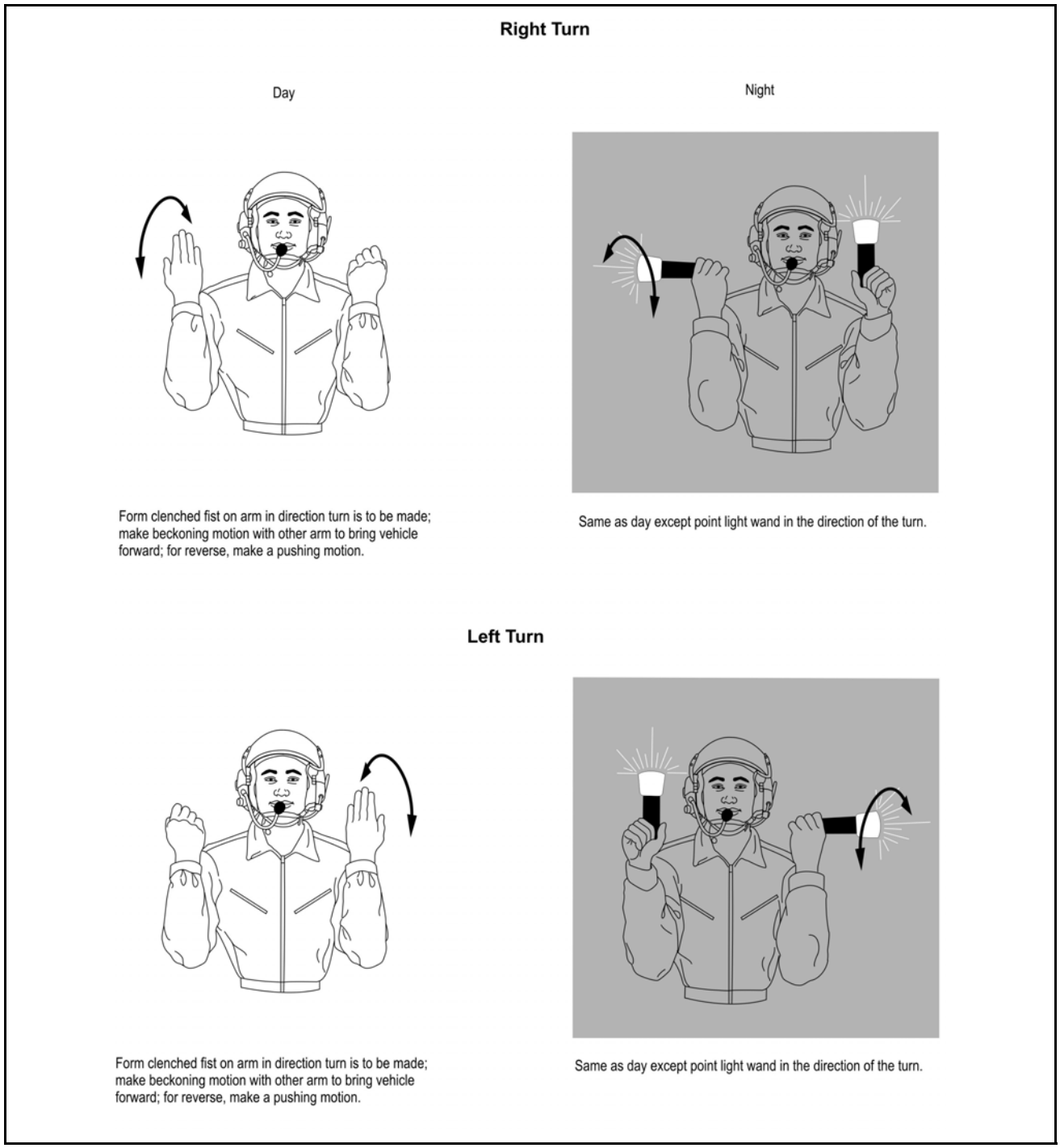


Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 18 of 20)

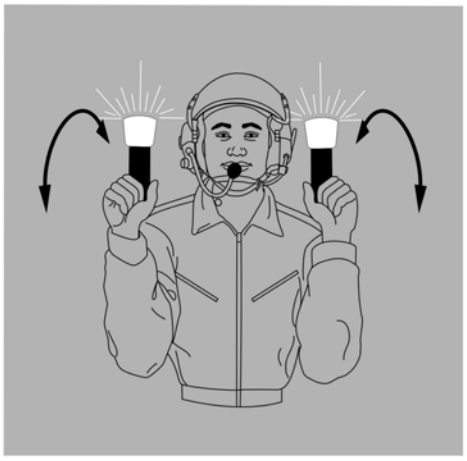
Move Forward

Day



Move the hands and forearms backward and forward with palms toward the chest as if pulling the vehicle.

Night

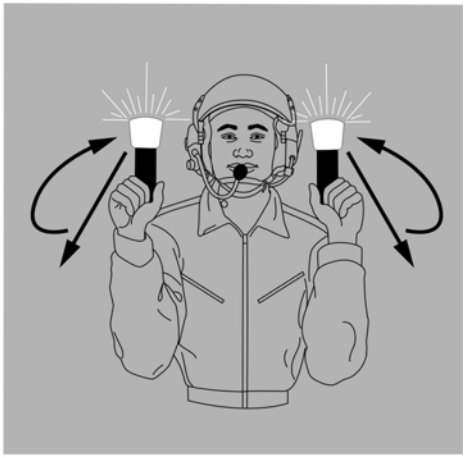


Same as day except with the use of light wands.

Move in Reverse



Move the hands and forearms backward and forward with palms toward the vehicle as if pushing the vehicle.



Same as day except with the use of light wands.

Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 19 of 20)

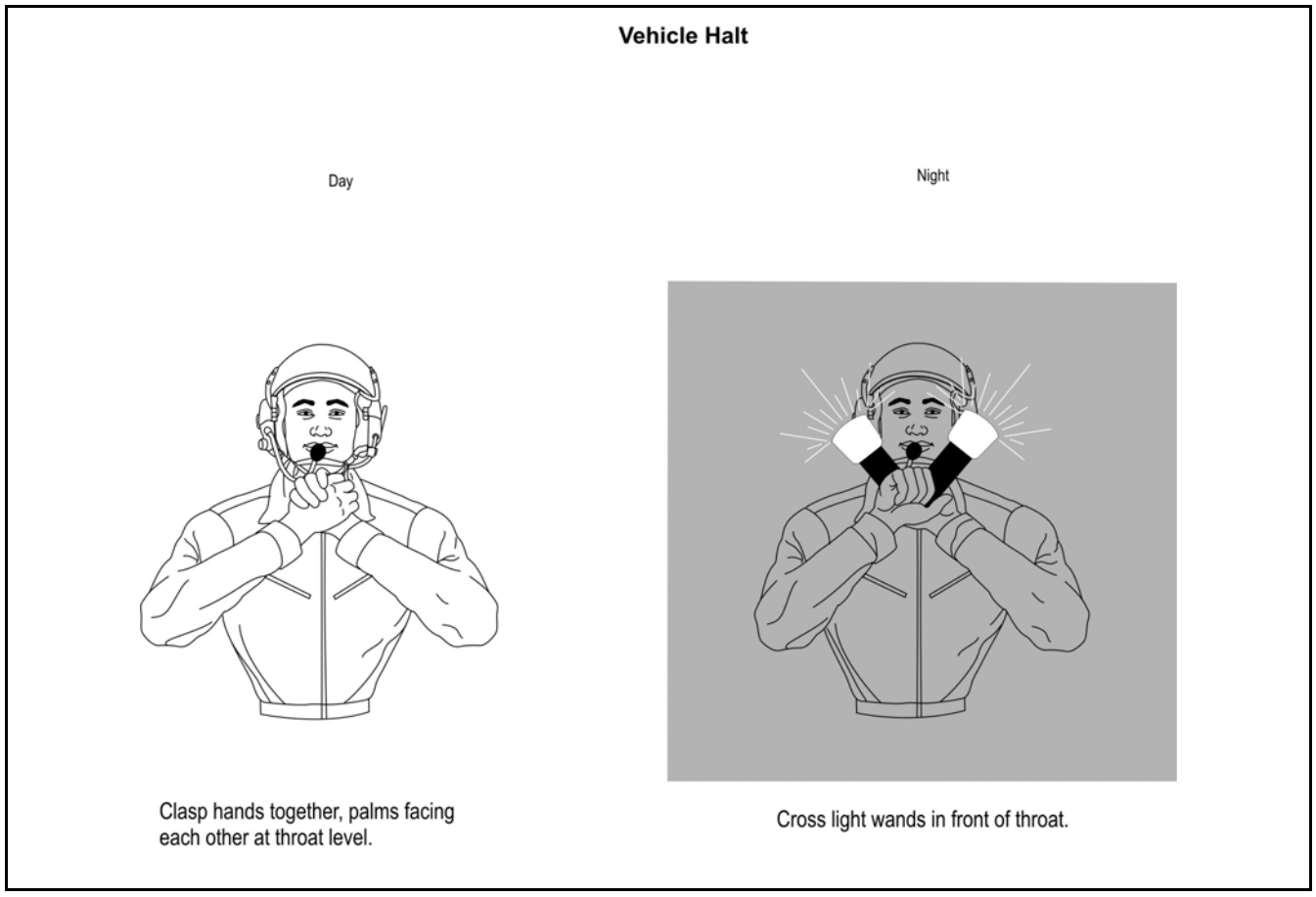


Figure A-2. Arm and Hand Control Signals – Displacement Landing Craft and Amphibious Vehicles (Sheet 20 of 20)

APPENDIX B

Identification Flags, Insignia, Markers, Lights, and Signals

B.1 STANDARD IDENTIFICATION

A variety of identification flags, insignia, markers, and lights are used in ship-to-shore movement to identify control ships, landing craft, and waves assigned to a landing beach. They are shown in Figures B-1 through B-9. Additionally, displacement landing craft, amphibious vehicle beaching signals, and LCAC maneuvering hand signals are shown in Figures B-10 and B-11. Landing craft remove from sight flags and insignia upon crossing the LD. Required flags and insignia are displayed again after the last scheduled wave has landed, or earlier if directed by the BP.

B.1.1 Flags and Insignia

Appropriate flags are displayed on the starboard bow of displacement landing craft. The BWC carries the ZERO, WHISKEY, and numeral flags for waves to facilitate substitution of one boat for another.

In large-scale, multicraft amphibious operations, boat teams may be identified by boat team paddles displayed in the forward section of amphibious vehicles and displacement landing craft prior to crossing the LD. These paddles facilitate the formation of landing craft into waves during static debarkation. They are optional for underway launches. An example of a boat team paddle and other insignia is shown in Figure B-9.

B.1.1.1 Boat Team Paddles

A boat team paddle is readable from any direct and marked with the boat team number obtained from the landing craft and amphibious vehicle assignment table.

The number on the paddle indicates the scheduled wave and position of the landing craft in that wave. The first digit(s) indicates the wave, the last digit(s), the position within the wave. For example, boat team paddle 7-3 indicates the third landing craft in the seventh wave.

Boat team paddles are made to the following specifications:

1. Three rectangular-shaped boards, 14 X 10 inches, nailed together to form a three-sided figure attached to a wooden staff, 6 feet X 2 inches X 2 inches

2. Black numerals, 7 inches high, on a white background.

B.1.2 Beach, Unloading Point, and Oceanographic Markers

Markers are positioned by the BP and used to indicate positions to which landing craft proceed, to mark channels or obstructions off beaches, and to provide information. Beach marker flags or panels for which no dimensions

are given are approximately the size of the number four signal flag. Fluorescent cloth is used whenever possible for easier identification under all weather conditions. Figures B-5 through B-8 depict these markers.

B.1.3 Cargo Identification

Displacement landing craft and amphibious vehicles carrying cargo display distinctive flags or lights so control and BP personnel may readily identify the type of cargo. The flags and lights used to identify various types of cargo are listed in Figure B-2. For example, a displacement landing craft carrying a nonscheduled unit serial, such as bulk cargo, flies a RED flag under a GREEN flag; or at night, shows two steady red lights under a steady green light.

B.1.4 Beach, Unloading Point, and Oceanographic Lights

At night and during periods of reduced visibility, colored lights are used in lieu of flags or markers. These lights and their purpose are shown in Figures B-5 through B-9. Beach marker lights are visible from the LD, and other lights are visible from at least 1,000 yards. Beach and unloading point lights are directional, with a maximum 10-point arc of visibility to seaward. Oceanographic lights are always all-around lights.

B.1.5 Night and Reduced Visibility Identification Lights for Amphibious Vehicles and Displacement Landing Craft

Identification lights displayed by the boat group are explained in Figures B-1, B-2, and B-9. Wake lights are displayed in addition to standard navigation lights. Based on the tactical situation, all may be darkened or displayed at normal brilliance. Prior to H-hour, identification lights are screened and visible only from astern. After H-hour, boat crews convert them to all-around lights.

B.1.6 Line of Departure Dispatching Signals

The LD dispatching signals are shown in Figure B-3. Visual signals may be paralleled on the Beach Boat Control circuit (Alfa Net) or by flashing light from the PCS. Signal lights are shielded and aimed only at the approaching wave.

Numeral flags are flown on the port and starboard halyards by the PCS. Waves with two-digit numbers are dispatched by a hoist using the numeral flag corresponding to the last digit of the wave number.

B.1.7 Beaching Signals

The visual signals used by the BP to direct displacement landing craft are shown in Figure B-10. The Beach Boat Operation circuit (Bravo Net) or loud hailers may be used in lieu of or parallel to these visual signals.

B.1.8 Visual Emergency Signals

Displacement landing craft and amphibious vehicles will identify emergency situations as follows:

1. Man overboard – OSCAR flag.

2. Breakdown – life jacket on a perpendicular boat hook.

3. Fire/flooding – BRAVO flag.

4. Loss of communications – ZULU flag.

5. AAV emergency – NOVEMBER flag:

   a. Red flare

   b. Spotlight or battle lantern held vertically.

At night, the Bravo Net may be used to identify emergency situations.

B.2 LANDING CRAFT AIR CUSHION LIGHTS

LCAC do not display any amphibious-unique flags, lights, or insignia because of the FOD these devices could cause. For night operations, LCAC display normal navigation lights, as the tactical situation permits. These lights include an ACV light. The ACV light is an amber, all-around, and 120-flashes-per-minute light. LCAC beach markers are depicted in Figure B-5, and maneuvering hand signals are shown in Figure B-11.

Table of Lights	
Light Locations	**Lights**
Ships and Landing Craft	
CCS	2 vertical, blinking white
ACCS	2 vertical, blinking, color to be designated
PCS	1 steady, directed seaward, same color as beach (all-around after first wave touches down)
SCS	1 blinking, same color as beach
Approach Lane Marker Ship (Note 1)	
Salvage Craft	3 wake lights, horizontal, 2 feet apart, red (convertible to all around)
Medical Craft	3 vertical, steady, 1 foot apart, green, all around
Floating Dumps	2 or 3 vertical (1 steady green over 1 to 2 cargo colors, 2 feet apart) (see Figure B-2)
Ocean Markers and Navigation Lights	
Obstruction	Blinking white over blinking red
Channel, Port Side	Blinking green
Channel, Starboard Side	Blinking red
Fairway	Blinking white
Screened Wake Lights	
1st Wave	1 red
2nd Wave	1 blue
3rd Wave	1 amber
4th Wave	1 green
5th Wave	2 red (Note 2)
6th Wave	2 blue (Note 2)
7th Wave	2 amber (Note 2)
8th Wave	2 green (Note 2)
Successive Waves	Repeat entire sequence

Notes:
1. If ship is available.
2. Two lights, horizontal, 3 feet apart.

Figure B-1. Table of Lights

Floating Dump Supplies	Day	Night
	GREEN Flag Over Cargo Flag	**Steady Green Light Over Cargo Color Light(s), 2 Feet Apart**
Rations	ONE flag	1 steady white light
Medical Supplies	TWO flag	1 steady green light
Water	FOUR flag	1 steady blue light
81-mm Ammunition	FIVE flag	1 steady amber light
Bulk Cargo	RED flag	2 steady red lights
Self-Propelled Vehicles	BLUE flag	2 steady blue lights
Cargo Requiring Prime Mover	YELLOW flag	2 lights, steady blue over steady amber

Figure B-2. Floating Dump Cargo Identification

Wave	Departure Time	Day	Night
Wave 1	5-min standby 2-min standby 1-min standby Departure time	One flag at dip One flag closeup - - - One flag hauled down to dispatch wave	Steady red light for 30 sec Flashing red light for 30 sec Flashing red light for 50 sec, then a 10-sec steady red light Extinguish 10-sec steady red light to dispatch wave
Wave 2			Flashing blue light for 30 sec Flashing blue light for 50 sec, then a 10-sec steady blue light Extinguish 10-sec steady blue light to dispatch wave
Wave 3e		Same as wave 2	Amber light is used
Wave 4		Same as wave 2	Green light is used
Wave 5		Same as wave 2	Red light is used
Wave 6		Same as wave 2	Blue light is used
Successive Waves		Continue using cycle outlined above for waves three through six	

Figure B-3. Departure Time Sequence

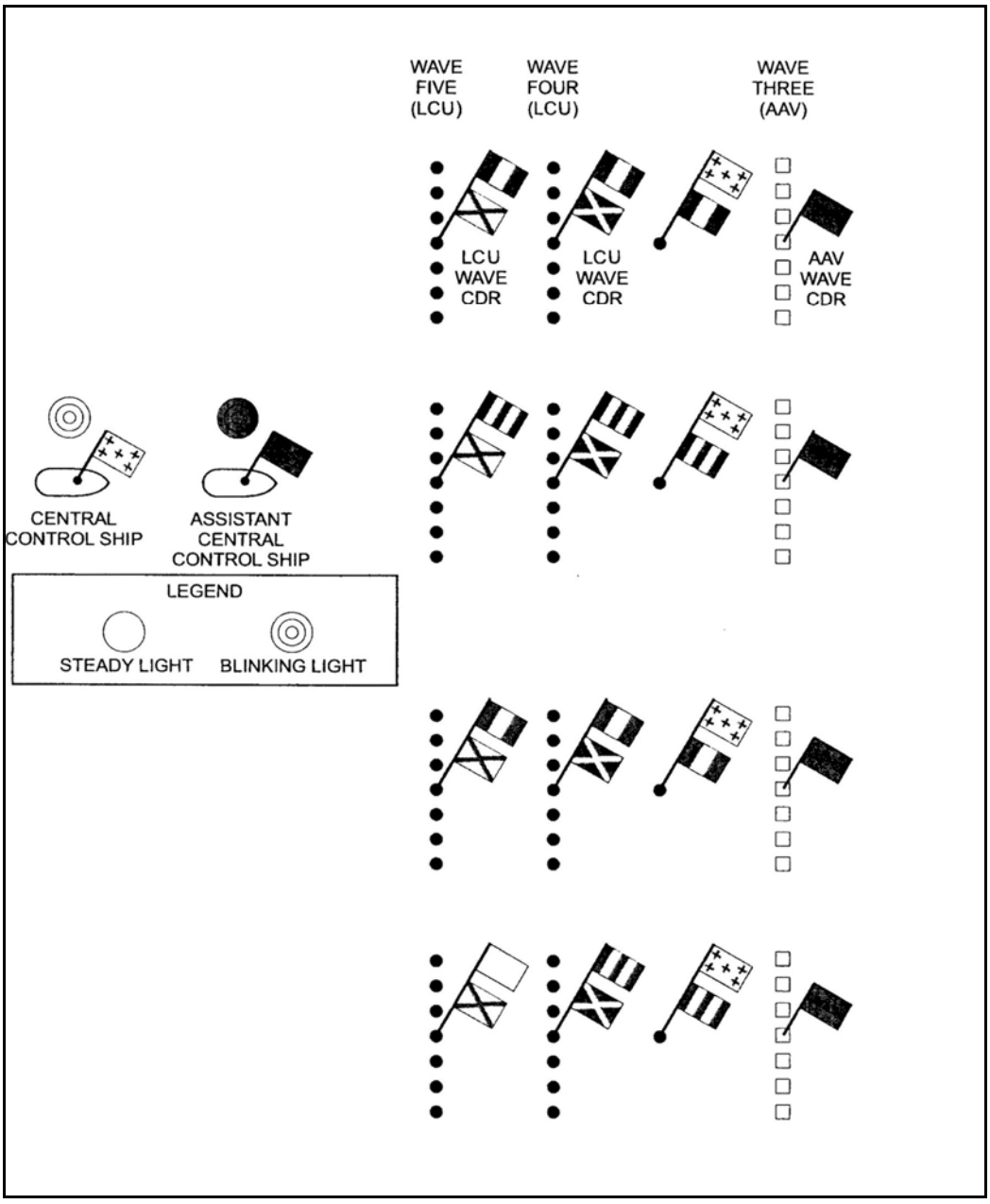


Figure B-4. Standard Flags and Identification Insignia (Sheet 1 of 2)

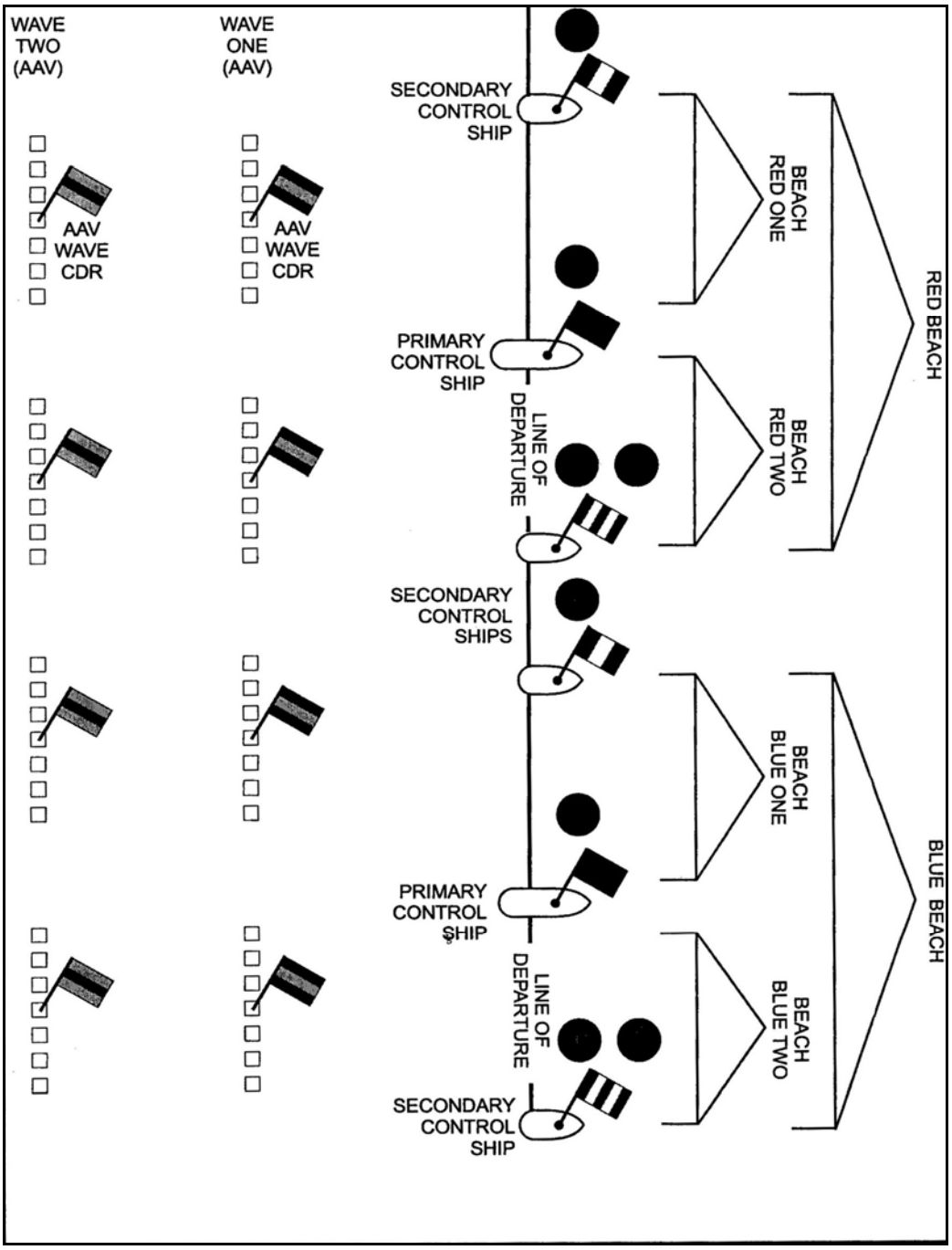


Figure B-4. Standard Flags and Identification Insignia (Sheet 2 of 2)

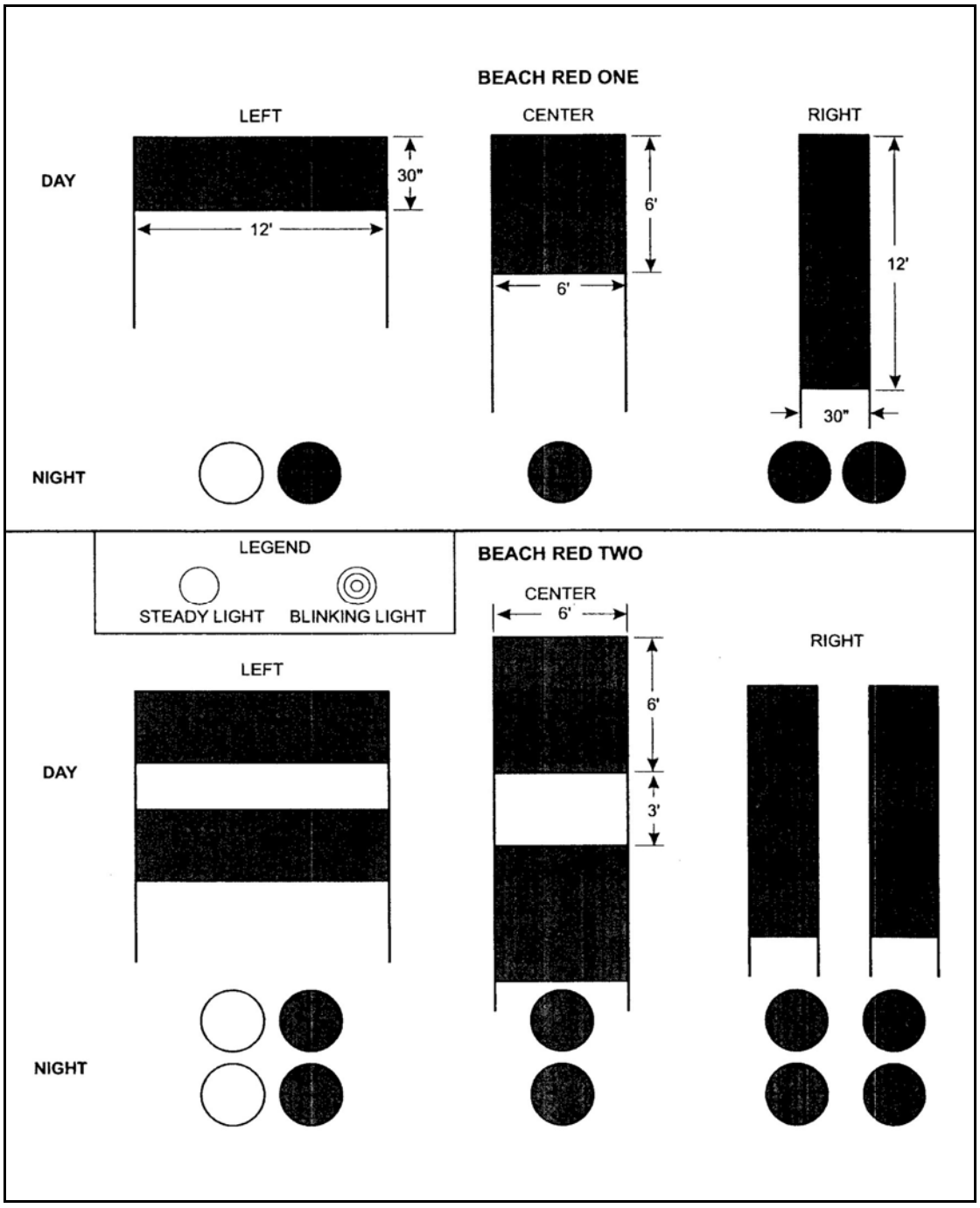


Figure B-5. Beach Markers (From Seaward) (Sheet 1 of 2)

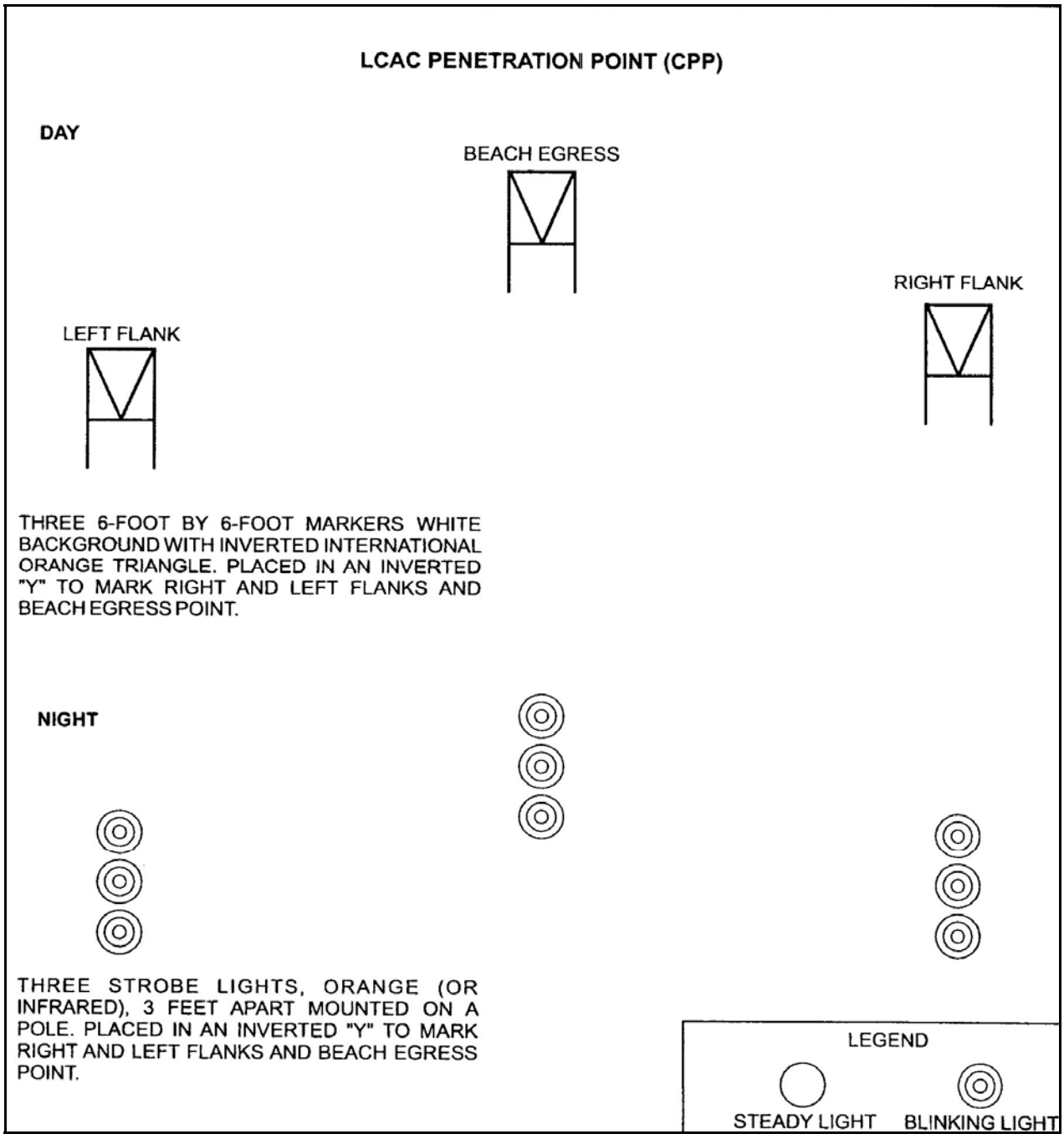


Figure B-5. Beach Markers (From Seaward) (Sheet 2 of 2)

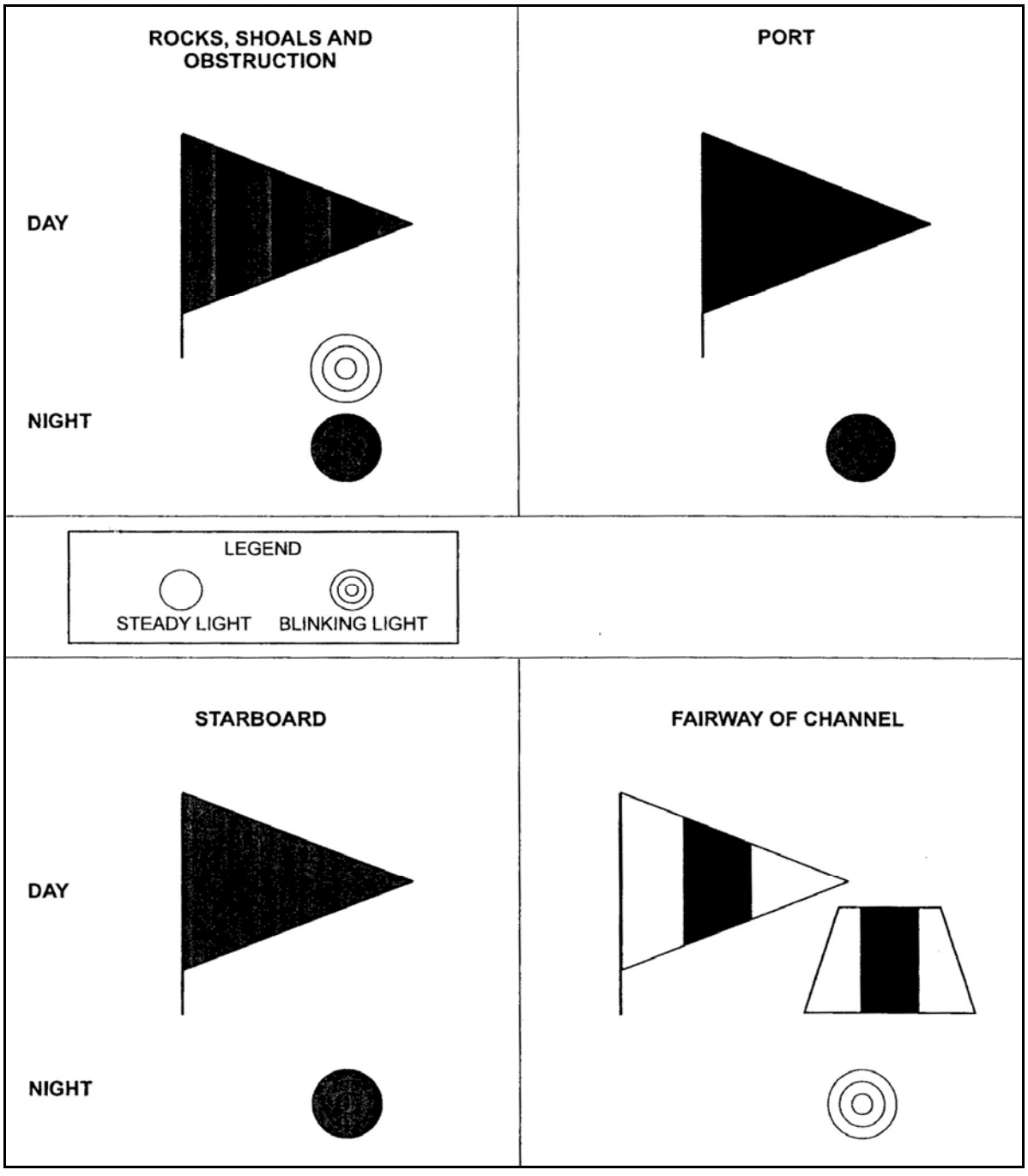


Figure B-6. Oceanographic Markers (From Seaward)

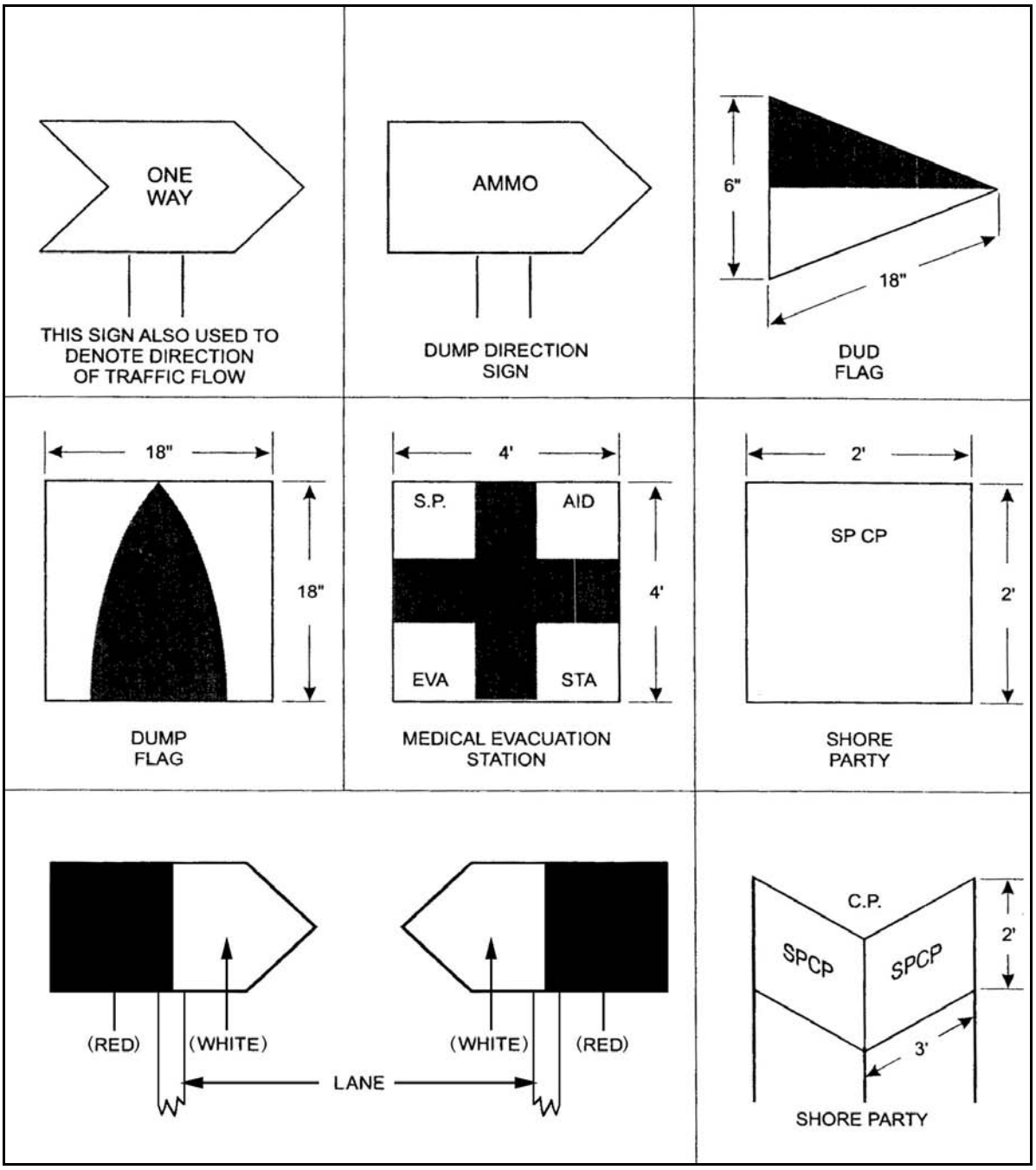


Figure B-7. Miscellaneous Beach Signs

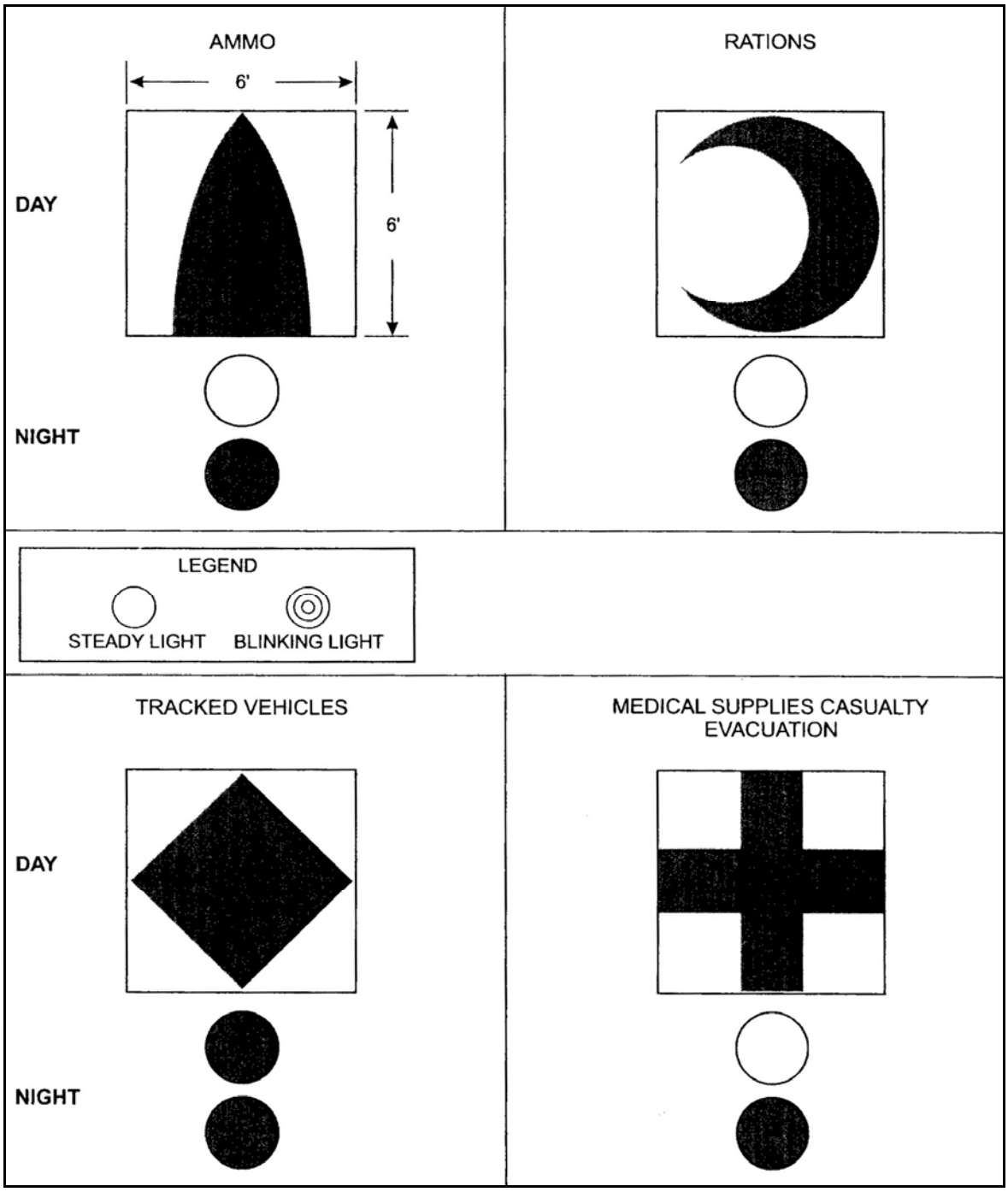


Figure B-8. Unloading Point Markers (Sheet 1 of 3)

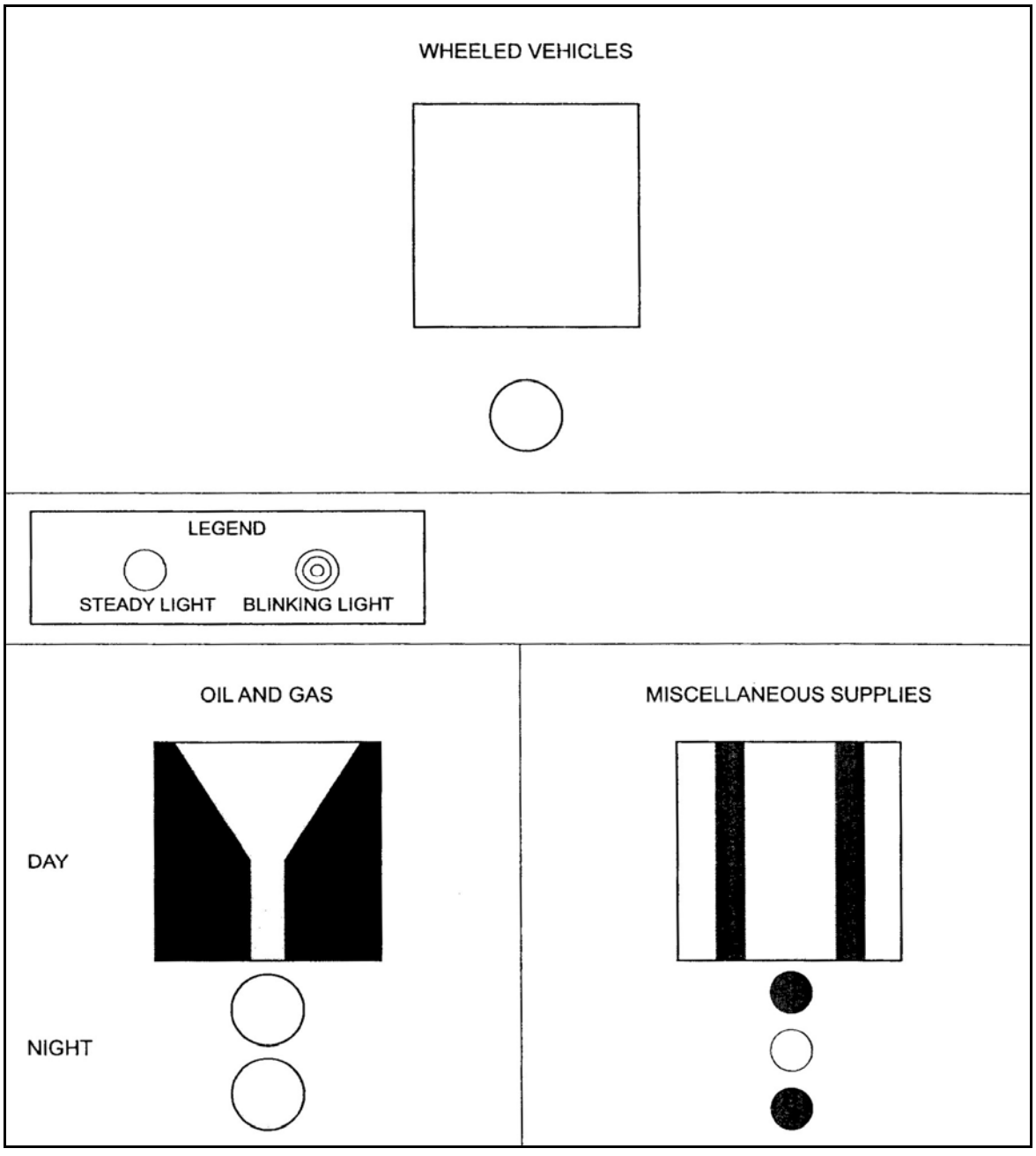


Figure B-8. Unloading Point Markers (Sheet 2 of 3)

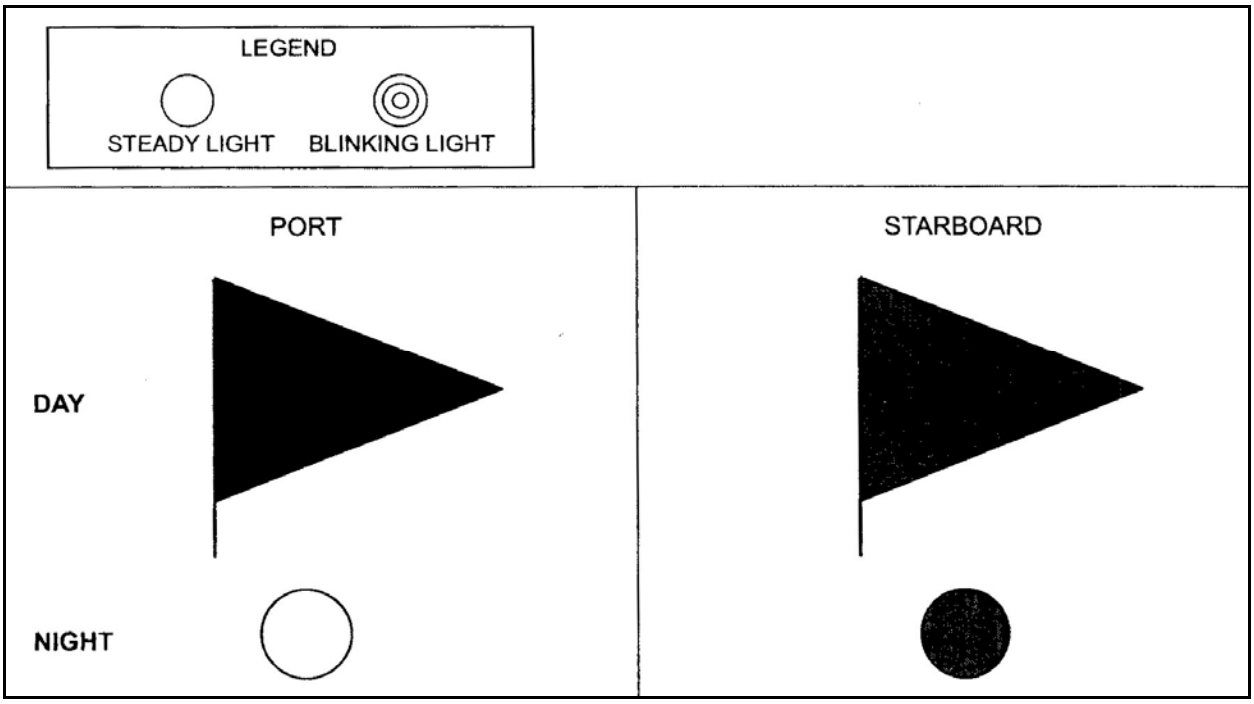


Figure B-8. Unloading Point Markers (Sheet 3 of 3)

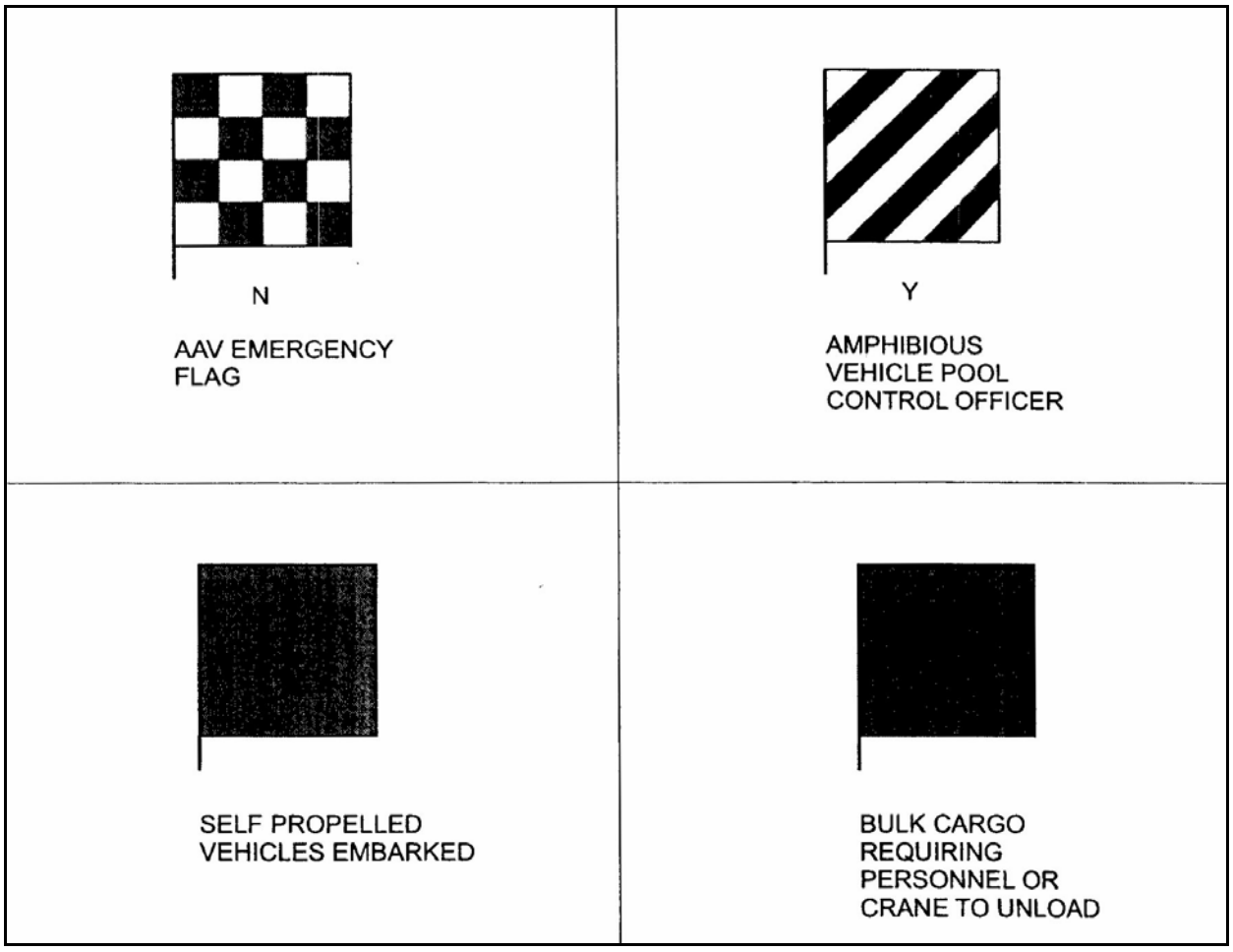


Figure B-9. Miscellaneous Flags and Identification Insignia (Sheet 1 of 3)

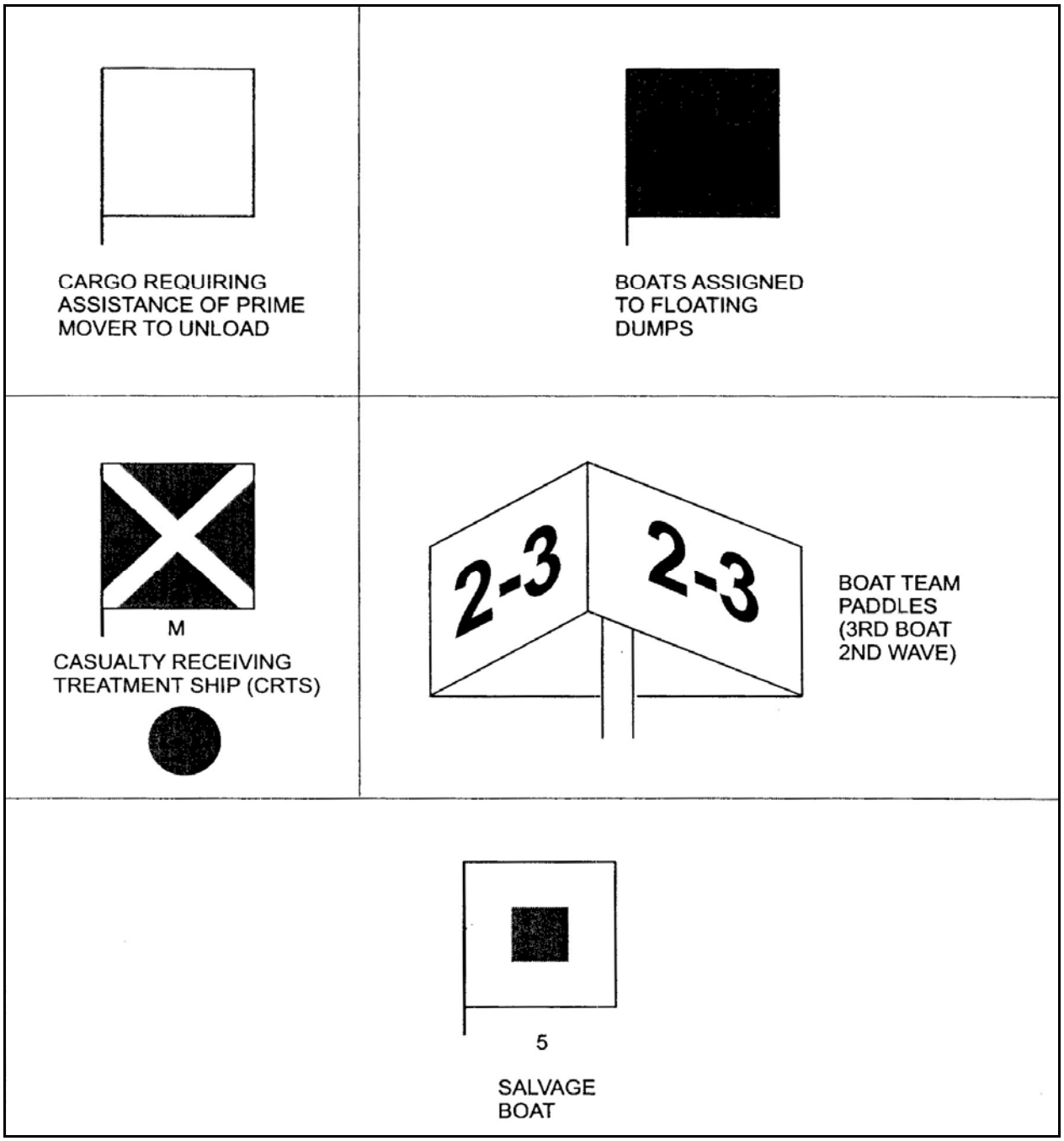


Figure B-9. Miscellaneous Flags and Identification Insignia (Sheet 2 of 3)

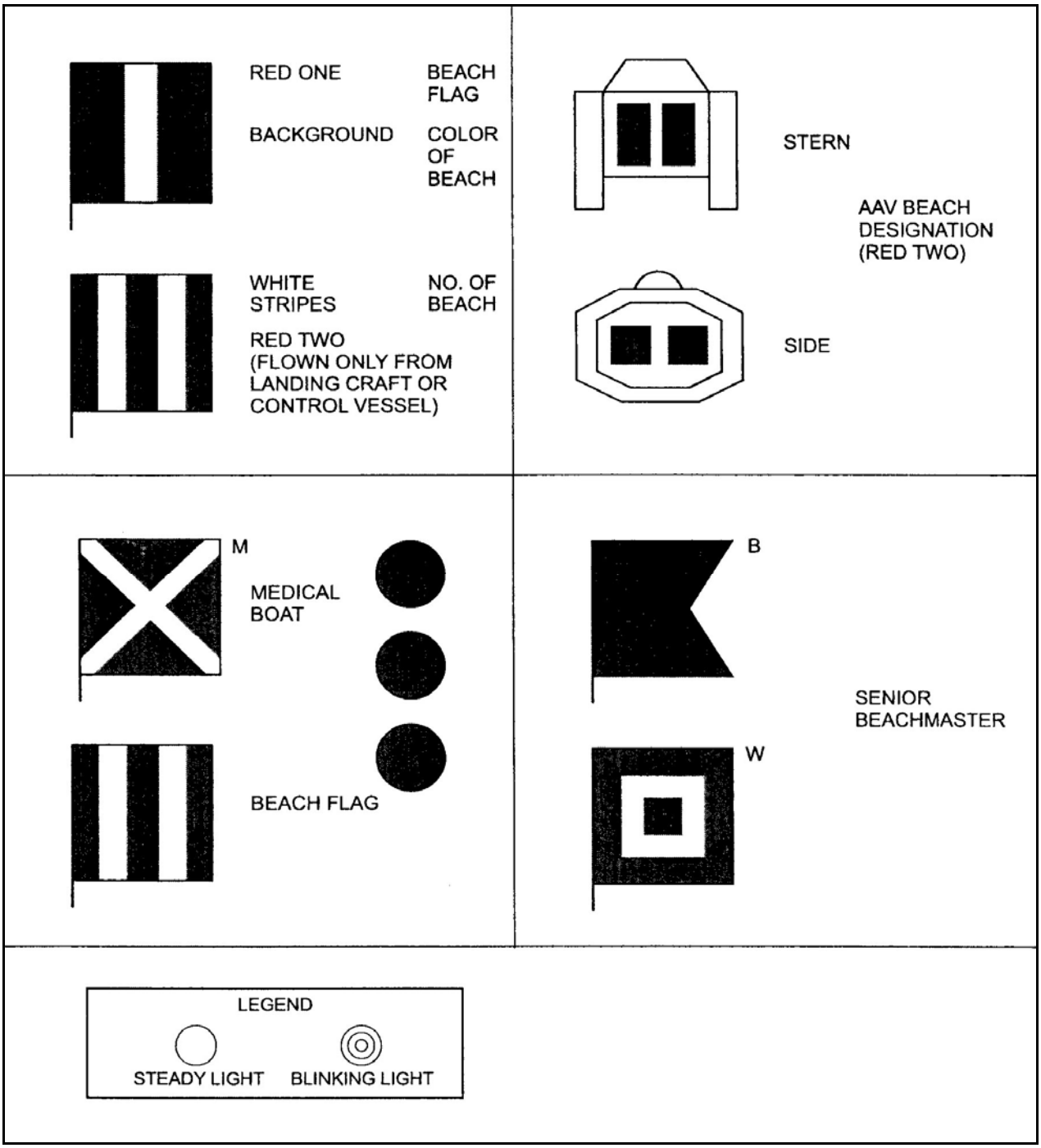


Figure B-9. Miscellaneous Flags and Identification Insignia (Sheet 3 of 3)

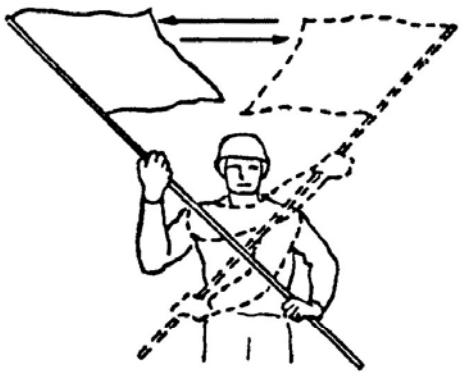
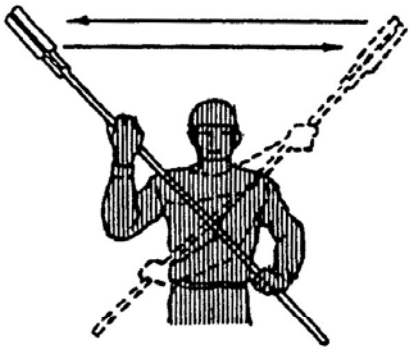
Meaning	Day	Night
Calling boats to the beach For small boats:	 QUEBEC flag held high in front of body and waved parallel to beach at which boat is to land.	 A steady white light (two wands tied together) mounted on a 6-foat pole waved over the head in the same manner as the QUEBEC flag.
For LCU:	 UNIFORM flag held high in front of body and moved as for LCM.	 Two white wands held at the sides of the body, then pointed straight up. Resume position and repeat, extinguishing light before each downward stroke.

Figure B-10. Day and Night Displacement Landing Craft and
Amphibious Vehicle Beaching Hand Signals (Sheet 1 of 5)

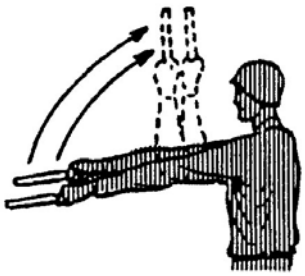
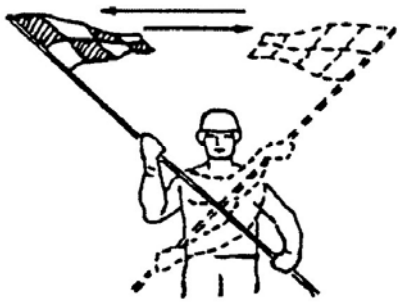
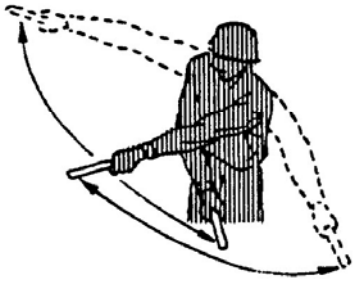
Meaning	Day	Night
Directing boats to the beach	 Arms are extended in front of body at a slight angle toward the deck with palms of hands facing upward; hands are then brought in toward body by bending arms at elbows.	 Arms are used in the same manner as day signal except white wands are used, extinguishing the light before the downward stroke.
Turn away (to keep landing craft and amphibious vehicles from beaching)	 NEGATIVE pennant is waved above head in same manner as QUEBEC flag.	 White wands are held straight out at shoulder height parallel to deck and waved horizontally so that they alternately are crossed and extended at arm's length

Figure B-10. Day and Night Displacement Landing Craft and Amphibious Vehicle Beaching Hand Signals (Sheet 2 of 5)

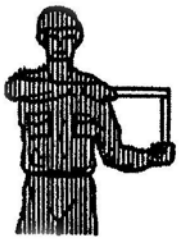
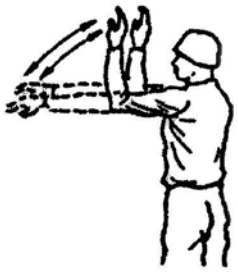
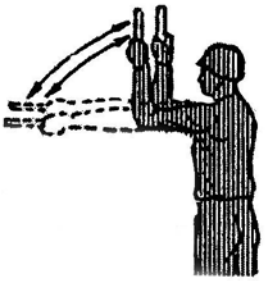
Meaning	Day	Night
Hold	 Left arm extended above head with fist.	 Arms extended with one while wand shining horizontally and the other vertically to form an inverted "L." May be at either side or in front of signalman.
Sending boats out	 Arms are held out, bent up at elbows and palms of hands facing away from body; then extend arms out horizontally. As breaker approaches stern of boat, an increase in engine speed may be indicated by more rapid movement of the arms.	 Arms are used in the same manner as day signal except white wands are used, extinguishing the light before the upward stroke.

Figure B-10. Day and Night Displacement Landing Craft and Amphibious Vehicle Beaching Hand Signals (Sheet 3 of 5)

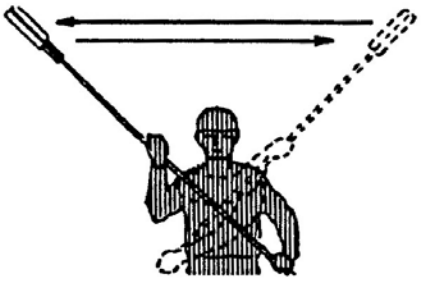
Meaning	Day	Night
Salvage (to call salvage boats to effect salvage)	 SIERRA flag is waved above head. Traffic controlman is seaward at base of disabled boat.	 A steady red light (two wands tied together) mounted on a 6-foot pole waved over the head from left to right.
Ramp up and dogged	 Arms held out parallel to deck with hands held open and pointed up. Coxswains acknowledge with same signal to inform traffic controlman ramp is up and dogged.	 Arms held out parallel to deck with wands pointed straight up (Coxswains acknowledge with same signal to inform traffic controlman ramp is up and dogged.)

Figure B-10. Day and Night Displacement Landing Craft and Amphibious Vehicle Beaching Hand Signals (Sheet 4 of 5)

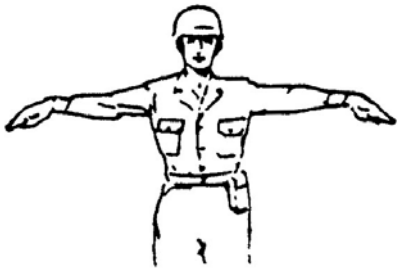
Meaning	Day	Night
Ramp down	 Arms held out parallel to deck with hands held open and pointed down. Coxswain acknowledge with same signal.	 Arms held out parallel to deck with wands pointed straight down. Coxswains acknowledge with same signal.
Stop towing	 BRAVO flag held high in front of body and waved parallel to beach in vicinity of boat being salvaged.	 Flashing red wand held front of body and waved parallel to beach in vicinity of boat being salvaged.

Figure B-10. Day and Night Displacement Landing Craft and Amphibious Vehicle Beaching Hand Signals (Sheet 5 of 5)

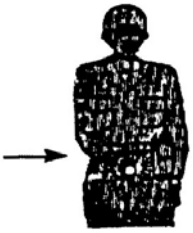
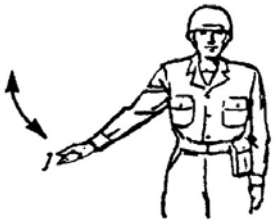
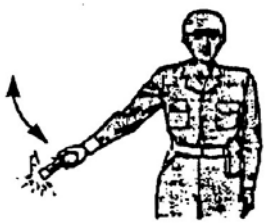
Meaning	Day	Night
This marshall	 Arms above head in vertical with palms facing forward.	 Same as day with addition of wands.
Proceed to next marshall	 Right or left arm down. Other arm moves across the body and extends to indicate direction to next marshall.	 Same as day with addition of wands.
Slow down	 Right arm extended downward. Move up to horizontal position. Repeat as necessary.	 Turn light on when arm is in the starting position. Turn light off when signal is completed. Arm does not move above the horizontal. Repeat as necessary.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 1 of 7)

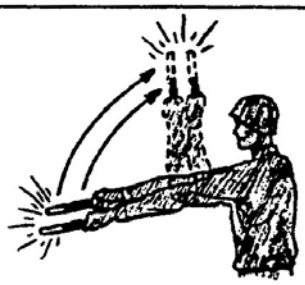
Meaning	Day	Night
Move ahead	 Arms are extended in front of body at a slight angle toward the deck with palms of hands facing upward. Hands are then brought in toward body by bending arms at elbows.	 Arms are used in same manner as day signal except wands are used, extinguishing the light before the downward stroke.
Turn to port	 One arm straight out to side on which turn is to be made. Other arm raised straight up. Palm forward.	 Lighted wand in each hand. One straight out. Other straight up.
Turn to starboard	 One arm straight out to side on which turn is to be made. Other arm raised straight up. Palm forward.	 Lighted wand in each hand. One straight out. Other straight up.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 2 of 7)

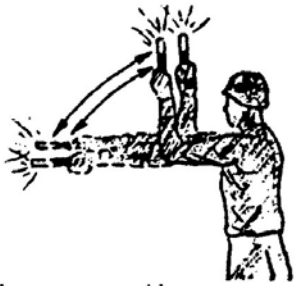
Meaning	Day	Night
Move back	 Arms are held out bent up at elbows and palms of hands facing away from body. Then extend arms out horizontally.	 Arms are used in same manner as day signal except wands are used, extinguishing the light before the upward stroke.
Landing direction	 Marshall turns and faces direction craft is to land. The arms are lowered from a vertical position to a horizontal position.	 Same as day with addition of wands.
Increase cushion	 Arms overhead slightly angled. Wave both arms at the same time in a circular motion.	 Same as day with addition of wands.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 3 of 7)

Meaning	Day	Night
Hover and stop	 Arms extended horizontally sideways palms downward and bring arms into chest repeating until craft hovers.	 Same as day with addition of wands.
Decrease cushion	 Arms down slightly angled apart. Move both arms in a circular motion at the same time.	 Same as day with addition of wands.
Translate to port	 Right arm extended horizontally sideways in direction of movement and other arm swung over the head in same direction in a repeating motion.	 Same as day with addition of wands.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 4 of 7)

Meaning	Day	Night
Translate to starboard	 Left arm extended horizontally sideways in direction of movement and other arm swung over the head in same direction in a repeating motion.	 Same as day with addition of wands.
Stop emergency	 Arms down and crossed in front and then swung out at an angle repeating until understood.	 Same as day with addition of wands.
Land	 Arms crossed and extended downwards in front of the body.	 Same as day with addition of wands.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 5 of 7)

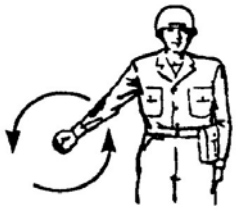
Meaning	Day	Night
Connect power	 Hands above head left fist partially clenched. Right hand moved in direction of left hand with first two fingers inserted into circle made by fingers of the left hand.	 Same as day with addition of wands.
Disconnect power	 Hands above head left fist partially clenched. Right hand moved away from left hand withdrawing first two fingers from circle made by fingers of the left hand.	 Same as day with addition of wands.
Start engines	 Right arm down at angle and moved in a clockwise circle until engines are started.	 Turn light on when arm is in starting position. Turn light off when signal is complete.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 6 of 7)

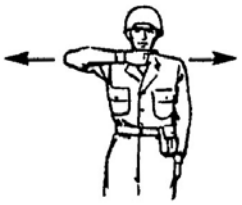
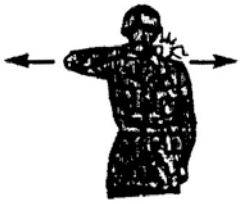
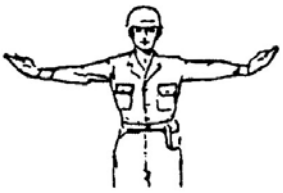
Meaning	Day	Night
Cut engines	 Either arm and hand level with shoulder moving across throat palm downward. The hand is moved sideways with arms remaining bent.	 Same as day with addition of wands.
Ramp down	 Arms held out parallel to deck with hands held open and pointed down.	 Same as day with addition of wands.
Ramp up and dogged	 Arms held out parallel to deck with hands held open and pointed up.	 Same as day with addition of wands.

Figure B-11. Landing Craft Air Cushion Maneuvering Hand Signals (Sheet 7 of 7)

APPENDIX C

Landing Craft and Amphibious Vehicle Control Procedures

C.1 GENERAL

The grid reference system is currently the only approved procedure available for a craft control team to conduct positive control of landing craft. However, new control procedures are under development that incorporate the use of existing technology such as GPS. The grid reference system relies on radar returns to determine the positions of the controlling ship and the landing craft. These returns are plotted using a DRT overlay composed of a series of boat lanes constructed from the LD to the beach. This allows tracking of the position and progress of the landing craft throughout their transit. The boat lanes are marked with the specific times and speeds applying to each wave. The system may also be used to control displacement landing craft en route to the rendezvous area, on-call waves, and nonscheduled unit during darkness or reduced visibility.

C.2 PURPOSE

The purpose of this appendix is to describe the grid reference system of landing craft/wave control, highlight the difficulties and inherent inaccuracies of that system, and introduce some alternative craft control procedures. Most of these new procedures rely on emerging technologies that are further described in the SWDG landing craft control procedures TACMEMO.

C.3 LANDING CRAFT CONTROL PROCEDURES

There are three types of landing craft control procedures: positive, advisory, and independent. The type of control exercised is dependant upon a number of factors: the nature of the mission; the tactical situation; capabilities of the craft being controlled, including its installed navigation and C2 suite; the C2 capabilities of the PCS or LCAC control ship, or designated alternates; and the assigned control team.

C.3.1 Positive Control

Positive control is that method of controlling landing craft during which the PCS or LCAC control ship control team provides near-constant direction to the craft commander (e.g., craft master, vehicle driver, or wave commander) based upon information obtained by monitoring the craft's position versus the intended track. Course and speed changes are made at the direction of the control team, except as determined by the craft commander for emergency conditions. The control team continuously updates the craft's position via an external control source, which may be electronic (e.g., radar), voice communications, data link, or a combination of sources. Positive control is rarely used in LCAC operations, because LCAC possess sufficient navigational capability to allow advisory control in most situations, and the LCAC SEAOPS recommends against it.

C.3.2 Advisory Control

When exercising advisory control, control teams provide craft commanders the launch position and a vector to the first CP. Upon reaching that first CP, the craft commander follows a prescribed navigation route per a specified timetable to complete the mission. Normally, the PCS/LCAC control ship tracks the craft and periodically provides the craft commander or wave commander a current position and "time early or late" based on position relative to the approach schedule. It is then incumbent upon the craft commander to modify course and speed, as necessary, to regain the control plan. Advisory control is the preferred method of control for LCAC, and due to C2 improvements in LCUs, this method of control is the preferred method for controlling these craft as well.



The EFV, discussed in Appendix I, is expected to be equipped with a navigational capability that will likely make advisory control its preferred control method.

C.3.3 Independent Control

Under independent control, the responsibility for the safe movement of the craft lies exclusively with the craft commander. This manner of control generally occurs when the craft is expected to travel over an extended distance, usually without PCS or LCAC control ship assistance. Independent control could also be used when EMCON conditions are in effect. When executing independent control, craft commanders rely on the use of installed navigation systems such as the GPS or the craft's radar to ensure safe navigation.

Figure C-1 provides the preferred control methods for landing craft and amphibious vehicles in different situations.

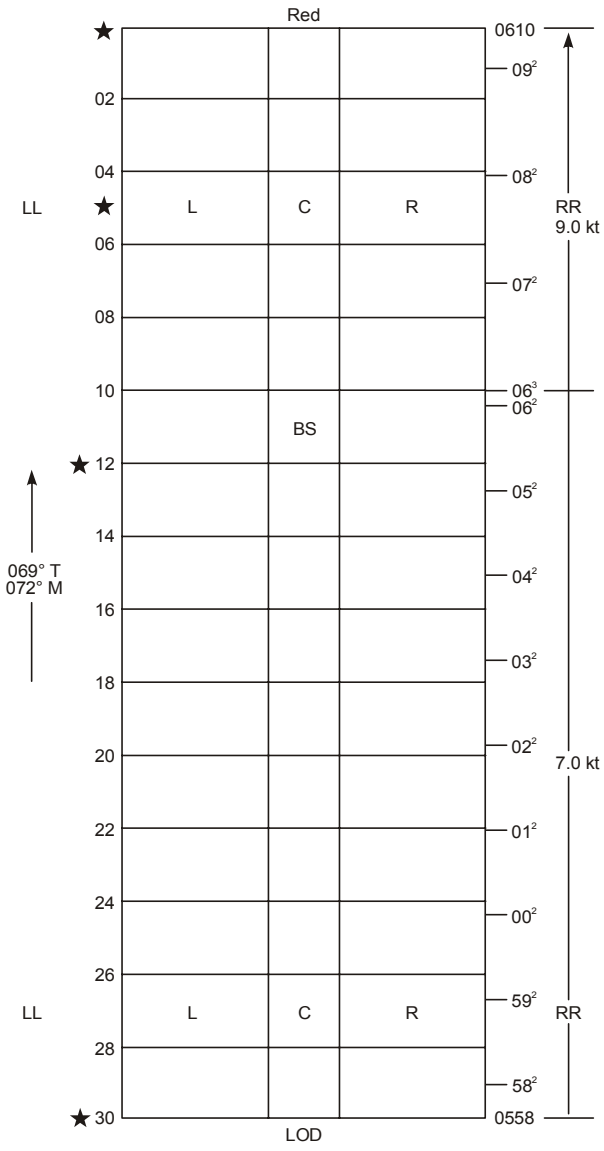
Preferred Methods of Control			
Craft	**Independent Operations**	**OTH Assault Operations**	**Operations Within 10 nm of Land**
LCAC	Independent	Advisory	Advisory
LCU	Independent	Advisory	Advisory
AAV	NA	NA	Positive

Figure C-1. Preferred Methods of Control for Landing Craft

C.4 GRID CONSTRUCTION

The PCS or LCAC control ship control team develops the grid diagram constructing one boat lane for each wave to be controlled. An example of this diagram is shown in Figure C-2.

Longitudinal lines in the gird diagram divide the lane into three sections: "L" (left), "C": (center), and "R" (right). The left and right sections are both 40 percent of the lane width, and the center section is 20 percent of the grid's width.



General Notes:

1. The boat lane shown is constructed for wave 3, 7, 9, or 11 to cross the LD at 0558 and touchdown at 0610.
2. The use of solid distance lines and dashed time lines is optional.
3. SOA is listed for each boat lane:
   a. LD to battle speed (7.0 kt)
   b. Battle speed to beach (9.0 kt).
4. Each boat lane is color coded.
5. Stars indicate required reports to the central control officer made on the control ship coordination net.
6. Distance (LD to beach) are shown on the left side and times (LD to beach) are shown on the right side of the grid diagram.
7. Control ships' stations are not fixed on the LD. They may be assigned underway sectors to avoid the shore-based threats. When an underway sector is used, a navigation point is designated to fix the PCS's position by radar prior to marking a wave's position.

Figure C-2. Example of a Grid Diagram of a Boat Lane

Lateral lines are drawn at 200-yard intervals along the grid diagram and numbered to indicate distance-to-go in hundreds of yards.

Wave positions on the grid diagram are described by a letter ("L", "C", or "R") followed by a number of one or two digits to indicate position in the boat lane and distance to the beach. Positions outside the lane are indicated by the double letter "RR" or "LL."

Timelines are plotted on the grid diagram by the following method:

1. Using Figure C-3, determine the SOA and BS for the wave.

2. For the final 1,000 of the transit, waves will be making BS; therefore, count backwards from touchdown time to the BS line accounting for the complete time (whole and fractions of minutes).

3. Divide the time from LD to BS using the SOA and accounting for the complete time.

4. Label the times on the boat lanes as shown in Figure C-2. This determines the time the LD will be crossed.

5. When controlling more than one wave, the time clock is divided into four primes:

Prime	Time (sec)	Waves Marked
0	52.5 to 7.5	1, 5, 9
1	7.5 to 22.5	2, 6, 10
2	22.5 to 37.5	3, 7, 11
3	37.5 to 52.5	4, 8, 12

By using the 15-second primes for grid construction (and grid position transmission), the complete time for the transit can be systematically accounted for.

C.4.1 Wave Control

Prior to debarkation, the craft commanders are issued a grid diagram of the boat lane. Displacement landing craft or waves in the rendezvous area are provided navigational assistance by the PCS to maintain proper distances from the LD. The control team begins marking the position of each landing craft or wave on the grid diagram as it departs the rendezvous area. As each wave or craft crosses the LD, the control team advises the craft commanders and/or LWCs of the wave's position via voice radio or flashing light. The grid positions may be supplemented with vectors and "early" or "late" information, as necessary. Upon receiving a grid position indicating the landing craft or wave is not in the center of the boat lane or not progressing according to schedule, the craft commander or the wave adjusts course and speed, as necessary, to regain schedule. If it is apparent to the control team that appropriate action is not being taken to regain the center of the boat lane, the control team may direct changes by ordering vector and speed changes.

Grid positions are transmitted every minute from the rendezvous area to 200 yards from the beach per the prime schedule from Paragraph C.4. If corrective action is required, grid positions may be transmitted more frequently. Grid positions are provided once each minute in periods of low visibility from the rendezvous area to the beach.

Craft Type	Battle Speed (kt)	SOA Good Weather (MSI less than 6)	SOA Foul Weather (MSI greater than 6)
LCU 1627	10.0	8.0	5.0
LCU 1646	11.0	10.0	4.5
AAVP7	8.0	6.5	5.0

Figure C-3. Table of Standard Planning Data for Ship-to-Shore Movement

To maintain an accurate track of their progress, craft commanders are generally directed to plot their position each time that the PCS transmits its location. This action helps to determine and correct the effects of the winds and seas, or any incorrect actions taken. Once reliable radio communications are established, receipt for grid positions is not required; however, radio or visual receipt for vector or speed changes is always required, and PCS should continue to "transmit in the blind" or request visual acknowledgment until receipt for the information is transmitted.

C.4.2 Voice Communications Procedures and Communications Circuits

Sample voice callups and typical net transmissions for wave or landing craft control are provided in the following paragraphs.

The communications circuits used to control surfaceborne ship-to-shore movement are described in the following paragraphs and Section 4.5.

C.4.2.1 Boat Control and Operations Nets

Two nets are designated for each colored beach. The Alfa Net is a directed net used by the PCS to pass grid positions and vectors to scheduled waves and landing craft from the rendezvous or underway launch area until touchdown. The Bravo Net is used by PCS and other ships to control boats prior to being dispatched to the beach and following wave or craft touchdown. Touchdown reports and administrative traffic between PCS and landing craft are also passed on this net.

C.4.2.1.1 Control Ship Coordination Net

If more than one colored beach is being used, a Control Ship Coordination Net is established for coordinating the assault or action and providing status reports between PCOs and the CCO. The PCO uses this net to report when each wave is crossing the LD; when wave one is 1,200 yards and 500 yards from the beach; and touchdown for each wave. See Paragraph C.4.2.3.

C.4.2.2 Net Procedures

The Control Ship Coordination Net and Bravo Net use daily changing call signs. The Alfa Net uses call signs from JANAP 119, Joint Voice Call Sign Book. The PCS and each craft commander use the JANAP 119 call signs in Figure C-4 on all nets to avoid confusion.

Unit	JANAP 119 Call Sign
CCO	Catskill
PCO/PCS Blue Beach	Blue Catskill
LCU 1627	Clapper Blue One
LCU 1646	Clapper Jr. Blue One

Figure C-4. Example of Net Procedures for Colored Beach

C.4.2.3 Sample Voice Transmissions for Blue Beach

Sample voice transmissions for blue beach are provided in Figure C-5.

Net	Voice Transmission
Wave/craft commander Two Blue One turnover from parent ship to PCS (Bravo Net)	"____ ____ ____(wave/craft commander), this is ____ ____ ____ (parent ship). Report to Blue Catskill for control and vector to blue beach. Over." (Note 1)
Wave/craft commander Two Blue One reporting to PCS (Bravo Net)	"Blue Catskill, this is ____ ____ ____ (wave/craft commander). Reporting for control and vector to the beach. Over."
Positive control (Bravo Net)	"____ ____ ____ (wave/craft commander), this is Blue Catskill. Have you under positive radar control. Steer course ____ speed ____ for the LD. Set and drift at the LD is (direction) and (speed in knots). Switch to channel Alfa. Over." (Note 2) "____ ____ ____ (AAV wave), this is Blue Catskill. Have you under positive radar control. Maintain present course and speed. Over." "____ ____ ____ (AAV wave), this is Blue Catskill. Have you under positive radar control. Steer course ____ speed ____ for the LD. Set and drift at the LD is (direction) and (speed in knots). My intention is to execute a left (right) flanking movement seaward of the LD. Shift to channel Alfa. Over." (Note 3)
Wave/craft commander dispatch from the LD (Alfa Net)	"Two Blue One, this is Blue Catskill. You are dispatched from the LD to the beach. Steer course ____ speed ____. Over." (Notes 4 and 5)
LD crossing report (Control Ship Coordination Net)	"Catskill (CCO), this is Blue Catskill. Two Blue One crossed LD late one-half. Over." (Note 6)

Figure C-5. Sample Voice Transmissions for Blue Beach (Sheet 1 of 2)

Net	Voice Transmission
Grid positions (Alfa Net)	"Two Blue One, this is Blue Catskill. Grid posit Romeo three eight. Out." (Wave Two Blue One is on the right side of the boat lane, 3,800 yards from the beach and on time.) "Two Blue One, this is Blue Catskill. Grid posit Romeo three zero early one. Out." (Wave Two Blue One is on the right side of the boat lane, 3,000 yards from the beach and early 1 minute.)
Vectoring waves (Alfa Net)	"Two Blue One, this is Blue Catskill. Grid posit Romeo three zero early one. Vector left ten. Over." (Note 7) "Two Blue One, this is Blue Catskill. Grid posit Romeo Romeo two seven early one. Vector left twenty. Over." (Note 8)
Speed changes (Alfa Net)	"Two Blue One, this is Blue Catskill. Grid posit Charlie two four early one. Slow down. Over." "Two Blue One, this is Blue Catskill. Grid posit Romeo two zero late two. Speed up. Over." (Note 9)
1,200/500 yard reports (wave one only) (Control Ship Coordination Net)	"Catskill, this is Blue Catskill. One Blue One at 1,200 (500) yards, on time, early or late. Over." (Note 6)
Battle speed (Alfa Net)	"Two Blue One, this is Blue Catskill. Grid posit Charlie one zero. Battle speed, battle speed! Over." (Notes 10 and 11)
Touchdown report (each wave craft commander on Bravo Net)	"Blue Catskill, this is ____ ____ ____ (wave/craft commander). Touchdown, touchdown, touchdown! Over."
Touchdown report (Control Ship Coordination Net)	"Catskill, this is Blue Catskill. Two Blue One touchdown. Late one-quarter. Over." (Note 6)

Notes:

1. ____ ____ ____, where appearing, indicates the daily changing call sign.
2. The shift to boat Alfa can be ordered by PCS when desired, but no later than when waves or craft cross the LD. If no shift order is given, waves or craft will automatically shift to the Alfa Net on crossing the LD.
3. AAV waves require an "intention" statement from PCS when PCS takes positive radar control.
4. A full callup is required for transmissions to ensure the proper wave or landing craft receives the information. Ordered courses to waves or craft are given in degrees magnetic.
5. Dispatch orders are not required of waves or craft to shift to the Alfa Net upon crossing the LD if provided for in the OPORD, PCO instructions, or prebriefed.
6. Reports to the CCO will include a time status. Fractions of minutes are spoken as one-quarter, one-half, three-quarters, etc.
7. Due to the general nature of magnetic boat compasses, after waves cross the LD, course change vectors are given in tens of degrees, not by course heading. Waves/landing craft should still compare their magnetic compass headings with PCS while transiting to the rendezvous area. Vectors may be given at any time and in any amount to maintain a wave or craft's position in the center of the boat lane. However, vectors should be limited to 10° in the surf zone for craft safety.
8. Waves or landing craft outside the boat lane must be expeditiously vectored to regain the boat lane.
9. Prior to the order for battle speed, speed changes may be given at any time to keep waves or landing craft on time. Speed changes must be ordered when waves are early or late 2 minutes or more.
10. Battle Speed will be ordered at the 1,000-yard mark. Even as a wave is making maximum speed prior to the 1,000-yard mark, the Battle Speed, Battle Speed order is still mandatory at that time. However, the Battle Speed order for wave one may be delayed to preclude landing early.
11. Information transmissions end in "out," and those directing waves or craft to perform a duty end in "over." If at any time it is desired for a wave to receipt for information, the transmission should end with "over."

Figure C-5. Sample Voice Transmissions for Blue Beach (Sheet 2 of 2)

C.4.3 Visual Procedures for Transmitting Grid Positions

Grid positions by flashing light will normally be preceded only by the wave number. However, if confusion would result from transmitting into different numbered boat lanes or different colored beach lanes, the callup is modified (e.g., to call the wave or craft commander of wave three, Blue Beach Two, the normal callup is numeral 3). If confusion would result, the callup is transmitted as numeral 3 blue numeral 2. The PCS, after establishing communications with the wave or craft commander, transmits the grid position.

The wave or craft commanders receipt for each group by flashing "T" and receipt for the message with "R."

Visual grid positions and information are transmitted by PCS using the following procedures:

1. After the wave or craft callup, insert the group "GP." This acts as a proword and alerts the receiver that a grid position is to follow.

2. Transmit the grid position using "L" for left, "C" for center, "R" for right, and "LL" or "RR" for being outside of the boat lane to the left or right, respectively. The distance from the beach is transmitted in hundreds of yards as a single or double numeral (e.g., "9" equals 900 yards, and "41" equals 4,100 yards from the beach).

3. Transmit "T" followed by two digits to indicate the time, in minutes, of the grid position. Knowing the time of the position, the wave or craft commander knows how early or late the wave is once the position is plotted. Knowledge of grid position time is important because, depending on the proficiency of the boat control team, receipt of the grid position can take up to 2 minutes from the time it was computed.

4. If necessary to order a Speed Up or Slow Down, the group "SS" or "TT," respectively, is sent.

5. If necessary to order a course change, a vector "V," in tens of degrees indicating the direction left, "L," or right, "R," is sent. For example, for a course change of 20° to the right, the group "V2R" is sent. Direction of the vector is always included.

6. The group "BS BS" is an order to go to battle speed.

7. If needed, the group "TA" is an order to turn away.

Examples of visual grid positions and information are provided in Figure C-6.

Upon touchdown each landing craft should send "TD, TD, TD" to PCS.

C.5 ALTERNATE LANDING CRAFT CONTROL PLOTTING PROCEDURES

Although the grid reference system discussed in the preceding sections is currently the only approved procedure for positive control of landing craft, the need for procedures to control landing craft without the use of radar and/or a DRT will become more and more necessary. The AN/KSQ-1/AADS systems will solve this problem in

Signal	Meaning
4GPL40T32	Wave/craft four grid posit is in the left portion of the boat lane 4,000 yards from the beach at time 32.
2GPR32T47SS	Wave/craft two grid posit is in the right portion of the boat lane 3,200 yards from the beach at time 47, and Speed Up is ordered.
1GPLL29T52V2R	Wave/craft one grid posit is outside of the boat lane to the left 2,900 yards from the beach at time 52, and Vector 20° to the right is ordered.
3GPC20T17TT	Wave/craft three grid posit is in the center of the boat lane 2,000 yards from the beach at time 17, and Slow Down is ordered.

Figure C-6. Examples of Visual Grid Positions and Information

the future, but is neither mature nor fully fielded on all landing craft or amphibious ships. Many factors contribute to the need for craft control without radar including:

1. STOM and seabasing concepts which call for landing craft transits of up to 25 nm.
2. Increased proliferation and technological advancement of adversary ASCMs.
3. FAC/FIAC threat.
4. Mini-subs and semi-submersibles.
5. Procurement of new systems, such as the EFV.
6. Radar tracking limitations such as the height of the radar above sea level, (i.e., radar horizon), the size of the object (craft) being tracked, and radar system sensitivity.
7. Contacts at, or near, landmasses seeming to merge into one contact.
8. Craft operating in close proximity to each other merge into one contact (i.e., the return from two or more contacts merge to form a single radar return, therefore, not providing a distinct return for craft identification).
9. New class designs that do not include a DRT or radar consoles for plotting landing craft.

C.5.1 Global Positioning System Craft Control

The accuracy of current GPS technology has led to the development of craft control procedures currently available in the landing craft control procedures TACMEMO. It is one avenue being considered to allow for craft and vehicle control without radar or outside the radar horizon. GPS units are a part of the standard navigational

equipment in all ships as well as LCAC and most LCUs. Also, hand-held portable GPS units can be used in AAVs, although usually, there are not enough units available for each craft to have one.

The use of PLGR in AAVs poses some restrictions to exclusively using GPS. AAV operators close the hatches of their vehicles prior to launching, so unless the craft is equipped with an external antenna, the PLGR sometimes loses its signal until the hatches are reopened. Therefore, once the AAV is launched and the hatches reopened, drivers/vehicle commanders must re-initialize the PLGR. As a result, the AAV is forced to navigate "blind" until the unit re-initializes and sufficient GPS track data is generated to obtain an accurate indication of what direction the AAV is actually heading.

Planners must therefore take the potential GPS system limitations into account when planning ship-to-shore movement involving landing craft and vehicles. Techniques for using GPS technology to control and guide landing craft have been proposed in the landing craft control procedures TACMEMO. A brief overview of GPS craft control procedures is provided in the paragraphs below.

C.5.1.1 Global Positioning System Techniques

Two techniques for GPS plotting are delineated as a part of this appendix. GPS Method 1 is intended for use when both the ship and each wave of landing craft have GPS, when operating outside the radar horizon and when radar is degraded or insufficient. Method 2 is intended for use on LPD 17 Class ships that do not possess a radar console for craft control plotting. Both methods can contribute to reductions in manning requirements for the craft control team.

C.5.1.1.1 Global Positioning System Method 1

GPS Method 1 allows for craft control outside the radar horizon or when radar is degraded or insufficient for craft control, and enhances the manner in which craft control teams can conduct positive control operations. In some instances, it can enable craft that routinely have been excluded from being tasked with completing advisory or independent control operations to being considered for such operations. To conduct operations using GPS Method 1, landing craft must be upgraded with an installed GPS unit or a hand-held GPS unit along with trained personnel.

The specific procedures for GPS Method 1 are dependent upon the equipment and space in CIC. The procedures described in the paragraphs below are used for GPS Method 1.

a. Global Positioning System Method 1, Bearing/Range

For CICs with space for a PMP arm, bearing/range to beach center or waypoint is used. All navigation points (i.e., beach center, LOD, CCA, CDP, CPP, CCPs) are entered into the landing craft's GPS and plotted on a large-scale chart or blank piece of paper on the control ship. The boat lane, if applicable, is constructed on the same chart or piece of paper. To maintain accuracy, all points are to the nearest second or two decimal places. As the landing craft proceeds along its track, at each designated mark time (1 minute, 2 minutes, 30 seconds), the wave commander reports the craft's position to the control ship by stating bearing and range to the beach center or designated waypoint (i.e., "327/3,300 yards"). The craft control team on the ship plots the craft on its chart, boat lane, or LCAC track and makes the appropriate recommendations (during positive or advisory control) (i.e., "Romeo 50 yards, on time. Recommend coming left to 210"). This process is repeated until touchdown.

b. Global Positioning System Method 1, Latitude/Longitude

For CICs without a PMP arm (LPD 17 Class), latitude/longitude is used. All navigation points (i.e., beach center, LOD, CCA, CDP, CPP, CCPs) are entered into the landing craft's GPS and plotted on a large-scale chart or blank piece of paper on the control ship. The boat lane is plotted on the same chart or piece of paper. A latitude/longitude grid to the nearest second or two decimal places is constructed on top of the same chart or piece of paper. To maintain accuracy, all points are to the nearest second or two decimal places. As the landing craft proceeds along its track, at each designated mark time (1 minute, 2 minutes, 30 seconds), the wave commander reports the craft's position to the control ship by stating the latitude/longitude (i.e., for latitude 35 23.75 and longitude 034 54.41, the craft would report, "375, 441"; or for latitude 35 23.78 and longitude 034 54.47, the craft would report, "378, 447"). The craft control team on the ship plots the craft on its chart, boat lane, or LCAC track and makes the appropriate recommendations (during positive or advisory control) (i.e., "Romeo 50 yards, on time. Recommend coming left to 210"). This process is repeated until touchdown.

C.5.1.1.2 Global Positioning System Method 2

GPS Method 2 is intended primarily for use during underway launches of landing craft that do not have installed or portable GPS units (e.g., LCM 8s and/or AAVs) or when their crews have not been trained on the use of GPS for controlling craft movement while waterborne. GPS Method 2 improves upon the legacy grid reference system by using the ship's GPS unit to ascertain the ship's position and to mark the ship's position relative to a waypoint that has been entered into its system to mark the center of the beach landing site or other geographic reference point. By using GPS to fix the ship's position, the overall accuracy of the navigational plot and the ship-to-shore movement control process is improved. In LPD 17 Class ships, SSDS accomplishes this method because the ship's position in SSDS is determined by GPS inputs to the ship's gyro, and craft symbology is ascertained from the ship's radar inputs to SSDS.

Fixing the ship's position via GPS and its bearing and range from the GPS waypoint, the craft control team, operating radar repeaters in CIC, is able to fix the position of the landing craft. Maintaining positive control, the craft control team then provides grid positions and vectors to ensure the craft remain within the assigned boat lanes and travel per the landing plan.

Central to this procedure is the technique of simultaneously marking the position of the ship and the landing craft at a specified time (i.e., mark time) and maintaining radar coverage of each wave and beach center. To accomplish this, the radar repeater operator may need to "freeze" the radar repeater screen (i.e., pause the image) and carefully report the position of the landing craft.

Because the ship's position has been determined using GPS and the position of the landing craft have been frozen momentarily to obtain all required information at the same relative time, the accuracy of the position of the landing craft should be significantly improved upon. Likewise, because the video presentation has been frozen, the need to mark each boat wave at different times (i.e., on the primes of each minute) can be eliminated and all waves marked simultaneously. Still, some craft control teams may find it easier to continue the practice of marking on the primes as a way to aid in contact management.

On LPD 17 Class ships, SSDS automatically performs the functions of GPS Method 2, and the craft symbology is sufficient to determine craft positions at each mark time.

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APPENDIX D

Salvage Operations

D.1 MISSION

Landing craft, boat, and vehicle casualties are inevitable during surfaceborne ship-to-shore movement. The mission of the salvage organization is to carry out operations, as necessary, to keep boat lanes and beaches clear of disabled boats and landing craft so that the ship-to-shore movement is not impeded or delayed.

NTRP 3-02.1.2 also contains information on salvage operations and equipment used during amphibious operations.

D.2 ORGANIZATION

The ATF commander is responsible for salvage operations during amphibious operations. The PCO is delegated the overall responsibility for conducting salvage operations at a landing beach to keep the beach, boat, and LCAC transit lanes, and CLZs unobstructed.

Once the BP is established ashore, the beachmaster or the BP commander becomes the senior salvage officer ashore and reports to the PCO for salvage operations conducted from the LD, or the three-fathom curve, to the HWM. The HWM is a mark left on the beach by wave wash at the preceding high water. It does not necessarily correspond to the HWL. Because it can be determined by simple observation, it is frequently used in place of the HWL, which can be determined only by a survey.

Seaward of the LD, the PCO directs salvage operations in coordination with the beachmaster, as necessary. If a landing craft or amphibious vehicle becomes disabled, the decision as to whether it is towed to the beach or to a ship is based primarily on the distance to each, weather and sea state, and the seaworthiness of the craft or vehicle.

When available, a salvage boat will be designated and remain outside the SZ to provide salvage assistance if required. Using this craft for salvage assistance of another craft in the SZ is the beachmaster's decision. The salvage boat will usually be an LCU; however, if MPF operations are being conducted simultaneously, an LCM 8 or SLWT may be used, if available.

If no salvage craft is available, and local craft are willing and capable, an alternative may be to contract for local tug or salvage craft services.

D.3 SALVAGE ASSETS

Organic assets available for salvage operations include the LARC V, an LCU, an AAVR7A1 (for other AAVs from surf to the HWM only), the D76/D79 bulldozer with a push pad and winch with wire rope, the MTVR, or the PWC. The AAVR7A1 is discussed in Paragraph 1.4.7.3, and the LARC V, D76/D79 bulldozer, MTVR, and PWC, in NTRP 3-02.1.2.

D.3.1 Salvage Teams

Salvage teams consist of the crew of the designated salvage asset, but may be augmented by personnel from amphibious ships in the transport group trained IAW qualification requirements established in the TYCOM's MTP and/or as required, appropriately trained and qualified LF personnel. One team member should be a qualified SAR swimmer.

D.3.2 Supplementary Salvage Equipment

Salvage equipment may vary depending on the magnitude of the salvage effort. Supplementary LCU equipment is covered in the craft's applicable allowance list and in the LCAC SEAOPS.

D.4 AFLOAT SALVAGE OPERATIONS

The manner in which a broached boat is freed or a disabled boat or amphibious vehicle is taken under tow varies according to the circumstances. LCAC salvage operations are covered in Section D.5 and the LCAC SEAOPS.

Before any salvage method is decided upon and executed, the BP commander takes into account all the variables to efficiently and safely execute the operation.

D.4.1 Salvage Procedures

This section describes boats in various situations and provides recommended salvage team procedures.

D.4.1.1 Broached Boat or Landing Craft

A broached boat or landing craft is one that is thrown broadside to the surf and grounded on the beach by some force acting on the stern.

During beaching and retracting operations, the BP commander coordinates with the LCU craftmaster to initiate positive measures to prevent broaching of the craft. Antibroaching procedures are detailed in Paragraph D.4.1.1.1. The decision to conduct antibroaching and possible follow-on salvage operations rests with the BP commander and depends on:

1. The possibility of blocking other boat lanes or damaging other craft
2. The possibility of damaging the craft on natural hazards
3. Environmental conditions such as tide, daylight, beach and surf hydrographics, and weather
4. Other beach operations being conducted
5. Salvage assets available.

The offload schedule generally dictates when salvage operations can commence on broached or stranded craft. The BP commander carefully assesses the overall operation and makes recommendations to higher authority to include beach conditions, shifting boat lanes, and any additional equipment requirements. During real-world operations, if salvage of a broached or stranded craft will prove disruptive to the mission and lead to significant

delays in building combat power ashore or landing essential troops and equipment for other amphibious missions, then it should be delayed, a new boat lane established, and the operation continued.

D.4.1.1.1 Antibroaching Procedures

Antibroaching procedures are:

1. LCU uses its stern anchor during the approach to the beach.

2. A bulldozer's winch wire is rigged to the LCU's bow. The beachmaster directs the bulldozer to the leeward side of the LCU's bow. The salvage team rigs the winch wire with a heaving line and passes it to the LCU. The winch wire is then connected to a 5-foot pendant on the LCU's leeward bow. The salvage PO then directs the bulldozer's operator to take up the slack and maintain a steady strain on the winch wire.

3. The bulldozer is positioned against the LCU ramp in conjunction with the use of the LCU's stern anchor. The beachmaster directs the bulldozer to the leeward side of the broached LCU's bow. The LCU raises its bow ramp to a 90° position, and the bulldozer is directed to place its pusher pad on the leeward corner of the ramp and hold. This allows the LCU to develop a pivot point, and as the LCU bow shifts, the beachmaster or salvage PO adjusts the bulldozer against the LCU's ramp. This procedure is repeated until the craft is perpendicular to the SZ.



If all above methods fail to prevent further broaching, the BP commander initiates seaward salvage procedures.

D.4.1.1.2 Stability

To minimize the difficulty of a salvage operation, particularly if the boat or landing craft is taking on water, the crew uses all means available to minimize the amount of water in the boat or craft.

D.4.1.1.3 Engine Use

If possible, the engine should be kept running with the drive in forward gear. Screw wash helps prevent incoming surf from carrying a towline into the screw or rudder. Also, as the engine accelerates in forward gear in combination with the receding wave, the screw discharge and wave current will wash sand away from the rudder and skeg. This serves to help prevent rudder damage and enables the boat to retract off the beach while keeping the stern from digging into the sand. It also helps minimize fishtailing on retraction, and helps maintain control of the boat as it ungrounds and tends to slingshot off the beach from the tension in the towline.

D.4.1.2 Passing a Towline to a Broached or Hard Aground Boat or Landing Craft

Paragraphs D.4.1.2.1 through D.4.1.2.4 delineate the procedures used to pass a towline in preparation for towing a broached or hard aground boat or landing craft off the beach.

D.4.1.2.1 Light Surf

In light surf, the salvage craft approaches close enough to pass the towline directly to the stranded craft. Depending on the salvage craft being used, if the beach is flat and the surf is breaking well to seaward, the salvage craft may beach and then pass the towline. If the surf is breaking close inshore or the beach is steep, the salvage craft may approach from windward or up current side, whichever is stronger, and throw a heaving line attached to the towline. As soon as the boat or landing craft has been freed and is at right angles to the surf, the salvage craft should maneuver to make the tow from a downwind position until clear of the SZ.

D.4.1.2.2 Heavy Surf

In heavy surf, the salvage craft remains outside the SZ and shoots a shotline or floats a messenger through the surf by attaching one end to a life ring or life jacket. The crew of the broached craft uses the messenger to haul in the towline.

D.4.1.2.3 Steep Beach Made Treacherous by a Heavy Backwash and Current

In this situation, the salvage craft beaches downwind and down current from the stranded boat or craft. The towline is carried by hand and secured to the stranded craft. This method decreases the danger of heavy backwash or current carrying the towline into the screws of either craft and is used only when recommended by the beachmaster.

D.4.1.2.4 Special Considerations

Should one or more of the following conditions exist, the BP commander and beachmaster should consider using the salvage procedure in which the towline is passed from the salvage craft outside the SZ to the broached craft before any other procedure is considered:

1. An MSI of 4 or greater (taking into account all variables in the overall calculation)

2. Ebb tide or a large tidal range in excess of 5 feet

3. Sand bars or holes exist in the vicinity of the broached craft.

D.4.1.3 Towing Displacement Landing Craft

There are three methods of towing displacement landing craft: bow to stern, stern to stern, and alongside. While preparing for one of the methods described in Paragraphs D.4.1.3.1 through D.4.1.3.4, the BP commander considers anchoring the LCU as described in Paragraph D.4.1.5. This will prevent the craft from traversing the beach, fouling other boat lanes, or grounding the craft and causing injury to personnel or damage to the craft.

D.4.1.3.1 Bow-to-Stern Method

The bow-to-stern method is the preferred towing method, particularly in bad weather. Providing the MSI and weather conditions are acceptable, the beachmaster or the BP commander directs the salvaging LCU to proceed seaward of the stranded craft clear of the SZ and drop anchor no closer than 500 feet from the broached craft. The salvaging craft then passes a messenger over by line throwing gun, bolo, or heaving line. If the messenger cannot be passed in this manner, the salvage PO directs a LARC V or PWC to pass a messenger from the broached craft

to the seaward salvage vessel or vice versa. Once the towline is passed and connected, the salvaging craft increases or decreases rpm in increments of 200. Once the broached craft is perpendicular to the SZ, it is towed clear at a safe speed. The broached craft informs the BP commander when clear of the SZ and in a safe operating area.

D.4.1.3.2 Stern-to-Stern Method

The stern-to-stern method is used during heavier weather. Taking the MSI, tide, wind, and surf conditions into account, the BP commander directs the salvage craft to approach as close as possible without entering the SZ. When within range, the messenger is passed in the manner described in Paragraph D.4.1.3.1. Once the towline is connected, the salvage craft maneuvers carefully to avoid fouling the either craft's screws while taking a strain. Once the broached craft is perpendicular to the SZ, it is towed clear at a safe speed. The broached craft informs the BP commander when clear of the SZ and in a safe operating area.

D.4.1.3.3 Alongside Method

The alongside method, also called the maneuvering tow, is the least desired method. It is used only for short distances in calm weather, very light surf, or a protected harbor. It may also be used when placing the towed craft alongside a ship or dock.

The salvage craft approaches on the leeward side of the broached craft and beaches as close as possible to the broached craft without endangering either craft. The salvage craft makes up alongside using stranded mooring lines or towlines and, if necessary, may also use its stern anchor for stability and assistance in backing seaward. Once made up securely, the salvage craft backs off the beach with the broached craft alongside. Upon clearing the SZ, the salvage craft tows the broached craft to a safe operating area.

D.4.1.3.4 Quick Tow

The quick tow is an expedient method to temporarily connect a towline, and minimize approach and line-handling times. It is used to get an inoperable craft out of the way of incoming waves as quickly as possible. A nylon strap is used between the forward inboard cleat of the salvage boat to the forward inboard cleat of the tow. The disadvantage of this method is reduced maneuverability of the tow.

D.4.1.3.5 Lighter, Amphibious Resupply, Cargo Mark V Method

As discussed in Paragraphs D.4.1.3.1 and D.4.1.3.2, the LARC V may be used to transfer the towline during the LCU bow-to-stern or stern-to-stern method. The procedures followed in using the LARC V are:

1. The assist LARC V proceeds to the salvage or broached craft making a downwind and retrieving the towline messenger.

2. After the towline messenger is aboard the LARC V, the crew takes the messenger to the Sampson post and secures it using two figure eights and leaving 10 feet of slack on deck.

3. The LARC V proceeds up away from the beached craft and parallel to the beach until all towline and the messenger line has been paid out.

4. The LARC V proceeds to the salvage craft's quarter and passes the towline messenger.

D.4.1.3.6 Personal Watercraft Method

The procedures followed by a PWC in transferring the towline during the LCU bow-to-stern and stern-to-stern method are:

1. Making a downwind approach, the PWC proceeds to the salvage craft to retrieve the towline messenger. The PWC then parallels the beach until all towline and messenger line are paid out.

2. The PWC proceeds to the beach, remaining in the leeward side of the craft, and passes the messenger to the BPT members.



- If surf conditions do not permit the LARC or PWC to come alongside, the crew will pass the towline to the broached craft by heaving line or bring the heaving line ashore and walk it down to the craft to be salvaged. The crew will not cast off until slack has been taken out of the heaving line.

- The towline messenger may also be floated to the broached craft, LARC V, or PWC by utilizing the current and a floating device (i.e., extra life jacket).

D.4.1.4 Towing Guidance

The following rules should be observed by the salvage crew and the crew of the towed craft during a towing evolution:

1. Extreme care is taken to avoid fouling the screws of either boat with the towline.

2. A broached craft is not towed from one quarter. Such procedure is inefficient and dangerous.

3. Slack in a towline is taken up slowly and smoothly, and full throttle is never used by the salvage craft. When the towline is taut, a steady strain should be maintained. Patience is required when towing a beached craft off the beach, as it will generally come off only a few inches at a time as the wave action raises the stern.

4. Once free from the beach, the broached craft should be towed clear of the SZ at a speed compatible with surf conditions.

5. When the towline is attached and under a strain, the salvage craft might respond sluggishly to rudder orders, as the stern drags the tow with it when turning. To correct this condition, the strain on the towline might need to be eased enough to allow the craft to turn in the desired direction. Again, caution is taken to preclude fouling the screws when slacking the towline.

D.4.1.5 Anchor Method

If seaward salvage is not immediately available due to surf conditions, or it is determined that the salvage effort can be deferred to a more convenient time, the anchor method may be employed. This method is recommended to preclude further damage to craft and prevent craft from traversing the beach.

The beachmaster and the BP commander direct that the craft to be salvaged be dead manned using a bulldozer. The LCU ramp is raised, and the bulldozer is placed at a 45° angle to the stern chock of the broached craft. The winch wire and an attached 5-foot pendant are run to the bitts and taken under a slight strain. If a second bulldozer is available, it may be secured with the winch wire at a 45° angle to the stern chock of the beached LCU. Then, by taking a strain on the wire, the broached craft is prevented from moving forward, and the dozer may be used as a pivot point if the LCU is still afloat or may assist the salvage craft using the stern-to-stern method.

D.4.1.6 Emergency Ramp Raising

A salvage operation routinely conducted by the beachmaster or landing craft crews is raising bow ramps that are inoperable due to such casualties as a parted cable or failed winch. The equipment used for this operation is required on each LCU and LARC V's allowance list.

D.4.1.7 Fire or Flooding

A P-100 pump is rigged on the salvage craft for fighting fires, controlling flooding, or dewatering a boat or another craft. A craft on fire is approached from upwind as firefighting efforts can be managed much more efficiently from this position. When the fire has been extinguished and the salvage craft comes alongside to inspect and dewater the damaged craft, a minimum number of lines are used to secure the craft to each other, and those that are rigged are rigged to be quickly cast off and slipped free from the salvage craft.

D.4.1.8 Structural Integrity

A hull inspection is made prior to freeing a broached craft from the beach. This inspection may be conducted by the broached craft's crew, if required, and any breach in hull integrity is repaired before the craft is towed free of the beach.

D.5 LANDING CRAFT AIR CUSHION SALVAGE OPERATIONS

An LCAC disabled seaward of the SZ is towed by another LCAC to an LCAC support ship (usually, an LSD class ship) for repairs. Inside the SZ, the beachmaster assists disabled LCAC or coordinates with the PCO for appropriate maintenance and repair teams.

D.5.1 Seaward of the Surf Zone

The PCO coordinates disabled LCAC salvage operations seaward of the SZ. Typically, an LCAC unable to operate on cushion is still able to provide its own propulsion in the displacement mode to return to an LCAC support ship for repairs. Towing equipment is carried aboard each LCAC, as specified in the CAL, and towing procedures are delineated in LCAC SEAOPS.

D.5.2 Inland of the Surf Zone

The beachmaster assists LCAC that have lost mobility due to terrain features exceeding operational parameters. Normally, a bulldozer with a towline is used to assist LCAC in this type of situation. The beachmaster also coordinates maintenance assistance for LCAC repairs beyond the crew's capability with the PCO on the Bravo Net. The LCO is then tasked to assemble the required organizational-level repair expertise or supply support and arrange transportation. Salvage operations conducted inland of the HWM on the beach are coordinated with the shore party.

APPENDIX E

Responsibilities for Loading, Stowing, and Offloading Landing Force Equipment

E.1 SCOPE

This appendix covers, in general terms, Navy and LF responsibilities for loading, stowing, and offloading LF equipment. More detailed information is found in JP 3-02.2, NTRP 3-02.1.2, and MCRP 4-11C.

E.1.1 Personnel

Although civilian personnel might perform some cargo loading and stowing functions, this appendix focuses on the responsibilities of Navy and LF personnel. These responsibilities are delineated in Paragraphs E.1.1.1 and E.1.1.2.

E.1.1.1 Landing Force Responsibilities

During loading operations, the LF provides personnel to:

1. Spot equipment and cargo on the pier or in landing craft
2. Rig slings
3. Hook on equipment and cargo
4. Spot equipment and cargo in holds or other stowage areas IAW the loading plan
5. Secure equipment and cargo for sea to the satisfaction of the ship's CO.

The LF provides personnel to supplement shipboard security forces and resecure cargo, as required. Also, during offloads, the LF provides personnel to work in holds or other stowage areas until cargo offloading and cleanup are completed. The LF personnel are referred to as the ship's platoon.

E.1.1.1.1 Ship's Platoon

The ship's platoon is part of the LF advance party, and comprises the working detail to assist ship's company in loading and unloading the ship. The size of the platoon varies depending on the type of ship, the number of cargo holds to be loaded, the type of cargo to be handled, the size and configuration of the holds, and the loading schedule. Where civilian stevedores are employed, members of the ship's platoon should be stationed in the holds to oversee and check the stevedore's work, as ultimately, platoon personnel will be required to unload the same cargo in the AOA.

E.1.1.2 Navy Responsibilities

The Navy provides landing craft crews, winch operators, hatch captains, and safety officers during loading and offloading operations. Navy personnel, assisted by LF ship's platoon members, also ensure equipment and cargo remain properly secured for sea by conducting frequent security checks.

E.1.2 Materiel

The LF and the Navy are responsible for providing specific materiel to handle or secure equipment and cargo.

E.1.2.1 Landing Force Materiel

The LF provides:

1. Dunnage and shoring
2. Waterproofing of equipment and materiel, as required
3. Full organic allowance of equipment (e.g., prime mover and trailer)
4. Special or additional items of MHE not included in the SLCP
5. Banding materiels and equipment for reconstituting damaged pallets.

E.1.2.2 Navy Materiel

Each SLCP lists cargo-handling capabilities and MHE, including:

1. Size and number of cargo nets
2. Size and number of gripes
3. Other available tiedown equipment
4. Number and capacity of forklifts and pallet jacks
5. Number of tractors and trailers
6. Size and number of slings, sheet metal plate hooks, rollers, pinch bars, lifts, and other cargo-handling devices.

E.1.2.3 Military Sealift Command Ships

When MSC ships participate in amphibious operations, special arrangements may be required to obtain sufficient nets, gripes, and tiedown equipment. These provisions are discussed in NWP 3-02.21.

Discharging cargo from MSC-chartered ships in an AO is discussed in Paragraphs 1.4.3 and 1.4.4.

APPENDIX F

The Tactical-Logistical Group

F.1 FUNCTION

The TACLOG group is a temporary LF organization that is collocated with the Navy ship-to-shore movement control organization. The group consists of representatives designated by troop commanders to assist Navy control officers aboard amphibious ships in the ship-to-shore movement of troops, equipment, and supplies. It advises the Navy and LF commanders, and Navy control officers of changes to the LF portion of the landing plan and their effects on the tactical situation ashore. The TACLOG group also facilitates the movement of tactical units and logistical support requested by LF tactical commanders ashore, and provides the LF commander with the status of ship-to-shore movement and a mechanism to influence the operations ashore.

F.2 BACKGROUND

In the ship-to-shore movement, the LF may establish three temporary organizations to facilitate the scheme of maneuver ashore, provide tactical and logistical support to LF units ashore, and keep the LF commander apprised of the operation's progress. These organizations are the TACLOG group, the LFSP, and the HST. These Navy and LF organizations provide the mechanisms for troops and equipment to move ashore, and receive tactical and logistical support without having to focus on these areas.

This appendix describes the organization and functions of the TACLOG group, and its relationship with the Navy control organization and, through the LFSP and the HST, LF units ashore. The LFSP is discussed in Appendix G and NTRP 3-02.1.2, and the HST is covered in Appendix H.

F.2.1 Navy and Landing Force Organizations

The Navy control organization provides positive centralized control of ship-to-shore movement. Close coordination among the waterborne and helicopterborne movements, and supporting, prelanding, and in-stride operations with the flexibility to change the landing plan is required. This coordination ensures maximum tactical effectiveness during the landing and subsequent buildup of infrastructure or combat power ashore. The LFSP lands with surfaceborne units to facilitate the flow of personnel, equipment, and supplies across the beach and beyond, and to establish a BSA to provide CSS to these units. Requests for on-call waves, prepositioned emergency supplies, nonscheduled units, and adjustments to the landing plan are made by tactical commanders through the LFSP to the TACLOG detachment for the required liaison with PCOs to provide the tactical units, CSS, or adjust the landing plan.

To facilitate required liaison between LF units ashore and the Navy control organization, the TACLOG detachment at each LF echelon is embarked in the same ship with the Navy control organization exercising OPCON over the ship-to-shore movement of that LF echelon. Figure F-1 depicts the TACLOG group organization for a MEF-sized MAGTF. Smaller MAGTFs do not require such an extensive TACLOG group structure.

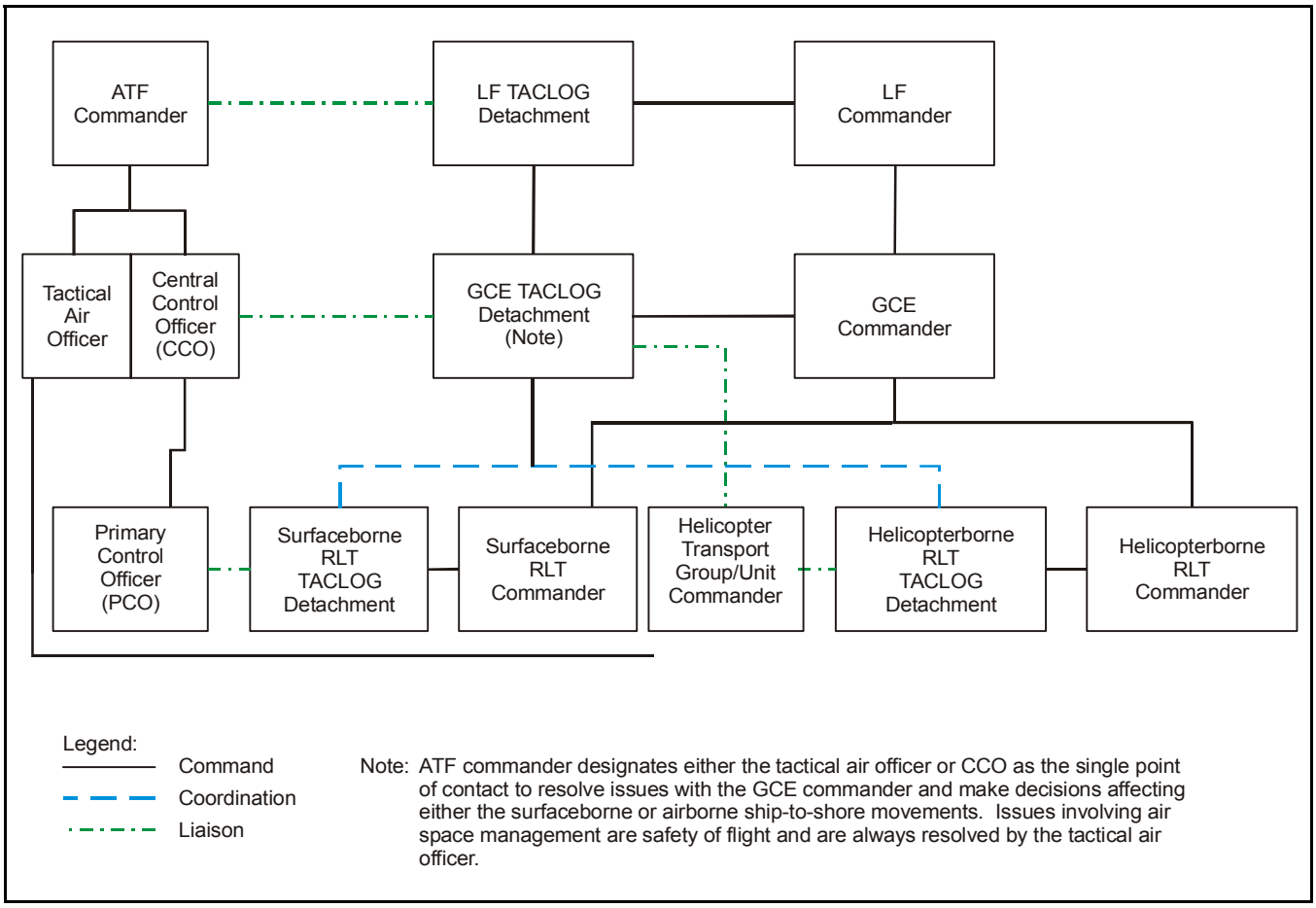


Figure F-1. Tactical-Logistical Group Organization for a Marine Expeditionary Force-Sized Marine Air-Ground Task Force

F.3 TACTICAL-LOGISTICAL DETACHMENT COMMON FUNCTIONS

Figure F-1 depicts the TACLOG group organization in a MEF-sized MAGTF. Common functions pertain to each TACLOG detachment depicted in this figure. These functions are:

1. Coordinate pre-D-day and pre-H-hour transfers.

2. Monitor the landing of scheduled waves and provide recommendations to the Navy control organization on landing on-call and nonscheduled serials.

3. Process tactical and logistical requests from LF tactical commanders.

4. Advise the Navy control organization of the location of serials and preferred transportation (landing craft or helicopter).

5. Advise the Navy control organization of the priority for landing serials if beach or LZ congestion, or shortages of transportation assets affect the tactical or CSS situation ashore.

6. Advise the Navy control organization of adjustments required to the LF landing plan and their effects on the tactical situation ashore.

7. Maintain a DTG record of scheduled, on-call, and nonscheduled serials showing when requested, dispatched, and received.

8. Coordinate the planned CSS buildup in CSSAs with the LFSP.

9. Represent the commander in decisions that may affect tactical operations ashore (within limits of the authority delegated).

10. Coordinate transport group movement with the Navy control organization.

11. Coordinate withdrawal and backload operations with the Navy control organization IAW priorities and preferred transportation modes established by the supported commander.

F.4 ORGANIZATION

The LF TACLOG detachment is the senior TACLOG group agency. It exercises a coordinating authority relationship with subordinate TACLOG detachments. The LF TACLOG detachment advises the LF commander on the progress of the ship-to-shore movement and changes being contemplated by subordinate commanders.

JP 1-02 defines a coordinating authority as a commander or individual assigned responsibility for coordinating specific functions or activities involving forces of two or more military departments or two or more forces of the same service. This individual has the authority to require consultation between the agencies involved, but does not have the authority to compel agreement. In the event that essential agreement cannot be obtained, the matter is referred to the appointing authority (in this case, the LF commander). Coordinating authority is a consultation relationship, not an authority through which command may be exercised.

As per the definition in the preceding paragraph, the LF TACLOG detachment does not have OPCON of subordinate TACLOG detachments. Therefore, the LF commander, if necessary, may direct changes to the LF landing plan or influence the decisions of subordinate commanders via a command authority relationship with the LF.

Alternate TACLOG detachments are formed and embarked in the Navy commander's alternate flagship, alternate primary AATCCs, and SCSs designated to replace PCSs in emergency situations.

F.4.1 Transitioning From a Tactical to Combat Service Support Focus

In a combat situation or potential combat situation, the initial unloading period is tactical and provides for the rapid buildup of combat power ashore and quick response to LF tactical and logistical requirements. The general unloading period is primarily logistical and emphasizes the rapid unloading of personnel and equipment remaining in amphibious shipping. During the tactical period, each TACLOG detachment is directed by a representative from the respective commander's operations staff (G/S-3). At specified times during the initial unloading period, the TACLOG group's focus transitions from a tactical to a logistical orientation. Each

detachment is then directed by a representative from the respective commander's logistics staff section (G/S-4). The decision to shift from a tactical to a logistics focus is made by the LF commander based on the tactical situation ashore. Subordinate TACLOG detachments make the decision to transition when various levels of tactical and CSS units, and capabilities are positioned ashore.

F.4.2 Tactical-Logistical Group Organizational Considerations

The LF commander prescribes the TACLOG group organization in the LF OPORD and also specifies which subordinate commanders form TACLOG detachments. Subordinate commanders have OPCON of their detachments and determine the composition of their respective TACLOG detachments based on the mission assigned and the type of force to be employed in a combat situation. The following are pertinent considerations for establishing a TACLOG group organization:

1. Personnel attached to a TACLOG detachment are from the commander's G/S-3 and G/S-4 sections, embarkation organization, communications section, and administrative section. Representatives from each LF unit embarked in scheduled and on-call waves may also be assigned.

2. Personnel having a detailed knowledge of operations and embarkation orders, the ships' role in the landing plan, tactical CSS plans and capabilities, and each document comprising the landing plan affect the establishment of a TACLOG group organization. These documents are discussed or depicted in Chapter 3.

3. Dedicated communications equipment is available on each control ship.

F.4.3 Landing Force Tactical-Logistical Detachment

The LCE commander forms and directs the LF TACLOG detachment. This commander is intimately involved with the operational and logistics aspects of the operation, and monitors the progress of establishing the LFSP. This detachment is collocated with the Navy and LF commanders, and is comprised of key personnel from the G-3 and G-4 staff sections.

F.4.3.1 Landing Force Tactical-Logistical Detachment Functions

The following functions are specific to the LF TACLOG detachment and are in addition to the common TACLOG group functions listed in Section G.3:

1. Advise the LF commander on the TACLOG group structure to support the landing plan.

2. Advise the Navy and LF commanders on changes to the LF landing plan or scheme of maneuver ashore, emphasizing their effects on the tactical situation.

3. Coordinate issues with subordinate TACLOG detachments that require resolution by the Navy or LF commander.

F.4.3.2 Landing Force Tactical-Logistical Detachment Focus Transition

During the initial unloading period, a representative from the LF G-3 staff assists the LCE commander in directing LF TACLOG detachment operations. When all scheduled and on-call waves have landed, this responsibility transfers from the LF G-3 representative to a representative from the LCE G-3 staff. The decision

on when to make this transition is made by the LF commander based on the tactical situation ashore. The LCE commander, when satisfied that sufficient CSS capability is ashore, recommends to the LF commander that the responsibility for CSS operations ashore shifts from the LFSP to the MLG, and the general unloading period commences. The LF commander directs that the LF TACLOG detachment transition from the LCE G-3 representative to a representative from the LF G-4, and the LCE commander and staff move ashore. The LF TACLOG detachment continues operations until a specified level of LF supplies is positioned ashore.

F.4.4 Ground Combat Element Tactical-Logistical Detachment

The GCE commander forms the GCE TACLOG detachment. Key personnel are provided from the GCE commander's G-3 and G-4 staff sections. It is collocated with the Navy control organization responsible for planning and conducting the waterborne and helicopterborne ship-to-shore movements. Separate Navy control organizations are established for both means of movement ashore, and the Navy commander designates either a CCO or TAO as the single POC for the GCE commander to resolve issues and make decisions affecting either waterborne or helicopterborne ship-to-shore movement.

F.4.4.1 Ground Combat Element Tactical-Logistical Detachment Functions

The following functions are specific to the GCE TACLOG detachment and are in addition to the common TACLOG group functions listed in Section G.3:

1. Monitor waterborne and helicopterborne ships-to-shore movements to ensure a balanced buildup of combat power or infrastructure ashore.

2. Coordinate changes to the LF landing plan with the surfaceborne and helicopterborne RLT TACLOG detachments to facilitate the balanced buildup of combat power or infrastructure ashore.

3. Periodically advise the LF TACLOG detachment on the progress of the operation.

F.4.4.2 Ground Combat Element Tactical-Logistical Detachment Focus Transition

GCE TACLOG detachment operations during the landing of scheduled and on-call waves are directed by a representative from the GCE G-3 staff. When scheduled, and on-call waves have landed, this responsibility transitions to a representative from the G-4 staff section. The GCE TACLOG detachment terminates operations when the GCE commander's HQ move ashore.

F.4.5 Surfaceborne Regimental Landing Team Tactical-Logistical Detachment

Each RLT commander conducting an assault or operation over a landing beach establishes a surfaceborne RLT TACLOG detachment. Key personnel are provided from the RLT commander's S-3 and S-4 staff sections, and are collocated with the PCO. Subordinate BLTs provide representatives to the surfaceborne RLT TACLOG detachment to resolve issues that affect their unit. Surfaceborne TACLOG detachments are not established below the RLT echelon, unless BLTs are conducting operations in landing areas widely separated from the RLT commander that precludes effective C2.

F.4.5.1 Surfaceborne Regimental Landing Team Tactical-Logistical Detachment Functions

The following functions are specific to the surfaceborne RLT TACLOG detachment and are supplemental to the common TACLOG group functions listed in Section G.3:

1. Monitor the landing of scheduled waves and provide recommendations to the PCO on landing on-call waves and nonscheduled units.

2. Advise the GCE TACLOG detachment and PCO of adjustments to the assault schedule and LF landing sequence table, and their effects on the tactical situation ashore.

3. Process tactical and logistical requests from surfaceborne units. These requests are received from the LFSP. The sequence of events and communications nets employed are described in Paragraphs 5.6.6 and 5.6.7.

4. Coordinate the maintenance of basic load supply levels in the LZ with the HST and a planned CSS buildup in an LZSA with the LFSP.

F.4.5.2 Surfaceborne Regimental Landing Team Tactical-Logistical Detachment Focus Transition

The commander's responsibility to direct the surfaceborne RLT TACLOG detachment and the procedure for transitioning from a representative of the S-3 section to the S-4 representative is the same as described for the GCE TACLOG detachment. The surfaceborne RLT TACLOG detachment terminates operations when all scheduled and on-call serials, and nonscheduled units assigned to the RLT have landed.

F.4.6 Helicopterborne Regimental Landing Team Tactical-Logistical Detachment

A helicopterborne TACLOG detachment is established when helicopterborne forces consist of a battalion-sized force or larger. For a MEF-sized MAGTF, the helicopterborne force is normally an RLT, and the helicopterborne RLT TACLOG detachment is collocated with the helicopter transport group/unit commander. If helicopterborne forces are landed in multiple LZs, a helicopterborne TACLOG is established with each primary AATCC to advise the helicopterborne unit commander and helicopter transport unit commander of the status of helicopterborne operations. Close coordination among the helicopterborne RLT TACLOG detachment, HCS detachment, primary AATCC, and HLSC is required during the landing of on-call waves and nonscheduled units because of the increased probability of change to the planned employment of helicopters to best take advantage of their speed and mobility. Chapter 5 provides more information on the HCS detachment, helicopter transport unit commander, primary AATCC, and HLSC, respectively.

F.4.6.1 Helicopterborne Regimental Landing Team Tactical-Logistical Detachment Functions

The following functions are germane to the helicopterborne RLT TACLOG detachment and are in addition to the common TACLOG group functions listed in Section G.3:

1. Monitor the landing of scheduled waves and provide recommendations to the HCS detachment and HLSC for landing on-call waves and nonscheduled units.

2. Advise the HCS detachment, HLSC, and GCE TACLOG detachment of adjustments to the helicopter HEALT and their effects on the tactical situation ashore.

3. Process tactical and logistical requests from helicopterborne units. Requests are normally received from the HST. The sequence of events and communications nets used are described in Chapter 5.

4. Coordinate the maintenance of basic load supply levels in the LZ with the HST and a planned CSS buildup in a landing zone support area (LZSA) with the LFSP.

F.4.6.2 Helicopterborne Regimental Landing Team Tactical-Logistical Detachment Focus Transition

The commander's responsibility to direct the helicopterborne RLT TACLOG detachment and the procedure for transitioning from an S-3 staff representative to the S-4 representative is the same as described for the GCE TACLOG detachment. The helicopterborne RLT TACLOG detachment terminates operations when all scheduled and on-call serials, and nonscheduled units assigned to the RLT have landed.

F.5 COMMUNICATIONS

The following paragraphs describe the communications connectivity between LF units and the TACLOG group. Additional information on communications nets used during ship-to-shore movement is found in Sections 4.5 and 5.8.

F.5.1 Landing Force Tactical Net

The LF TACNET facilitates C2 and coordination of the action among the LF commander and the GCE, ACE, and LCE commanders. The LF commander influences the decisions of subordinate commanders on this net.

F.5.2 Supported Unit Tactical Net

The supported unit TACNET is monitored by the shore party, HST, and the appropriate TACLOG detachment to anticipate logistics requests.

F.5.3 Helicopterborne Unit Command Net

The Helicopterborne Unit Command Net is an internal net used by the helicopterborne unit to request logistics support from its HST.

F.5.4 Helicopter Support Team Control Net

The HST Control Net is used to request on-call air serials, resupply missions, and to facilitate the exchange of information among the HST, helicopterborne RLT TACLOG detachment, and the LFSP. Multiple HST Control nets may be required depending on the scope of helicopter operations.

F.5.5 Landing Force Support Party Command Net

The LFSP Command Net is used by the LFSP to direct CSS operations with the surfaceborne and helicopterborne RLT TACLOG detachments, shore party, and HST.

F.5.6 Landing Force Support Party Control Net

The LFSP Control Net is used by surfaceborne units to request support from the LFSP and by the LFSP to coordinate logistics requests from surfaceborne units with the surfaceborne RLT TACLOG detachment.

F.6 REPORTS

Periodic reports from TACLOG detachments are required to keep the LF commander, LF tactical commanders, and Navy control officers informed of the status of the ship-to-shore movement. The periodicity and format for these reports are contained in LF OPORDS.

APPENDIX G

Landing Force Support Party

G.1 GENERAL

The LFSP is the forward echelon of the LCE formed to facilitate ship-to-shore movement and the LF landing plan. It is task organized and generally contains a Marine Corps surface assault support element (shore party), a MAGTF HSE, and an NBG element. The LF commander establishes the LFSP through a formal activation order.

In large-scale operations, the LCE, ACE, and various Navy assets normally provide CSS functions. The concept of LFSP operations centralizes the responsibility for CS under a single organization. As the LF establishes itself ashore, the LFSP organization matures to provide an expanding range of support and services to the LF until relieved by the LCE.

G.2 PURPOSE

The primary mission of the LFSP is to provide initial CS and CSS for the LF during ship-to-shore movement. It ensures an organized and uniform flow of personnel, equipment, and supplies over and into beaches and LZs in support of the LF CONOPS ashore. A detailed discussion of LFSP missions and functions is found in NTRP 3-02.1.2.

This appendix describes the LFSP organization, functions, and operational and command relationships; post-LFSP CS and CSS operations; and the planning process for establishing the LFSP.

G.3 LANDING FORCE ELEMENT FUNCTIONS

Functions carried out by LF elements of the LFSP are C2, BSA development, beach throughput, support for airborne units, and other functions as required. These functions are described in the following paragraphs.

G.3.1 Command and Control

C2 functions are:

1. Establish and operate information centers and maintain current situation maps.
2. Control of traffic in the BSA.
3. Maintain communications with surfaceborne and airborne commanders, and their TACLOG detachments.
4. Establish lateral communications between beaches and LZs.
5. Maintain a record, by category, of units, equipment, and amount of supplies landed.

6. Establish, as part of the overall LF warning system, a system to warn of attacks within the BSA.

G.3.2 Beach Support Area Development

BSA development functions are:

1. On the recommendations of the Navy element, select locations for landing craft to beach.
2. Mark beach unloading sites.
3. Construct helicopter landing sites.
4. Establish multiclass supply dumps, including ABLTS'.
5. Mark and remove obstacles.
6. Construct and maintain beach lateral and exit roads.
7. Provide local security and coordinate the overall defense of the BSA and naval elements.

G.3.3 Beach Throughput

Beach throughput functions are:

1. Assist units in landing and moving across beaches.
2. Coordinate the movement of amphibious vehicles carrying supplies, unload supplies from landing craft and helicopters, and move supplies to LF units or dumps as required.
3. Operate ABLTS facilities.
4. Provide emergency maintenance and dewaterproofing facilities for equipment landed in surfaceborne operations.
5. Load helicopters with supplies for further delivery inland.

G.3.4 Support for Helicopterborne Units

Support for helicopterborne units are:

1. Provide training to tactical units and HSTs.
2. Provide landing support and HCEs to the helicopter unit's HST. (See Appendix H.)
3. Provide CSS to helicopterborne units when HST operations cease. (See Appendix H.)

G.3.5 Other Functions

Other functions performed are:

1. Process requests from LF units ashore for supplies or pass requests to the appropriate TACLOG detachment.
2. Establish and operate FARPs.
3. Establish and operate BESs.
4. Establish and operate EPW holding facilities.
5. Evacuate casualties and EPWs to designated locations.
6. Provide mortuary affairs services.
7. Initiate civil affairs and military government activities.
8. Provide personnel for ships' platoons, when directed. (See Paragraph E.1.1.1.1.)
9. Provide CSTs to the BP.
10. Provide support to countermine operations in the beach zone to include attached LF EOD personnel.
11. Construct CLZs.

G.4 NAVY ELEMENT FUNCTIONS

The naval component of the LFSP is under the OPCON of the LFSP commander. As the BP, it is comprised primarily of BMU personnel. A list of the Navy element functions and a figure depicting the manner in which it is incorporated in the LFSP are contained in NTRP 3-02.1.2.

G.5 LANDING FORCE SUPPORT PARTY ORGANIZATION

The LCE commander normally establishes the LFSP organization. The LF's organization and mission dictate the specific structure of the LFSP, which is task organized to accommodate the number of LZs and landing beaches to be used, the size of the unit landing, and the mission assigned.

The LFSP nucleus is comprised of personnel from the LCE, the ACE, and the PHIBCBs. Augmentees may also be assigned from other LF units, the NBG, SEALs, and EOD units. These personnel provide the requisite expertise that is not normally available from each organization in the initial stages of a large-scale amphibious operation. When the LFSP is dissolved, or if one is never established, the various functions performed by the LFSP transition to each component organization, and the individuals that comprised the LFSP perform those functions for their parent organizations. The specific task organization for the LFSP may vary depending on the specific mission; however, the basic LFSP structure is depicted in Figure G-1.

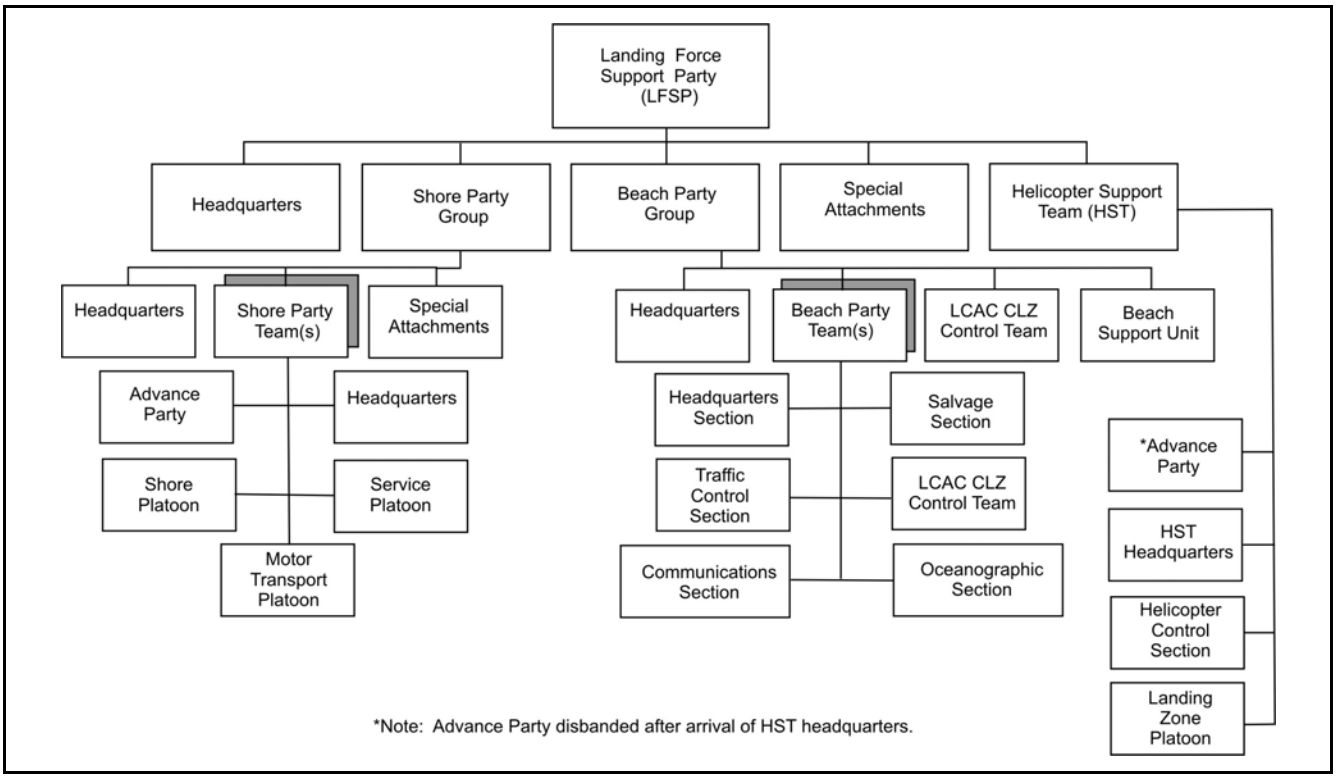


Figure G-1. Basic Landing Force Support Party Organization

The LFSP consists of:

1. LFSP HQ
2. Shore party group
3. BP group
4. Special attachments
5. HST.

G.5.1 Landing Force Support Party Headquarters

The LFSP commander and staff control landing support operations within the landing area. In smaller scale operations when an LFSP may not be established, the BP performs this function. The LFSP matures from a small cell in the early stages of ship-to-shore movement when beach and LZ operations are decentralized, to a robust organization as LFSP operations expand and become more centralized.

In the initial stages of ship-to-shore movement in a large-scale operation, the LFSP HQ is afloat. LFSP operations are decentralized to the shore party and the BP teams on colored landing beaches. A shore party team and BP

team are among the first waves of troops ashore to each colored landing beach. The shore party team commander has OPCON of the BPT until the BP group commander comes ashore. As the ship-to-shore movement progresses, the shore party group commander will be phased ashore. Once established ashore, the shore party group commander will take OPCON of the shore party team(s), BP group, and BPT(s) and coordinate with the BPG commander for employment of units in the BSA. The LFSP HQ will then be established ashore, and the LFSP commander assumes OPCON of the shore party group, BPG, special attachments, and all other LFSP units ashore. If the LFSP HQ is not established ashore concurrently with the BP and shore party HQs, OPCON for landing support operations in the landing area resides with the shore party group commander until the LFSP commander assumes OPCON.

The size and capability of the shore party and BP expand with the magnitude of the LFSP activity and responsibility. However, the increase in size is small relative to the growth of the shore party and BP HQs, because the shore party and BP HQs perform the majority of LFSP C2 tasks.

G.5.2 Shore Party Group

The shore party group is the LF component of the LFSP. The shore party group supports a beach over which an RLT, or equivalent Army force, lands. Upon landing, the shore party group commander assumes control of the landing support activities of subordinate shore party teams and BPTs, and begins consolidating activities at the group level. The shore party group consists of the following elements:

1. Shore party HQ
2. Shore party team(s)
3. Special attachments.

Shore party group functions include:

1. Allocating shore party personnel and equipment as required
2. Establishing shore party group communications and consolidating shore party team communications
3. Providing liaison personnel to supported unit's HQs
4. Coordinating defensive measures
5. Coordinating with the BPG commander
6. Submitting reports and records.

G.5.2.1 Shore Party Group Headquarters

The shore party group HQ consists of the following sections:

1. The command section is formed from the LSC, CLR (F), within the MLG.

2. The evacuation section is formed from the medical battalion. This section facilitates the evacuation of casualties for the LFSP, supervises operation of BESs and HESs, and prepares evacuation reports to higher HQ.

3. The MP section is formed from the division or MLG MP company. This section supervises shore party MP activities, establishes the LF EPW stockade, and directs EPW evacuation operations.

4. The communications section is formed from the communications company, CLR (F), within the MLG.

5. The motor transport and equipment section is sourced primarily from the MLG, with augmentation from the division combat engineer battalion.

6. The replacement pool consists of personnel waiting to be drafted into operation and are referred to as replacement units. Replacement units may be used to augment the ship's platoon, which is a working party of LF personnel assigned by the COT on each ship with LF troops and equipment embarked used for general loading and unloading. Replacement units assist the shore party in accomplishment of general tasks, and can be assigned to unloading during the general unloading phase, and as longshoremen at beach unloading points and dumps within the BSA.

7. The BPT is also listed as part of the shore party group HQ. As mentioned in Paragraph G.5.1, the LFSP is moved ashore in phases. Once the shore party group HQ is established ashore, it takes OPCON of the BPTs and shore party teams, and coordinates with the BPG commander.

G.5.2.2 Shore Party Team

Shore party teams support a specific beach and provides support to a BLT across a designated beach. The nucleus for shore party team personnel is the LSP, landing support company, with augmentation for specific missions. Figure G-2 depicts basic shore party team organization. The shore party team consists of:

1. The advance party, which is comprised of command, liaison, communications, and BP sections. The advance party is a temporary group formed from the various shore party and BPT elements. It conducts early beach RECON, establishes communications, and marks landing sites and dumps. The advance party reverts to current parent units in the shore party and BPTs when it comes ashore, except for liaison sections operating with the BLT. The liaison section remains with the BLT and becomes part of the shore party team HQ.

2. The shore party team HQ, which is composed of command, evacuation, MP, communications, and security sections. Command section personnel are provided from the HQ, LSP; evacuation section personnel are from the medical battalion, CLR (GS), of the MLG. The MP section normally consists of a detachment from the MLG; the communications section normally consists of a detachment from the communications company, CLR (F), and other communications units as required; security section personnel are augmented specifically for this duty.

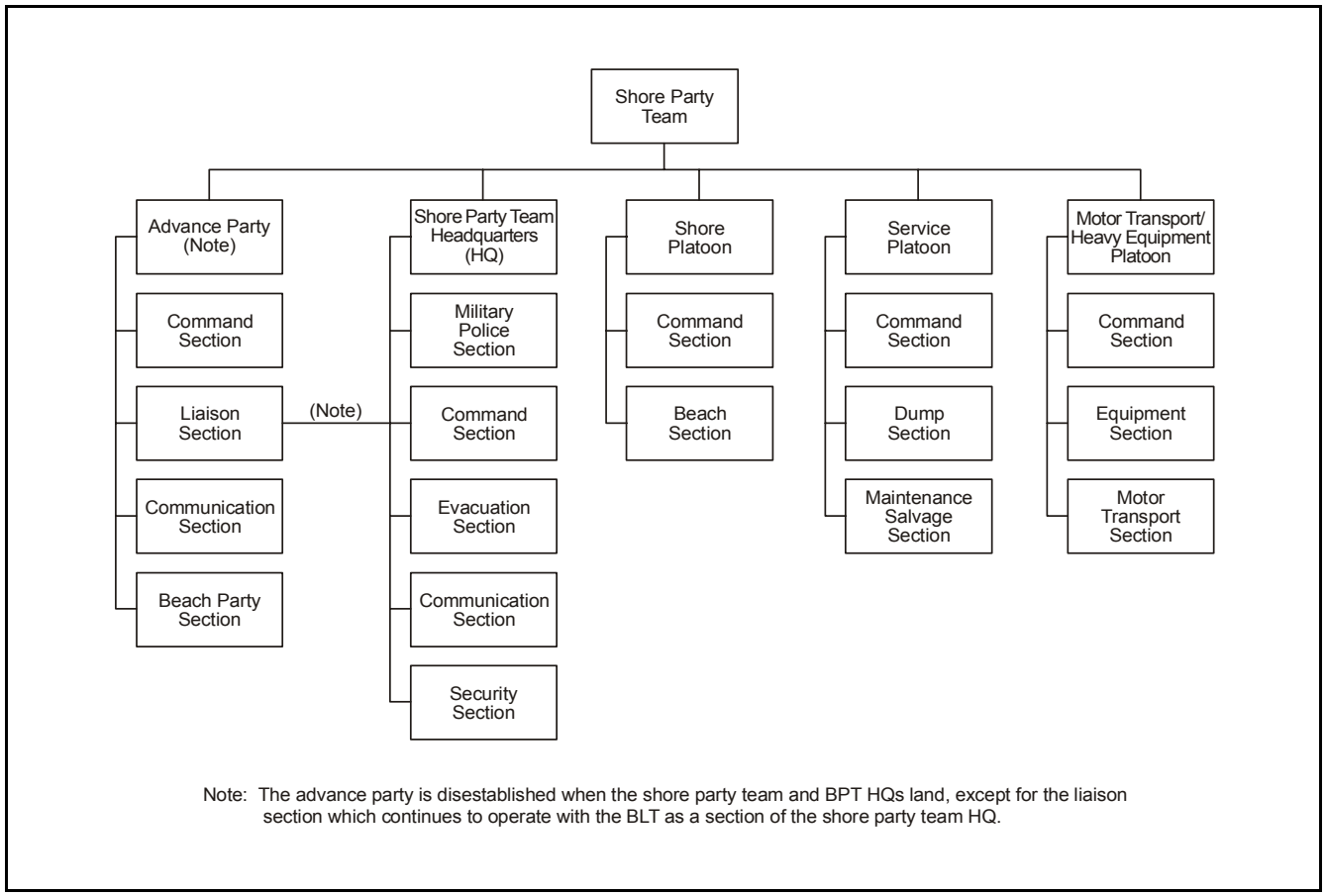


Figure G-2. Basic Shore Party Team Organization

3. The shore platoon, which consists of a detachment from the shore party platoon, augmented as necessary. The shore platoon establishes and operates facilities for unloading supplies and equipment from landing craft or aircraft, and moving that materiel to dumps, staging areas, or out of the BSA.

4. The service platoon, consisting of command, dump, and maintenance salvage sections. It is organized from the shore party platoon augmented by specialized personnel from other MLG units. The service platoon organizes and operates dumps, maintenance, and salvage areas in the BSA.

5. The motor transport and heavy equipment platoon, whose vehicles are provided by the MLG and the combat engineer battalion of the division. This platoon provides engineer equipment and motor transport support to the shore party team.

Shore party team functions include:

1. Marking limits of the beach unloading points, and designating and marking landing sites for helicopters

2. Assisting the BP in the landing and moving of units across the beach

3. Marking and removing obstacles in the BSA that are hazardous to personnel or may impede operations
4. Controlling traffic in the BSA
5. Maintaining communications with troop commanders of waterborne and airborne assault units, and the TACLOG for control purposes
6. Establishing lateral communications between beaches and HLZs
7. Loading/unloading supplies from landing craft, ships, and helicopters
8. Providing emergency maintenance for equipment in the waterborne assault
9. Maintaining continuous records, by category, of units, equipment, and amounts of supplies landed
10. Establishing FARP, as required.

G.5.2.2.1 Shore Party Team Headquarters

The shore party team HQ sections are ally established by shore party teams when established ashore. Attachments that do not have a service or task to perform as part of one of the shore party teams are normally retained under the control of the shore party group HQ. The shore party team HQ sections are:

1. The command section, whose nucleus is the landing support company HQ.
2. The MP section that consists of the CO of the attached MP unit, and administrative and communications personnel as required. This section coordinates MP element operations attached to shore party teams.
3. The evacuation section that maintains records of personnel evacuated and ensures CASEVAC procedures are functioning effectively.
4. The communications section that provides communications for the shore party team HQ and consolidates communications established by the shore party teams. When communications of shore party teams are no longer required, they return to the shore party group's communications section.
5. The security section that provides security and coordinates defense of the BSA.

G.5.2.3 Shore Party Special Attachments

In addition to ATF units that are provided to the LFSP as special attachments, units may be attached to the shore party to facilitate throughput operations within the BSA (e.g., EOD and combat engineers may be attached to conduct countermine operations). The FARP support team is another common special attachment to the shore party.

G.5.2.3.1 Forward Arming and Refueling Point Support Team

When required, FARPs are established ashore as early as the tactical situation permits to provide arming and fueling capabilities ashore. FARPs decrease aircraft response and turn around times, and reduce LF dependence

on Navy ships form arming and refueling services. Personnel, equipment, and supplies to establish and operate a FARP are in on-call waves or nonscheduled units (the latter category being the norm). The FARP support team is task organized like the helicopter control section of the HST with personnel and equipment provided from the same organizations. The FARP support team organization differs from the helicopter control section as follows:

1. The FARP support team has an aviation ordnance section to prepare and load ordnance on aircraft. This section is formed from rotary wing and AV-8 squadrons, and from the rotary wing MALS and fixed-wing MALS.

2. The helicopter maintenance and refueling section includes AV-8 squadron maintenance personnel.

When called ashore, a FARP support team, its supplies, and equipment move by vertical-lift aircraft or landing craft. Criteria required in establishing a FARP include:

1. Distance and stability of the FEBA

2. Aircraft time on station

3. Physical security requirements

4. Susceptibility of the FARP location to indirect fire

5. C2 requirements

6. Coordination of logistics efforts, adequacy of road networks, and level of aircraft support necessary to sustain FARP operations.

Anticipated duration of FARP operations at a specific location before movement to a new location is necessary to support the LF CONOPS ashore.

G.5.3 Beach Party Group

The BPG is the Navy complement to the shore party group within the LFSP and is under the OPCON of the LFSP commander. Figure G-1 depicts the organizational relationship among the BPG, shore party group, and LFSP. Paragraph G.5.1 describes the command relationship between the BPG and shore party until each respective HQ and the LFSP HQ are established ashore. The BPG consists of the following:

1. BP HQ

2. BPT(s)

3. BSU

4. LCAC CLZ control team.

Detailed information on and functions performed by the BPG, BPTs, and the BSU are found in Chapter 2 and Appendix A of NTRP 3-02.1.2. Figure G-3 depicts the BPG organization.

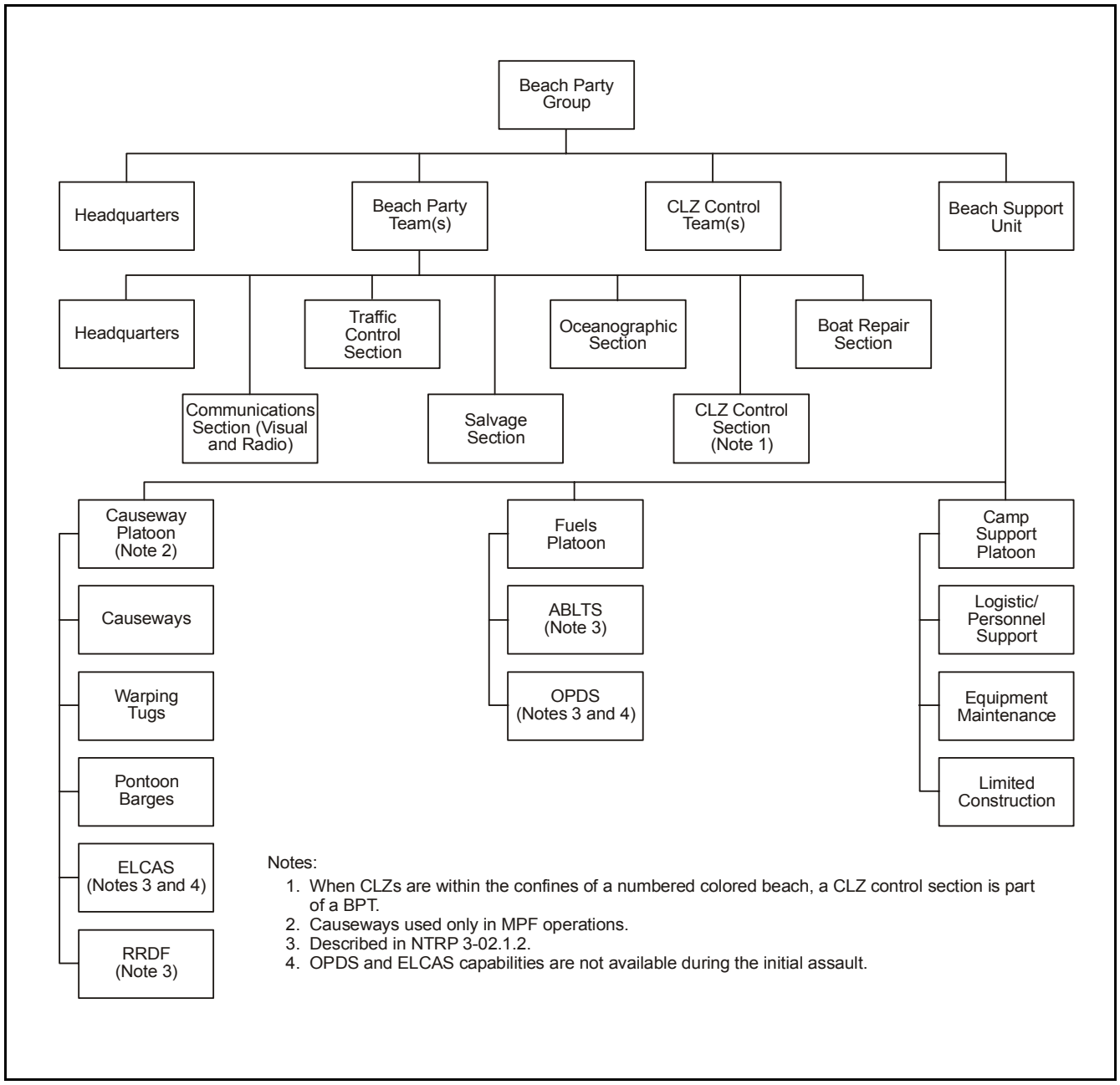


Figure G-3. Basic Beach Party Group Organization

The BPG primary functions include:

1. Installing navigational aids and removing obstacles
2. Marking beach landing sites
3. Advising shore party commanders of suitable sites for beaching amphibious craft
4. Directing landing craft and amphibious vehicles to and from beach landing sites
5. Observing and reporting surf conditions to the PCO
6. Conducting salvage operations and making emergency repairs to amphibious craft.

G.5.3.1 Beach Party Team(s)

The BPT is the Navy counterpart of the Marine Corps shore party team and is the basic unit of the BP. The BPT will remain under OPCON of the shore party team commander until the BPG commander moves ashore. The BPT supervises and coordinates Navy activities on a designated colored beach. The BPT commander is a Navy officer who lands with the shore party team. The BPT HQ is broken down into:

1. The BPT HQ section, which is formed by the BPT commander, communications and various administrative support personnel.
2. The communications section, which maintains connectivity with the PCO, other naval control units, and adjacent beaches. Radio and visual communications are the primary means of communicating among the BP and with approaching landing craft.
3. The traffic control section, which controls traffic from the SZ to the HWM. This section also directs the beaching and retraction of landing craft, and ensures the safe and expeditious landing of craft, informs the BPT commander of unsafe conditions, and supervises the embarkation and debarkation of troops and equipment.
4. The salvage section, which is comprised of personnel from the PHIBCBs and BMU detachments, and tasked with salvaging broached or stranded landing craft and vehicles in the SZ. Equipment used by the salvage section includes the D76/D79 bulldozer and the LARC V.
5. The oceanographic section contains Navy SPECWAR personnel who mark and remove obstacles in the beach approaches up to the HWM, conduct hydrographic surveys, prepare the BSIRs, and mark channels and navigational hazards.
6. The CLZ control section, which lands with the first wave after the initial LCAC assault. The CLZ control section controls LCAC from seaward of the CPP to a CLZ. When a CLZ falls within the confines of a designated colored beach, the CLZ control section is placed under OPCON of the BPT commander. If the CLZ is not contained within a colored beach, the CLZ control will be separate from the BPT and be under the OPCON of the BPG commander. See Paragraph G.5.3.3 for a discussion of the CLZ control team.

7. The boat repair section, which provides emergency repairs to landing craft on the beach. Any craft that require extensive repairs are towed to amphibious ships previously designated as boat havens.

G.5.3.2 Beach Support Unit

The BSU is formed from the PHIBCB detachment and consists of a causeway platoon, fuels platoon and camp support platoon. The BSU contains:

1. The causeway platoon, which launch, assemble and secure to the beach causeway sections to facilitate rapid movement ashore of wheeled and tracked equipment.

2. The fuels platoon, which establishes and operates AABFS/ABLTS for bulk fuel and water distribution.

3. The camp support platoon, provides limited beach construction capabilities to the BP.

G.5.3.3 Landing Craft Air Cushion Landing Zone Control Team

The CCT controls LCAC from seaward of the CPP to a CLZ for unloading, and then back to the SZ. The CCT organization is depicted in Figure G-4. It contains:

1. An advance party that lands as early in the operation as possible. Task organized from other CCT elements, the advance party surveys the CLZ, and ingress and egress routes to determine their suitability for LCAC operations.

2. The CCT HQ that is comprised of BMU personnel. It provides C2 for LCAC operations between the CPP and CLZ to include operations within the CLZ.

3. The terminal guidance section that is composed of BMU personnel who employ visual signals to direct LCAC through the CPP, in ingress routes, into the CLZ, and in egress routes back to the SZ.

4. The communications section, composed of BMU personnel who provide connectivity among the CCT, LCAC, BPG, and the LCO.

5. The CST comprised of BPG personnel providing support functions in the CLZ. These functions and the command relationship with the BP are covered in Paragraph G.5.3.3.1.

G.5.3.3.1 Cushion Landing Zone Support Team (Cushion Support Team)

The CST is under the OPCON or in direct support of the CLZ control team. The OPCON or direct support relationship is normally based on the proximity of the CLZ to other shore party and BP activities. The CST performs the same functions in the CLZ as the shore platoon within the shore party team performs ashore. The team unloads, personnel, equipment, and supplies from LCAC and facilitates their movement out of the CLZ. CST activities are normally confined to the CLZ itself, while supply dumps, staging, or assembly areas adjacent to or associated with a CLZ are normally run by another shore party element. When a section of the CCT is

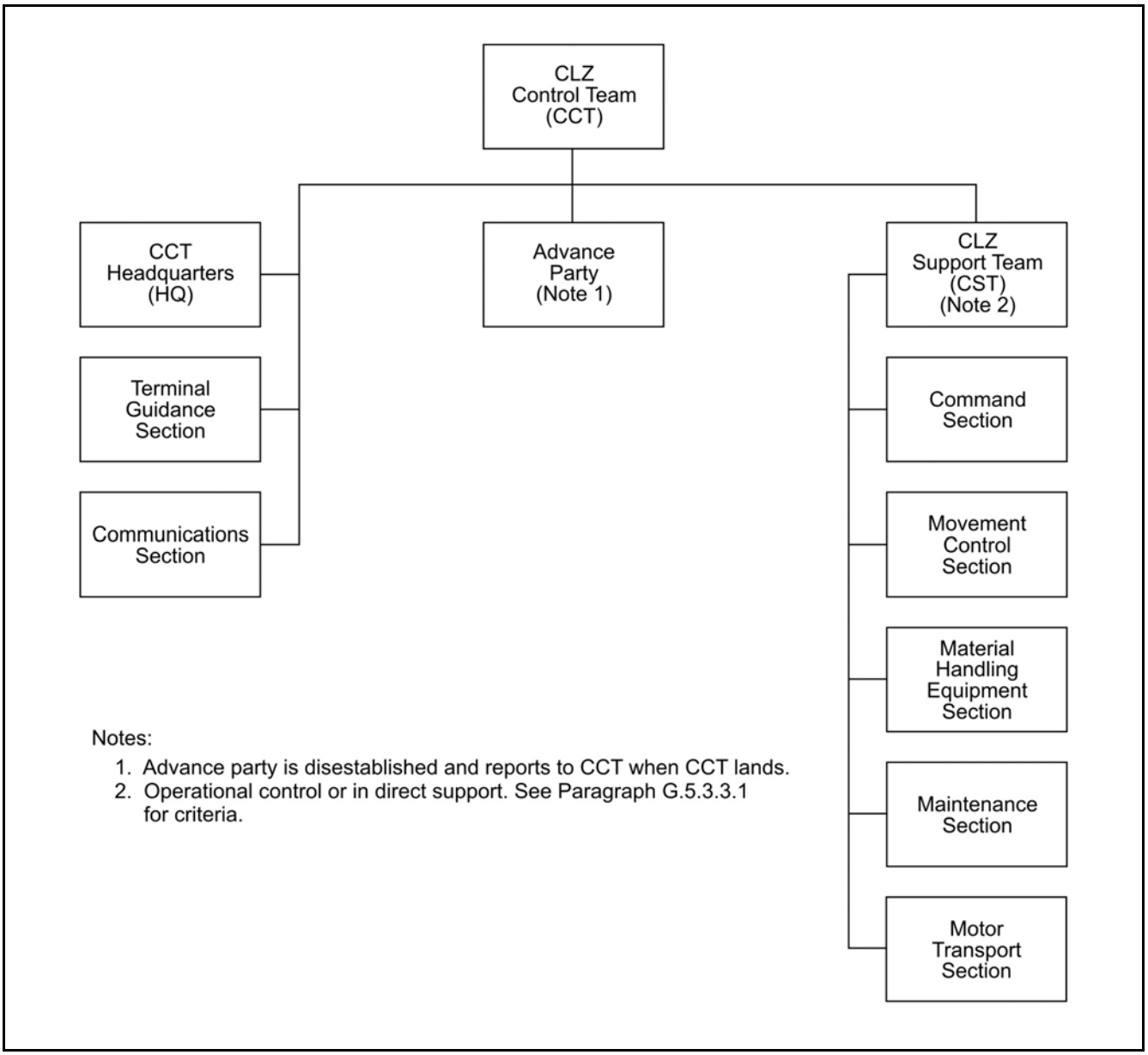


Figure G-4. Basic Landing Craft Air Cushion Control Team Organization

assigned to a BPT, a CST section is task organized to accompany the CCT section. The LFSP commander specifies the resultant command or support relationship. A CST is organized as follows:

1. The command section coordinates activity associated with unloading and moving personnel, equipment, and supplies from LCAC and the CLZ.

2. The movement control section directs the flow of personnel, equipment, and supplies.

3. The MHE section provides and operates equipment used to unload LCAC and clear the CLZ.

4. The maintenance section provides maintenance support for equipment, exclusive of electronics, and vehicles used by the CST and CCT, including inoperative vehicles and equipment in serials coming ashore, to clear them from LCAC and the CLZ.

5. The motor transport section provides the CST vehicles.

G.5.3.4 Beach Party Special Attachments

Special attachments to the BP perform tasks or provide capabilities not normally included in the BP organization. These attachments may provide support to the LF, but generally perform tasks that are more effective when performed from the beach. Examples of such tasks are clearing obstacles and mines in the SZ (HWM to 10-foot water depth) and the VSW zone (10-foot to 40-foot water depth) by SEALs and Navy EOD. As the amphibious operation progresses, additional units not directly associated with the LFSP, such as the ACU and NCHB, may phase ashore and be attached to the BP.

G.5.4 Landing Force Support Party Special Attachments

Units may be specially attached to the LFSP for defense of the BSA to provide liaison personnel and for other specialized tasks. For example, AD units may be attached to establish facilities to detect and counter air attacks in the BSA. During the early stages of the operation, the LF commander may deploy tactical units to counter threats to the BSA or LZSA, or the LFSP commander, in an emergency situation, may assume OPCON of any units operating in the BSA or LZSA to provide area defense. If Army, allied, or coalition units are included in the LF, representatives from these organizations are attached to the LFSP for coordination and liaison. The NCR may also be attached to the LFSP. The NCR is an LF- controlled unit used for construction tasks beyond the capabilities or focus of Marine Corps engineer units. The NCR, also known as the PHIBCBs, constructs or repairs aviation and CSS facilities for LF use, and develops or improves the transportation network within the BSA or LZSA.

G.5.5 Helicopter Support Team

The HST is described in detail in Appendix H of this publication. The HST is comprised of personnel from the LFSP and helicopter unit(s). The HST commander is designated by the LFSP commander and exercises overall control of the HST.

G.6 POST-LANDING FORCE SUPPORT PARTY OPERATIONS

When the LF commander directs that LFSP operations cease, the LFSP dissolves based upon the recommendation of the LCE commander. This recommendation is made when the LCE commander determines there are sufficient LCE units and C2 capabilities in place ashore. Once the LCE is established ashore, LFSP operations are terminated if no longer deemed necessary. It does not signal the end of ship-to-shore movement, the amphibious operation, or the existing command relationship between the ATF and LF organizations.

G.6.1 Aviation Combat Element and Naval Construction Regiment Organization

When LFSP operations cease, if the ACE and NCR HQs are not yet established ashore, those ACE and NCR units that were a part of the LFSP change OPCON to the LCE commander until their respective HQs are established and ready to assume OPCON of their units.

G.6.2 Beach Party and the Naval Beach Group

The BPG shifts OPCON to the LCE if the NBG HQ is not ashore when LFSP operations terminate. However, this change does not alter the BPG commander's latitude to carry out Navy functions not directly related to LF operations. The LCE commander retains OPCON of the NBG until sufficient Navy capability is established ashore or until the CNBG assumes OPCON of landing support operations in the beach or port area.

G.7 PLANNING

LFSP planning begins upon receipt of the order initiating the amphibious operation. LFSP planners plan concurrently with other LF and ATF organizations in the parallel chains of command within the AF. The amphibious planning process is discussed in detail in JP 3-02, NTRP 3-02.1.2, and Chapter 3 of this publication.

G.7.1 Landing Force Support Party Missions and Tasks

During the planning process, an analysis of LFSP missions and tasks is performed. They are grouped type, time, intensity, location, and duration. This process develops an LFSP concept for support, and based on the notional LFSP organization, an LFSP organization best suited for combat or the specific mission is created.

LFSP missions and tasks are also discussed in JP 3-02; NTRP 3-02.1.2; and MCWP 4-11.3, Transportation Operations.

G.7.2 Landing Force Support Party Concept for Support and Organization for Combat

When the LFSP concept for support and organization for combat or a specific mission is completed, LFSP planners consult with the various organizations that contribute units or personnel to the LFSP to refine that concept and organization. Concurrently, planning by other LF and ATF units proceed, and information such as the CONOPS ashore, fire support requirements, required supply levels, casualty estimates, EPW forecasts, and NCR information, enables LFSP planners to determine task organizations and control requirements. When the initial requirements are completed, they are forwarded to the LF commander for review and approval. The LF commander tasks various LF organizations to provide the necessary support to the LFSP and submits a request to the ATF commander for Navy support. As the LF plans mature, the ATF commander establishes the ship-to-shore movement control organization as a parallel chain of command to the levels of command employed by the LF to execute the ship-to-shore movement. When the Navy control organization is established, the LF establishes a TACLOG group (see Appendix F) to assist, advise, and coordinate with the Navy control organization.

G.7.3 Landing Force Support Party Operation Plan

The LFSP OPLAN is published and disseminated concurrently with LF subordinate commanders' OPLANs. Concurrent and parallel planning continues after the OPLAN is published and on through operation execution.

G.8 COMMUNICATIONS

The following paragraphs describe the connectivity required by the LFSP to carry out its mission. Chapters 4 and 5 include additional information on communications connectivity during ship-to-shore movement.

G.8.1 Landing Force Support Party Command Net

The LFSP Command Net is used to direct CSS operations with the TACLOG group, shore party, and HST.

G.8.2 Landing Force Support Party Control Net

The LFSP Control Net provides a means for surfaceborne units to request support from the LFSP and for the LFSP to coordinate logistics requests from surfaceborne units with the TACLOG group.

G.8.3 Helicopter Request Net

The HR Net is used to coordinate vertical lift support for LF units ashore. The LFSP guards this net for surfaceborne units.

G.8.4 Helicopter Support Team Control Net

The HST Control Net is used to exchange logistics information among the HST, TACLOG group, and the LFSP.

G.8.5 Supported Unit Tactical Net

The supported unit TACNET is monitored by the LFSP and appropriate TACLOG detachment to anticipate requests for logistics support.

APPENDIX H

The Helicopter Support Team

H.1 PURPOSE

The HST is introduced in Chapter 5; however, this appendix provides more indepth coverage of HST fundamentals, organization, and responsibilities as well as the operational aspects of providing CSS to the helicopterborne unit. The AF's tactical plans support the helicopterborne unit in a rapidly changing environment over potentially widespread areas. The CSS to support and sustain this unit is tailored to the special considerations inherent in helicopterborne operations.

H.2 FUNDAMENTALS

Airborne operations are dynamic and potentially dangerous if not planned carefully with attention to detail. Therefore, elements of the MAGTF and Navy control agencies coordinate planning and shall be familiar with the role that each plays during amphibious operations.

H.2.1 Helicopterborne Operations

The following factors are fundamental to helicopterborne operations:

1. The helicopter borne unit commander is responsible for all aspects of the operation in the HLZ. The helicopterborne unit receives support and augmentation from other LF organizations, but the helicopterborne unit commander retains OPCON.

2. An HST is formed for helicopter operations. The helicopterborne unit commander determines the HST composition based on the mission and whether a CSS buildup is planned.

3. HST members shall be appropriately trained and familiar with all aspects of landing operations.

4. The LFSP's role in supporting a helicopterborne operation depends on the nature of the mission and the LF task organization. The level of support provided to the helicopter unit by the LFSP may vary from HLZ operations training to providing personnel and equipment to form the HST HQ and HLZ platoon if a CSS buildup is planned.

5. The helicopterborne unit and its attachments are responsible for preparing, rigging, and hooking up the organic equipment and supplies for external helicopter lift.

6. Within the LF, the LFSP centrally controls and manages slings and cargo nets used for external helicopter lifts. Landing support units provide training in the use of this equipment to helicopterborne units.

7. During a helicopter operation, the helicopterborne unit commander has OPCON over all units. The LF OPORD will specify when or under what conditions OPCON of these units passes back to their parent organization.

8. Division or force RECON units normally provide ITG for helicopter waves. Once established in the HLZ, the HST assumes responsibility for helicopter terminal guidance from the RECON unit. The helicopterborne unit commander or higher HQ, as appropriate, assigns follow-on missions the RECON unit.

9. HST operations are normally terminated when the helicopterborne unit is no longer dependent on helicopters as the primary means of support or when a planned CSS buildup commences in the HLZ.

H.2.2 Planned Combat Service Support Buildup

Based on the mission, a helicopterborne unit moves ashore with the necessary personnel, equipment, and a basic load of consumable supplies. The LF commander prescribes the basic load carried by the helicopter unit.

The basic load for all classes of consumable supplies, except ammunition, is expressed in DOS. DOS is the amount of supplies a unit requires to sustain itself in combat for 1 day. A DOS for many supply items is specified in a unit's SOP.

The basic load for ammunition has two parts; a BA and DOA(s). The BA is the quantity of ammunition mandated in Marine Corps orders to be maintained for each weapon employed in combat. DOA is the total ammunition, calculated at standard consumption rates, for each organic and attached weapon employed in combat. In addition to the BA, a unit carries sufficient ammunition and consumable supplies to sustain itself in combat for 2 days without resupply.

The basic load is issued to, controlled, and carried to the objective area by the helicopterborne unit. The basic load is considered helicopterborne unit organic supplies. The helicopterborne unit commander specifies the portions of the basic load carried by subordinate units and what portions remain under central control at the HLZs run by the HST.

Movement of the helicopterborne unit's basic load to the objective area and resupply of the basic load to maintain the specified supply level is not considered a CSS buildup. A CSS buildup takes place when additional supplies above and beyond the basic load are moved to the objective area. When a CSS buildup in an HLZ commences, control of the HLZ transitions from the helicopter unit HST to the LFSP, and the HLZ is redesignated an LZSA.

The LF OPORD specifies when control of the HLZ passes from the HST to the LFSP. Passing control can be specified by time, event (i.e., when sufficient LFSP C2 capability is established in the HLZ), or on order of the LF commander.

With increased emphasis on seabasing, a concept that specifies that most CSS be provided by afloat amphibious and MPF ships, there will be a significant impact on the requirement for CSS buildups ashore.

H.3 HELICOPTER SUPPORT TEAM ORGANIZATION AND RESPONSIBILITIES

The helicopterborne unit is supported by organic, attached, and external CSS elements organized to move supplies, materiel, and ammunition forward by air. The primary CSS unit within the helicopterborne unit is the HST. The HST's mission is delineated in Paragraph 5.3.2.1.

H.3.1 Helicopter Support Team Organization

The HST is task organized and composed of personnel and equipment from the helicopterborne force, and the LFSP, augmented from other LF units as required. Normally, an HST is assigned to each HLZ and provides support to units operating in and around that zone. HST organization is based on the scope of the planned operation. The HST normally consists of an advance party, HQ, HCE, and HLZ platoon, as shown in Figure H-1. The following paragraphs explain the basic functions of each of these elements.

H.3.1.1 Advance Party

The advance party is normally 8 to 10 members with hand-carried equipment. It includes representatives from all HST elements that land early, select suitable areas for establishing CSS installations, establishes the advance party command post (CP), and marks the unloading point. These markers, identical to those depicted for the LFSP in Appendix B, indicate where various categories of supplies are landed and where casualties are evacuated. The advance party's OIC assumes OPCON over the division or force RECON unit that provided the ITG into the HLZ. The OIC retains OPCON of the RECON unit until the HCE of the HST assumes responsibility for helicopter air traffic control.

H.3.1.2 Helicopter Support Team Headquarters

The HST HQ element assumes control of the HLZ from the advance party as rapidly as possible. Its members may come from the service platoon of the helicopterborne unit when no CSS buildup is planned or required, or from the landing support platoon of the LFSP when a CSS buildup is planned. Providing LFSP landing support personnel to form the HST HQ when a CSS buildup is planned simplifies transfer of control of the HLZ to the LFSP when that buildup commences. Depending on focus of effort for the action ashore and the geographical suitability of the location, if feasible, the HST commander establishes the CP in an area readily accessible by surface and helicopter transportation means. The HST HQ consists of:

1. A CE provided by the appropriate HQ and augmented as required

2. A C4I section comprised of appropriate personnel from the helicopterborne unit or the LFSP, as appropriate

3. An MP section from the MP company, division HQ battalion, or the LFSP, as appropriate

4. A security section provided by the helicopterborne unit

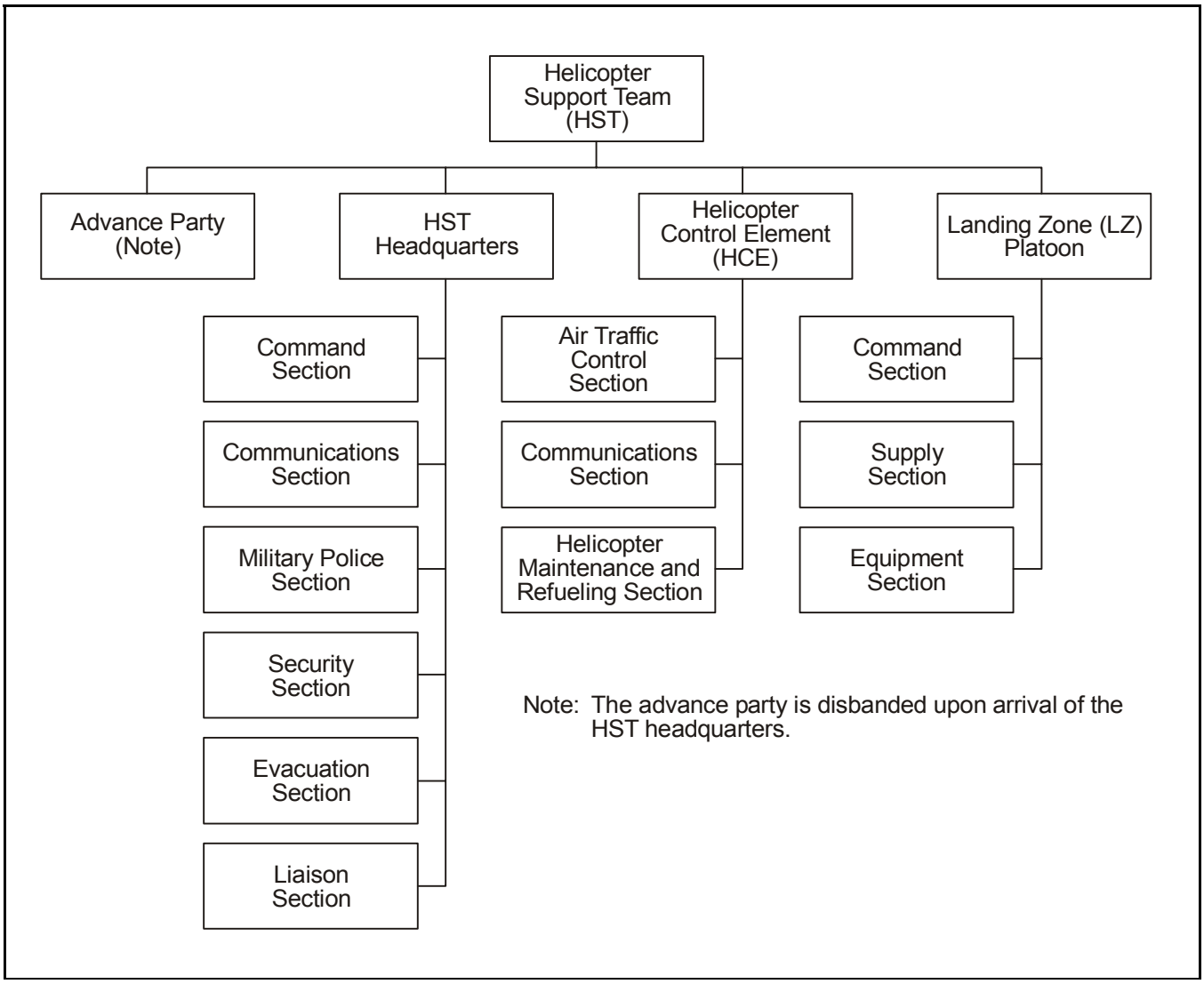


Figure H-1. Helicopter Support Team Organization

5. An evacuation section provided by the medical section of the helicopterborne unit

6. A liaison section provided by the helicopterborne unit.

H.3.1.3 Helicopter Control Element

The ACE commander, through the HCE, is responsible for air traffic control ashore of aircraft within the MAGTF. During the initial stages of an amphibious operation, the MACCS is not established, so the ACE provides HCEs to the LFSP, and the HCEs become part of the LFSP's HSE. The LFSP in turn task organizes the HSE, including HCEs, to meet mission requirements and provide task-organized HCEs to the helicopterborne unit for inclusion in the HST.

The HCE consists of:

1. An air traffic control section provided by the MACCS
2. A C4I section provided by the MACS of the MWCG
3. A helicopter maintenance and refueling section provided as follows:
   a. Maintenance personnel from the helicopter squadron making the lift
   b. Crash, fire, and rescue personnel and equipment from an MWSS (rotary wing) of the MWSG
   c. Fuel team personnel and equipment from an MWSS (rotary wing).

H.3.1.4 Helicopter Landing Zone Platoon

The HLZ platoon is provided by the unit providing the HST command section and is organized into command, supply, and equipment sections. If MHE is required in the HLZ or if helicopter slings and related equipment are required for external lifts, the equipment section is provided by the LFSP. This type of equipment is not organic to combat units. This is valid whether a CSS buildup is planned or not.

H.3.2 Helicopter Support Team Responsibilities

The HST performs tasks within an HLZ similar to those performed by the LFSP in the BSA. An HST:

1. Establishes communications with the HQ element of the helicopterborne unit, supporting helicopters, TACLOG groups, and the primary AATCC/ASC(A)
2. Prepares and marks landing sites, removes or marks obstacles, and sets up wind direction indicators
3. Selects areas for supply dumps and other CSS installations, the HST CP, HESs, and defensive positions to provide HLZ security
4. Controls helicopter operations within the HLZ and supports helicopterborne units landing in the HLZ
5. Unloads helicopters to include external lifts
6. Loads cargo nets, pallets, slings, and casualties onboard helicopters
7. Establishes dumps, issues supplies, and maintains necessary records of supplies received, issued, and available
8. Maintains the helicopterborne unit's basic load at the prescribed level
9. Passes requests for the replenishment if the basic load, supplies not contained in the HLZ dumps, or on-call serials to the helicopterborne RLT TACLOG detachment collocated with the primary AATCC and HLSC

10. Provides personnel and vehicle ground control
11. Maintains a situation map and information center
12. Performs emergency helicopter repair and refueling
13. Evacuates EPWs and casualties
14. Performs firefighting duties in the HLZ.

H.4 HELICOPTER SUPPORT TEAM OPERATIONS

The following paragraphs discuss HST embarkation, organization for landing, and operations ashore in a large-scale operation.

H.4.1 Embarkation

If the helicopterborne unit is embarked aboard several amphibious ships, the HST may be spread loaded; however, this is not mandatory.

H.4.2 Organization for Landing

The HST is formed into heliteams for ship-to-shore movement. These teams and their equipment are normally landed in scheduled waves. The position of the HST serials in scheduled waves of the helicopterborne unit is determined by the landing plan of that unit. Heliteams normally land in the following sequence:

1. Advance party
2. HCE
3. HST HQ (first echelon)
4. HLZ platoon (first echelon)
5. HST HQ (second echelon)
6. HLZ platoon (second echelon)
7. HST equipment.

H.4.3 Operations Ashore

HST operations ashore parallel those of the LFSP. Communications are established among the HST and the AATCC/ASC(A), helicopterborne RLT TACLOG detachment, helicopterborne unit CE, helicopters arriving subsequently in the HLZ, and when a CSS buildup is planned, the LFSP commander. The helicopterborne unit, through the HST commander, requests emergency supplies and troop serials required out of sequence.

H.4.3.1 Helicopter Landing Zone Organization

The HLZ is organized to effectively accomplish assigned helicopter support tasks. The development of the HLZ is generally the same as the development of the BSA, and the HCE's role is analogous to that of the BP. HST equipment is limited to that which is transportable by helicopter.

H.4.3.2 Control of Aircraft

The ACE commander provides the LFSP with sufficient trained personnel and equipment for HSTs to establish air traffic control facilities in each HLZ. The ITG teams from division or force RECON provide initial helicopter control required in the HLZ. When the HST is established, these control functions are assumed by the HCE. The HCE's air traffic control section directs traffic in, out, and around the LZ by coordinating with the ASC(A). The ASC(A) deconflicts helicopter traffic with fixed-wing air activities by coordinating with the TAC(A). During ship-to-shore movement, the ASC(A) is an extension of the TADC, and the TAC(A) is an extension of the ATF TACC.

H.4.3.3 Aircraft Maintenance and Refueling

Aircraft maintenance and refueling facilities in the HLZ are minimal. Personnel are available to make minor repairs and adjustments to aircraft, and provide emergency refueling. However, if major repairs are required, and the aircraft cannot return safely to amphibious shipping, additional maintenance personnel, parts, and equipment are flown from ships to the LZ.

The requirement for conducting aircraft repairs and refueling in the HLZ will be minimized as the seabasing concept matures.

H.4.3.4 Supply Movement

Troops, equipment, and supplies are kept clear of HLZ landing points to prevent interference with helicopter operations. Transportation and MHE within the HLZ is minimal; however, adequate equipment and personnel must be available to move incoming supplies rapidly to prevent the HLZ from becoming fouled.

H.4.4 Termination of Helicopter Support Team Operations

HST operations normally terminate when the helicopter unit is no longer dependent on helicopter support as the primary means of CSS or when a planned CSS buildup commences in the HLZ. When HST operations end, and if there is no longer a requirement for an HLZ, the HCE terminates operations and reverts to LFSP OPCON.

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APPENDIX I

Transformational Concepts, Equipment, and Capabilities

I.1 GENERAL

Today's Navy and Marine Corps have been placed clearly on a course of change, reform, and the need to adapt warfighting strategies, concepts, processes, and doctrine to meet the challenges of today's post-9/11 world and the evolving challenges of the future. The information contained in this appendix is just a portion of a larger concept for sizing, shaping, operating, and continuously transforming the Navy and Marine Corps to meet those evolving challenges. The items addressed herein relate directly or indirectly to the changes taking place within amphibious operations, particularly ship-to-shore movement, and their impact or potential impact, on the overall transformation within the Navy, the Marine Corps, and the joint community.

I.2 PURPOSE

The purpose of this appendix is to provide a brief description and overview of some of the transformational concepts, equipment, and capabilities that will impact amphibious operations, specifically ship-to-shore movement. Some are several years from being realized or becoming operational, but some are being tested and may be incorporated into Navy and Marine Corps doctrine within the next 10 to 15 years.

I.3 SEABASING

Seabasing, along with sea strike and sea shield, is one of the three fundamental concepts underlying Sea Power 21, the Navy's vision for how it will operate in the 21st century. Seabasing is also a principal enabling concept supporting Marine Corps expeditionary concepts, including STOM, which is discussed in Paragraph I.6. Therefore, it is one of the key operational concepts the Navy and Marine Corps will use to fight and win the littoral conflicts of the future. It is an evolving transformational concept in which certain LF support functions are shipboard-based as opposed to being established ashore. This concept will significantly reduce, and possibly eliminate, the requirements for the beach and inland areas discussed in Paragraphs 3.6 through 3.6.6. The overall intent of seabasing is to make use of the flexibility and protection provided by the sea base while minimizing the presence of the MAGTF ashore.

Today, seabasing enhances ship-to-shore movement by significantly reducing the need to move major C2 elements, heavy fire support systems, or logistic stockpiles to the beach. It will eventually allow for the creation of a sea base to facilitate maneuver to conduct operations ashore and from which all operations ashore can be sustained without the need for ground logistics bases. Enhancing this capability will free future joint forces from the requirement to obtain host nation air and seaports of debarkation. Forces ashore will be sustained and maintained afloat and replenished in stride from ships arriving from CONUS or from support bases located near the operation.

I.4 MARITIME PREPOSITIONING FORCE (FUTURE)

Today's MPF is discussed in Chapter 1, and in greater detail in NTTP 3-02.3/MCWP 3-32 and NTRP 3-02.1.2. The MPF (F) will be similar to today's MPF, but it will be loaded with all of the equipment to support a MEB, and will be able to transport and then offload that equipment anywhere in the world. In short, this concept will significantly increase the strategic agility and mobility of operating forces. The MPF (F) and seabasing concepts are linked to each other, and the success or shortfalls of one will significantly impact the other.

A key aspect of MPF (F) is that to accomplish the mission, MPF (F) vessels will not have be to concerned with obtaining access to secure ports and airfields. Today's MPF groups are ships of like capabilities; however, MPF (F) will consist of groups of ships of several types and capabilities. Specifically, additional functions performed by MPF (F) will be:

1. At-sea arrival and assembly of units

2. Direct support of the AE of a MAGTF

3. Indefinite sea-based sustainment of the LF

4. At-sea reconstitution and redeployment of the force.

The MPF (F) will have the ability to unload import or over a beach via JLOTS operations.

Per the logistics concepts inherent in seabasing, MPF (F) will support MAGTFs transported ashore as part of the STOM concept. Maintenance, repair, medical treatment, and supply operations will be conducted primarily from sea-based platforms that are replenished, as described in Paragraph I.3.

The MPF (F) will also provide a sea-based staging area and added maneuver space that will better allow the JFC to execute the operational aspects of STOM. Specifically, by using the MPF (F) as an at-sea staging and logistics base of operations and eliminating the need for establishing a large logistics and operational base of support ashore, the JFC and MAGTF commander can execute a key STOM objective. This will place combat forces in the adversary's rear area, where an attack will be least expected.

MPF (F) sea-based operations will take advantage of the increased maneuverability provided by the sea. MPF (F) ships will be able to conduct at-sea arrival, assembly of units, and selective offloading of equipment. The MPF (F) will also be important in enhancing joint operation responsiveness through the ability to assemble a MEB or joint force at sea that has arrived in theater via airlift or sealift from CONUS, or from forward support bases.

Today's amphibious operations are planned and executed in phases. However, the accelerated employment and deployment times associated with MPF (F) will permit elements of the MAGTF and MPF (F) to arrive in the AOA already integrated and prepared for operations. This has the potential to decrease the time to prepare for and carry out an operation to days as opposed to weeks or months.

The ability to conduct RSO&I at sea will also provide added FP. The sea can be used as an added layer of defense against small boat or aircraft attack, and the increased distance from shore will afford accompanying combatant ships a greater opportunity to detect and engage incoming threats.

For the LF, MPF (F) ships will be the platforms that sustain in-theater logistics, communications, and medical treatment capabilities. The MPF (F) located offshore will essentially become the LF commander's surfaceborne BSA. The MPF (F) will not only support the combat or action elements of the MAGTF, MEB, or JTF, but also the CS and LCEs assigned to it for direct support.

I.4.1 Maritime Prepositioning Force (Future) Challenges

While the MPF (F) offers the potential for significantly enhanced sustainability and support for MAGTF forces, it is at this point only a concept that still faces several challenges. Some of these are:

1. MPF (F) must be able to function well with both legacy and future transportation and logistics systems. Ultimately, reliable high-speed surface craft and aircraft must be designed and become operational to provide requisite logistics support.

2. Although the ability to remain at sea may add another layer of defense, because the MPF (F) will be so vital to the success and survivability of any amphibious operation and LF troops engaged in combat or other operations ashore, combatant ships operating in support of such operations will be increasingly called upon to provide FP.

3. Supply and CSS systems will have to keep pace with the enhanced capability of the warfighter to quickly arrive in the AOA, commence operations, and maneuver and advance rapidly once ashore. The process of resupply, from ordering to delivery, must become faster and take advantage of current and future technologies to remain cost effective and above all, reliable enough to ensure just-in-time and sustained logistics support continues without significant interruption.

I.5 HIGH-SPEED VESSEL

The HSV 2 (Swift) was delivered to the Navy in 2003 and currently supports GCC operational requirements and DON experimentation. GCC support includes TSCPs, HADR operations, and GWOT missions. DON experimentation support centers on the use of HSV 2 as a surrogate for future ship development. Foremost in its role as a surrogate ship is the support provided to the LCAC control ship and the JHSV programs. Although not generally envisioned as a direct participant during ship-to-shore operations in a hostile environment, the HSV has a proven ability to support the previously mentioned missions and such other operations as NEO, MCM, and responsive logistics support and sustainment.

The HSV is an aluminum-hulled, wave-piercing catamaran whose general characteristics are depicted in Figure I-1.

1.5.1 High-Speed Vessel Characteristics

Other than its speed, shallow draft, and ability to get into small and austere ports unassisted, the HSV characteristic that may prove most useful in the future is its flexibility to modularly adapt and reconfigure to support a number of missions. This craft can accept and integrate existing modular suites, containers, and mission-specific mobile transportable mission modules into its cargo spaces. Simultaneously, the HSV can embark the requisite staff required to accomplish the assigned missions.

Characteristics	
Length (ft)	321½
Beam (ft)	88.6
Draft (ft)	11.15
Payload Capacity	≈ 605 tons
Cargo Capacity	Cargo deck has approximately 28,740 ft^2, and the vessel has greater than 40,000 ft^2 of stowage space overall
Propulsion	Caterpillar 3,618 marine diesel engines
Maximum Speed (kt)	45+
Operating Speed (kt)	30+
Range (nm)	3,500+
Self-Defense Capability	Mk 96 with 25-mm/40-mm stabilized gun; Mk 45 Snake Eyes with Mk 19 grenade machinegun
Normal Crew Size	≈ 42
Maximum Crew Size	Berthing for 107 with additional temporary berthing for 87 when seating is converted

Figure I-1. HSV 2 (Swift)

The following items add to the HSV's exceptional versatility:

1. Helicopter deck
2. Multifunction C2 suite
3. Vehicle ramps and vehicle stowage areas for RO/RO containers and vehicles
4. Mission planning cells
5. Crew quarters
6. Embarked force living spaces.

The HSV's potential for integrating into the Navy's seabasing concept as a logistics, supply, and resupply vessel has been endorsed by the DOD with the standup of the JHSV program. Moreover, the Navy and Marine Corps have already experimented with the HSV's ability to transport troops and their equipment together at high speeds in one trip. In one exercise, the HSV Westpac Express transported a 400-ton load that included 370 Marines and their gear, 5 Cobra helicopters, 2 Huey helicopters, and aviation ground support equipment at far less time and cost than currently employed airlift. This is just one example of the capability of the HSV to embark large

payloads; transport them at high speeds; and deliver a credible and potent force, and then rapidly shift to a sustainment role.

I.6 SHIP-TO-OBJECTIVE MANEUVER

STOM is a future concept that will have a significant impact on current amphibious operations as well as this publication. At present, there is no definitive published doctrine for STOM; however, MCCDC has been charged with developing the STOM TTP and has initiated that process in the draft document, MCWP 3-31, Ship-to-Objective Maneuver (STOM). Because there is presently no definitive STOM doctrine, discussion of it appears only in this appendix, and its terminology and tenets are not incorporated in the body of this NTTP/MCWP.

Before describing the STOM concept, there are two terms that require definition to facilitate an understanding of the principles that comprise STOM: EMW and OMFTS.

I.6.1 Expeditionary Maneuver Warfare

EMW is the Marine Corps' capstone concept behind its transformation and preparation as a "total force in readiness" to meet the challenges of a rapidly changing world. It focuses on the Marine Corps' core competencies, evolving capabilities, and innovative concepts to ensure the Marine Corps provides the JFC with forces optimized for forward presence, engagement, crisis response, and warfighting.

Specifically, EMW focuses on:

1. Joint and multinational enabling. Marine forces are ready to serve as lead elements of a joint force, act as joint enablers and/or serve as JTF or functional component commanders: JFLCC, JFMCC, or JFACC.

2. Strategic agility, or rapidly and fluidly transitioning from precrisis state to full operational capability in a distant theater.

3. Operational reach, or projecting and sustaining relevant and effective power across the depth of the battlespace.

4. Tactical flexibility, or operating with tempo and speed and bringing multirole flexibility (air, land, and sea) to the joint team.

5. Support and sustainment, or providing focused logistics to enable power projection independent of host-nation support against distant objectives across a theater of operations.

I.6.2 Operational Maneuver From the Sea

OMFTS applies across the range of military operations, from MTW to smaller scale contingencies. It applies maneuver warfare to expeditionary power projection in naval operations as part of a joint or multinational campaign. OMFTS reflects the Marine Corps' EMW concept in the context of amphibious operations from a sea base, as it enables the force to:

1. Shatter the adversary's cohesion

2. Pose menacing dilemmas

3. Apply disruptive firepower
4. Establish superior tempo
5. Focus efforts to maximize effect
6. Exploit opportunity
7. Strike unexpectedly.

OMFTS is based on six principles. They are:

1. Focus on the operational objective
2. Use the sea as maneuver space
3. Generate overwhelming tempo and momentum
4. Pit friendly strength against adversary weakness
5. Emphasize intelligence, deception, and flexibility
6. Integrate all organic, joint, and combined assets.

I.6.3 Ship-to-Objective Maneuver Concept

STOM is the tactical implementation of OMFTS by the MAGTF to achieve the JFC's operational objectives and is the application maneuver warfare to amphibious operations at the tactical level. It involves the rapid employment of a MAGTF by air and surface means from amphibious shipping or from a sea base to objectives in the littoral and beyond. This affords vastly increased FP, operational mobility, and tactical flexibility, in addition to the opportunity to achieve speed and surprise not usually possible in today's expeditionary or amphibious operations.

STOM treats the sea as maneuver space using the sea as a both protective barrier and an unrestricted avenue of approach. While the goal of ship-to-shore movement is to secure a beachhead, STOM thrusts MARFOR ashore at multiple points to concentrate at the decisive place and time in sufficient strength to enable success. The objective is to create multiple dilemmas too numerous for the adversary commander to respond, thereby disrupting his cohesiveness and diminishing the adversary's will or capacity to resist. This concept focuses the force on the operational objective providing increased flexibility to strike at the opposition's critical vulnerabilities. Seabasing of some of the fire support and much of the logistics support reduces the footprint of forces ashore while maintaining the tempo of operations. Emerging C2 capabilities will allow commanders to control the maneuver of their units the moment they cross the LD at sea to include changing the axis of advance or points where they cross the beach during the operation.

It would appear that in many ways the STOM concept dovetails well with the tenets of the Navy's Sea Power 21 initiatives. However, before STOM can be put into operational practice, significant coordination and discussion remain between Navy and Marine Corps planners and doctrine developers.

APPENDIX J

Planning Diagrams and Tables

J.1 PURPOSE

Figures J-1 through J-27 are various diagrams and tables that were originally included in Chapter 3. Generally used in MEB- /MEF-sized ship-to-shore operations, they have been moved to this appendix so they can be preserved for use, as necessary, in future large-scale operations. In some cases, these diagrams, figures, and tables can be modified for use in smaller scale amphibious operations.

Ship	LCAC	LCU	Remarks
Total Landing Craft Onboard			
USS WHIDBEY ISLAND (LSD 41) USS WASP (LHD 1) USS TRENTON (LPD 14)	4	 2 1	
Total Less 10% for spares	4	3	
Total available for employment	4	3	
Landing Craft for Naval Use			
Safety boats/wave guides Boat group commander/assistant boat group commander Causeway tender boats Salvage boats Medical boats	 1	 1	1 LCU rigged for heavy salvage and 1 LCAC rigged for medical use (when available)
Total for naval use	1	1	
Total available for landing force use	3	2	

Figure J-1. Example of an Expeditionary Strike Group-Sized Landing Craft Availability Table

Part 1 – Beach Red One **A. Pre-H-Hour Transfers**

No. of Craft	Type	From Parent Ship	Report to	Time of Arrival	Period Attached	Remarks
1	LCU	LHD 1	LHD 1	H-90	One trip	Transfer serials 1081, 1082, and 1088 to LPD 4
1	LCU	LPD 14	LPD 14	LTLF	One trip	Transfer serials 1100 and 1102 to LSD 41

General Notes:
1. Pre-H-hour transfers should be avoided whenever possible.
2. Pre-H-hour transfers to well deck ships should be executed in the well deck.
3. LTLF as used here means "land the landing force."

Part 1 – Beach Red One **B. Scheduled and On-Call Waves**

No. of Craft	Type	From Parent Ship	Report to	Time of Arrival	Period Attached	Remarks
4	LCAC	LSD 41	LSD 41	H-30	Until released	Beach Red One CLZ
1	LCU	LPD 14	LPD 14	H-50	One trip	Wave 4; one boat officer embarked
2	LCU	LHD 1	LHD 1	H-60	One trip	Wave 3; one boat officer embarked
1	LCU	LHD 1	LHD 1	H-Hour	One trip	Serials 1090, 1091 on-call

Part 1 – Beach Red One **C. Naval Use**

No. of Craft	Type	From Parent Ship	Report to	Time of Arrival	Period Attached	Remarks
1	LCU	LPD 14	LPD 14	H-90	Until released	Salvage crew and full salvage equipment allowance embarked (if LCU is available)
1	LCU	LHD 1	LPD 14	LTLF	Until released	Floating dumps (if available)

Part II – Beach Red Two (same format as above)

Part III – Beach Blue Two (same format as above)

Figure J-2. Example of a Landing Craft Employment Plan

Landing Craft Order	Well Deck
1st	LCU 3-1 BT 3-1
2nd	LCU 3-2 BT 3-2
3rd	LCU 3-3 BT 3-3
4th	LCU 3-4 BT 3-4
5th	LCU 3-5 BT 3-5

Figure J-3. Example of a Debarkation Schedule

H-Hour 0530 Local
Beach Red One

Wave	Leave Rendezvous/ Launch Area	Cross Line of Departure	Land
1	H-23 min	H-20 min	H-Hour
2	H-20 min	H-17 min	H+3 min
3	H-17 min	H-14 min	H+6 min
4	H-14 min	H-11 min	H+9 min
5	H+2 min	H+5 min	H+16 min

H-Hour 0530 Local
Beach Red CLZ

Wave	Leave CDP	CCP 1	CCP 2	CPP	CLZ
1	H-24 min	H-12 min	H-6 min	H-3 min	H-Hour
2	H-14 min	H-2 min	H+4 min	H+7 min	H+10 min
3	H-4 min	H+8 min	H+14 min	H+17 min	H+20 min
4	H+6 min	H+18 min	H+24 min	H+27 min	H+30 min

Figure J-4. Example of an Approach Schedule (Sheet 1 of 2)

H-Hour 0530 Local
Beach Red Two

Courses: a. Rendezvous area to LD:	140° T, 144° magnetic
b. LD to beach:	174° T, 178° magnetic
Primary Control Officer:	CAPT D. Jones, USN, embarked in LPD 14
LCAC Group Commander:	LT O.P. Hatch, USN

General Notes:

1. Wave composition is shown in Figure J-5.
2. Distances used to compute times shown:
   a. AAVs are launched 300 yards seaward of LD.
   b. Rendezvous area to LD is 1,000 yards.
   c. LD to beach is 4,000 yards.
3. Speeds used to computer times shown:

	AAV	LCU	LCAC
a. Launch/rendezvous area to LD	4	10	
b. LD to 1,000 yards from beach	5	10	
c. From 1,000 yards to beach (battle speed)	6	12	
d. LCAC from CDP to CPP			40

4. Adjustment to supporting fire plan is required for LCAC to cross the beach prior to H-hour.

Figure J-4. Example of an Approach Schedule (Sheet 2 of 2)

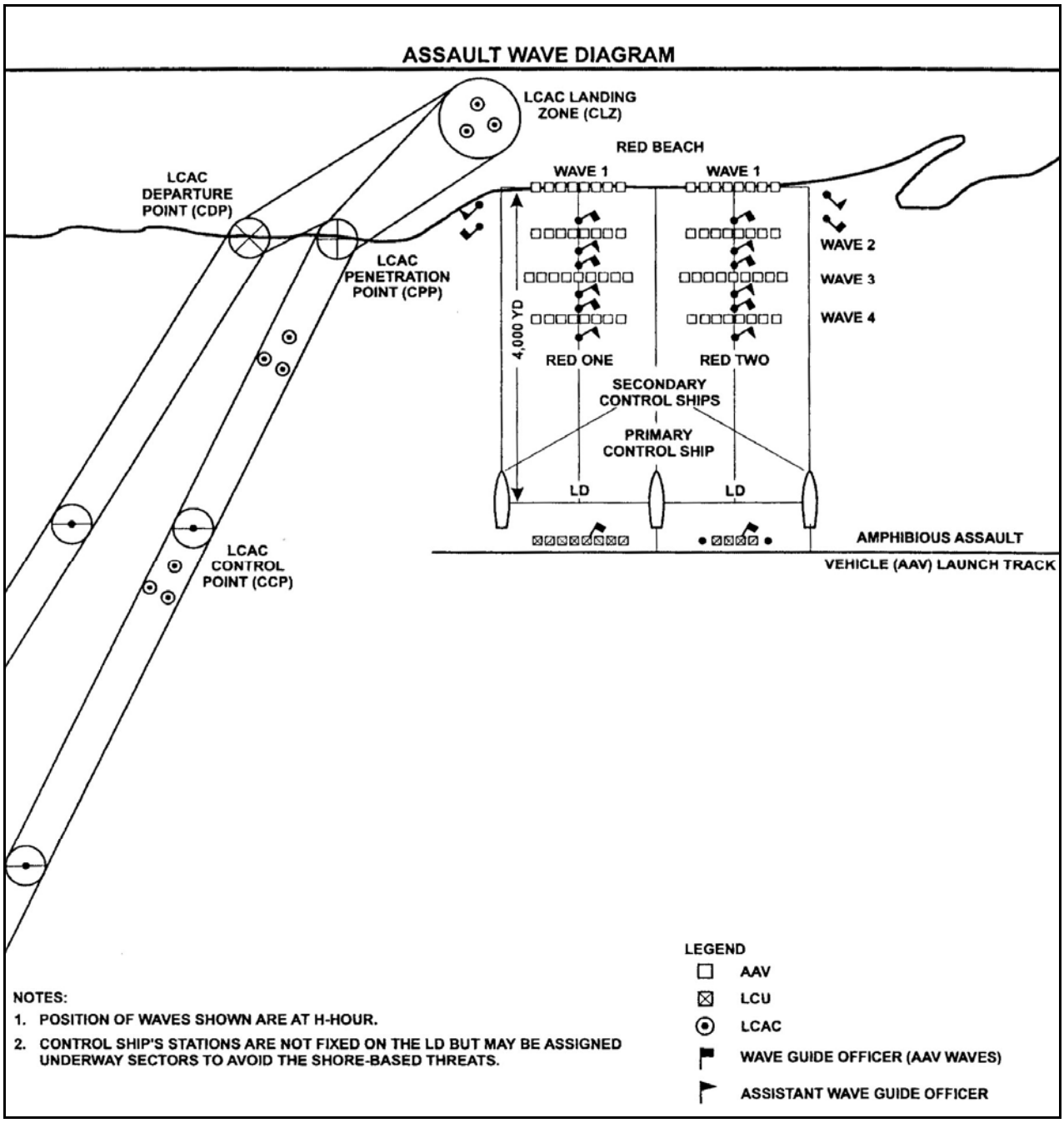


Figure J-5. Example of an Assault Wave Diagram

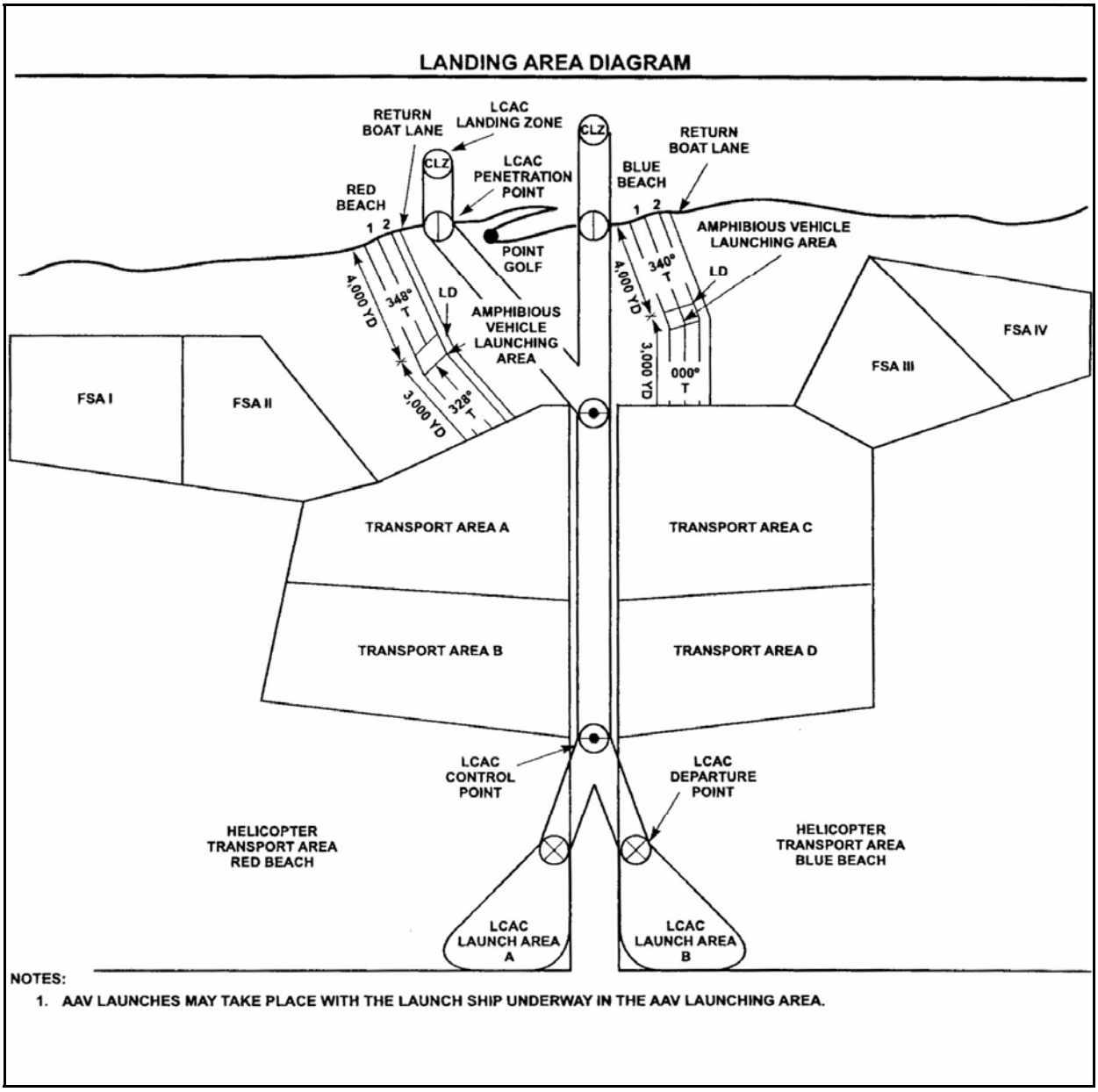
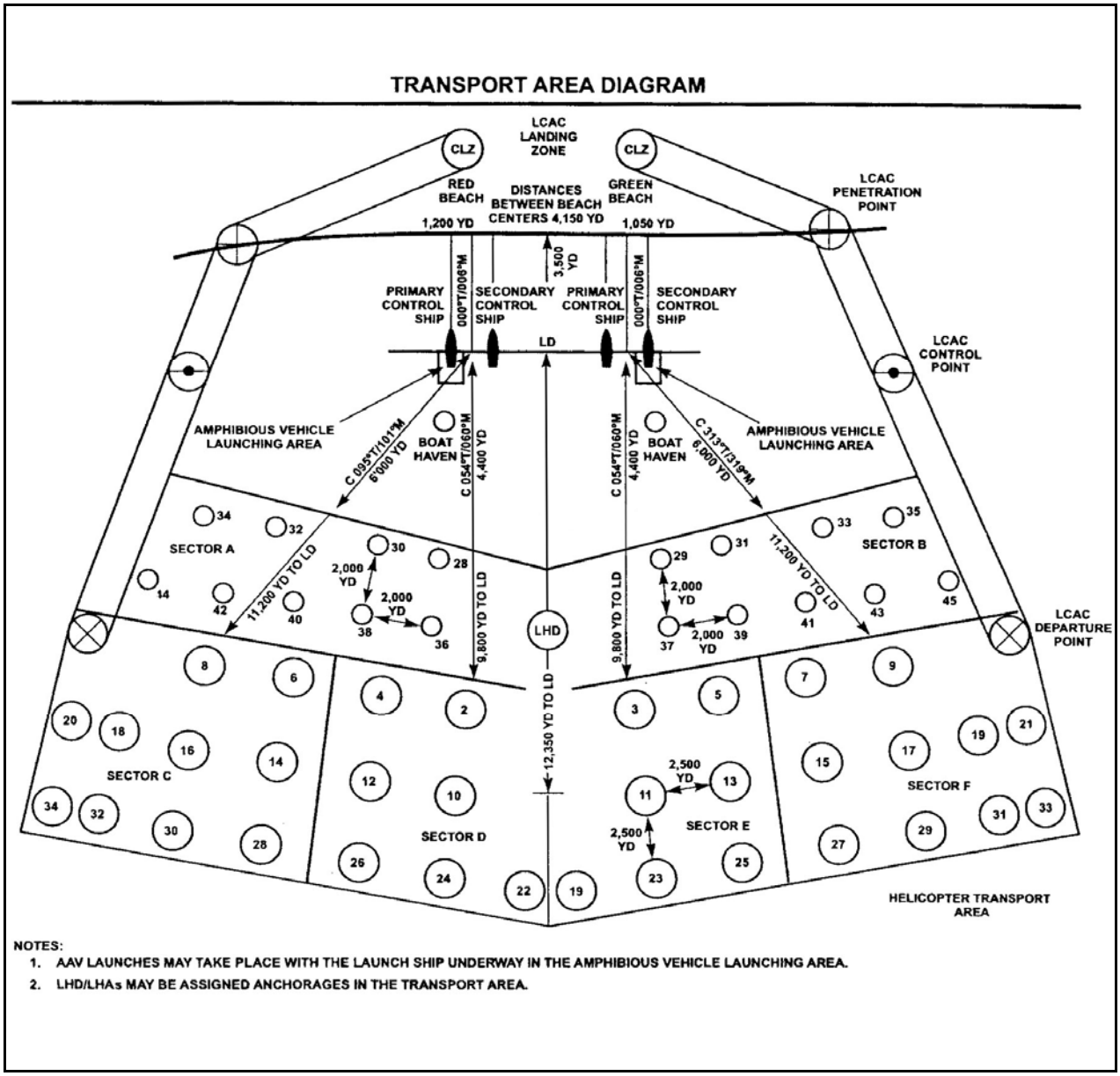


Figure J-6. Example of a Landing Area Diagram



NOTES:
1. AAV LAUNCHES MAY TAKE PLACE WITH THE LAUNCH SHIP UNDERWAY IN THE AMPHIBIOUS VEHICLE LAUNCHING AREA.
2. LHD/LHAs MAY BE ASSIGNED ANCHORAGES IN THE TRANSPORT AREA.

Figure J-7. Example of a Transport Area Diagram

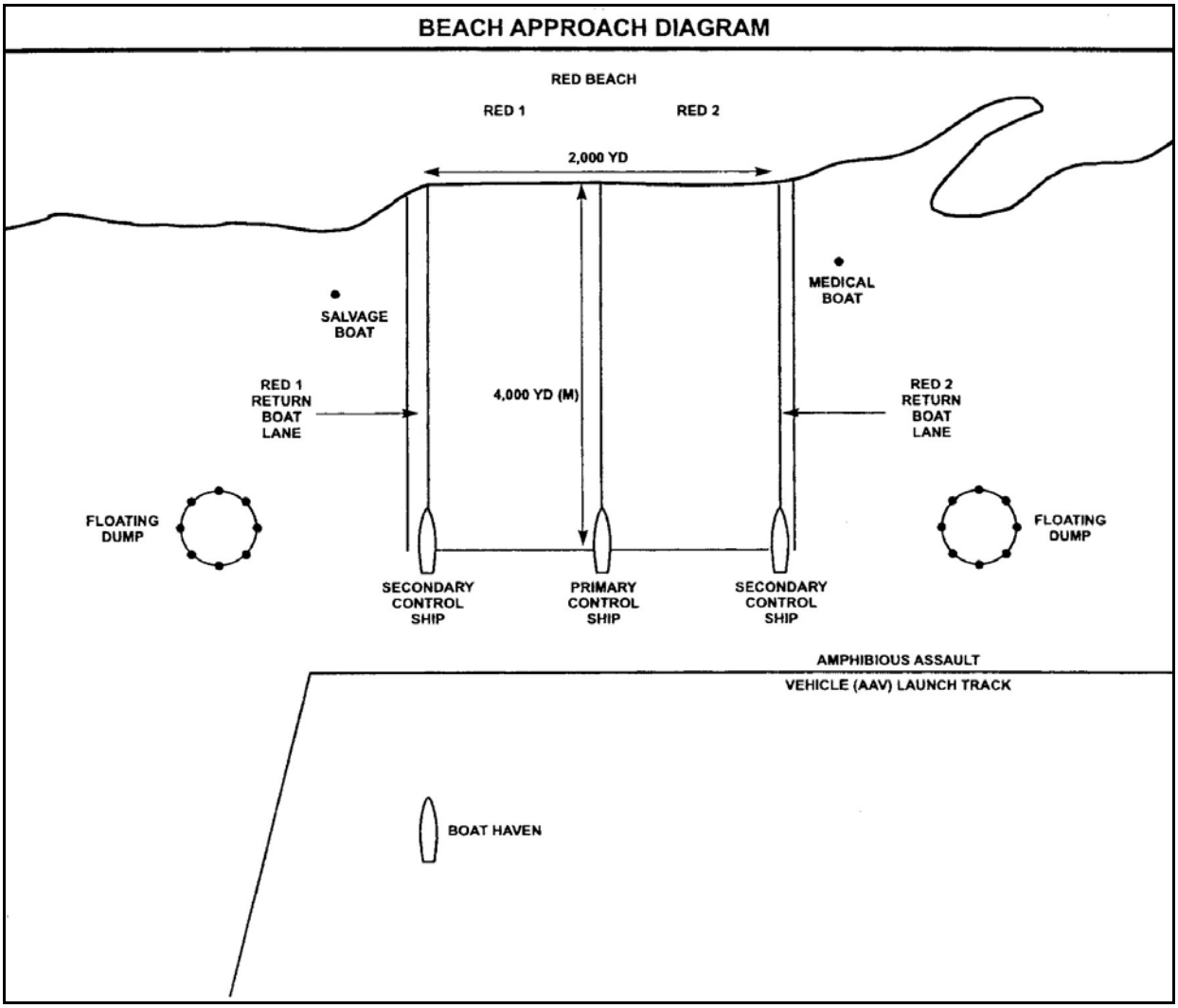


Figure J-8. Example of a Beach Approach Diagram

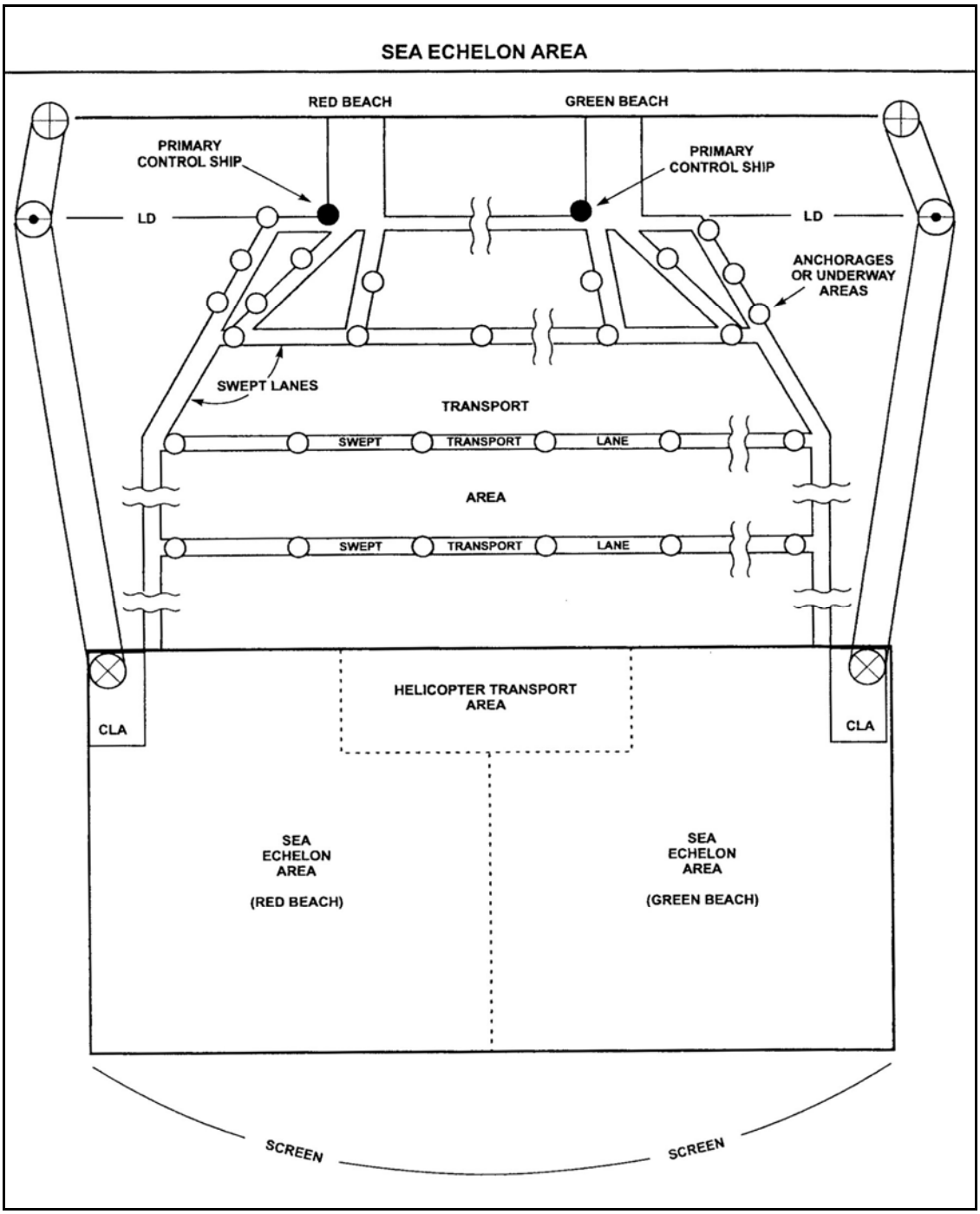


Figure J-9. Example of a Sea Echelon Area

Ship	Amphibious Vehicle Unit	AAVP7	AAVC7	AAVR7
LPD 4	1st Plat, Co A, 2nd Assault Amphibious Battalion	8	1	
LPD 4	2nd Plat, Co A, 2nd Assault Amphibious Battalion	8	1	
LPD 14	4th Plat, Co A, 2nd Amphibious Battalion	10	2	
LPD 14	HQ Plat, Co A, 2nd Amphibious Battalion	3	2	1
LSD 41	4th Plat, Co A, 2nd Assault Amphibious Battalion	6		
Total		35	6	1

Figure J-10. Example of an Amphibious Vehicle Availability Table

Craft No.	Personnel and Materiel	Boat Spaces	Beach Red One Formation
	First Wave		
1-1	1st Squad, 1st Platoon, Co B	11	
AAV	Corpsman	1	X 1-1
	MG Tm, 1st MG Squad	3	
	81-mm FO Tm with 1 Radio Operator	2	X 1-2
		17	
			X 1-3
1-2	Platoon Cmdr, 1st Platoon, Co B	1	
AAV	Messenger Radio Operator	1	
	2d Squad	11	X 1-4
	Corpsman	1	
	1st and 2d Tm, 3d Arm Security Weapons Co	4	X 1-5
		18	
			X 1-6
1-3	Squad Leader 3rd Squad, 1st Platoon, Co B	1	X 1-7
AAV	1st Fire Tm, 3d Squad, Co C	5	X 1-8
	Corpsman	1	X 1-9
	MG Tm and Squad Leader, 1st MG Squad	4	X 1-10
	60-mm Sec Leader	1	
	1st Squad, 60-mm Sec	3	
	1st Assault Team	2	
		17	
	Third Wave		
3-1	Sec Leader, 1st TK sec	1	-0-
LCU	Tank crew, 1st TK sec	3	3-1
	1 M60A	68	-0- -0-
		72	3-2 3-2
	On-Call		
Serial	VTR	68	
LCU	Det 1st Plat (rein), Co A, 2d Tank Battalion	5	
		73	
00-1	BLT Command Group		
AAV	Battalion Commander	1	
	S-3	1	Legend: X – AAV O = LCU
	Asst S-3, ALO	1	
	NSFS LNO	1	
	Weapons Company Commander	1	
	Artillery LNO	1	
	Radio Operator	4	
		10	
	Floating Dump		
LCU	Floating dump		

Figure J-11. Example of a Landing Craft and Amphibious Vehicle Assignment Table

Beach Red One **H-Hour 0530 Local**								
Scheduled Waves	**Wave Composition**							
	Assault Platoons, Co A and Co B							
Wave 1 H-Hour	1-1 (X)	1-2 X	1-3 X	1-4 X	1-5 X	1-6 X	1-7 X	1-8 X
	Co A (-) and Co B (-)							
Wave 2 H+3 min	2-1 (X)	2-2 X	2-3 X	2-4 X	2-5 X	2-6 X	2-7 X	2-8 X
	Leading Plats, Co C and 81 Mort Plat							
Wave 3 H+6 min	3-1 (X)	3-2 X	3-3 X	3-4 X	3-5 X	3-6 X	3-7 X	3-8 X
	Co C (-)							
Wave 4 H+9 min	4-1 (X)	4-2 X	4-3 X	4-4 X	4-5 X	4-6 X	4-7 X	4-8 X
	Weapons Company							
Wave 5 H+16 min		5-6 U	5-4 U	5-2 U	5-1 (U)	5-3 U	5-5 U	

General Note: Prepared and promulgated concurrently with the landing craft and amphibious vehicle assignment table.

Legend:
X – AAV
U – LCU
( ) – Wave Commander

Figure J-12. Example of a Landing Diagram

Serial No.	Unit	Personnel	Materiel Equipment Vehicles	Number of Craft and Type	Parent Ship	Remarks
1401	Assault Platoon, Co A (rein)	98	Normal Combat	4 AAVs	LPD 14	1st Wave Beach Red One
1402	Co A (-) (rein)	91	Normal Combat	4 AAVs	LPD 14	2nd Wave Beach Red One
1403	Assault Platoon, Co B (rein)	96	Normal Combat	4 AAVs	LPD 14	1st Wave Beach Red One
1500	Assault Platoon, Co D	15	Normal Combat	1 CH46	LHA 1	1st Wave LZ Hawk
2101	1st Platoon, Co A 1st Tank Battalion	12	3 M60A1s Tanks	1 LCU	LHD 1	3rd Wave Beach Red One
2251	B Battery	34	2 M198s 4 M923s 2 M105s 4 HMMWVs	1 LCU	LHD 1	On-Call Beach Red One

Figure J-13. Example of a Landing Force Serial Assignment Table

Allocation of Blocks by LF	
H&S Co, 1 MEF	0 to 500
1st MLG	501 to 1200
NMCB	1200 to 1300
1st MARDIV	1301 to 2200
2nd MARDIV	2201 to 3100
1st MAW	3101 to 4000
Allocation of Blocks by Division	
HQBN	1301 to 1350
1st MAR	1401 to 1500
Allocation of Blocks by Wing	
MWHS	3101 to 3200
MAG-13	3201 to 3300
Allocation of Blocks by Regiment	
HQ CO 1st MAR	1401 to 1425
1st BN 1st MAR	1426 to 1475

Figure J-14. Serial Number Allocation

Unit	Day of Anticipated Landing									Remarks
	D	D + 1	D + 2	D + 3	D + 4	D + 5	D + 6	D + 7	D + 8	
RLT 5	X - - - - - - - - - - ▸									Red Beach
RLT 7	X - - - - - - - ▸									Green Beach
3rd MAW	X - - - - - - - - - -

Unit	Element	Serial No.	Carrier No. and Type	Ship	Beach
1st and 2nd Plat A Co		905	3 LCUs	LPD	Red
A Co (-) 2nd Tank Battalion (FMF)		906	3 LCUs	LHD	Red
1st and 2nd Plat B Co		907	3 LCUs	LHD	Blue
2nd Tank Battalion B Co (-) 2nd Tank Battalion		908	3 LCUs	LPD	Blue
1/10	A/B/C	1013	7 LCUs	LHA	Red
1/10	Battery	1014		LHA	Red
		1015		LHA	Red
Division Tactical CP	HQ Battery	1016 1023	3 LCUs 8 LCUs	LPD LPD	Red Blue
A Co (-) 2nd Engineers	D Battery	401 105	4 LCUs 4 LCUs	LHA LHA	Blue Red

Figure J-16. Example of a Landing Force Landing Sequence Table

Wave	Time	Beach			
		Red		Blue	
		1	2	1	2
		Craft/VEH Unit Serial	Craft/VEH Unit Serial	Craft/VEH Unit Serial	Craft/VEH Unit Serial
1	H-hour	8 AAVs Aslt Plats E&F Cos (+), BLT 2/6 604/704	8 AAVs Aslt Plats A&B Cos, BLT 1/6 203/303	8 AAVs Aslt Plats A&B Cos, BLT 2/2 606/706	8 AAVs Aslt Plats A&B Cos, BLT 1/2 1801/1802
2	H+3 min	6 AAVPs E&F Cos (+) 605/705	6 AAVPs A&B Cos (+) 204/304	6 AAVPs E&F Cos (+) 607/707	6 AAVPs A&B Cos (+) 1802/1803
3	H+7 min	4 AAVPs Leading Plats C Co 803	4 AAVPs Leading Plats C Co 405	4 AAVPs Leading Plats G Co 804	4 AAVPs Leading Plats C Co 406
Rept PCS at H+90 min		2 AAVPs RECON Party 2/10 10 AAVPs 1&K Cos (+), BLT 3/6 1302/1502/2001		2 AAVPs RECON Party 1/10 8 LCMs 1&K Cos (+), BLT 3/2 2201/2301/1602	
Rept PCS at H+50 min		10 AAVPs Leading Plats L & Wpn Cos, BLT 3/6 1701/1803		10 LCUs Leading Plats L & Wpn Cos, BLT 3/2 1703/1804	
Rept PCS at H+55 min		10 AAVPs L & Wpn Cos (+) BLT 3/6 1702/1805		2 LCUs L & Wpn Cos (+), BLT 3/2 2401/2501	
Rept PCS at H+60 min		6 AAVPs BLT Hq 3/6 2101		6 LCUs BLT Hq 3/2 2601	
Rept PCS at H+90 min				10 LCUs Division Adv CP 1901	

Figure J-17. Example of an Assault Schedule

Ship	AAVP7	AAVC7	AAVR7	LARC	Wave	Destination	Remarks
LPD 4	5		1		1	Beach Red One	Assault Plat, Co B
LPD 14	5					Beach Red One	Assault Plat, Co F
LSD 48	5		1		1	Beach Red Two	Assault Plat, Co A
LPD 4	5				2	Beach Red One	Co E (-), BLT 2/6
LPD 14	5	1				Beach Red One	Co F (-), BLT 2/6
LSD 42	5				2	Beach Red Two	Co A (-), BLT 1/6
LPD 15	10	1			3	Beach Red One	Co G, BLT 2/6
LSD 46	10	1			3	Beach Red Two	Co C, BLT 1/6
Etc. for the Entire First Trip of Vehicles							
LPD 4				2		PCS Red Beach	Land BPT
LPD 14				2		PCS Red Beach	Land BPT
Etc. for Subsequent Employment							

Figure J-18. Example of an Amphibious Vehicle Employment Plan

Helicopter Unit and Call Sign	Number of Helicopters	Number of Helicopters Available		Model	Parent Helicopter Transport	Maximum Deck Launch Spots Available	Tentative Load Capacity		Remarks (as appropriate)
		First Trip 90% (Note 1)	Other Trips 75% (Note 1)				Troops	Cargo (lb)	
HMM-163 (Anvil)	12	10	9	CH-46E	LHA 1	9	16	4,080	All external lift capable
HMM-164 (Ripper)	12	10	9	CH-46E	LHD 1	9	16	4,080	All external lift capable
HMA-266 (Scarface)	12	10	9	AH-1W	LHD 2	9	NA	NA	Escort and CAS
HMA-267 (Hobo)	12	10	9	UH-1N	LHA 1	9	8	3,000 (Note 2)	All external lift capable
HMH-465 (Hauler)	16	14	12	CH-53E	LHA 3	9	32	32,000	Equipped with aircraft recovery sling
HMH-466 (Elvis)	16	14	12	CH-53D	LHA 2	9	32	14,000	All external lift capable

Notes:
1. These percentages may vary from operation to operation.
2. The UH-1N has only 220 cubic feet of cargo space and would normally exceed available volume before exceeding weight limitation.

General Note: Sea level at 90°F.

Figure J-19. Example of a Helicopter Availability Table

Wave	Heliteam Flight Serial (Note 1)	Personnel Troop Unit	Number	Supplies and Equipment	Lift Capability XXXX (Note 2) Personnel	Equipment	Total Weight
	Anvil 101 100-1	1st Sqd, 1st Plat Co A Aslt Tm, 1st Aslt Sqd, Wpns Plat	13 3 16	IM Dragon (31 lb) 2 Dragon rd (50 lb)	3,840	81	3,921
	Anvil 102 100-2	2nd Sqd, 1st Plat, Co A Aslt Tm, Aslt Sqd, Wpns Plat	13 3 16	IM Dragon (31 lb) 2 Dragon rd (50 lb)	3,840	81	3,921
	Anvil 103 100-3	Plat Cdr, 1st Plat, Co A Msgr. 3d Sqd (-) Corpsman Sqd Ldr, 1st MG Sqd MG Tm, 1st MG Sqd, Wpns Plat	1 1 9 2 2 4 17	1AN/PRC-77 (22 lb)	4,080	22	4,102
	Anvil 110 100-10	Elms 3d Sqd, 1st Plat, Co A MG Tm, 1st MG Sqd, Wpns Plat LZ Cntl Tm (RED) Plat Sgt, 1st Plat, Co A Msgr Plat guide, 1st Plat, Co A Corpsman Sqd Ldr, 1st Aslt Sqd, Wpns Plat	5 4 3 1 1 1 1 1 17	1 MG (24 lb)	4,080	24	4,104

Notes:
1. The heliteam flight serial is:
   - Anvil — Heliteam squadron radio call sign
   - 1 — Heliteam wave number
   - 01 — Heliteam position in the wave
   - 100 — Troop unit serial assignment
   - -1 — Troop unit heliteam number
2. Lift capability is computed by the ACE based on helicopter model and expected environmental conditions.

Figure J-20. Example of a Heliteam Wave and Serial Assignment Table

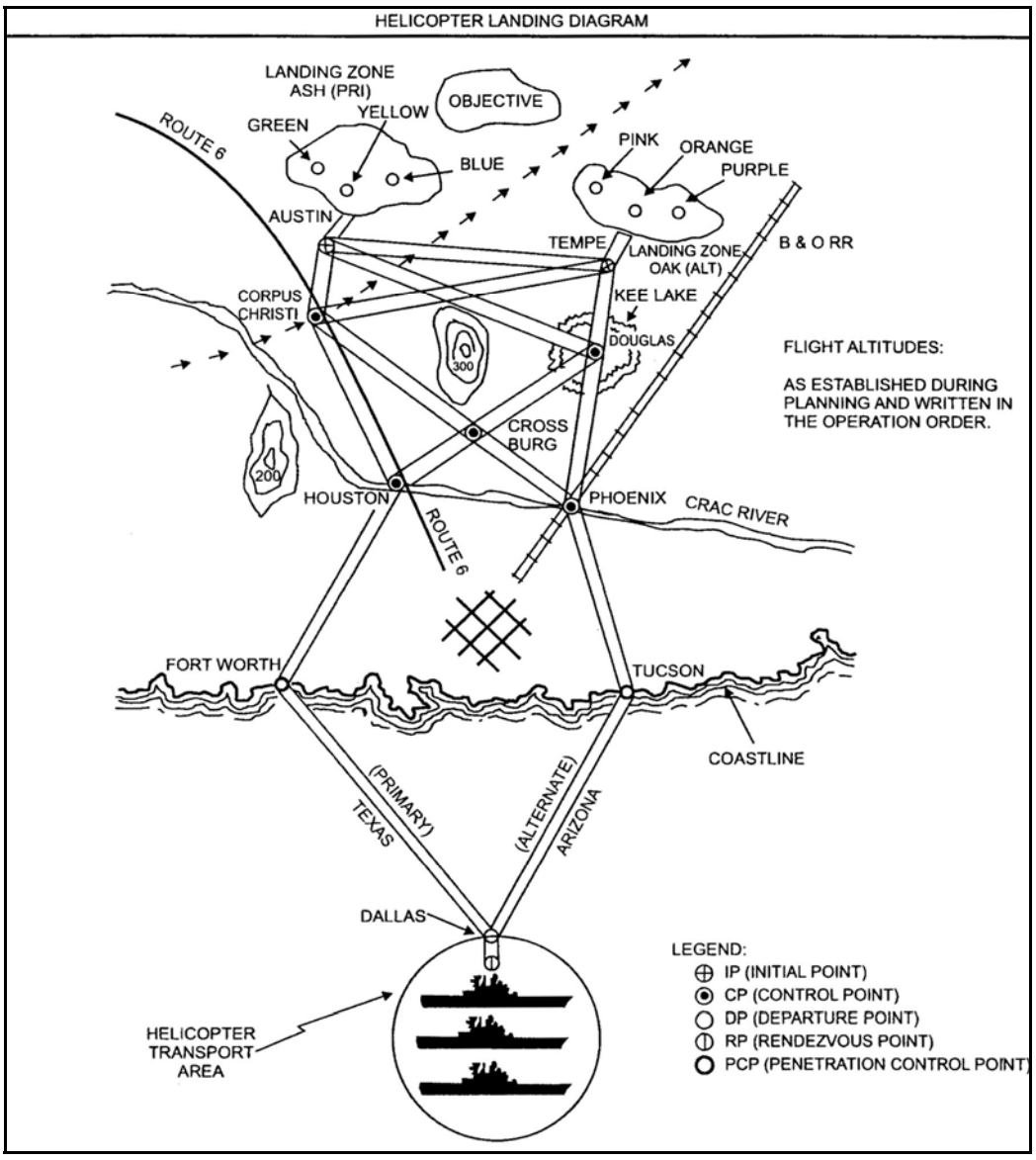


Figure J-21. Example of a Helicopter Landing Diagram

Wave	Aircraft Unit and Flight Number	Aircraft Number and Model	From Aircraft Transport	To Aircraft Transport	Time Load	Time Launch	Time Land	Destination LZ	Destination LS	Troop Unit Serial Remarks
1st	Anvil One	10 CH-46E	LHA 1	LHA 1	Preload	H-21	H-Hr	Hawk	Blue	Co A (-) (rein) Ser 101
	Ripper One	7 CH-46E	LHD 3	LHD 3	Preload	H-21	H-Hr	Hawk	Green	Co B (-) (Rein) Ser 105
	Elvis One	4 AH-1W	LHA 4	LHA 4	Preload	H-21	NA	NA	NA	Escort
	Hobo One	1 UH-1N	LHA 1	LHA 1	Preload	H-21	NA	NA	NA	HCA (refuel as necessary)
2nd	Anvil Two	9 CH-46E	LHA 1	LHA 1	Preload	H-11	H+10	Eagle	Red	Elms Co A Ser 107
	Elvis Two	7 CH-46E	LHD 3	LHD 3	Preload	H-11	H+10	Eagle	Green	Elms Co B Ser 103
	Hobo Two	4 AH-1W	LHA 4	LHA 4	Preload	H-11	NA	NA	NA	Escort
3rd	Anvil Three	9 CH-46E	LHD 1	LHD 1	H+39	H+45	H+71	Robin	Red	Co C (-) (Rein) Ser 211
	Elvis Three	6 CH-46E	LHD 3	LHD 3	H+39	H+45	H+71	Robin	Blue	Elms Co C Ser 212
	Hobo Three	4 AH-1W	LHA 4	LHA 4	NA	H+45	NA	NA	NA	Escort
On-call	Rock	16 CH-53E	LHA 3	LHA 3	TBA	On Order	TBA	TBA	TBA	Resupply
	Lava	16 CH-53D	LHD 1	LHD 1	TBA	On Order	TBA	TBA	TBA	Resupply

Figure J-22. Example of a Helicopter Employment and Assault Landing Table

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Appendix 3 (Ship-to-Shore Movement) to Annex R (Amphibious Operations) to Order 1 – [ ].

Ref: (a) NTTP 3-02.1M/MCWP 3-31.5
(b) CTF XX OPORD 1 – ( )
(c) MARFOR Order_____ (CSS SOP)

Time Zone: H

1. GENERAL

a. This plan calls for:
(1) Landing 1 RLT in 2 separate LZs.
(2) Landing 1 RLT over beaches, with 2 BLTs abreast.
(3) Landing of scheduled waves to be accomplished by helicopter, LCAC, and AAVs.
(4) The early landing of tanks to be accomplished by LCU as soon as scheduled waves have landed in assigned zones and on assigned beaches, secured dominating terrain, cleared beach obstacles.
(5) The landing of the division reserve by helicopter and landing craft.

b. For detailed instructions see enclosures in TABS.

2. CONTROL MEASURES

a. Ship-to-shore control in accordance with References (a) and (b).
b. TACLOG detachments organize, embark, and function IAW References (b) and (c).

BY COMMAND OF MAJOR GENERAL ZULU

W.X.YANKEE
Colonel, U.S. Marine Corps
Chief of Staff

TABS:
A. Surfaceborne ship-to-shore movement
B. Airborne ship-to-shore movement

Distribution: IAW Annex Z to CTF XX OPORD 1 – [ ]

Page Number **CLASSIFICATION**

Figure J-23. Example of a Ground Combat Element Landing Plan Format

Approach Schedule

(1) Scheduled Waves

Wave No.	Composition	Unit	Source	Leave Launch Area or Ship	Leave LOD	Land	Remarks
1	8 AAVPs	Aslt Plats Co A	LPD 14	H-13	H-7	H-hour	Ser 1401, 1405
2	8 AAVPs	Aslt Plats Co B	LPD 4	H-11	H-5	H+3	Ser 1402, 1406
3	3 LCUs	1st Plat Co B, 3rd Tank Bn	LSD 41	H-5	H-2	H+6	Ser 0103, 1403, 1407
4	2 LCUs	1st Plat Co B, 3rd Tank Bn	LHD 1	H-hour	H+3	H+9	Ser 0104, 1404, 1408

(2) On-Call Waves

Serial	Composition	Unit	Source	Report To	Time	Remarks
0905	1 LCU	BPT A	LSD 37	PCS	H-hour	Follow wave 4
1007	1 LARC	BPT A	LPD 14	PCS	H-hour	Follow wave 4

Figure J-24. Example of a Consolidated Landing and Approach Plan

(1) Scheduled Waves						
No.	**Type**	**From**	**To**	**Time of Arrival**	**Period Attached**	**Remarks**
3 2	LCU LCU	LSD 41 LHD 1	LSD 41 LHD 1	LTLF LTFL	One trip One trip	Wave 3 ser 0103 Wave 4 ser 0104
(2) On-Call Waves						
1 2	LCU LARC V	LHD 1 LPD 13	LHD 1 LPD 13	H-15 H-15	One trip Until released	Ser 0905 Ser 1007
Landing Craft for Naval Use						
1	LCU	LHD 1	PCS	LTLF	Until released	Salvage boat follow wave 4

Figure J-25. Example of a Landing Craft and Vehicle Employment Plan

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Appendix 3 (Ship-to-Shore Movement) to Annex R (Amphibious Operations) to OPORD 1 – [ ]

Ref: (a) LFM-02
(b) MCRP XX.X
(c) NTTP 3-02.1/MCWP 3-31.5

Time Zone: H

1. GENERAL
   a. This plan calls for:
      (1) Landing 1 MASS, 2 MACS, and detachments of 2 MABS (VH) with the assault elements of the 1st division by helicopter, landing craft, and amphibious vehicles.
      (2) Landing 1 MACS, MHWS, and 2 MABS (VH/VA) (-) and detachments of 2 H&MS (HR) by serials over beaches to be designated.
      (3) Landing the remainder of the wing less 1 MAG (VF/VA) in sequence during unloading by surface means and air.
      (4) Landing 1 MAG (VF/VA) after follow-up shipping arrives in the AOA about D+10.
      (5) For detailed instructions, see appendixes.

2. SHIP-TO-SHORE CONTROL
   a. Wing TACLOG officers embark and function with TACLOG detachments IAW CTF XX OPORD 1 – [ ].
   b. Wing units in scheduled, on-call, and nonscheduled waves land under division and MLG control.

3. AIRFIELDS, PONTOON CAUSEWAYS, FUEL HANDING SYSTEMS, AND ENGINEER OPERATIONS
   a. Airfields and operating bases

Designation	Time Estimated Operational
V-1	D+4
V-2	D+7
V-3 (HR)	D+3
V-4 (HR)	D+4
V-5	D+5
V-6	D+10

   b. Pontoon causeways to be installed Red One and Blue Two after D+2 for landing aviation and engineering equipment.
   c. 1st MLG (Bulk Fuel Company) installs ABLTS from Red One and Blue One to airfields and OPAREAs commencing D+2 IAW priorities in Wing Administrative Plan 1 – [ ].
   d. Two naval construction battalions and one force engineer battalion land commencing D+2 for repair and construction of airfields and operating bases.

BY COMMAND OF MAJOR GENERAL WILD
U.R. BOLD
Colonel, U.S. Marine Corps
Chief of Staff

Tabs:
A. Division Assault Schedule (extract)
B. Division Landing Sequence Table (extract)
C. Division HEALT (extract)
D. Division Heliteam Wave and Serial Assignment Table (extract)
E. LF Landing Sequence Table (extract)
F. LF Serial Assignment Table
G. Wing Echelons for Landing

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Figure J-26. Example of an Aviation Combat Element/Landing Force Aviation Landing Plan Format

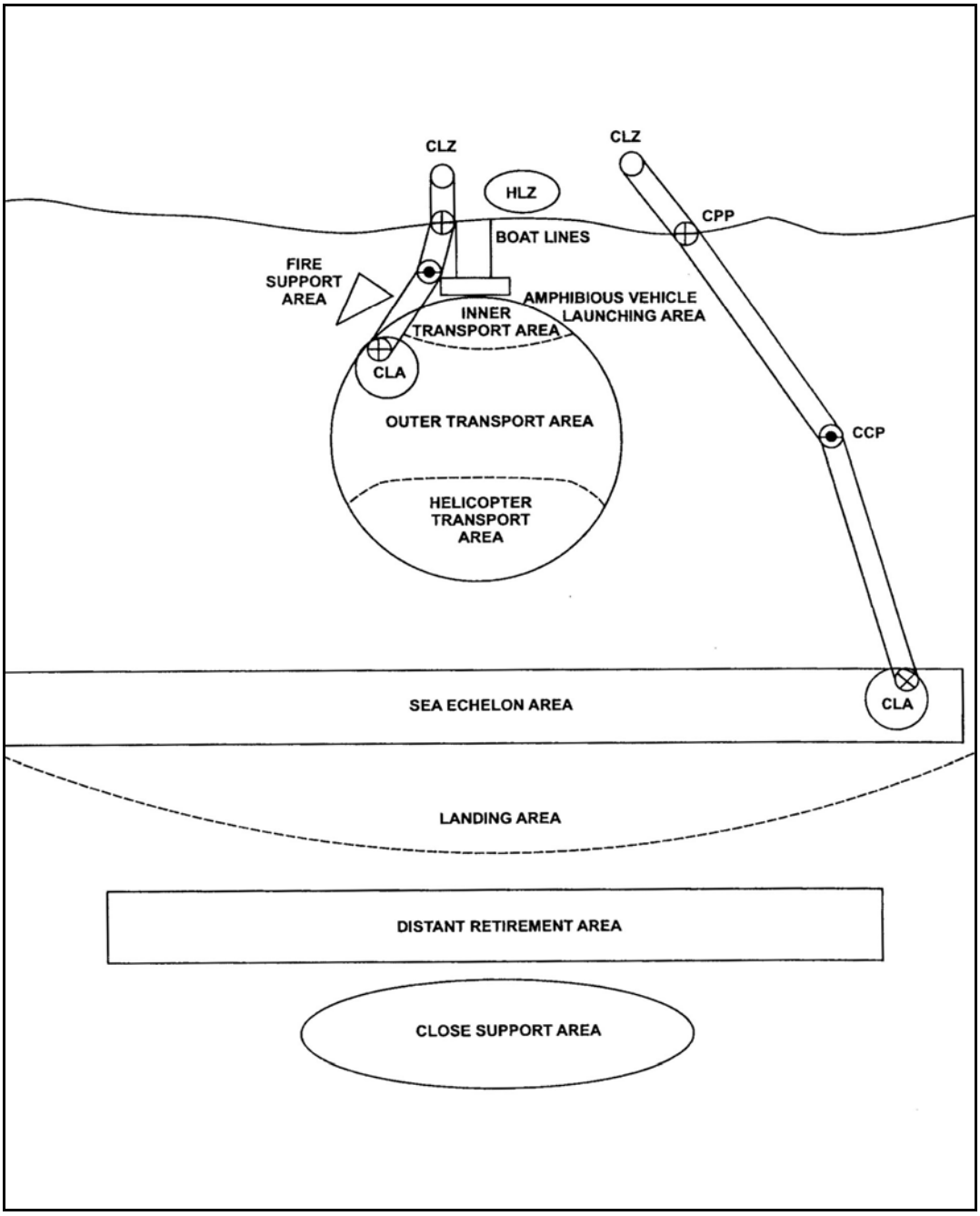


Figure J-27. Ocean Operating Areas and Sea Areas in the Amphibious Objective Area

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LIST OF ACRONYMS AND ABBREVIATIONS

AA	assault amphibian
AABFS	amphibious assault bulk fuel system (NTRP 1-02)
AABWS	amphibious assault bulk water system (JP 1-02)
AATCC	amphibious air traffic control center
AAV	amphibious assault vehicle (JP 1-02)
AAVC7A1	amphibious assault vehicle personnel carrier
AAVP	assault amphibious vehicle, personnel (NTRP 1-02)
AAVP7	amphibious assault vehicle command and control
AAVR7	amphibious assault vehicle recovery
AAW	antiair warfare (JP 1-02)
ABLTS	amphibious bulk liquid transfer system (JP 1-02)
ABS	American Bureau of Shipping
ACA	airspace control authority (JP 1-02)
ACCO	assistant central control officer
ACCS	assistant central control ship
ACE	aviation combat element Marine air-ground task force (MAGTF) (JP 1-02)
ACMREQ	airspace control means request (JP 1-02)
ACU	assault craft unit (JP 1-02)
ACV	air cushion vehicle (JP 1-02)
AD	air defense (JP 1-02)
ADC	air defense commander (JP 1-02)
AE	assault echelon; aeromedical evacuation (JP 1-02)
AF	amphibious force (JP 3-02)
AFL	assault flight leader (NTRP 1-02)
AFOE	assault follow-on echelon (JP 1-02)

ALO	air liaison officer (JP 1-02)
Alt ACE TACC	alternate air combat element tactical air command center
Alt ACE TADC	alternate air combat element tactical air direction center
Alt TACC	alternate tactical air command center
Alt TADC	alternate tactical air direction center
AMC	airborne mission commander (JP 1-02)
AMCM	airborne mine countermeasures (JP 1-02)
AMW	amphibious warfare (JP 1-02)
ANGLICO	naval gunfire liaison team
AO	area of operations (JP 1-02)
AOA	amphibious objective area (JP 1-02)
AOI	area of interest (JP 1-02)
APB	air planning board
APF	afloat prepositioning force (JP 1-02)
APP	allied procedures publication (NTRP 1-02)
AREC	air resource element coordinator (JP 1-02)
ARP	airborne relay platform; air refueling point (JP 1-02)
ASC	air support coordinator (NTRP 1-02)
ASC(A)	air support coordinator (airborne)
ASCM	antiship cruise missile (NTRP 1-02)
ASE	air support element (NTRP 1-02)
ASLT	air support landing team
ASR	assault support request (NTRP 1-02)
ASUW	antisurface warfare
ATACS	amphibious tactical air control system (NTRP 1-02)
ATF	amphibious task force (JP 1-02)
ATO	air tasking order (JP 1-02)
BA	basic allowance

BDA	battle damage assessment (JP 1-02)
BES	beach evacuation station
BGEN	brigadier general
BLT	battalion landing team (JP 1-02)
BMU	beachmaster unit (JP 1-02)
BP	beach party
BPG	beach party group (JP 1-02)
BPT	beach party team (JP 1-02)
BS	battle speed
BSA	beach support area (JP 1-02)
BSIR	beach survey intelligence report
BSU	beach support unit
BUMED	Bureau of Naval Medicine
C2	command and control (JP 1-02)
C3	command, control, and communications (JP 1-02)
C4	command, control, communications, and computers (NTRP 1-02)
C4I	command, control, communications, computers, and intelligence (JP 1-02)
CAL	craft allowance list
CAP	crisis action planning (JP 1-02)
CAS	close air support (JP 1-02)
CASC	communication air support central
CASEVAC	casualty evacuation (JP 1-02)
CAT	crisis action team (JP 1-02)
CATF	commander, amphibious task force (JP 1-02)
CBR	chemical, biological, and radiological (JP 1-02)
CCA	landing craft air cushion collection area
CCO	combat cargo officer; central control officer (JP 1-02)
CCP	cushion (LCAC) control point (NTRP 1-02)

CCS	central control ship (JP 1-02)
CCT	landing craft air cushion control team
CDP	landing craft air cushion departure point (JP 1-02)
CE	command element (MAGTF) (JP 1-02)
CF	causeway ferry (JP 1-02)
CHA	landing craft air cushion holding area
CIC	combat information center (JP 1-02)
CIVMAR	civilian mariner
CJTF	commander, joint task force (JP 1-02)
CLA	landing craft air cushion launch area (JP 1-02)
CLB	combat logistics battalion
CLF	commander, landing force (JP 1-02)
CLR	combat logistics regiment
CLR (F)	combat logistics regiment (forward)
CLR (GS)	combat logistics regiment (general support)
CLS	landing craft, air cushion (LCAC) landing site
CLZ	cushion landing zone (JP 1-02)
CNBG	commander, naval beach group
CO	commanding officer (JP 1-02)
COA	course of action (JP 1-02)
COL	colonel
CONOPS	concept of operations (JP 1-02)
CONUS	continental United States (JP 1-02)
COT	commanding officer of troops (JP 1-02)
CP	control point; command post (JP 1-02)
CPI	coastal patrol and interdiction (NTRP 1-02)
CPP	landing craft, air cushion (LCAC) penetration point
CRRC	combat rubber raiding craft (JP 1-02)

CRTS	casualty receiving and treatment ship (JP 1-02)
CS	combat support (JP 1-02)
CSG	carrier strike group (NTRP 1-02)
CSNP	causeway section, nonpowered (JP 1-02)
CSP	causeway section, powered (JP 1-02)
CSS	combat service support (JP 1-02)
CSSA	combat service support area (JP 1-02)
CST	landing craft, air cushion (LCAC) support team
CTF	commander, task force (NTRP 1-02)
CTM	class tactical manual
CTP	class tactical publication
DA	direct action (JP 1-02)
DAS	direct air support; deep air support (JP 1-02)
DASC	direct air support center (JP 1-02)
DASC(A)	direct air support center (airborne) (JP 1-02)
DDP	detachment deployable package
DDS	dry deck shelter (JP 1-02)
DET	detachment (JP 1-02)
DFSC	Defense Fuel Supply Center (JP 1-02)
DIRLAUTH	direct liaison authorized (JP 1-02)
DLA	Defense Logistics Agency (JP 1-02)
DOA	day(s) of ammunition
DOD	Department of Defense (JP 1-02)
DOS	days of supply (JP 1-02)
DP	departure point
DR	dead reckoning (JP 1-02)
DRT	dead reckoning tracer (JP 1-02)
DTG	date-time group (JP 1-02)

DZ	drop zone (JP 1-02)
EFV	expeditionary fighting vehicle
EFV (C)	expeditionary fighting vehicle – command and control variant
EFV (P)	expeditionary fighting vehicle – personnel variant
ELCAS	elevated causeway system (JP 1-02)
EMCON	emission control (JP 1-02)
EMW	expeditionary maneuver warfare (NTRP 1-02)
EOD	explosive ordnance disposal (JP 1-02)
EPW	enemy prisoner of war (JP 1-02)
ESF	expeditionary strike force (NTRP 1-02)
ESG	expeditionary strike group (NTRP 1-02)
EW	early warning; electronic warfare (JP 1-02)
EW/C	early warning/control
FAC	forward air controller (JP 1-02); fast attack craft
FAC(A)	forward air controller (airborne) (JP 1-02)
FARP	forward arming and refueling point (JP 1-02)
FCT	firepower control team (JP 1-02)
FDC	fire direction center (JP 1-02)
FEBA	forward edge of the battle area (JP 1-02)
FFCC	force fires coordination center (JP 1-02)
FIAC	fast inshore attack craft
FID	foreign internal defense (JP 1-02)
FMF	Fleet Marine Force (JP 1-02)
FMFM	Fleet Marine Force Manual
FO	forward observer (JP 1-02)
FOB	forward operations base (JP 1-02)
FOD	foreign object damage (JP 1-02)
FP	force protection (JP 1-02)

FPO	fleet post office (JP 1-02)
FRAGORD	fragmentary order (JP 1-02)
FSA	fire support area (JP 1-02)
FSC	fire support coordinator (JP 1-02)
FSCC	fire support coordination center (JP 1-02)
FSS	fast sealift ships (JP 1-02)
FST	fleet surgical team (JP 1-02)
GBAD	ground-based air defense
GCC	geographic combatant commander
GCE	ground combat element (MAGTF) (JP 1-02)
GPS	global positioning system (JP 1-02)
GWOT	global war on terror
HA	Helicopter Administrative (net); humanitarian assistance (JP 1-02)
HADR	humanitarian assistance and disaster relief
H&S	headquarters and support
HAU	helicopter advisory unit
HC	Helicopter Command (net)
HCE	helicopter control element (NTRP 1-02)
HCO	helicopter control officer (JP 1-02)
HCS	helicopter coordination section (JP 1-02)
HCSO	helicopter coordination section officer
HCU	helicopter coordination unit
HD	Helicopter Direction (net)
HDC	helicopter direction center
HEALT	helicopter employment and assault landing table
HEC	helicopter element coordinator (JP 1-02)
HES	helicopter evacuation site
HF	high frequency (JP 1-02)

HLC	helicopter logistics coordinator
HLSC	helicopter logistic support center
HLZ	helicopter landing zone (JP 1-02)
HMMWV	high mobility multipurpose wheeled vehicle (JP 1-02)
H&MS	headquarters and maintenance squadron
HOSTAC	helicopter operations from ships other than aircraft carriers (JP 1-02)
HQ	headquarters (JP 1-02)
HQBN	headquarters battalion
HR	helicopter request (JP 1-02)
H&S	headquarters and service (JP 1-02)
HSE	helicopter support element
HSS	health service support (JP 1-02)
HST	helicopter support team (JP 1-02)
HSV	high-speed vessel
HTA	helicopter transport area
HWL	high water line
HWM	high water mark (JP 1-02)
HWSAT	heliteam wave and serial assignment table
IAW	in accordance with (JP 1-02)
IFR	instrument flight rules (JP 1-02)
IMA	intermediate maintenance activity (NTRP 1-02)
IMC	instrument meteorological conditions (JP 1-02)
INLS	improved Navy lighterage system
IO	information operations (JP 1-02)
IP	initial point (JP 1-02)
ITA	inner transport area
ITG	initial terminal guidance (NTRP 1-02)
IVCN	Interior Voice Communication Net

JANAP	Joint Army, Navy, Air Force Publication (JP 1-02)
JAOC	joint air operations center (JP 1-02)
JCC	joint command center (JP 1-02)
JETB	joint effects targeting board
JFACC	joint force air component commander (JP 1-02)
JFC	joint force commander (JP 1-02)
JFEC	joint forcible entry commander
JFLCC	joint force land component commander (JP 1-02)
JFMCC	joint force maritime component commander (JP 1-02)
JHSV	joint high speed vessel
JIC	joint intelligence center (JP 1-02)
JLOTS	joint logistics over-the-shore (JP 1-02)
JP	joint publication (JP 1-02)
JTAR	joint tactical air strike request (JP 1-02)
JTF	joint task force (JP 1-02)
JTTP	joint tactics, techniques, and procedures
LAAD	low altitude air defense (NTRP 1-02)
LAADS	low altitude air defense system (JP 1-02)
LAR	light armored reconnaissance
LARC	lighter, amphibious resupply, cargo (JP 1-02)
LAV	light armored vehicle (JP 1-02)
LCAC	landing craft air cushion (JP 1-02)
LCC	launch control center (JP 1-02)
LCE	logistics combat element
LCO	landing craft air cushion control officer
LD	line of departure (JP 1-02)
LF	landing force (JP 1-02)
LFOC	landing force operations center

LFSP	landing force support party (JP 1-02)
LMSR	large, medium speed roll-on/roll-off (JP 1-02)
LNO	liaison officer (JP 1-02)
LOD	line of departure
LSC	landing support company
LSP	landing support platoon
LTLF	land the landing force
LWC	landing craft air cushion wave commander
LZ	landing zone (JP 1-02)
LZSA	landing zone support area
MABS	Marine air base squadron
MACCS	Marine air command and control system (JP 1-02)
MACG	Marine air control group (JP 1-02)
MACS	Marine air control squadron
MAG	Marine aircraft group (JP 1-02)
MAGTF	Marine air-ground task force (JP 1-02)
MALS	Marine aviation logistics squadron (NTRP 1-02)
MARAD	Maritime Administration (JP 1-02)
MARDIV	Marine division (JP 1-02)
MARFOR	Marine Corps forces (NTRP 1-02)
MARFORES	Marine forces reserve
MASS	Marine air support squadron (NTRP 1-02)
MATCD	Marine air traffic control detachment
MAW	Marine aircraft wing (JP 1-02)
MCCDC	Marine Corps Combat Development Command (JP 1-02)
MCDP	Marine Corps Doctrinal Publication (JP 1-02)
MCDS	modular cargo delivery system (JP 1-02)
MCM	mine countermeasures (JP 1-02)

MCMC	mine countermeasures commander
MCRP	Marine Corps reference publication (JP 1-02)
MCWP	Marine Corps warfighting publication (JP 1-02)
MEB	Marine expeditionary brigade (JP 1-02)
MEDEVAC	medical evacuation (JP 1-02)
MEDREG	medical regulating (NTRP 1-02)
MEF	Marine expeditionary force (JP 1-02)
MEU	Marine expeditionary unit (JP 1-02)
MEU(SOC)	Marine expeditionary unit (special operations capable) (JP 1-02)
MG	machinegun
MGEN	major general
MHE	materials handling equipment (JP 1-02)
MILVAN	military van (container) (JP 1-02)
MIO	maritime interception operations (JP 1-02)
MIW	mine warfare (JP 1-02)
MLB	Marine logistics battalion
MLG	Marine logistics group
MOM	man-on-the-move
MP	military police (JP 1-02)
MPF	maritime prepositioning force (JP 1-02)
MPF (F)	maritime prepositioning force (future)
MPP	mission planning procedure
MPS	maritime prepositioning ship (JP 1-02)
MPSRON	maritime prepositioning ships squadron (JP 1-02)
MRC	mobile radio communications
MRCC	medical regulating control center
MRCO	medical regulating control officer
MRT	medical regulating team

MRU	military radar unit
MS	master station (NTRP 1-02)
MSC	Military Sealift Command (JP 1-02)
MSE	major subordinate element
MSI	modified surface index (JP 1-02)
MSOC	Marine special operations company
MST	mission support team (JP 1-02)
MTACS	Marine tactical air command squadron
MTF	medical treatment facility (JP 1-02)
MTP	master training plan
MTVR	medium tactical vehicle replacement
MTW	major theater war (JP 1-02)
MWCG	Marine wing control group
MWCS	Marine wing communications squadron
MWHS	Marine wing headquarters squadron
MWSG	Marine wing support group (JP 1-02)
MWSS	Marine wing support squadron (JP 1-02)
NATOPS	Naval Air Training and Operating Procedures Standardization (JP 1-02)
NAVAIR	Naval Air Systems Command (JP 1-02)
NAVEDTRA	Naval Education and Training
NAVREGS	Navy regulations
NBG	naval beach group (JP 1-02)
NC2AO	naval control of air operations
NCHB	Navy cargo handling battalion (JP 1-02)
NCO	noncommissioned officer (JP 1-02)
NCR	naval construction regiment (JP 1-02)
NDRF	National Defense Reserve Fleet (JP 1-02)
NECOS	net control station

NEO	noncombatant evacuation operation (JP 1-02)
NFAF	Naval Fleet Auxiliary Force (NTRP 1-02)
NFO	naval flight officer (JP 1-02)
NMC	North Atlantic Treaty Organization message catalogue
NSE	Navy support element (JP 1-02)
NSFS	naval surface fire support (JP 1-02)
NSW	naval special warfare (JP 1-02)
NTACS	Navy tactical air control system (JP 1-02)
NTRP	Navy tactical reference publication
NTTP	Navy tactics, techniques, and procedures
NWDC	Navy Warfare Doctrine Command (JP 1-02)
NWP	naval warfare publication (JP 1-02)
OAS	offensive air support (JP 1-02)
OIC	officer in charge (JP 1-02)
OMFTS	operational maneuver from the sea (NTRP 1-02)
OPCON	operational control (JP 1-02)
OPDS	offshore petroleum discharge system (JP 1-02)
OPLAN	operation plan (JP 1-02)
OPORD	operation order (JP 1-02)
OPTASK	operation task (JP 1-02)
OPTASK AMPHIB	operational tasking amphibious (NTRP 1-02)
OPTEMPO	operating tempo (JP 1-02)
OR	operating room (NTRP 1-02)
OTA	outer transport area
OTH	over the horizon (JP 1-02)
PCO	primary control officer (JP 1-02)
PCP	penetration control point (NTRP 1-02)
PCS	primary control ship (JP 1-02)

PCVT	personnel, cargo, and vehicles unloaded, and estimated time to complete offload
PDF	portable document format
PEO	patient evacuation officer
PERMA	planning, embarkation, rehearsal, movement, and assault
PET	patient evaluation team
PHIBCB	amphibious construction battalion (JP 1-02)
PHIBRON	amphibious squadron (JP 1-02)
PLGR	precise lightweight global positioning system (GPS) receiver (JP 1-02)
PLRS	position location reporting system (JP 1-02)
PO	petty officer (JP 1-02)
POC	point of contact (JP 1-02)
POL	petroleum, oils, lubricants (JP 1-02)
PRIFLY	primary flight control (JP 1-02)
PWC	personal watercraft
R2P2	rapid response planning process (JP 1-02)
RCC	regional combatant commander (NTRP 1-02)
RECON	reconnaissance (JP 1-02)
RIB	rubberized inflatable boat (JP 1-02)
RLT	regimental landing team (NTRP 1-02)
ROE	rules of engagement (JP 1-02)
RO/RO	roll-on/roll-off (JP 1-02)
ROS	reduced operating status (JP 1-02)
RP	rendezvous point (NTRP 1-02)
RRDF	roll-on/roll-off (RO/RO) discharge facility (JP 1-02)
RRF	Ready Reserve Force (JP 1-02)
RSO&I	reception, staging, onward movement, and integration (NTRP 1-02)
RTF	return to force (JP 1-02)
SA	situational awareness (JP 1-02)

SAAWC	sector antiair warfare commander
SAC	supporting arms coordinator (JP 1-02)
SACC	supporting arms coordination center (JP 1-02)
SALT	supporting arms liaison team (JP 1-02)
SAR	search and rescue (JP 1-02)
SAT	serial assignment table
SCO	secondary control officer (NTRP 1-02)
SCS	secondary control ship
SEAL	sea-air-land team (JP 1-02)
SEAOPS	safe engineering and operations
SLCP	ship's loading characteristics pamphlet (JP 1-02)
SLWT	side loadable warping tug (JP 1-02)
SMCM	surface mine countermeasures (JP 1-02)
SNCO	staff noncommissioned officer (JP 1-02)
SOA	speed of advance (JP 1-02)
SOC	special operations capable; special operations craft (NTRP 1-02)
SOF	special operations forces (JP 1-02)
SOP	standard operating procedure (JP 1-02)
SPECWAR	special warfare (JP 1-02)
SPMAGTF	special purpose Marine air-ground task force (JP 1-02)
SR	special reconnaissance (JP 1-02)
SRIG	surveillance, reconnaissance, and intelligence group (JP 1-02)
SRP	seaward recovery point (JP 1-02)
SSDS	ship's self-defense system (NTRP 1-02)
STOM	ship-to-objective maneuver
SURF	surface
SUROBS	surf observation (JP 1-02)
SUW	surface warfare (JP 1-02)

SUWC	surface warfare commander (JP 1-02)
SW	shallow water (NTRP 1-02)
SWDG	Surface Warfare Development Group
SWH	significant wave height
SZ	surf zone (JP 1-02)
TAC	tactical air commander; tactical air controller (NTRP 1-02)
TAC(A)	tactical air coordinator (airborne) (JP 1-02)
TACAIR	tactical air (JP 1-02)
TACC	tactical air control center (USN); tactical air command center (USMC) (JP 1-02)
TACCWO	tactical air control center watch officer
TACGRU	tactical air control group (NTRP 1-02)
TACLOG	tactical-logistical (JP 1-02)
TACMEMO	tactical memorandum (NTRP 1-02)
TACNET	tactical net
TACON	tactical control (JP 1-02)
TACP	tactical air control party (JP 1-02)
TACRON	tactical air control squadron (JP 1-02)
TACTOM	tactical Tomahawk
TAD	temporary additional duty (JP 1-02)
TADC	tactical air direction center (JP 1-02)
TAO	tactical actions officer (JP 1-02)
TAOC	tactical air operations center (JP 1-02)
TAR	tactical air request (JP 1-02)
TBD	to be determined (JP 1-02)
TBM	theater ballistic missile; tactical ballistic missile (JP 1-02)
TK	track (NTRP 1-02)
TMD	theater missile defense (JP 1-02)
TRAP	tactical recovery of aircraft and personnel (JP 1-02)

TSCP	theater security cooperation plan (JP 1-02)
TTP	tactics, techniques, and procedures (JP 1-02)
TYCOM	type commander (JP 1-02)
UHF	ultrahigh frequency (JP 1-02)
UNAAF	Unified Action Armed Forces (JP 1-02)
UNREP	underway replenishment (JP 1-02)
USNS	United States naval ship (JP 1-02)
USW	undersea warfare (JP 1-02)
USWC	undersea warfare commander (JP 1-02)
VBSS	visit, board, search, and seizure (JP 1-02)
VFR	visual flight rules (JP 1-02)
VHF	very high frequency (JP 1-02)
VLZ	vertical landing zone
VMC	visual meteorological conditions (JP 1-02)
VMGR	Marine aerial refueler and transport squadron (JP 1-02)
V/STOL	vertical and/or short takeoff and landing aircraft (JP 1-02)
VSW	very shallow water (JP 1-02)
VTOL	vertical takeoff and landing (JP 1-02)
WARNORD	warning order (JP 1-02)
WEZ	weapon engagement zone (JP 1-02)
WGO	waveguide officer

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COMNAVSURFLANTINST/ COMNAVSURFPACINST 3340.3	Wet Well Operations Manual
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COMNAVSURFPACINST 6000.1	Shipboard Medical Procedures Manual
JANAP 119	Joint Voice Call Sign Book
JP 0-2	Unified Action Armed Forces (UNAAF)
JP 1-02	Department of Defense Dictionary of Military and Associated Terms
JP 3-02	Joint Doctrine for Amphibious Operations
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JP 3-02.2	Joint Doctrine for Amphibious Embarkation
JP 3-04.1	Joint Tactics, Techniques, and Procedures for Shipboard Helicopter Operations
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JP 3-15	Joint Doctrine for Barriers, Obstacles, and Mine Warfare

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JP 4-01.6	Joint Logistics Over-the-Shore
JP 4-02	Health Service Support
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JP 5-00.2	JTF Planning Guidance and Procedures
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MCWP 3-25.4	Marine Tactical Air Command Center Handbook
MCWP 3-25.5	Direct Air Support Center Handbook
MCWP 3-25.7	Tactical Air Operations Center Handbook
MCWP 3-31	Ship-to-Objective Maneuver (STOM)
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